

MLP training

Grid search

Fixed hyperparameters:

Warmup: 500 | **Patience:** 100 | **Max. Epochs:** 5000

Search space

- **Activation:** ReLU | Sigmoid
- **Dropout:** 0.1 | 0.3 | 0.5 | 0.7
- **Loss:** L1 | MSE
- **Architecture:** 6 hidden-layer configurations:
16 | 128 | 32 - 16 | 128 - 32 | 32 - 16 - 16 | 128 - 64 - 32

BEST COMBINATION

Activation	Dropout	Loss	Architecture
ReLu	0.7	MSE	128 - 64 - 32

PERFORMANCE

Split	PCC mean	PCC std
20 stringent	0.354	0.117
500 random	0.456	0.055

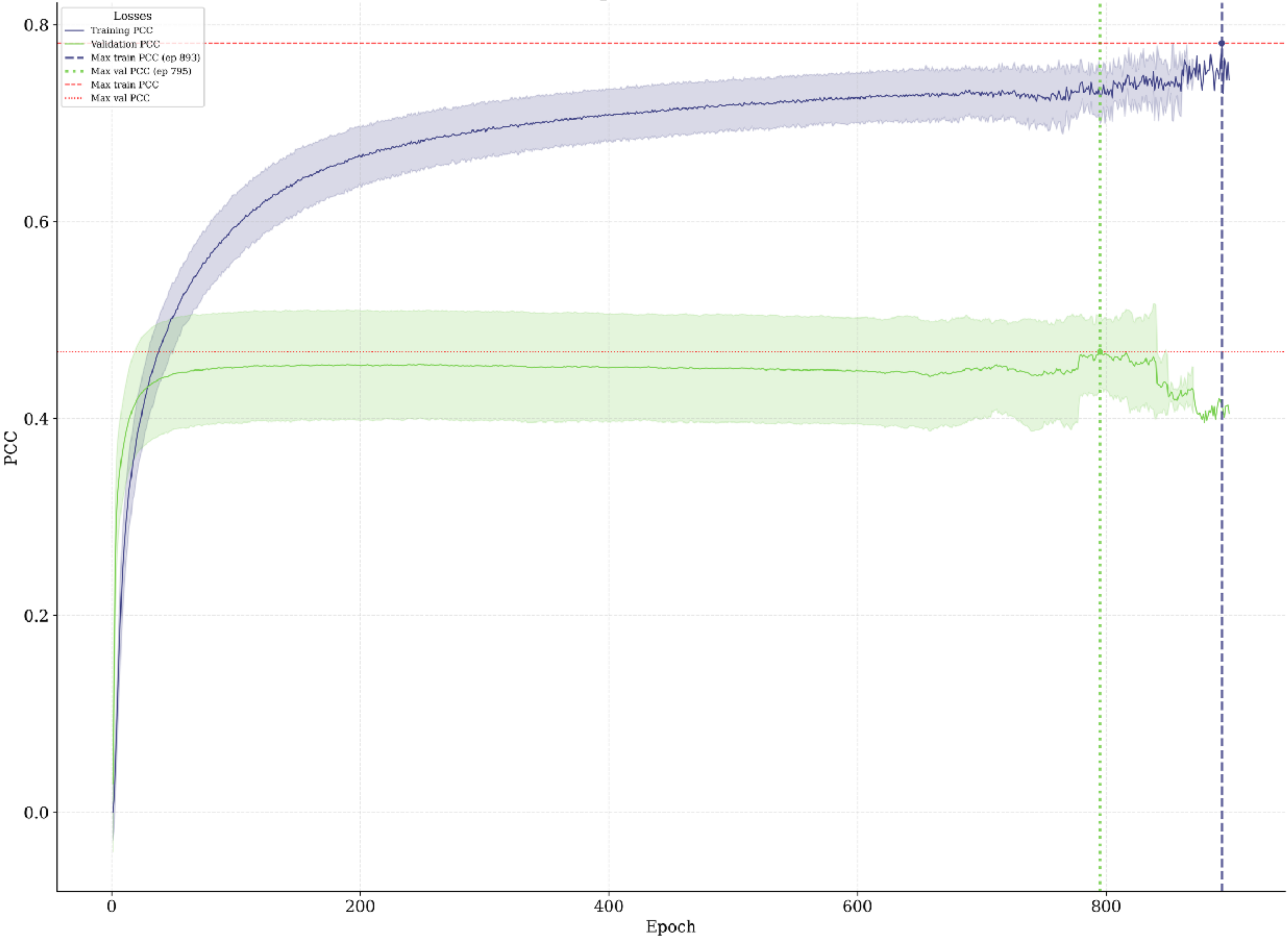
20 rigorous



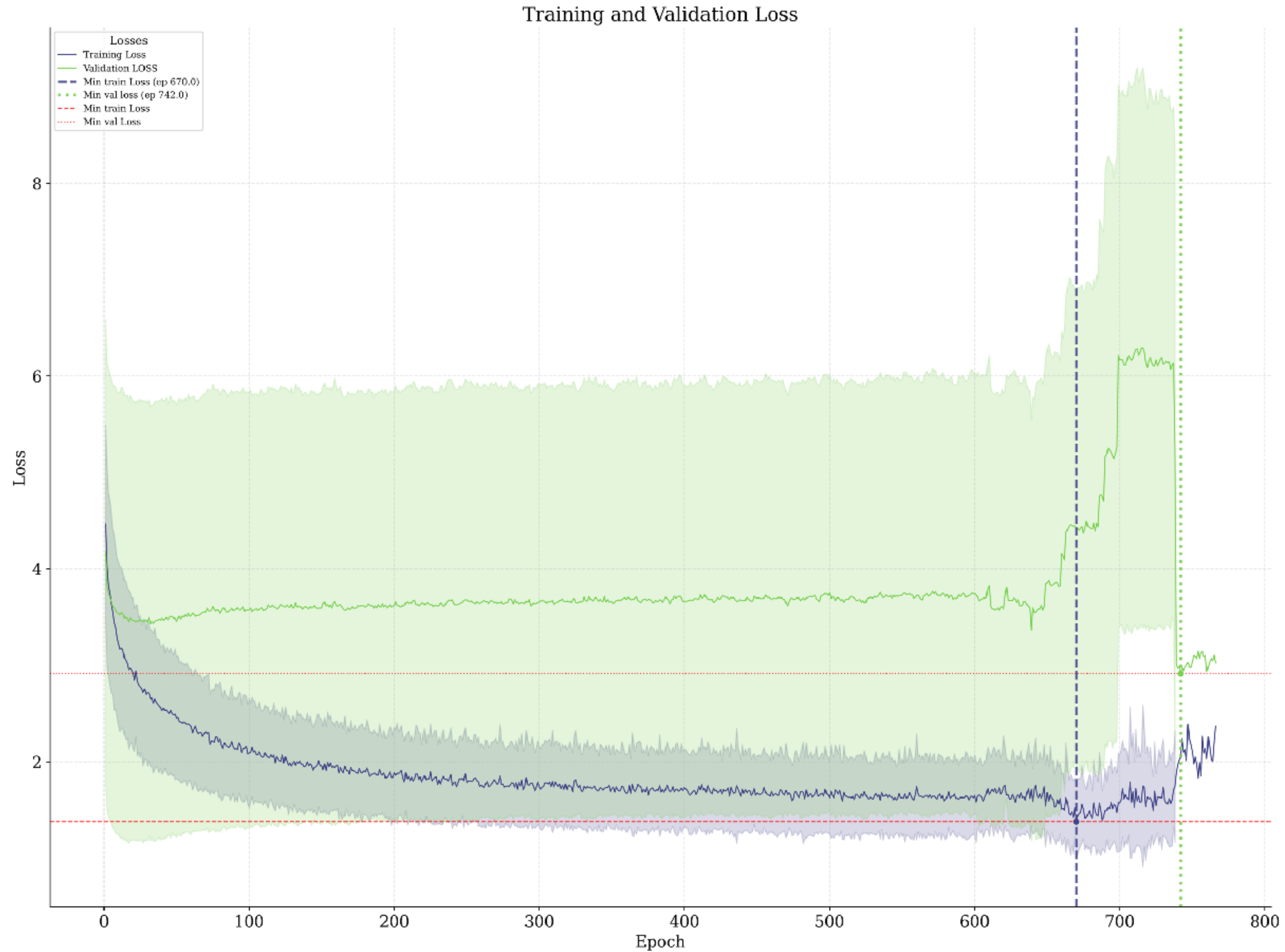
[rigorous_split_plots](#)

500 random

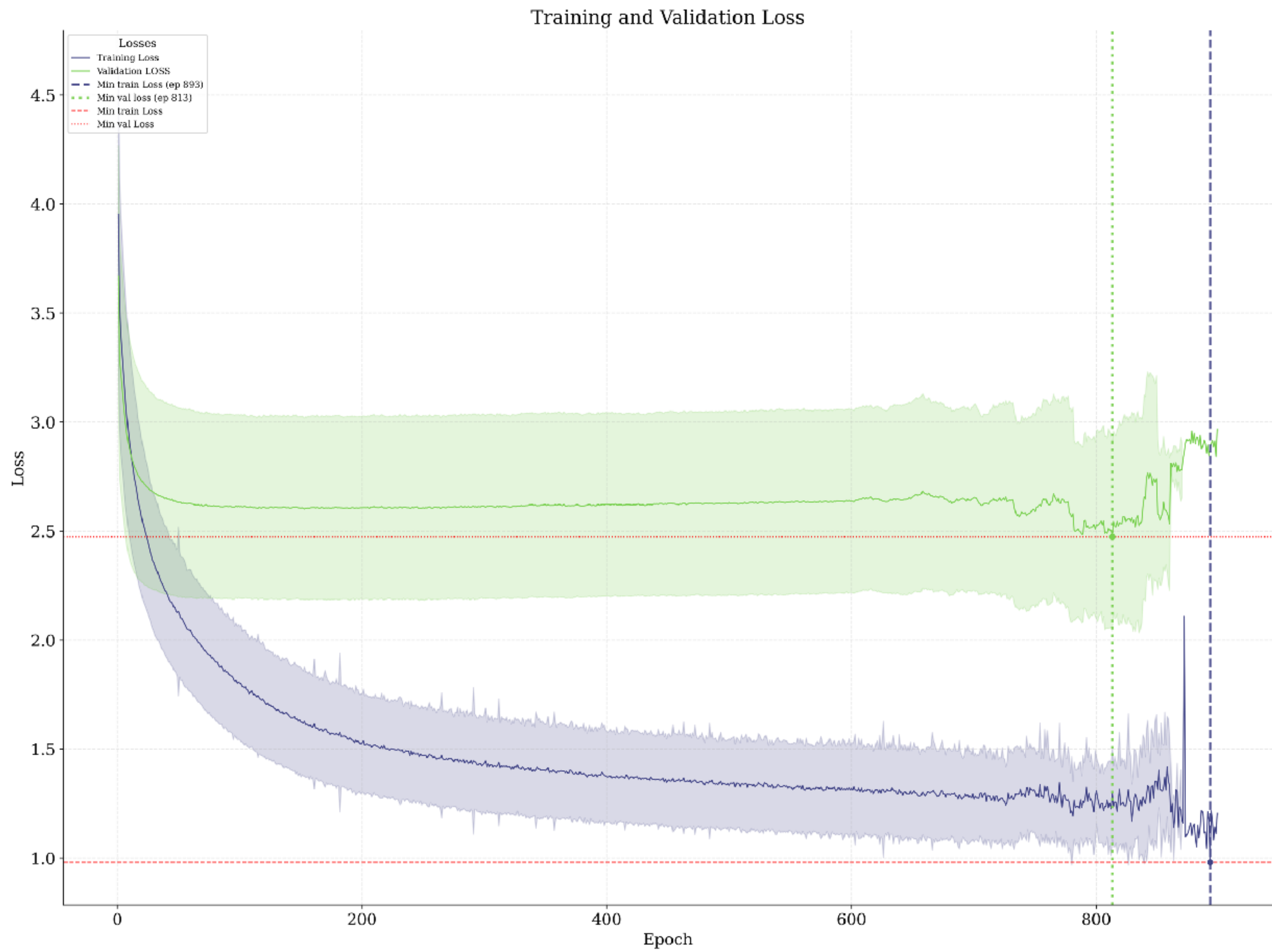
Training and Validation PCCs



20 rigorous

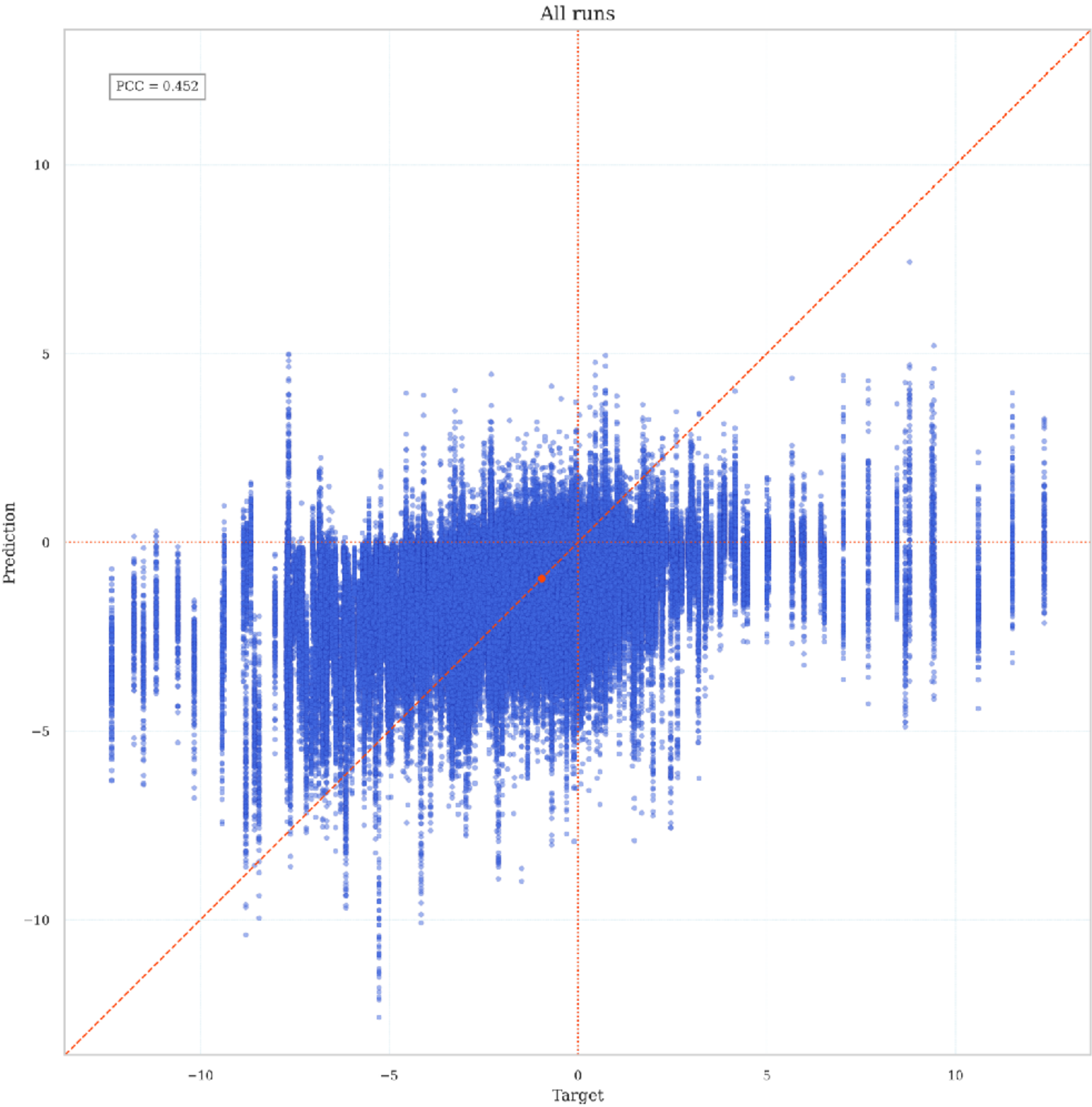


500 random



500 random splits

	<i>Mean</i>	<i>Std</i>
PCC	0.456	0.055
MSE	2.565	0.440



20 rigorous CV splits

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6
BTS	422	7	116	1	98	68
Total	4169	358	444	63	218	94

Cross-Validation Performance:

Best Validation Epoch

	Training	Validation	Test
PCC (all runs)	0.561	0.360	0.286
PCC (20 runs mean)	0.573	0.390	0.354
std (20 runs)	0.120	0.088	0.117
MSE (all runs)	2.542	2.827	3.055
MSE (20 runs mean)	2.540	3.270	3.501
std (20 runs)	0.969	2.120	2.242

Cross-Validation Performance:

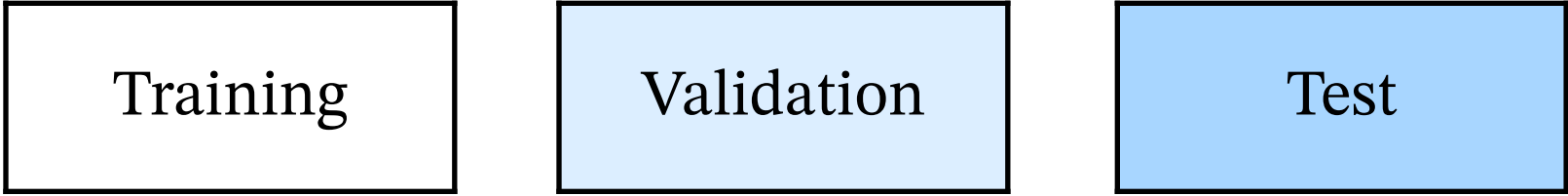
Last Validation Epoch

	Training	Validation	Test
PCC (all runs)	0.748	0.253	0.244
PCC (20 runs mean)	0.748	0.313	0.307
std (20 runs)	0.021	0.099	0.109
MSE (all runs)	1.664	1.664	3.371
MSE (20 runs mean)	1.660	3.707	3.781
std (20 runs)	0.411	2.170	2.297

	Train PCC (Best epoch)	Train PCC (Last epoch)	Validation PCC (Best epoch)	Validation PCC (Last epoch)	Test PCC (Best epoch)	Test PCC (Last epoch)
Max	0.799 (Run 14)	0.784 (Run 17)	0.521 (Run 6)	0.440 (Run 1)	0.498 (Run 15)	0.425 (Run 5)
Split	Tr: 1 2 3 Val: 4 Test: 0	Tr: 1 3 4 Val: 0 Test: 2	Tr: 0 1 4 Val: 3 Test: 2	Tr: 0 1 2 Val: 3 Test: 4	Tr: 1 2 4 Val: 0 Test: 3	Tr: 0 1 4 Val: 2 Test: 3
Min	0.371 (Run 5)	0.700 (Run 6)	0.215 (Run 14)	0.072 (Run 4)	0.068 (Run 1)	0.075 (Run 13)
Split	Tr: 0 1 4 Val: 2 Test: 3	Tr: 0 1 4 Val: 3 Test: 2	Tr: 1 2 3 Val: 4 Test: 0	Tr: 0 1 3 Val: 4 Test: 2	Tr: 0 1 2 Val: 3 Test: 4	Tr: 1 2 3 Val: 3 Test: 4

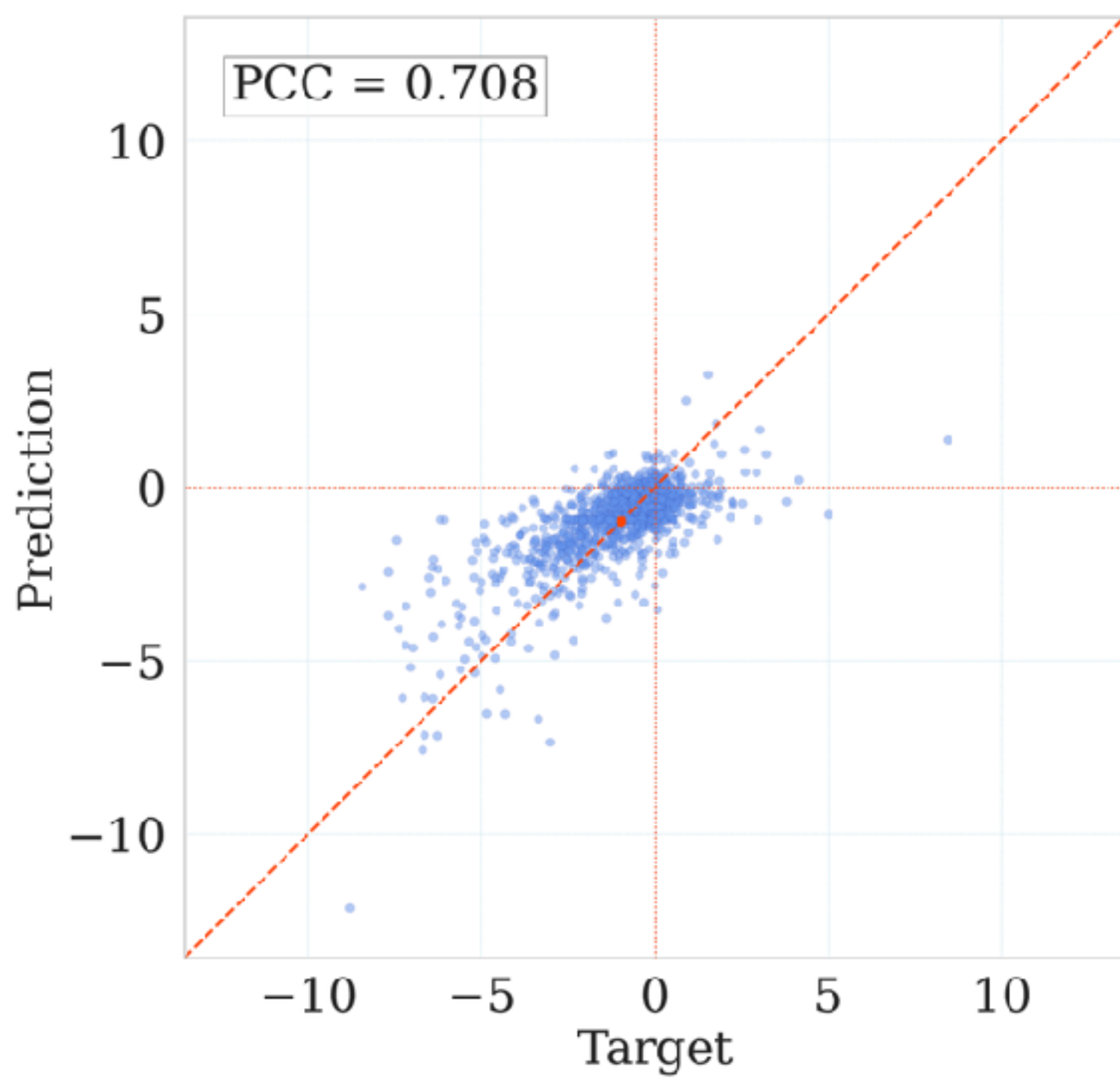
Run 1

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

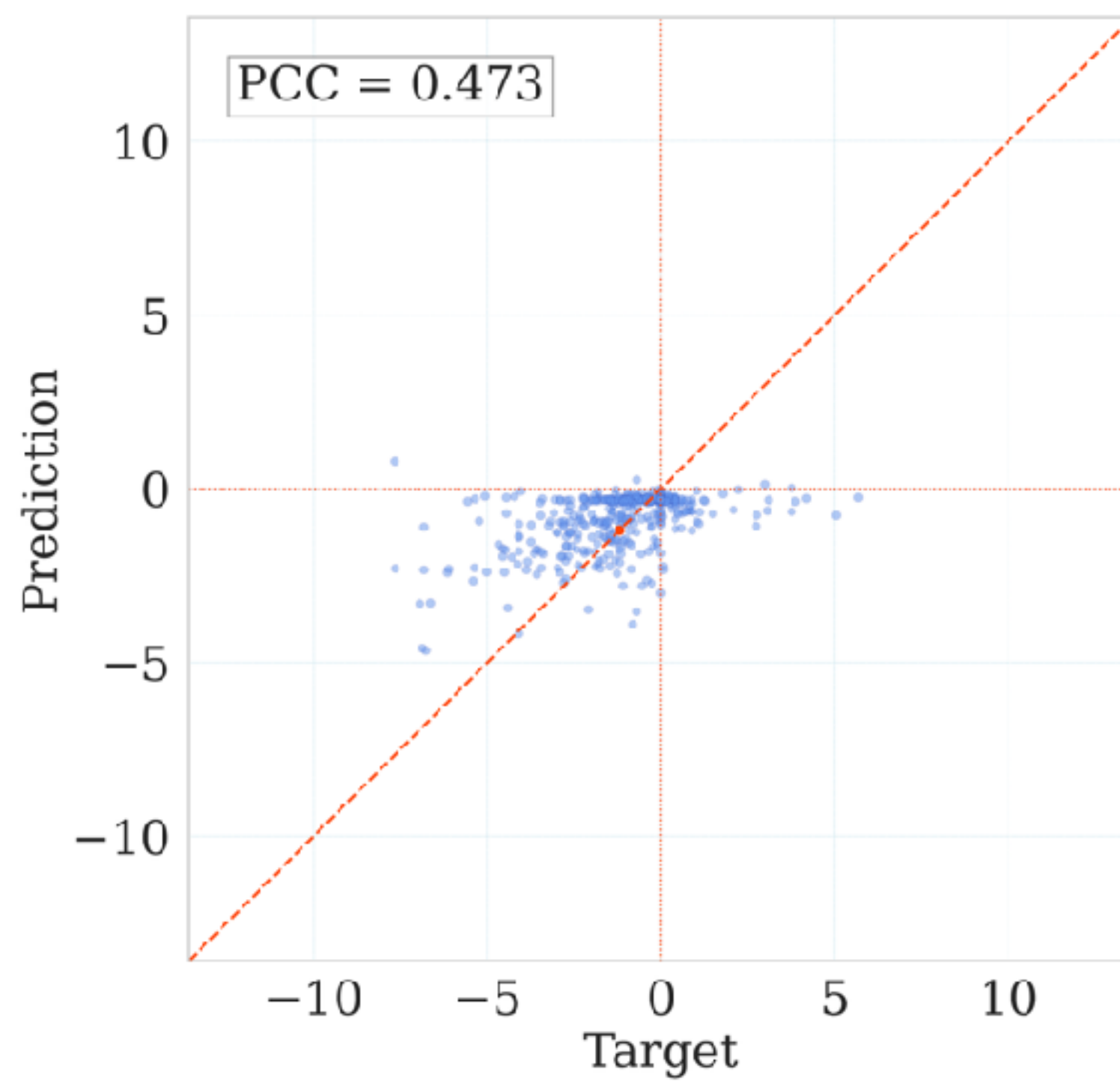


Run 1
Best Epoch: 589

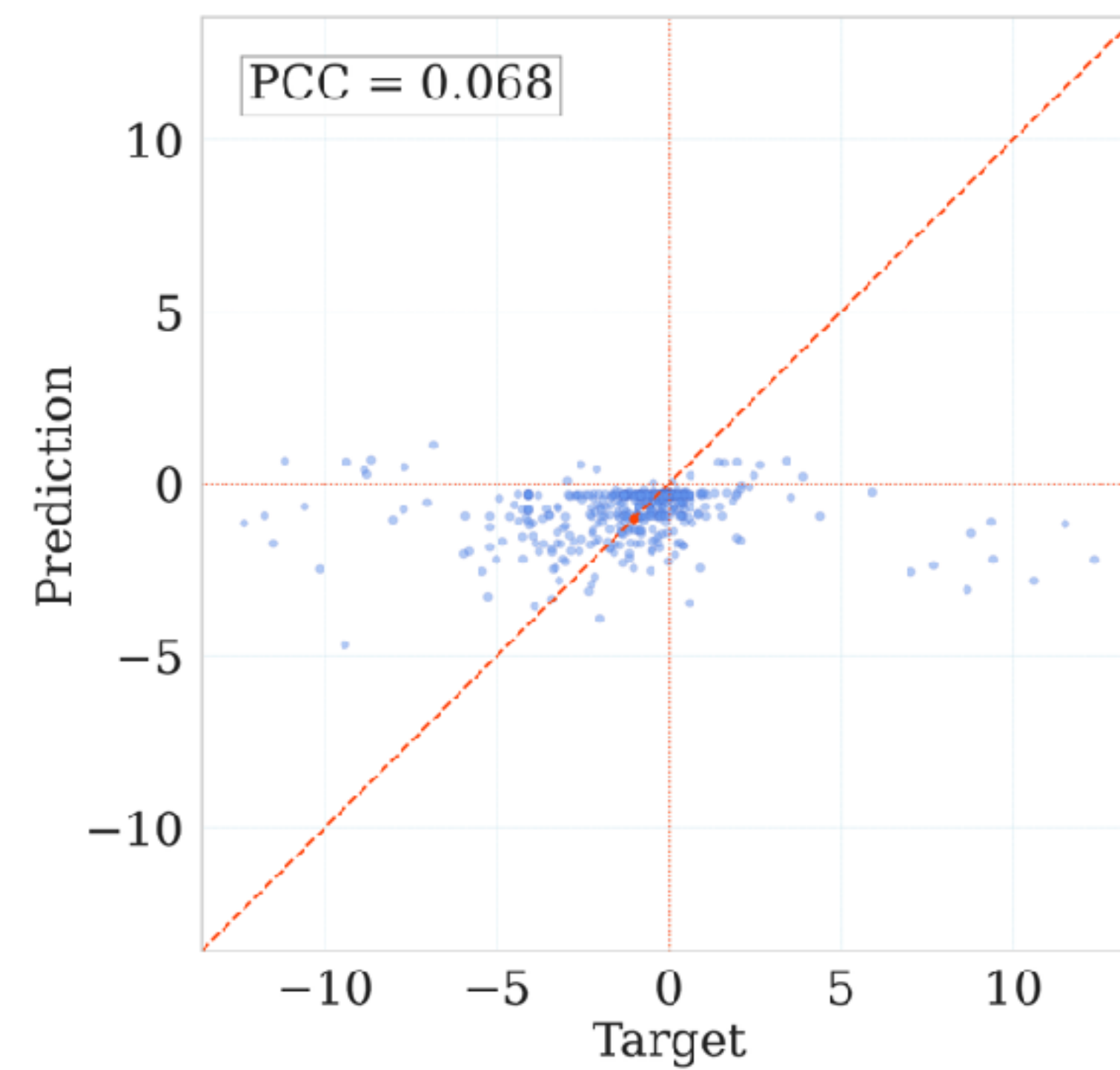
Training: 0 1 2



Validation: 3



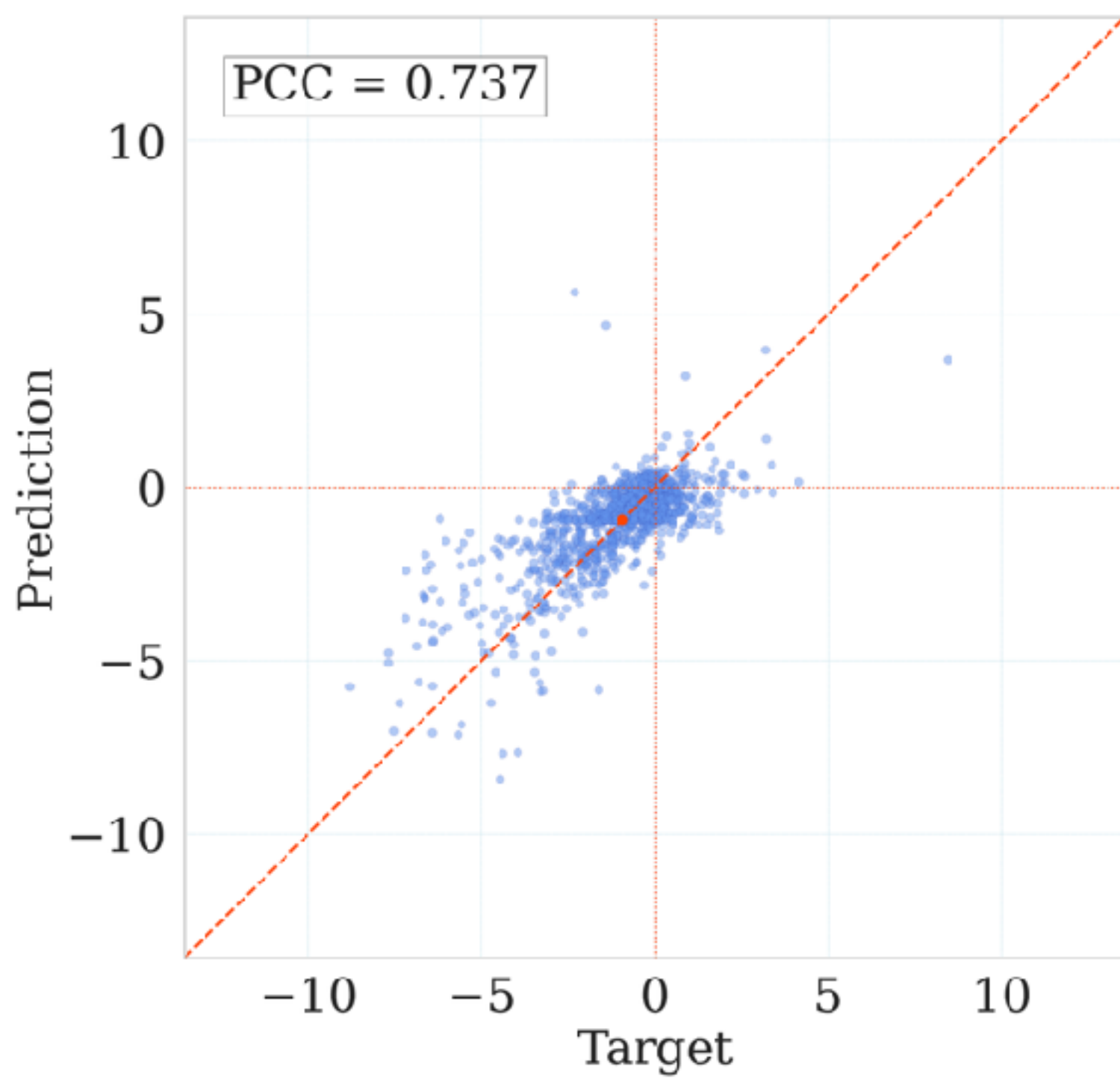
Test: 4



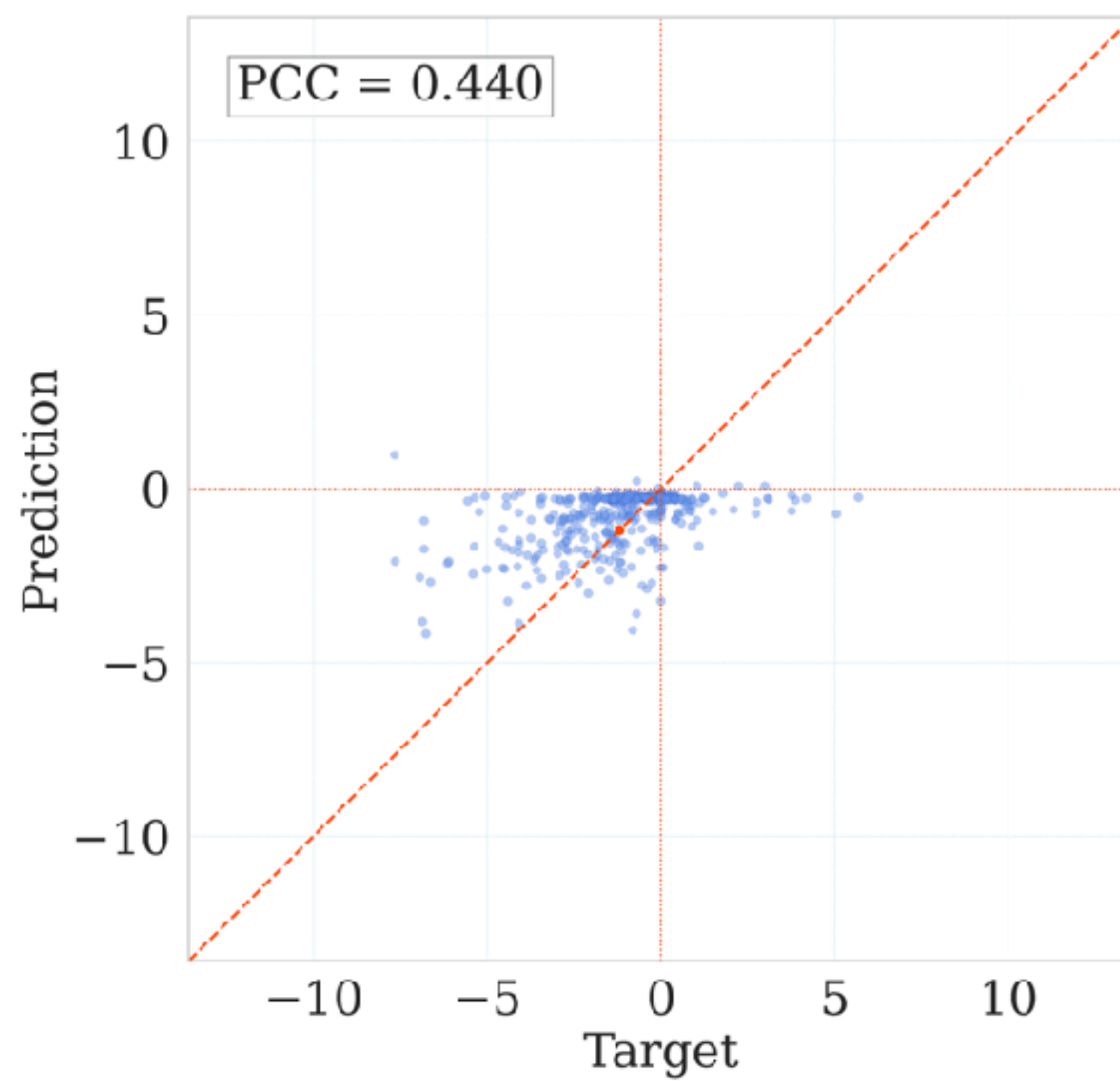
WORST

Run 1
Last Epoch: 689

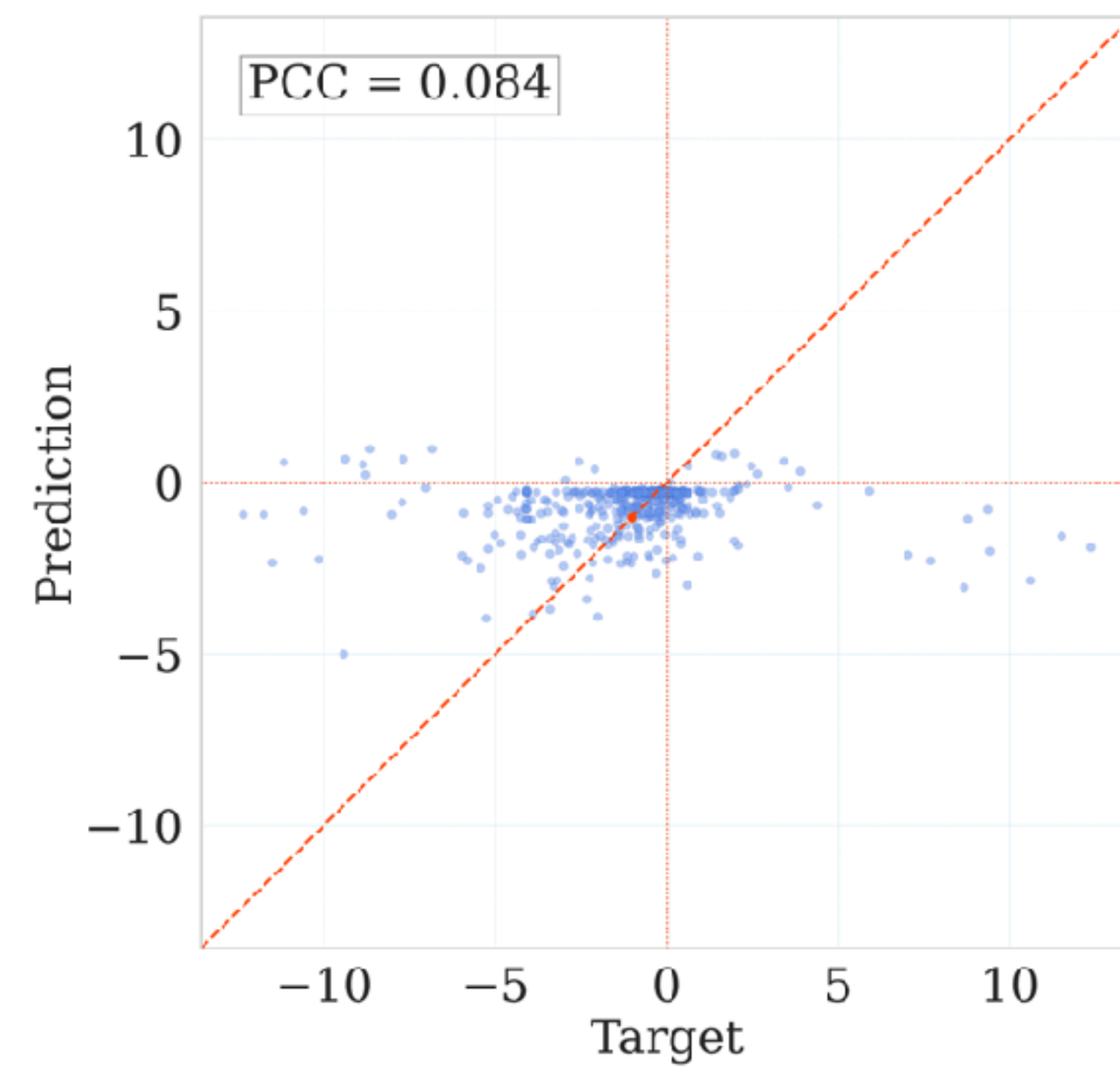
Training: 0 1 2



Validation: 3



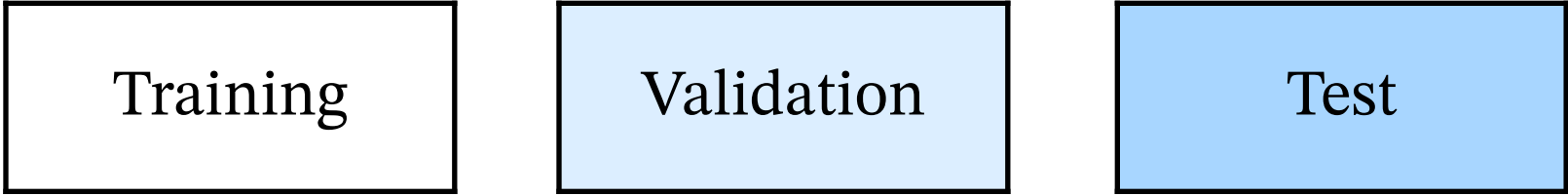
Test: 4



BEST

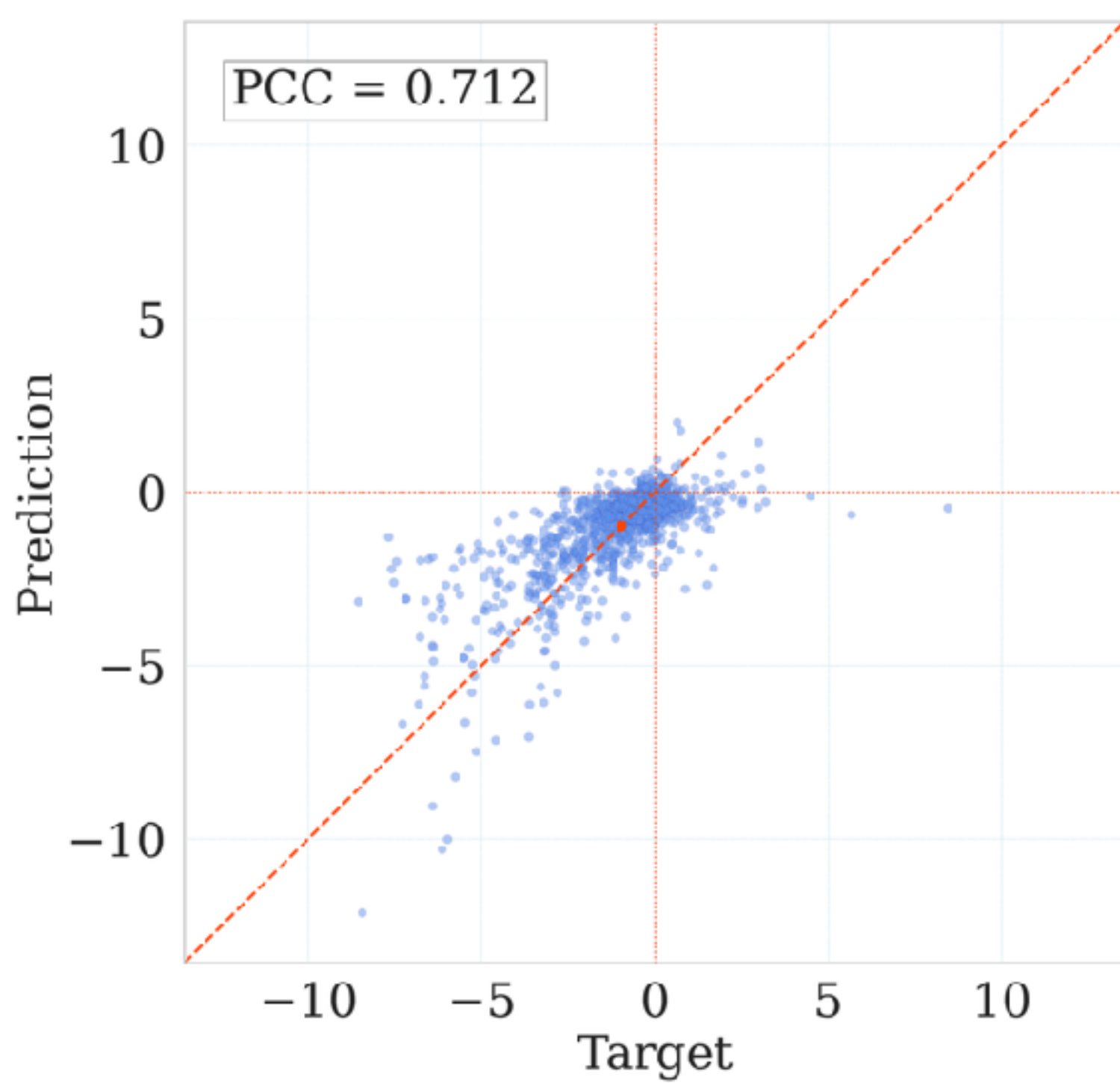
Run 2

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

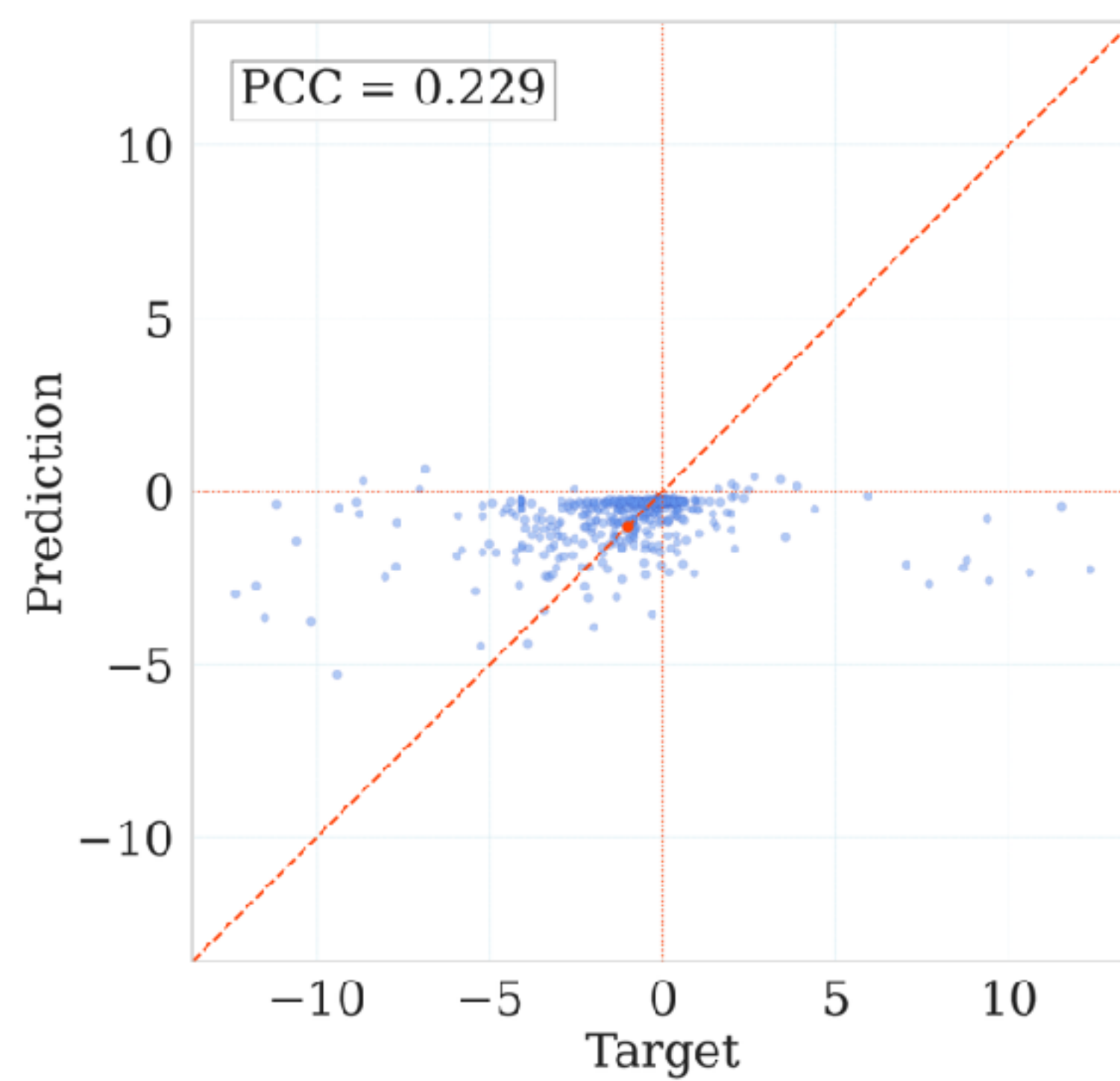


Run 2
Best Epoch: 327

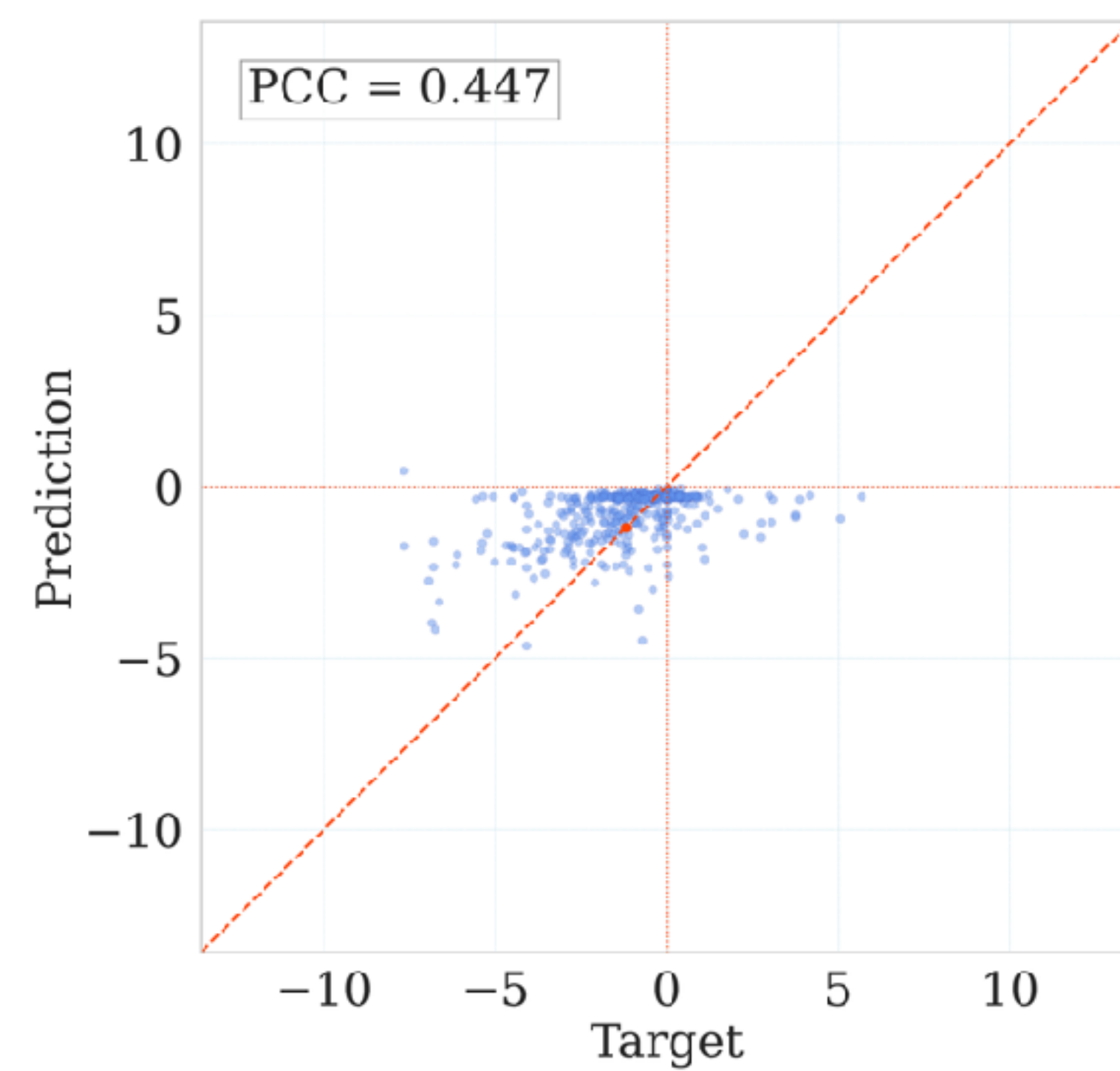
Training: 0 1 2



Validation: 4

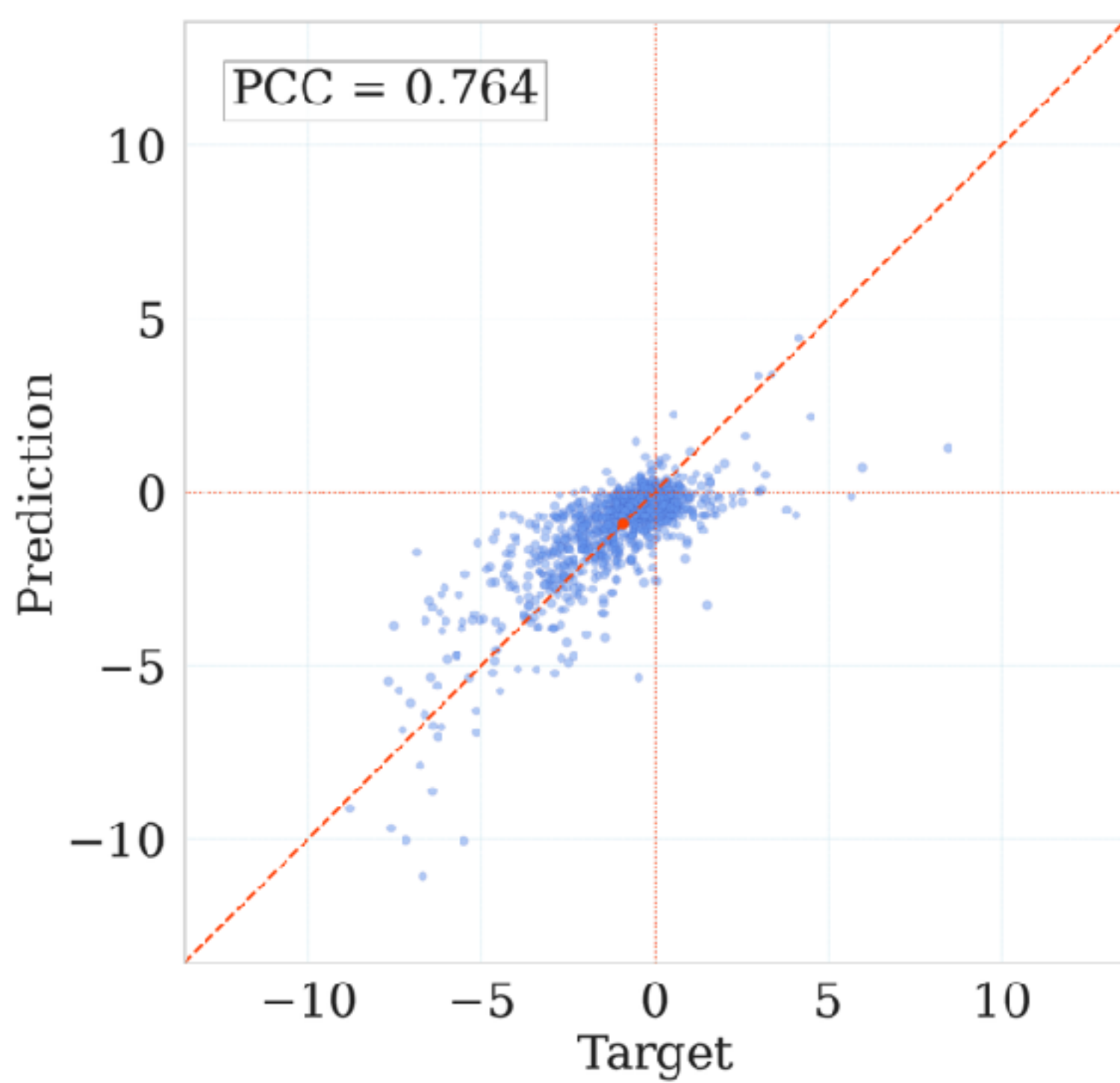


Test: 3

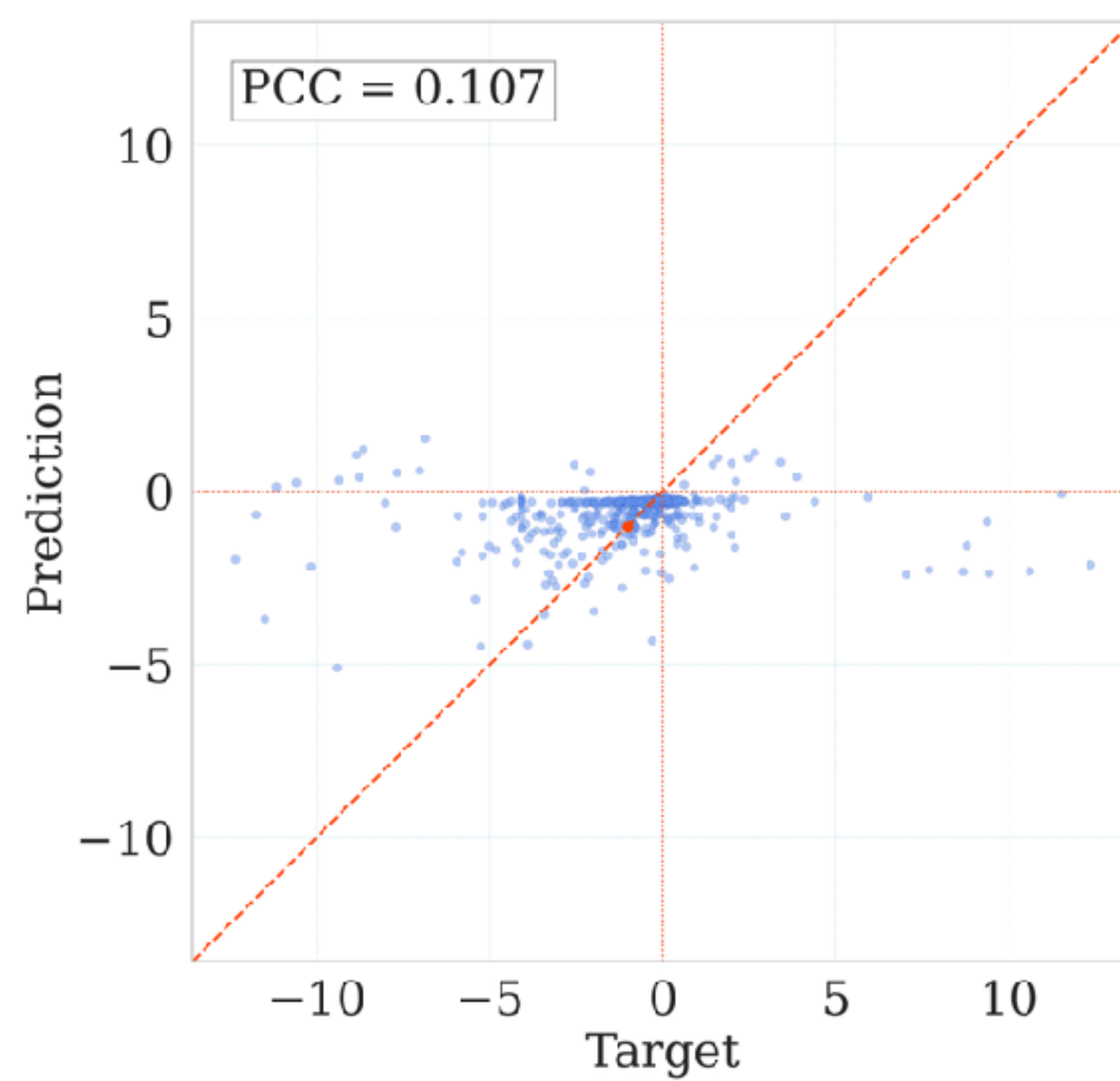


Run 2
Last Epoch: 610

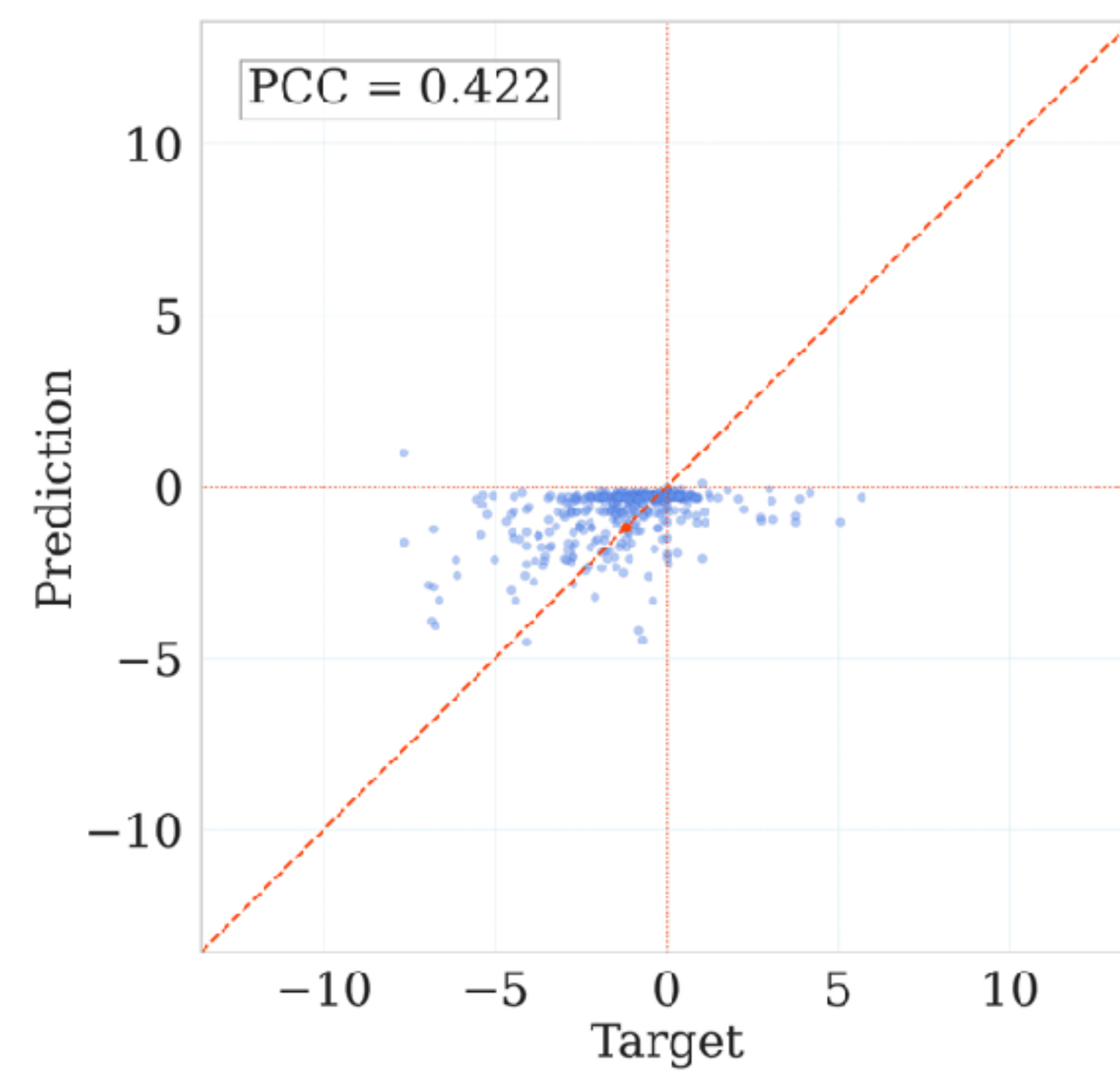
Training: 0 1 2



Validation: 4

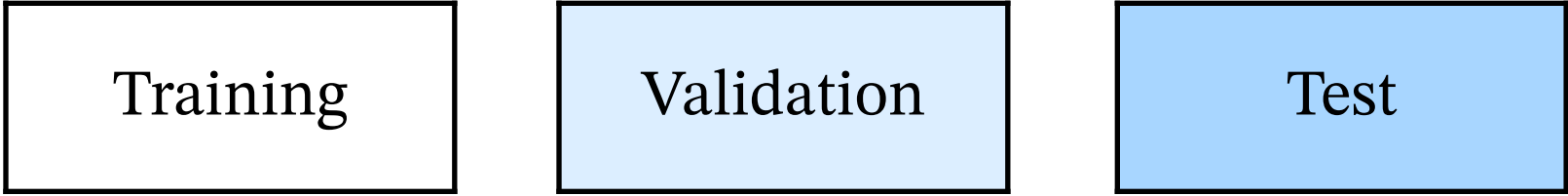


Test: 3



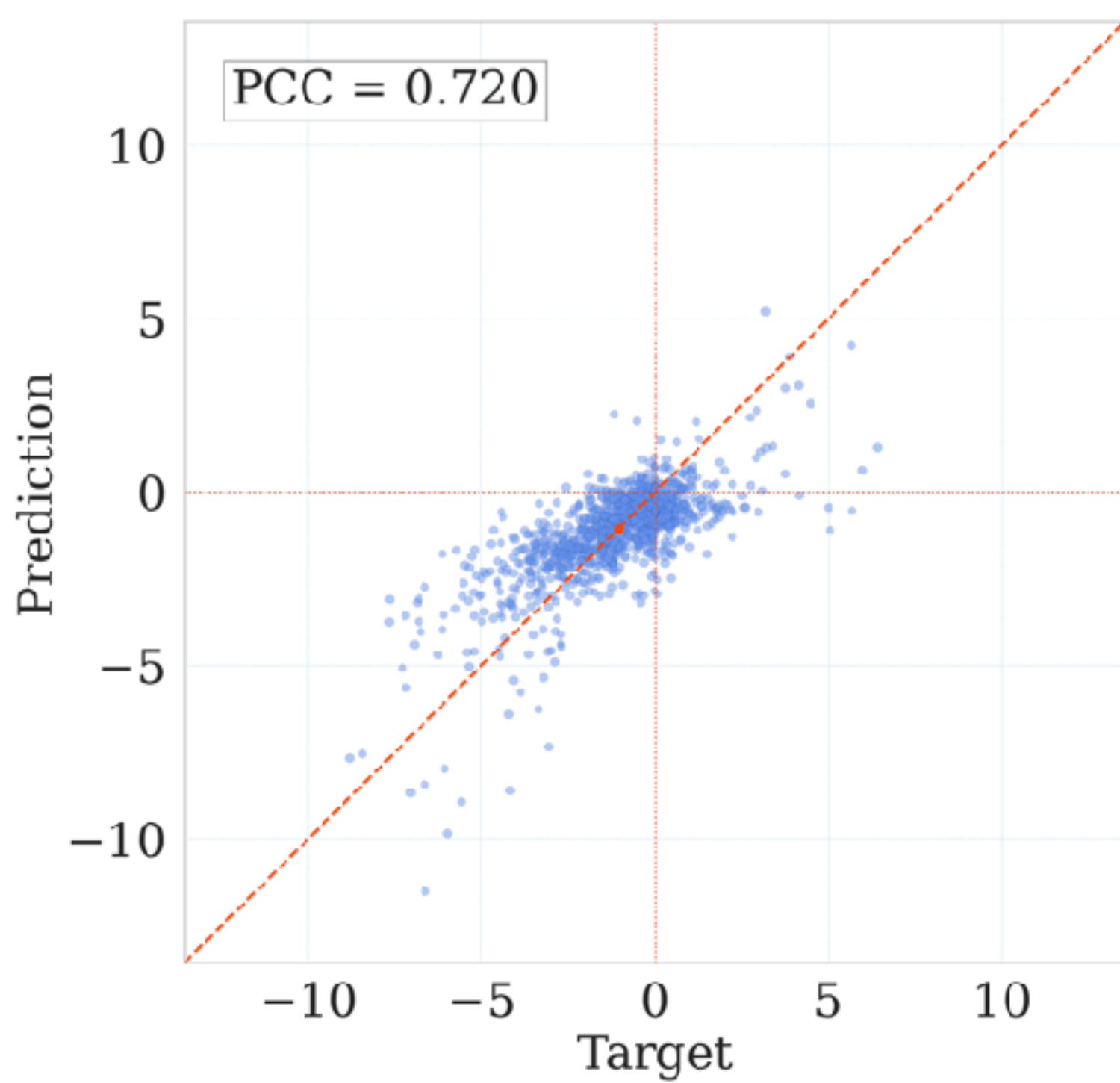
Run 3

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

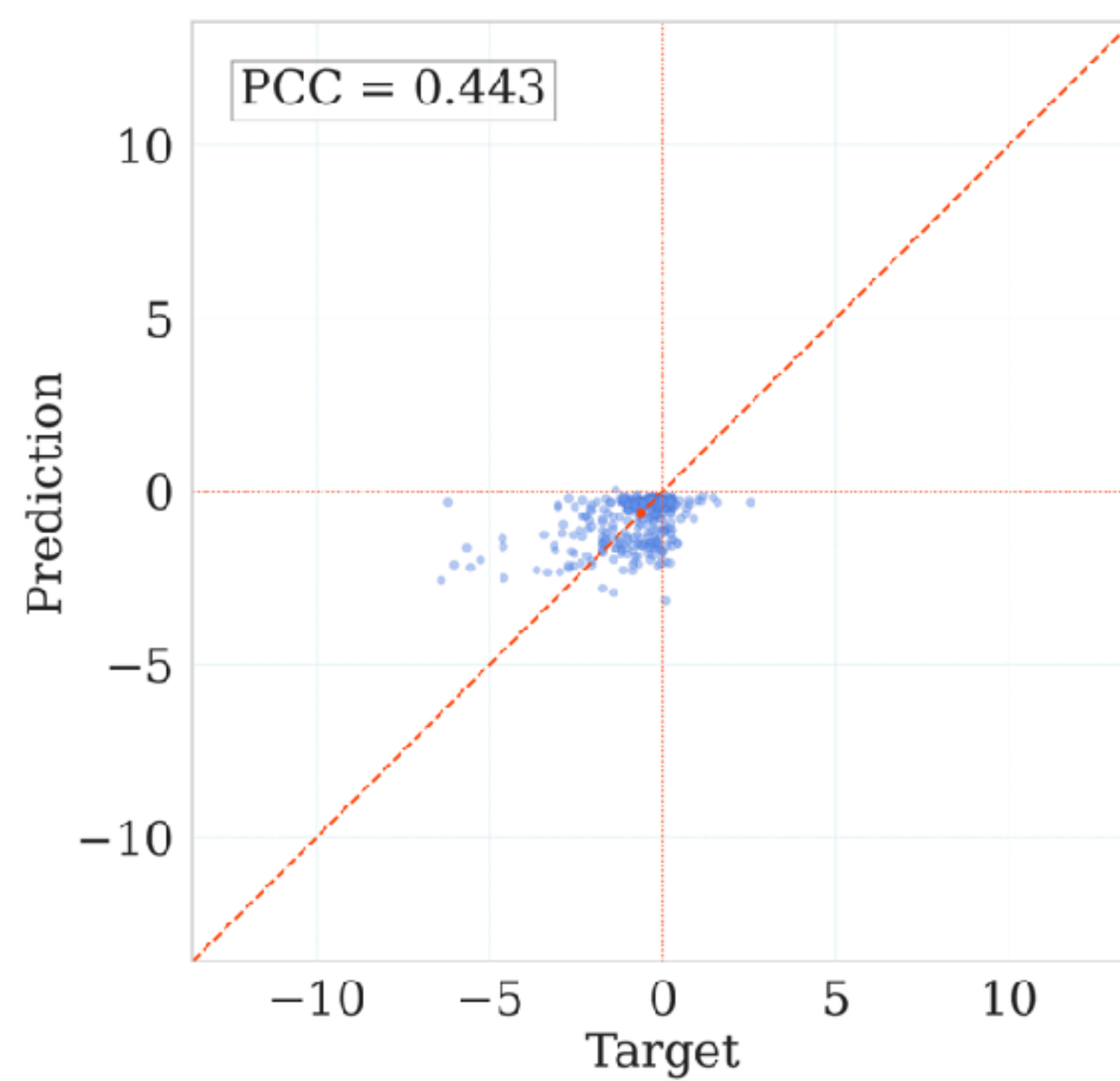


Run 3
Best Epoch: 352

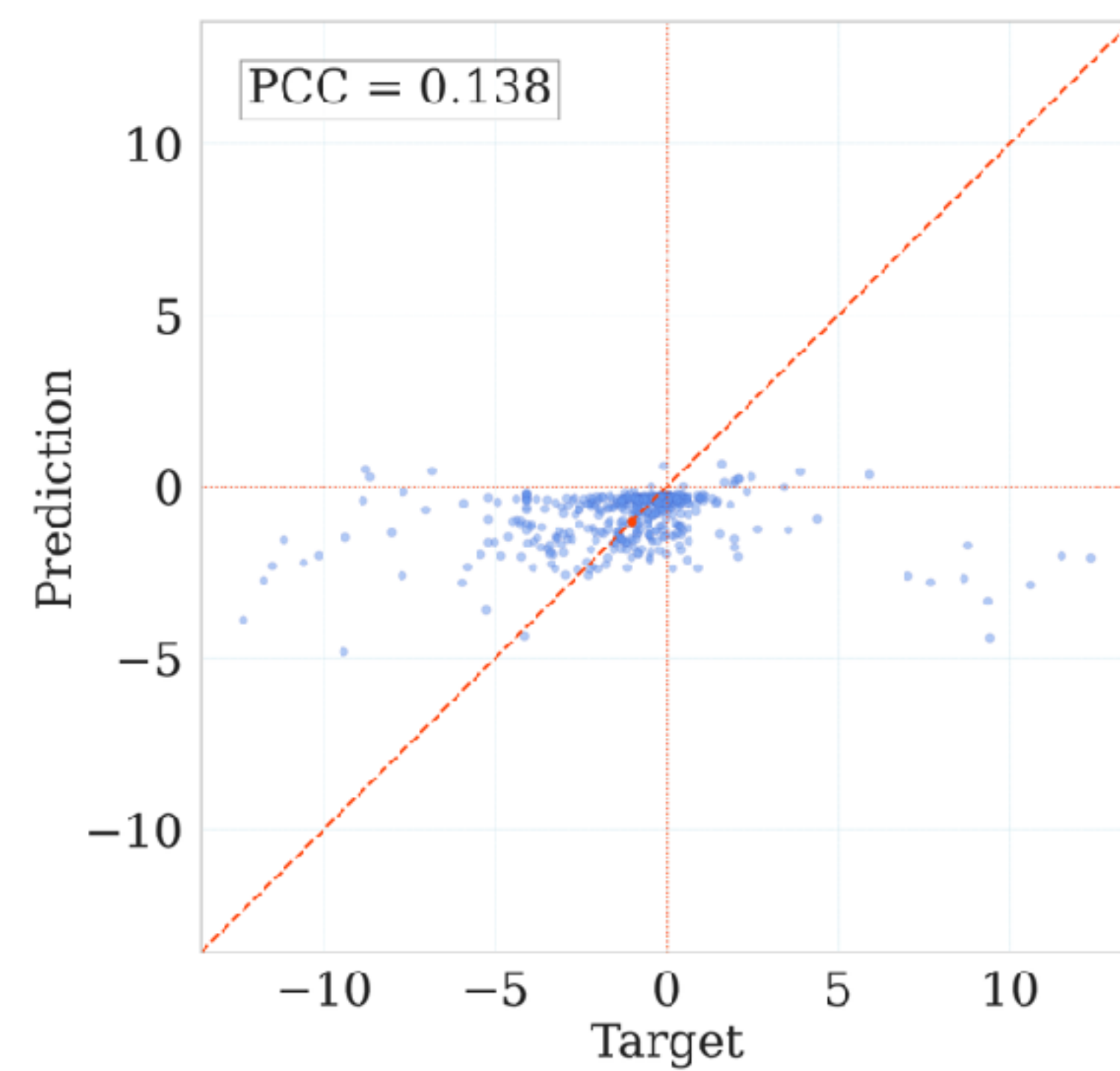
Training: 0 1 3



Validation: 2

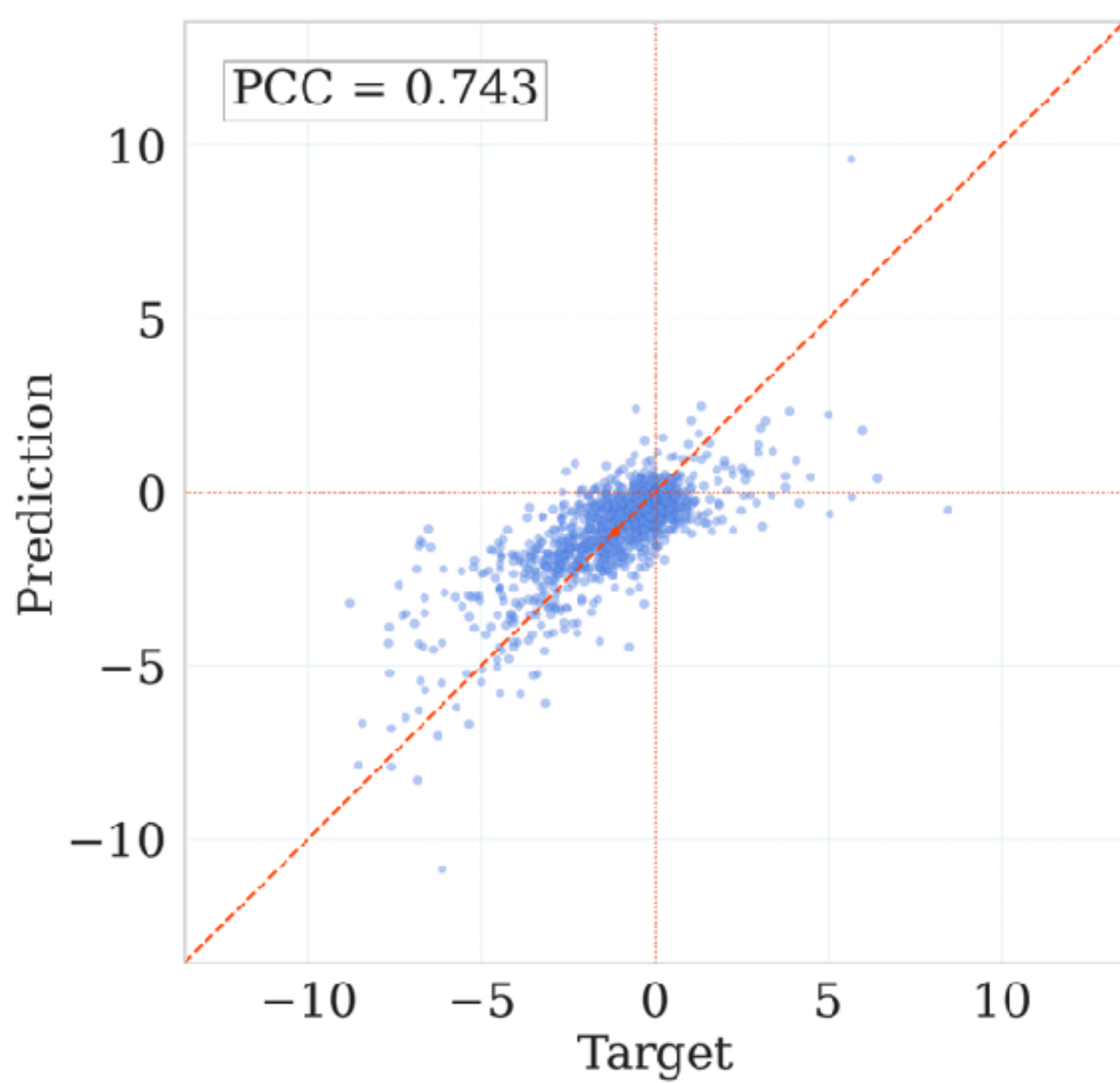


Test: 4

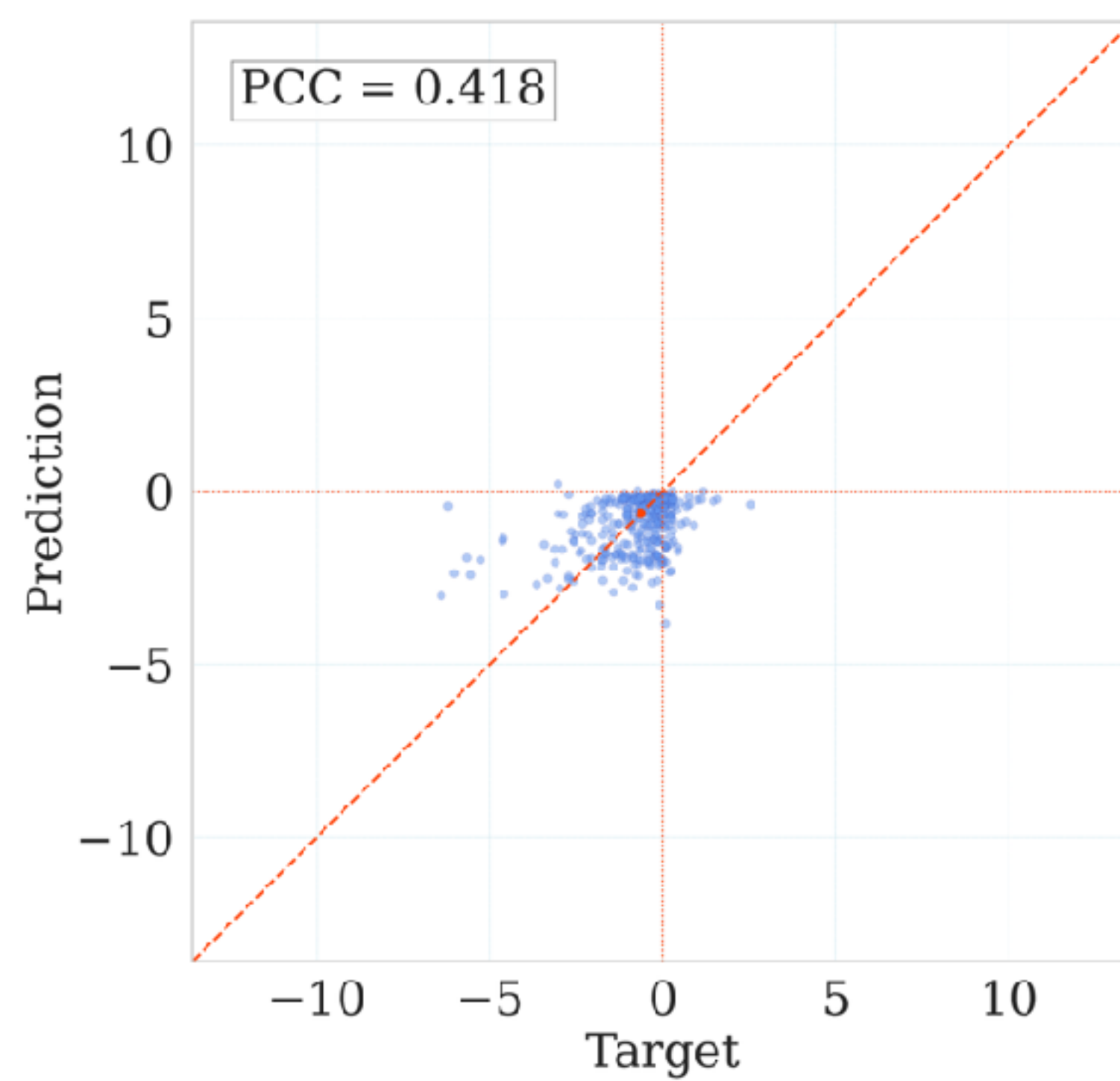


Run 3
Last Epoch: 620

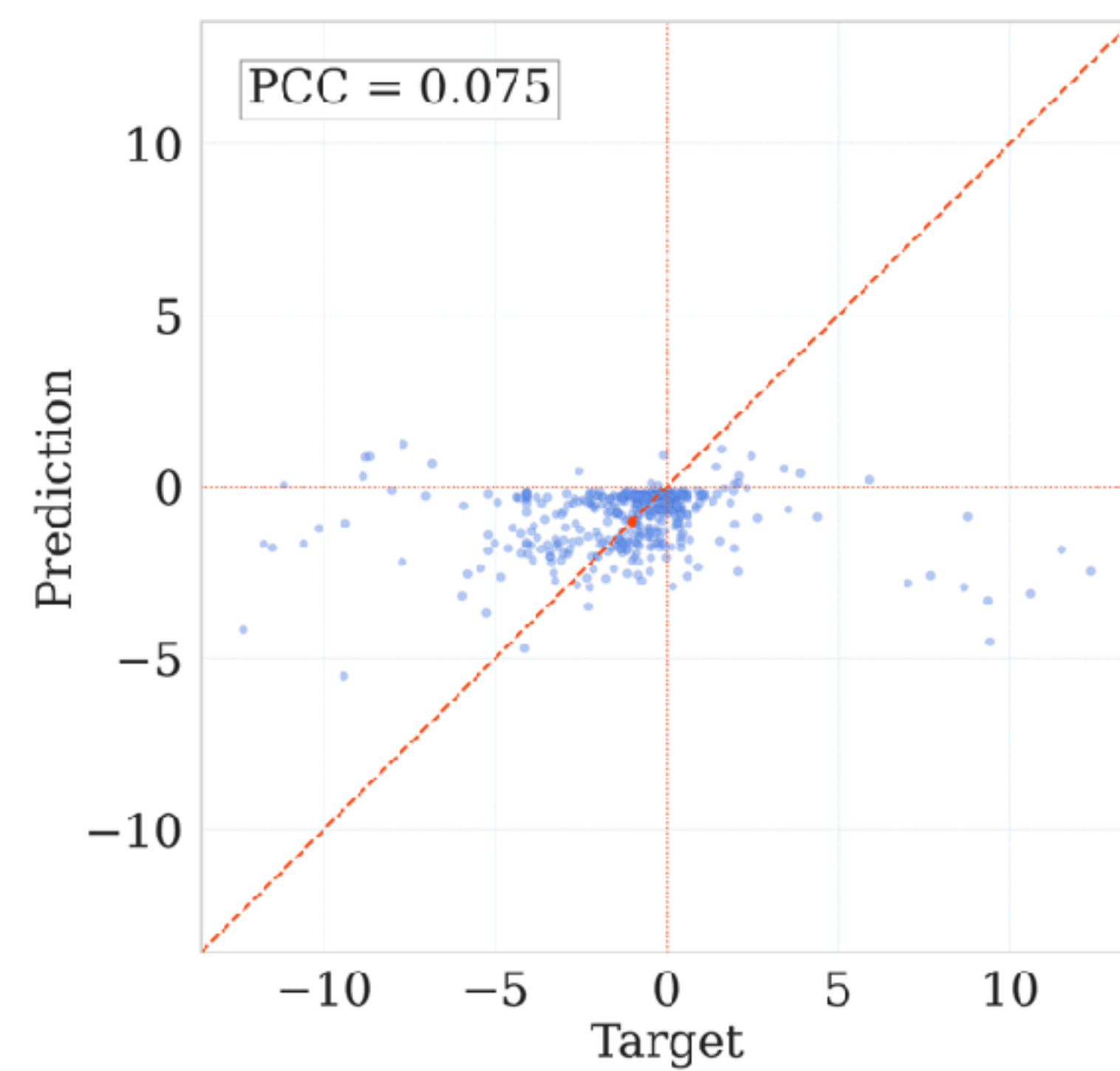
Training: 0 1 3



Validation: 2

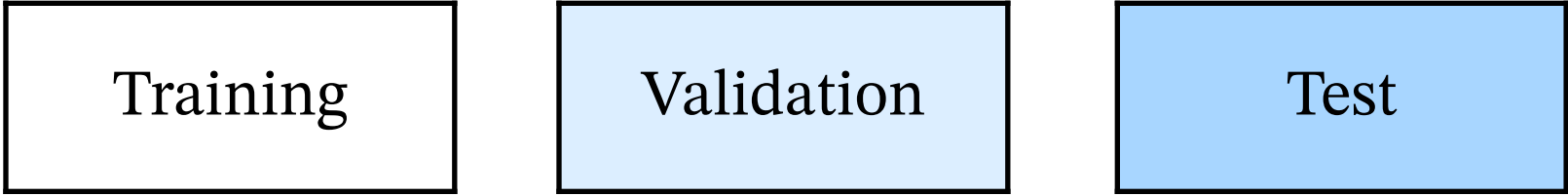


Test: 4



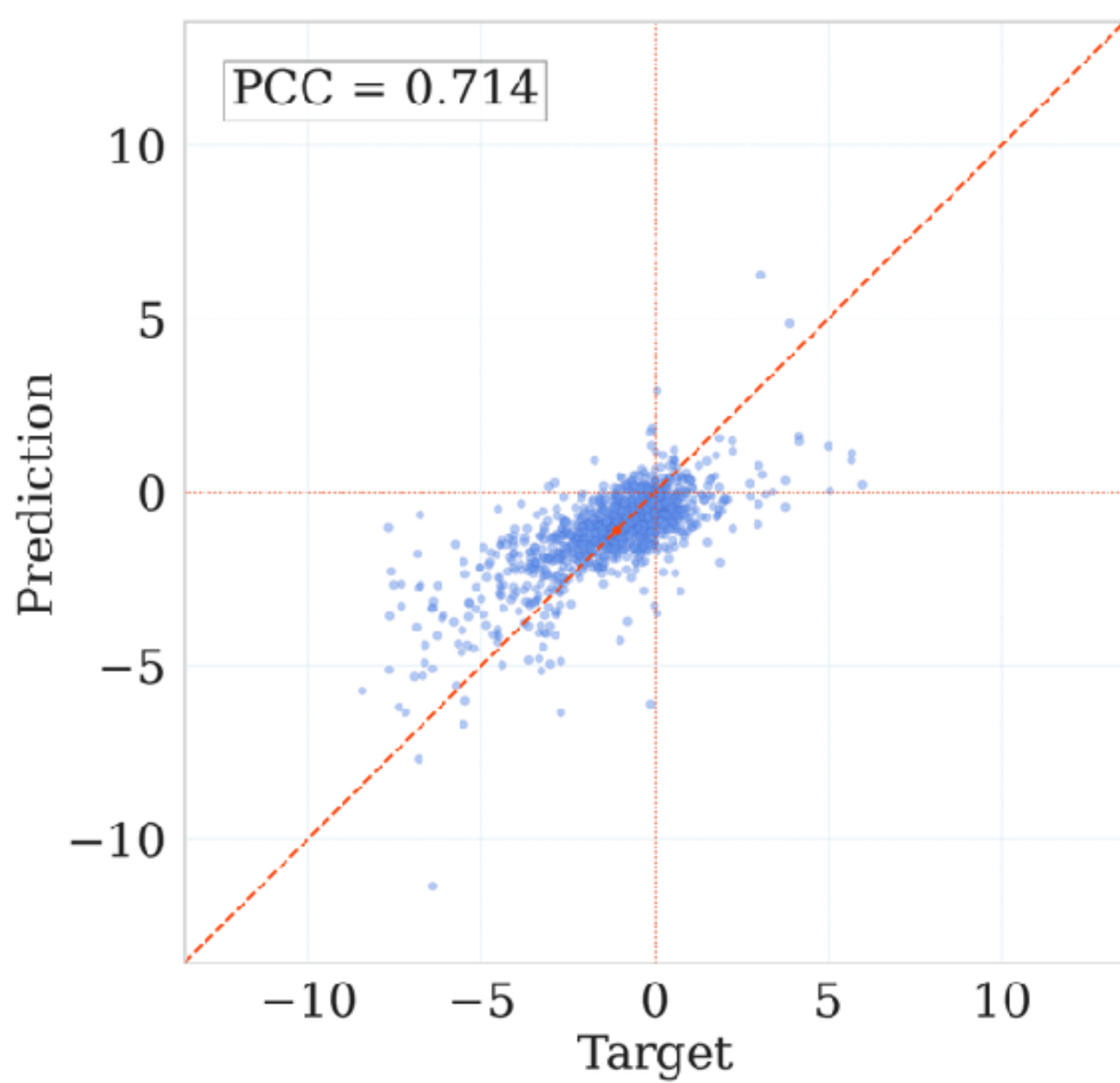
Run 4

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

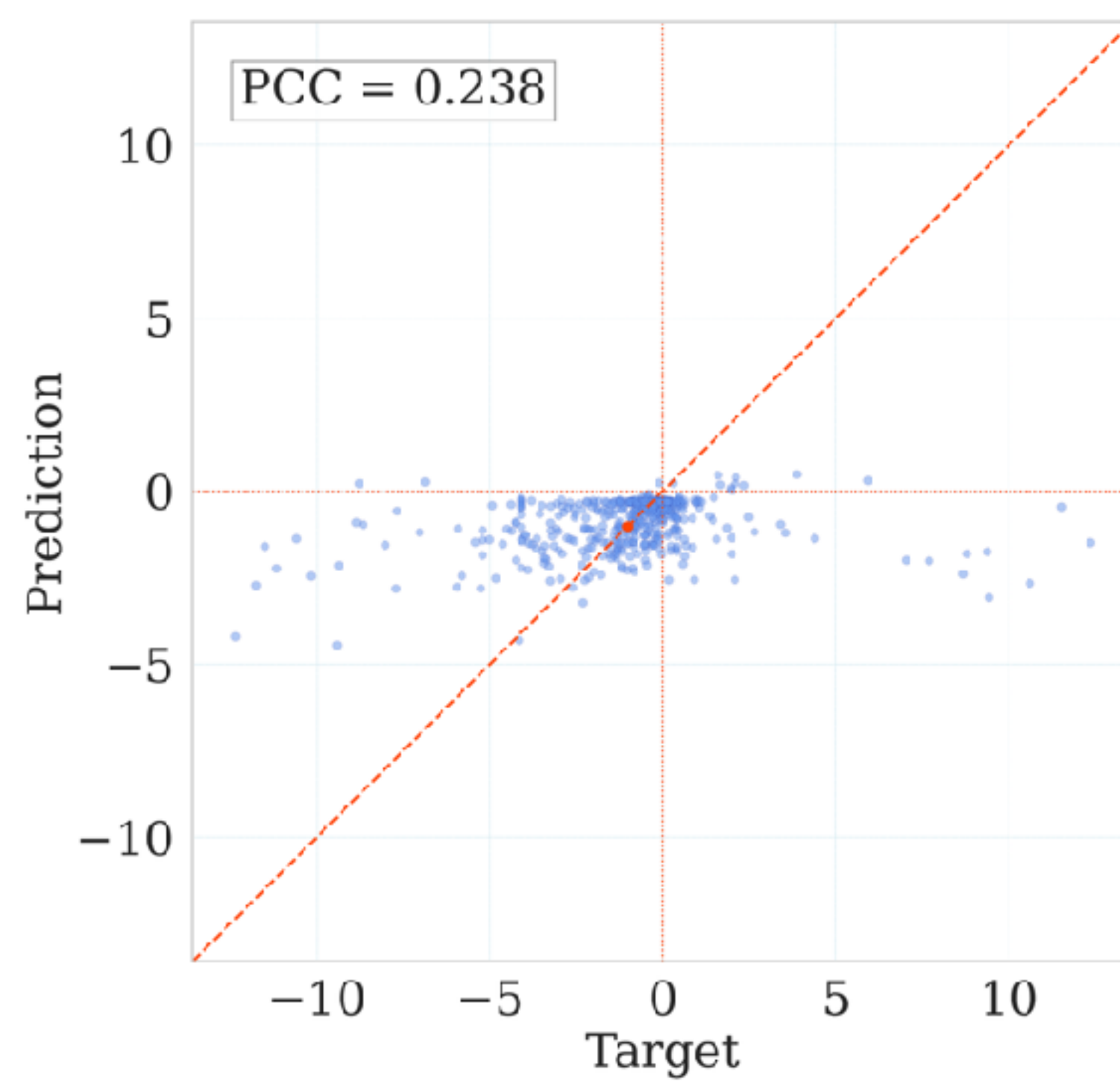


Run 4
Best Epoch: 219

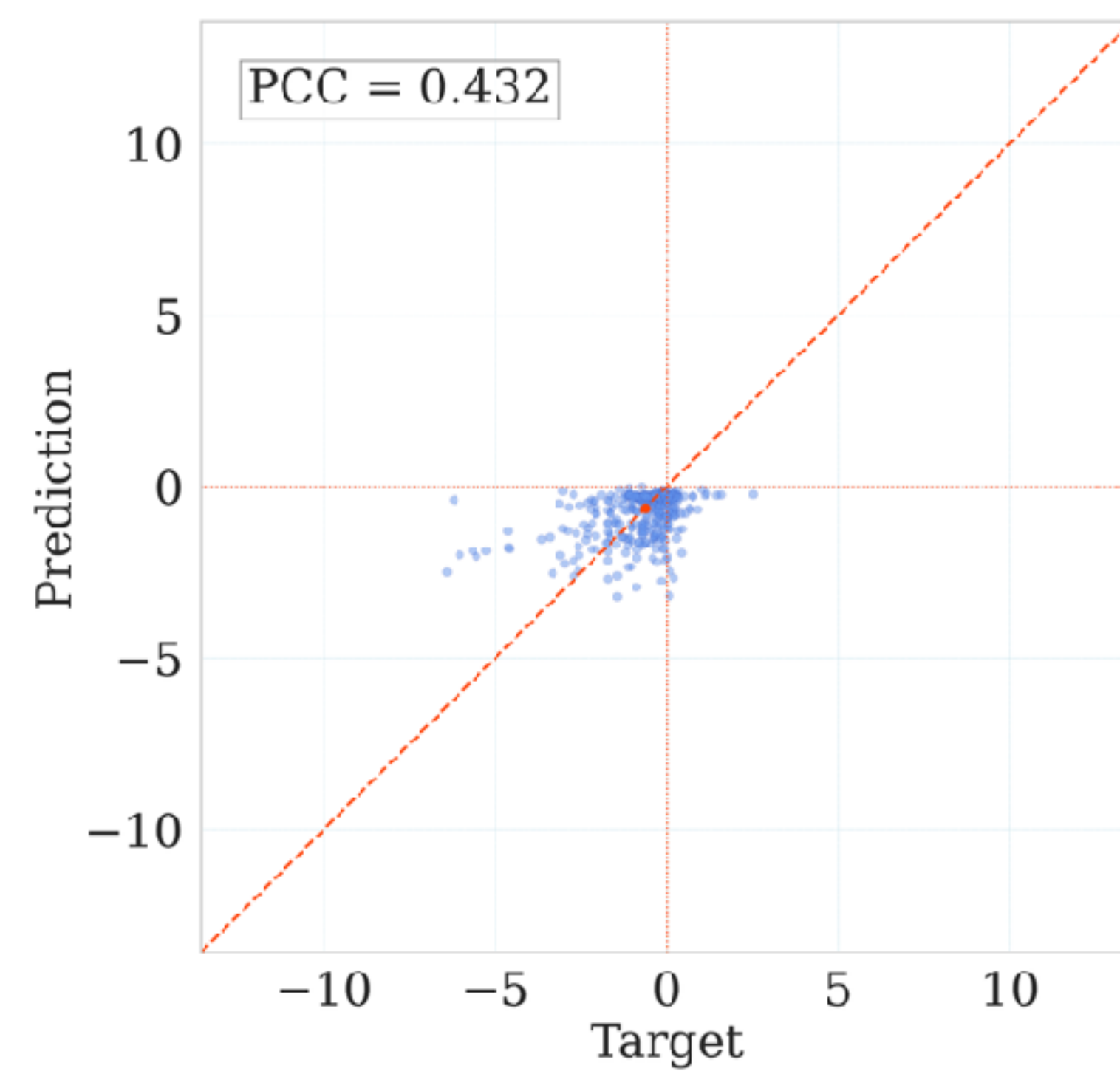
Training: 0 1 3



Validation: 4

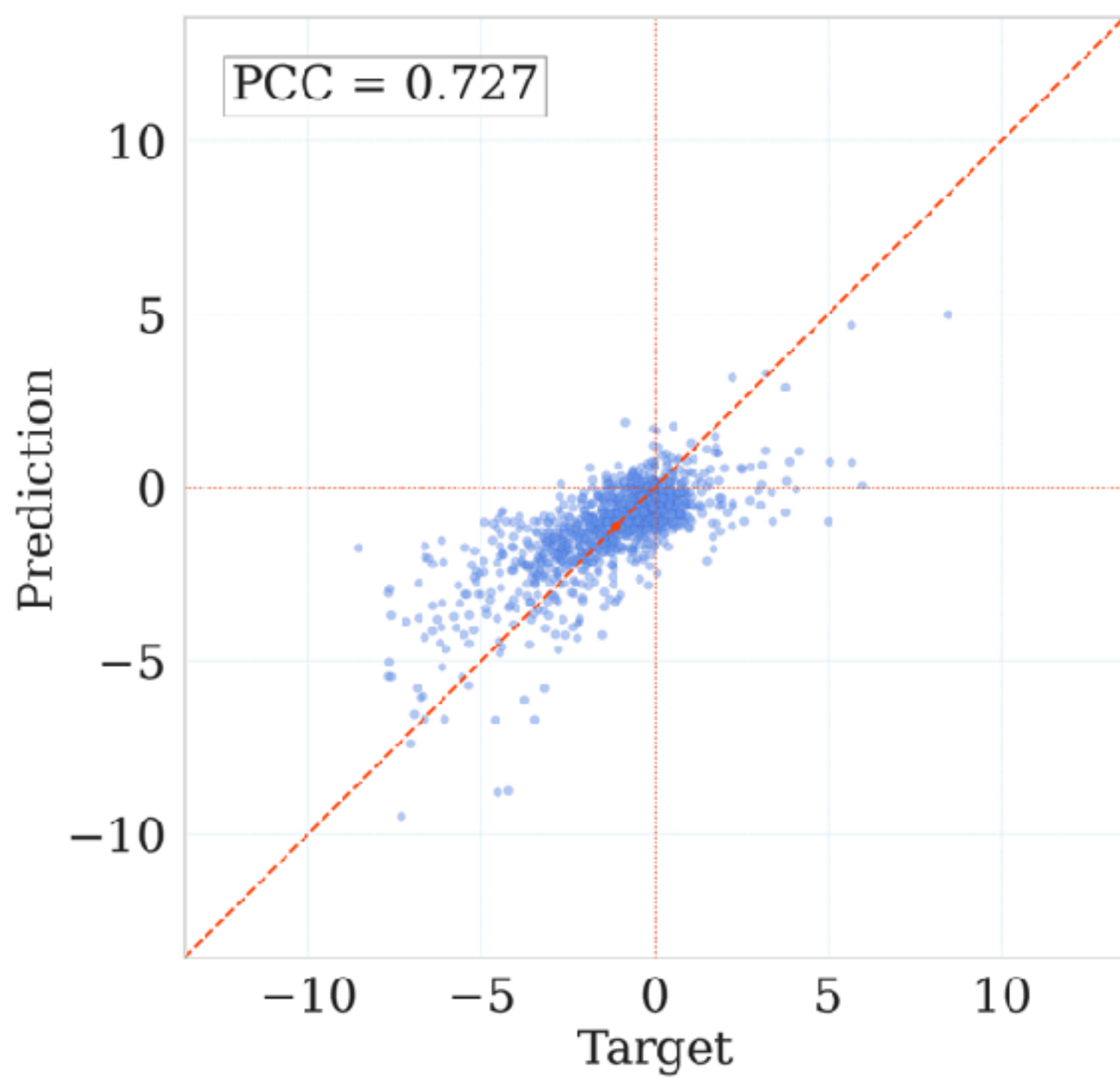


Test: 2

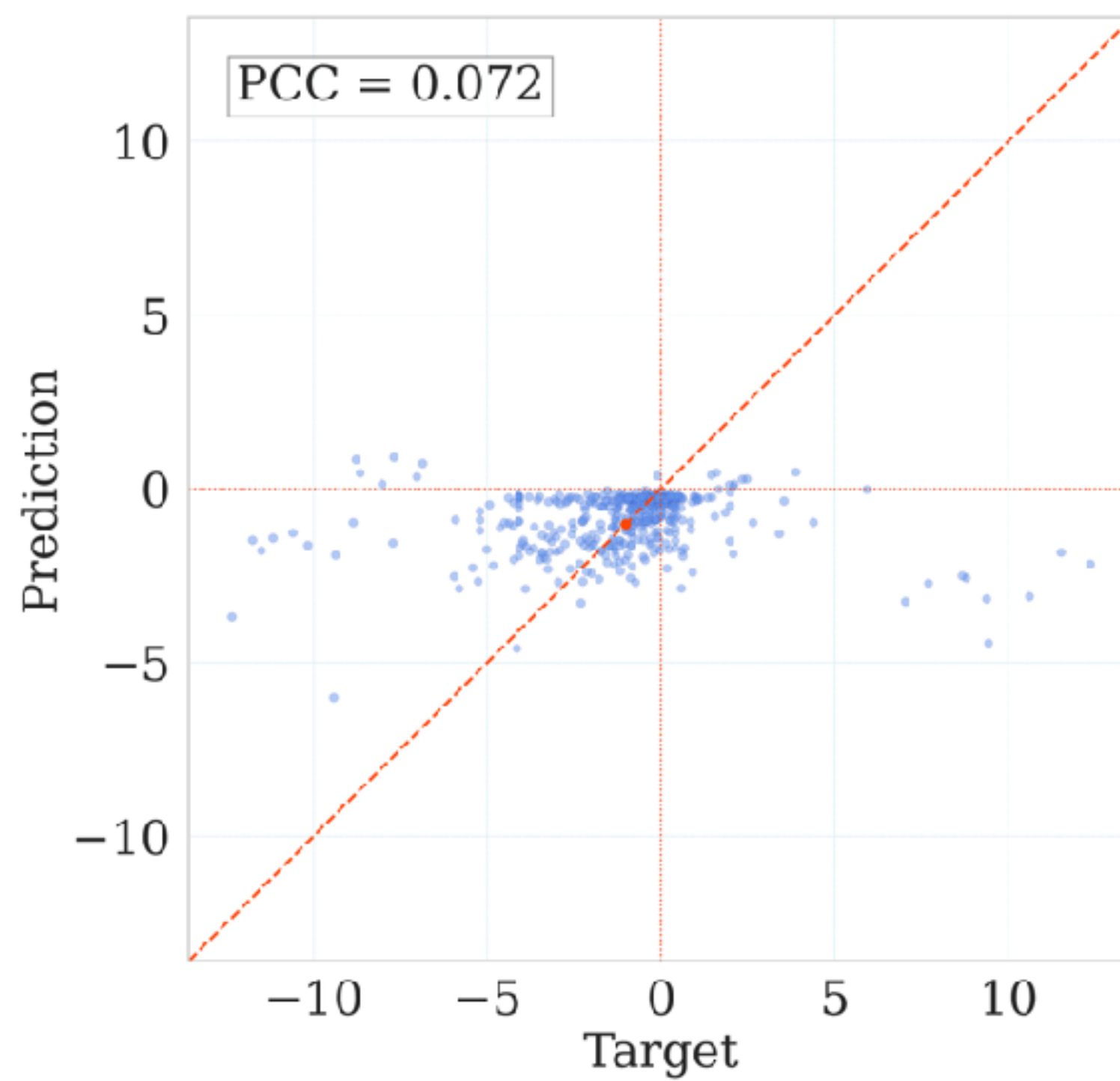


Run 4
Last Epoch: 737

Training: 0 1 3

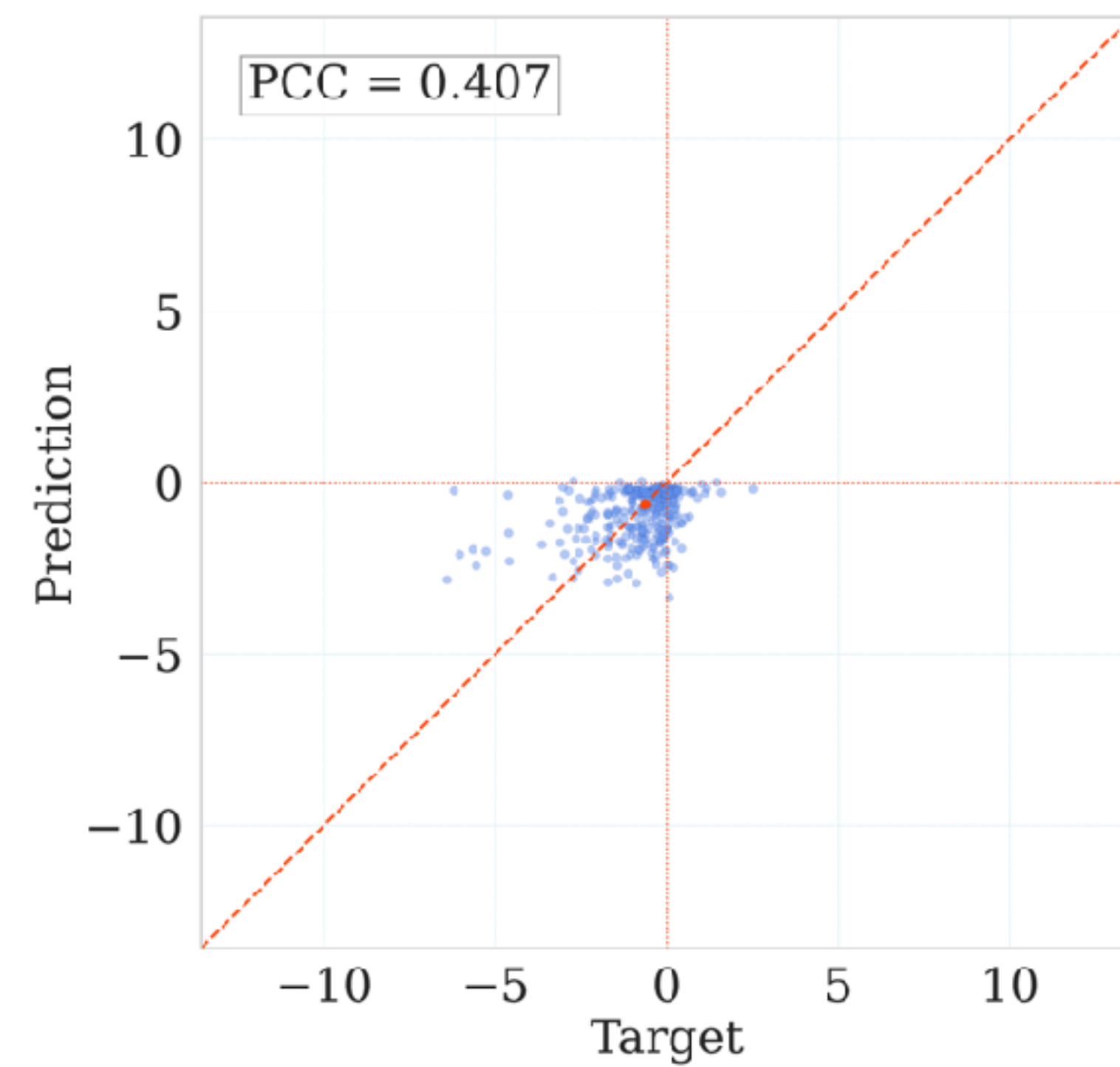


Validation: 4



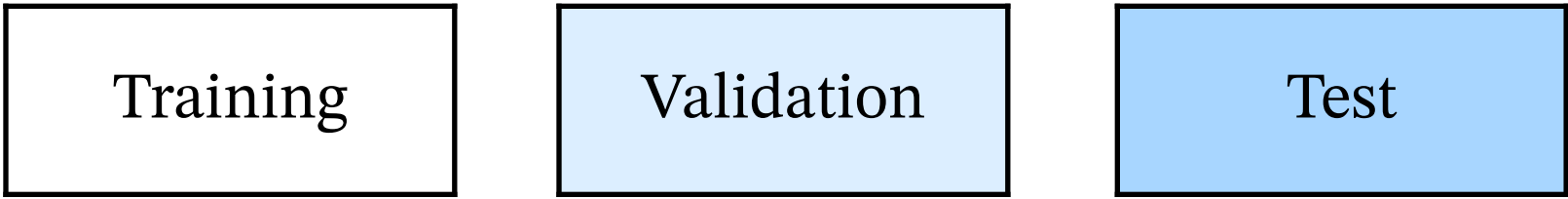
WORST

Test: 2



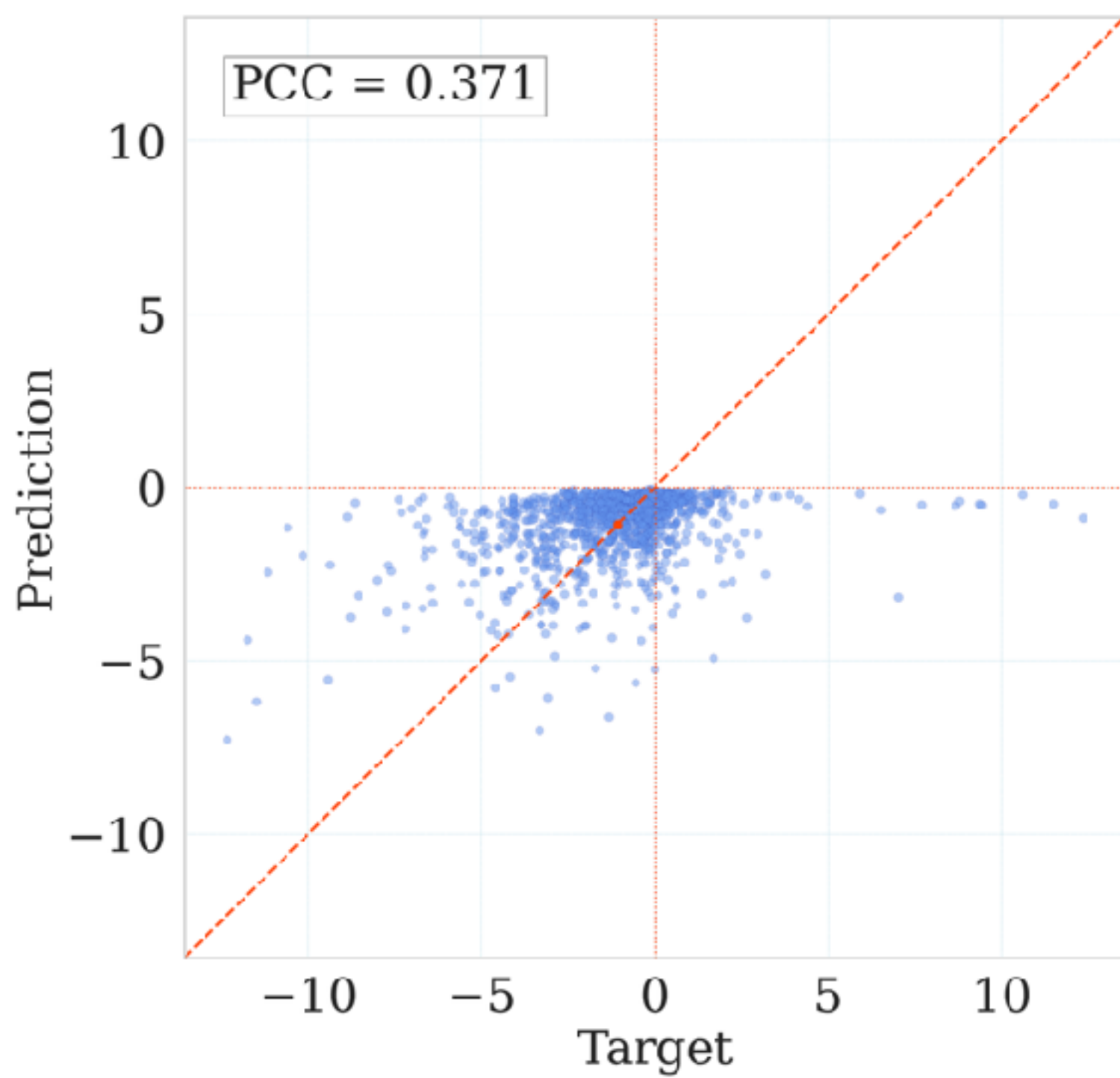
Run 5

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6



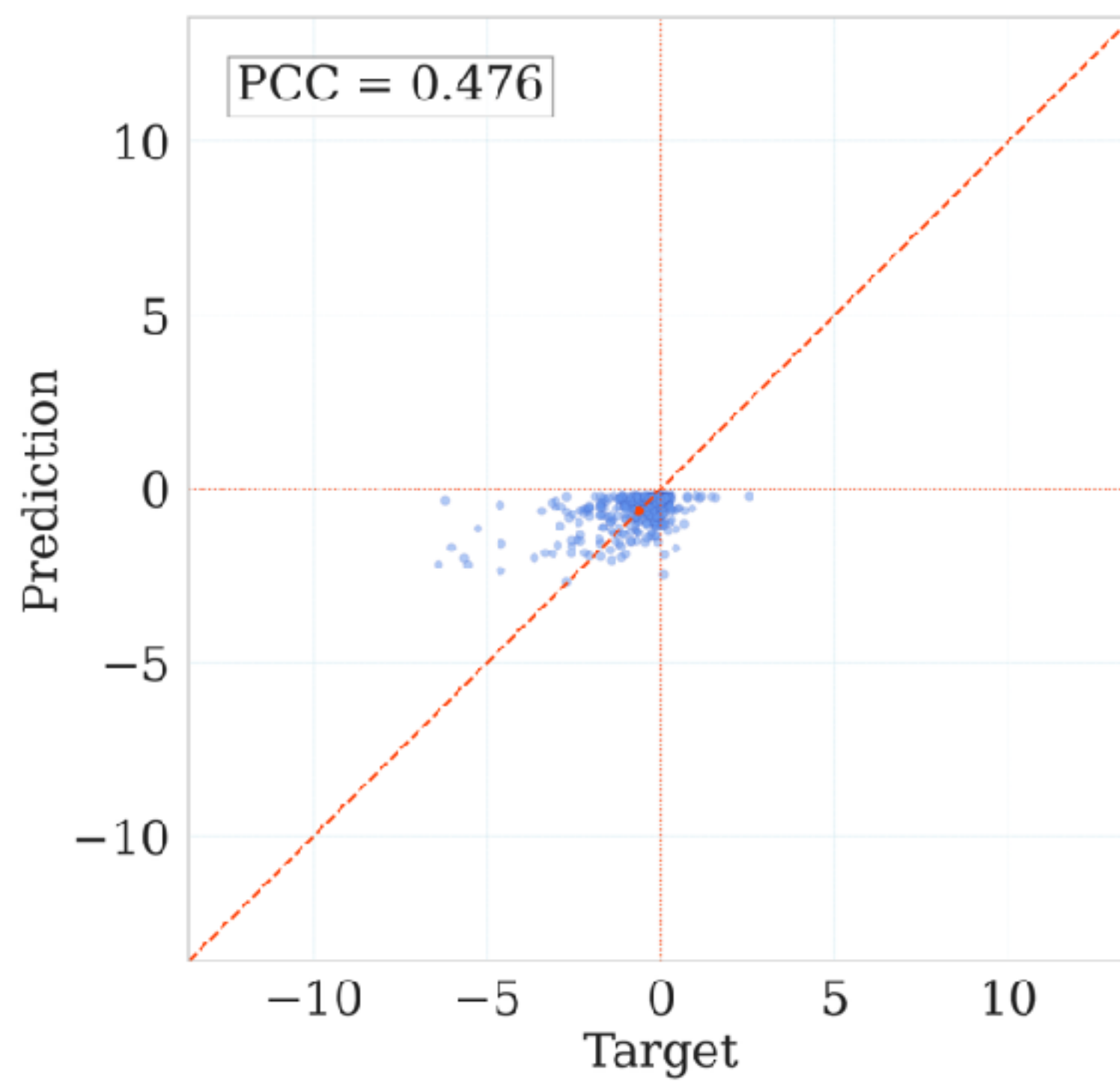
Run 5
Best Epoch: 18

Training: 0 1 4

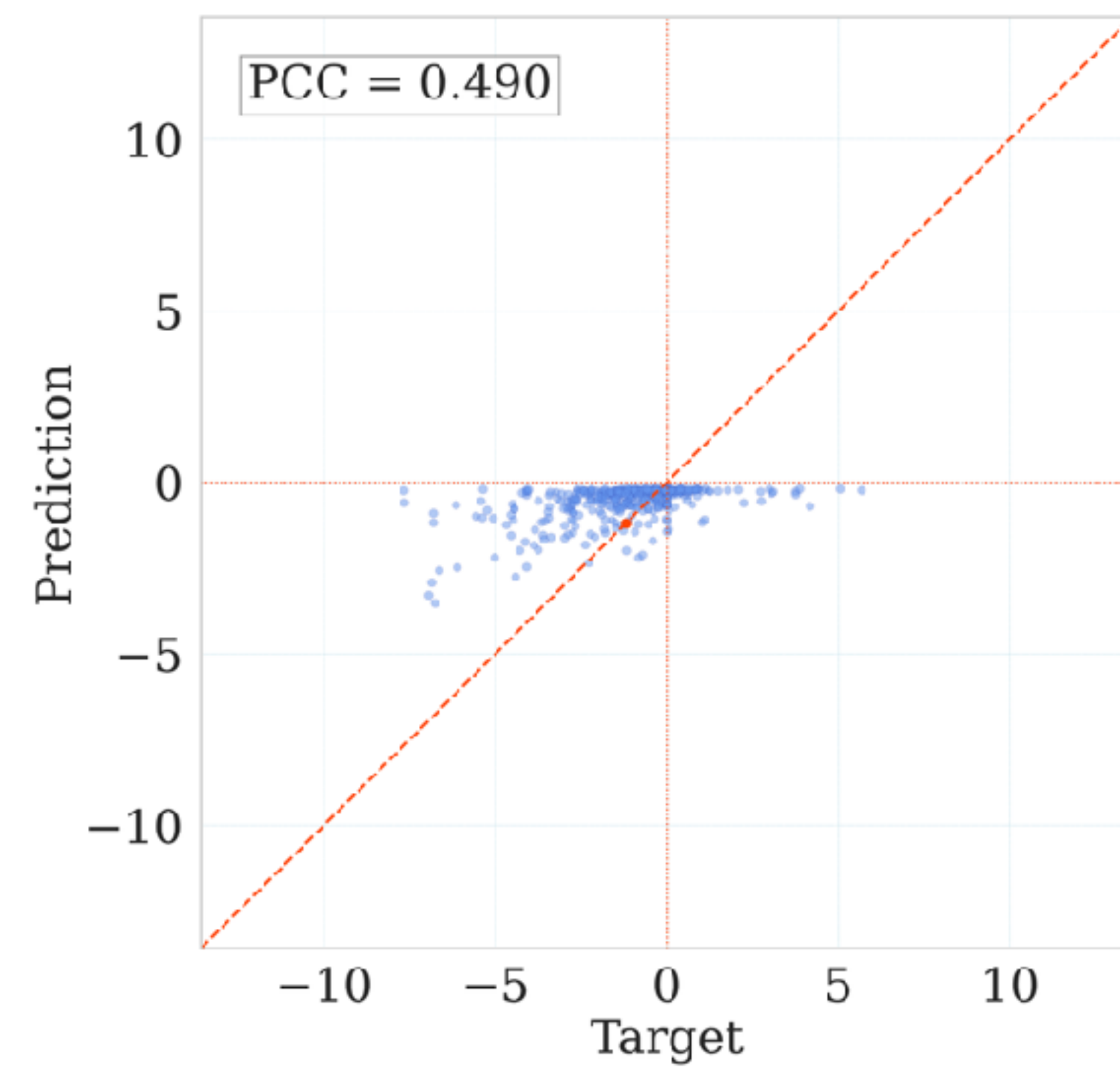


WORST

Validation: 2

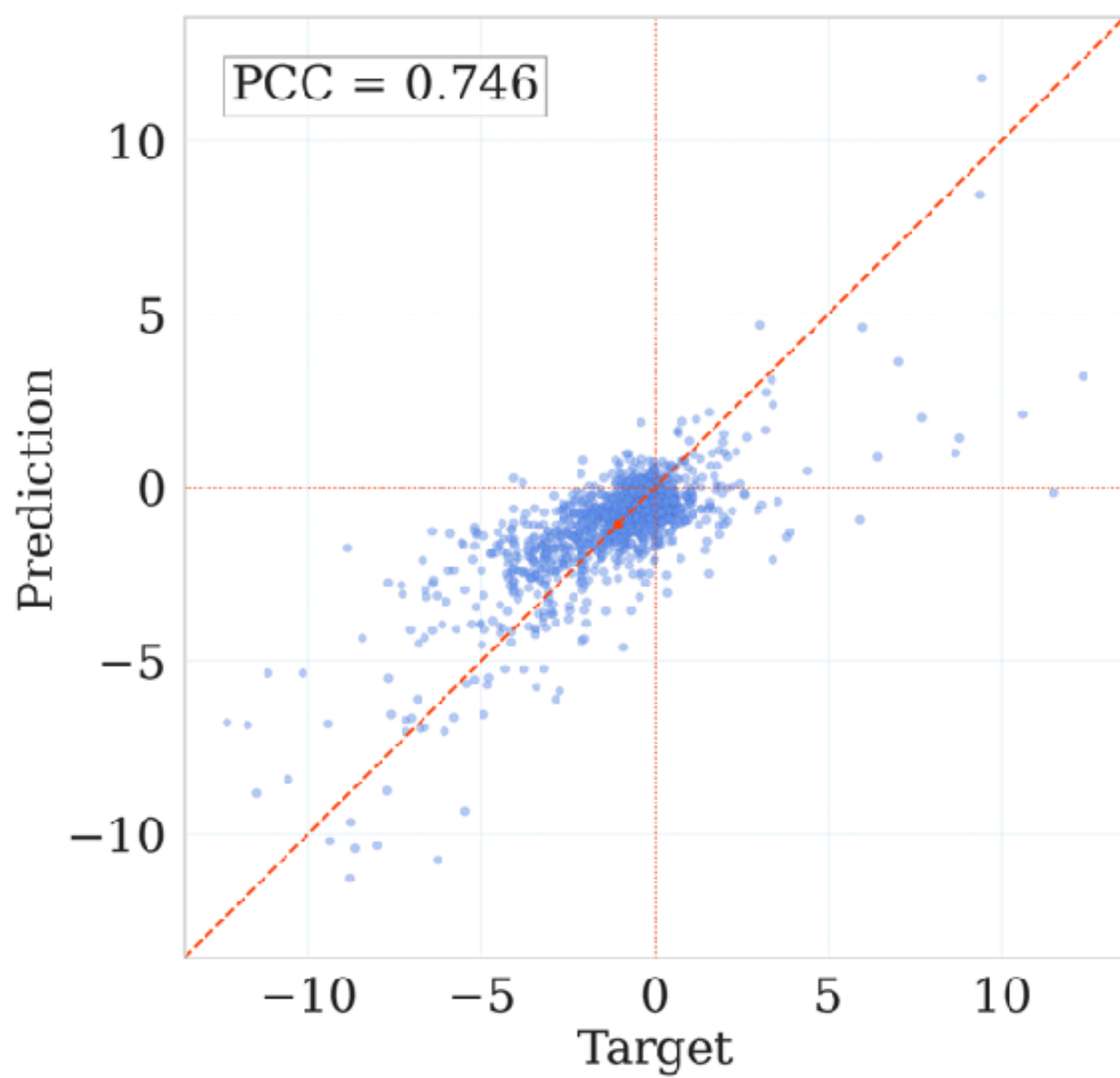


Test: 3

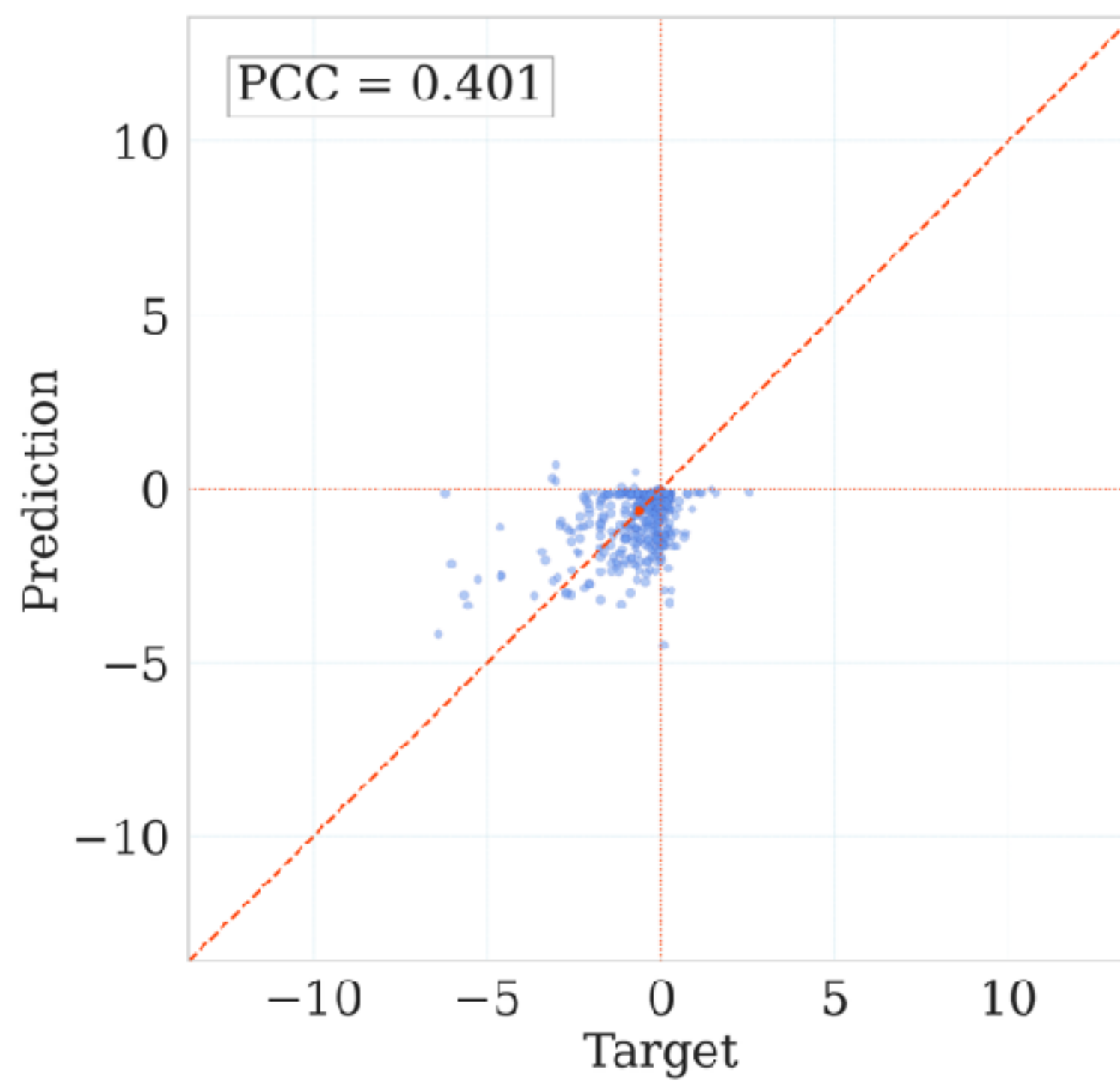


Run 5
Last Epoch: 659

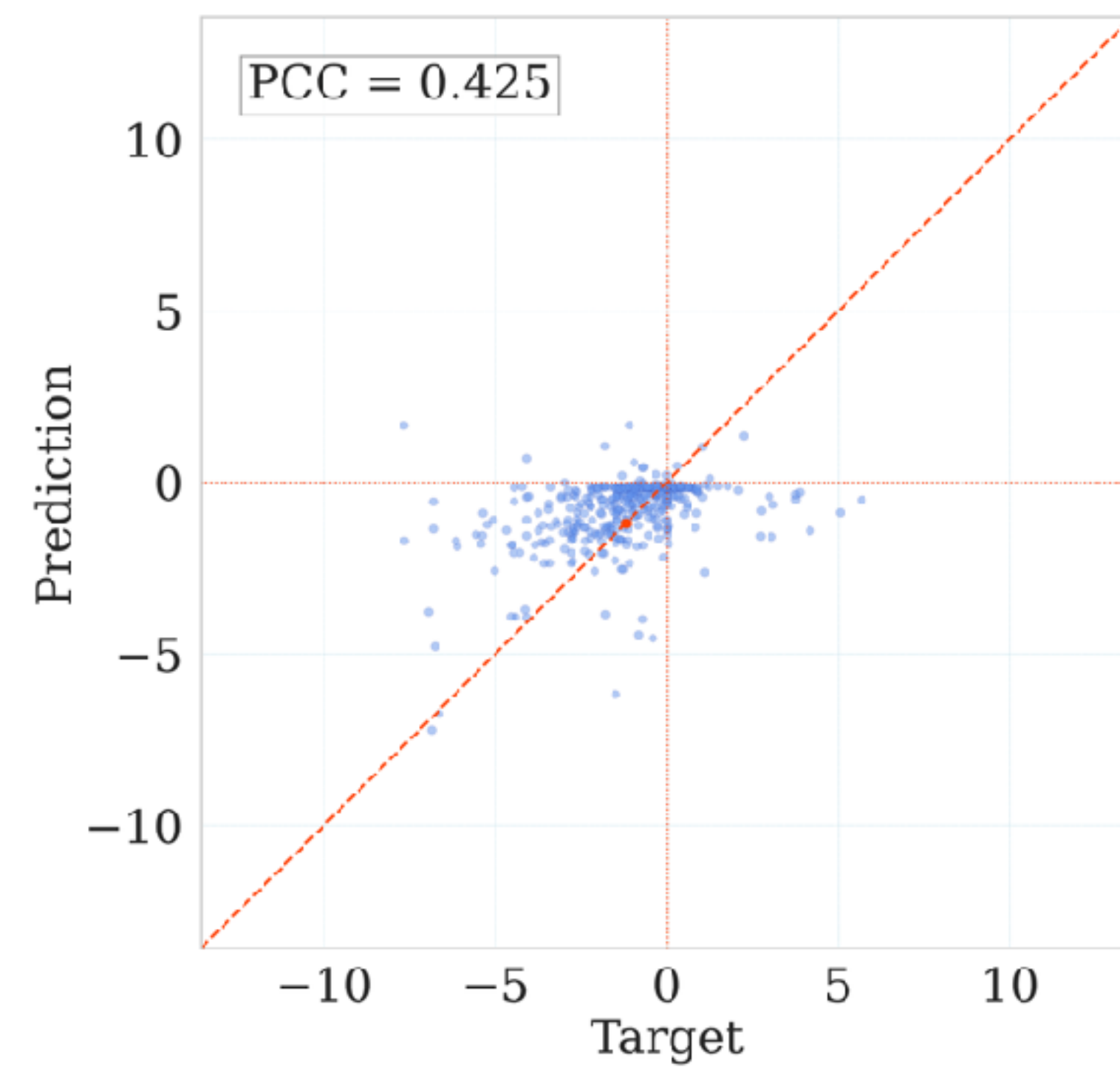
Training: 0 1 4



Validation: 2



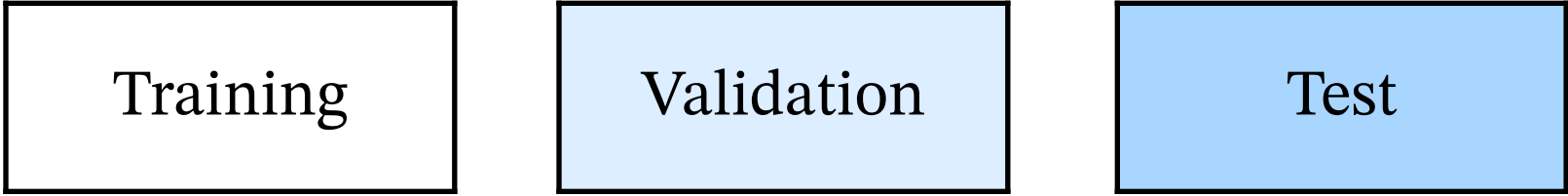
Test: 3



BEST

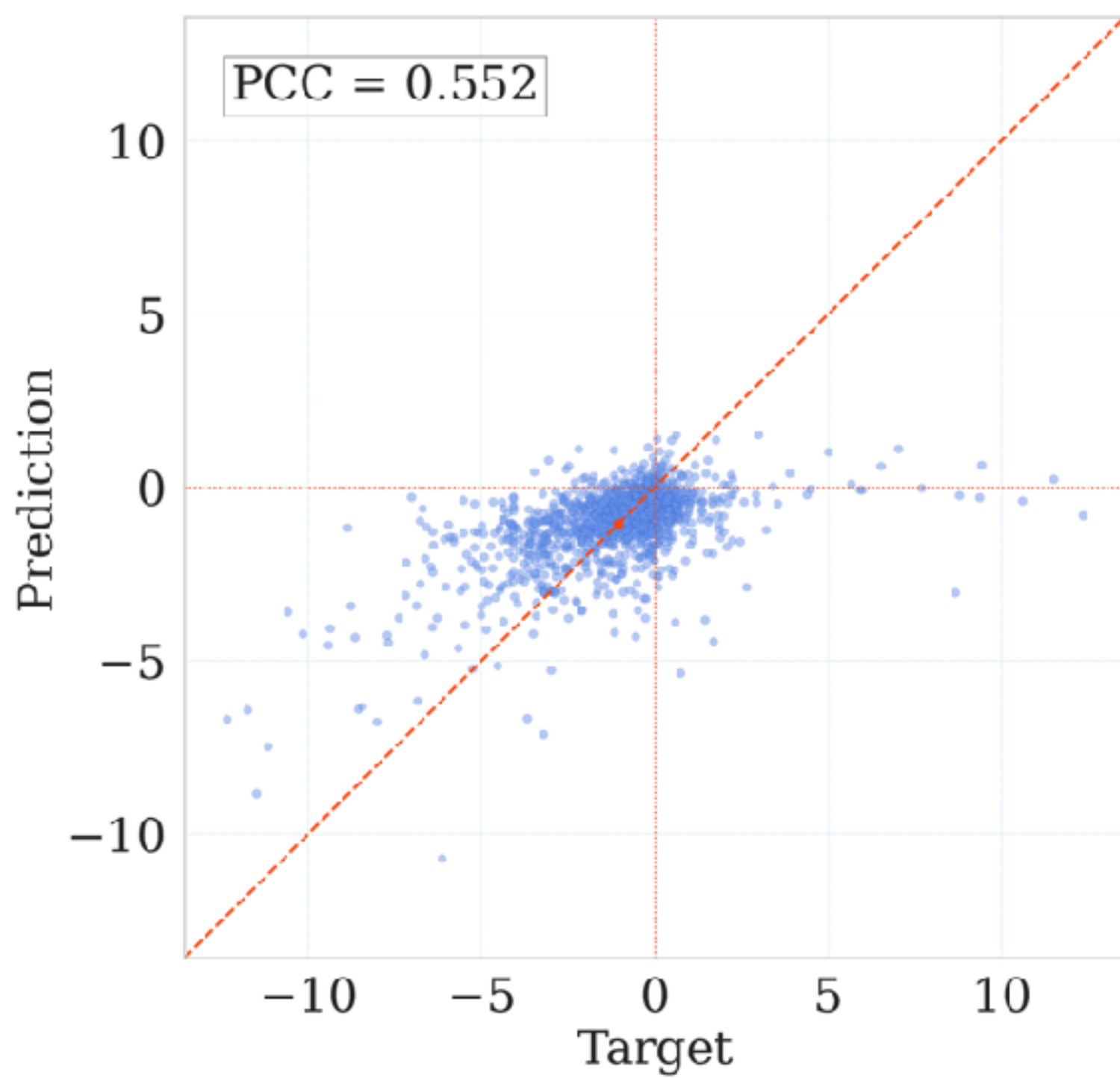
Run 6

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

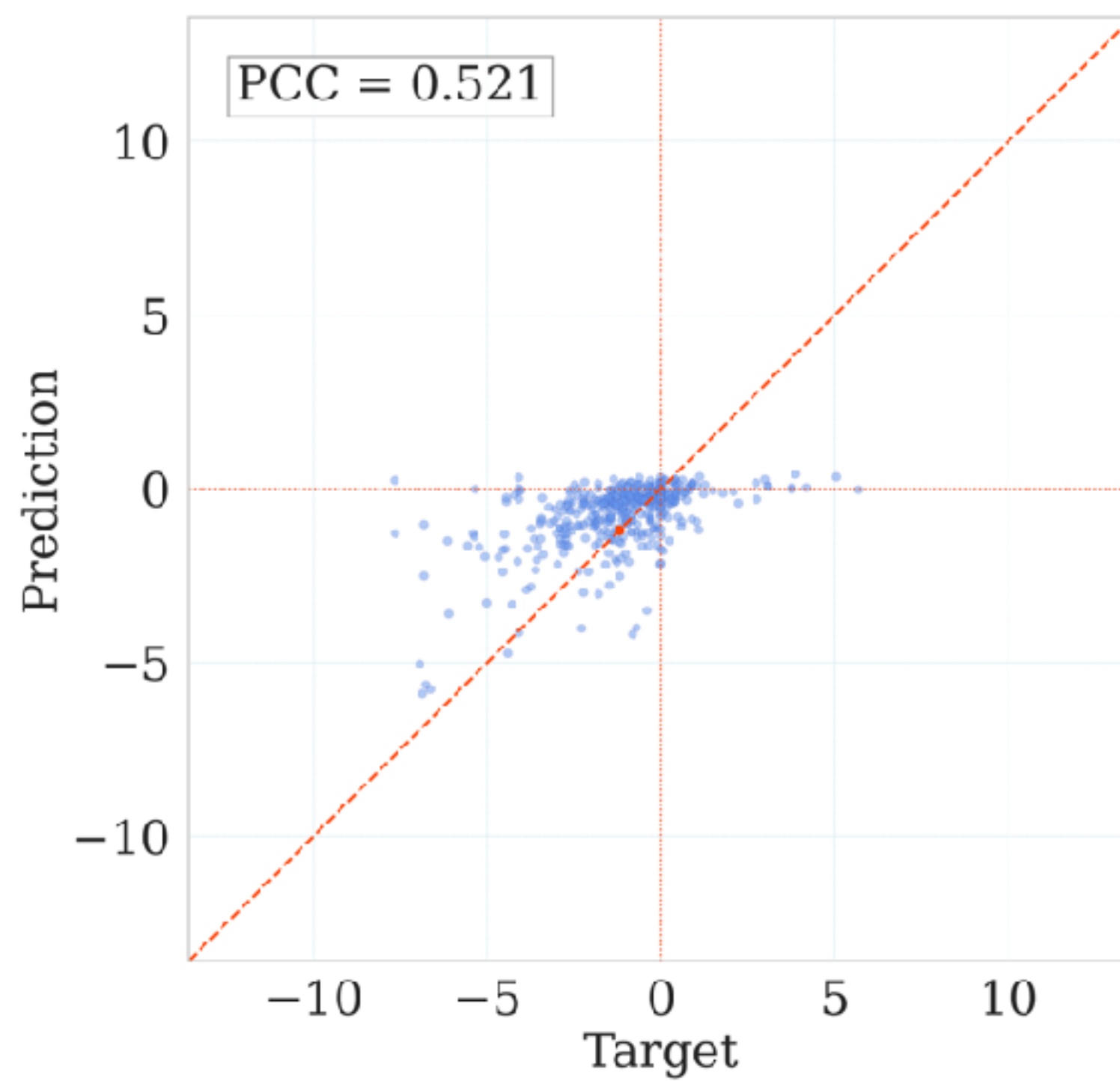


Run 6
Best Epoch: 46

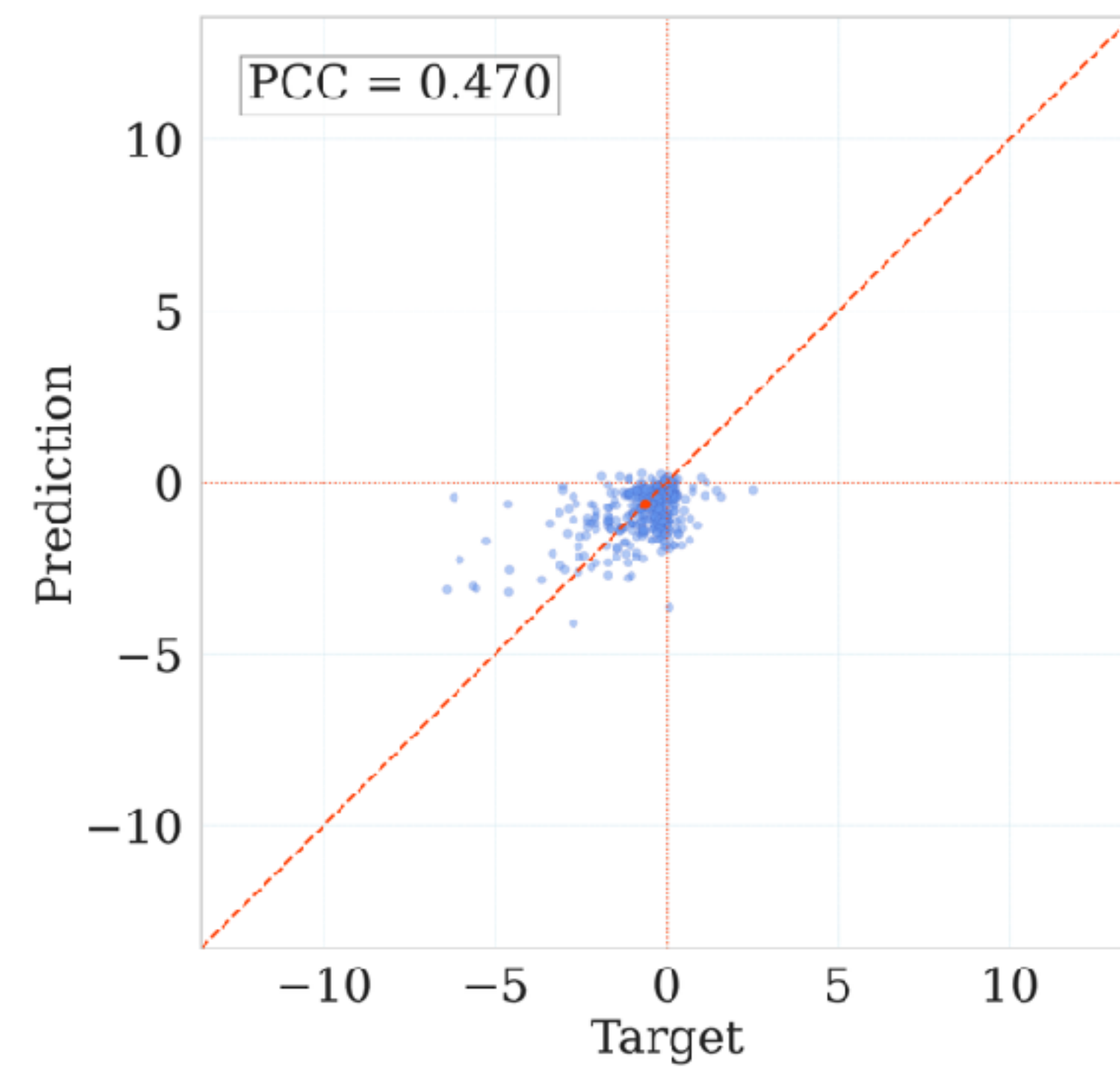
Training: 0 1 4



Validation: 3



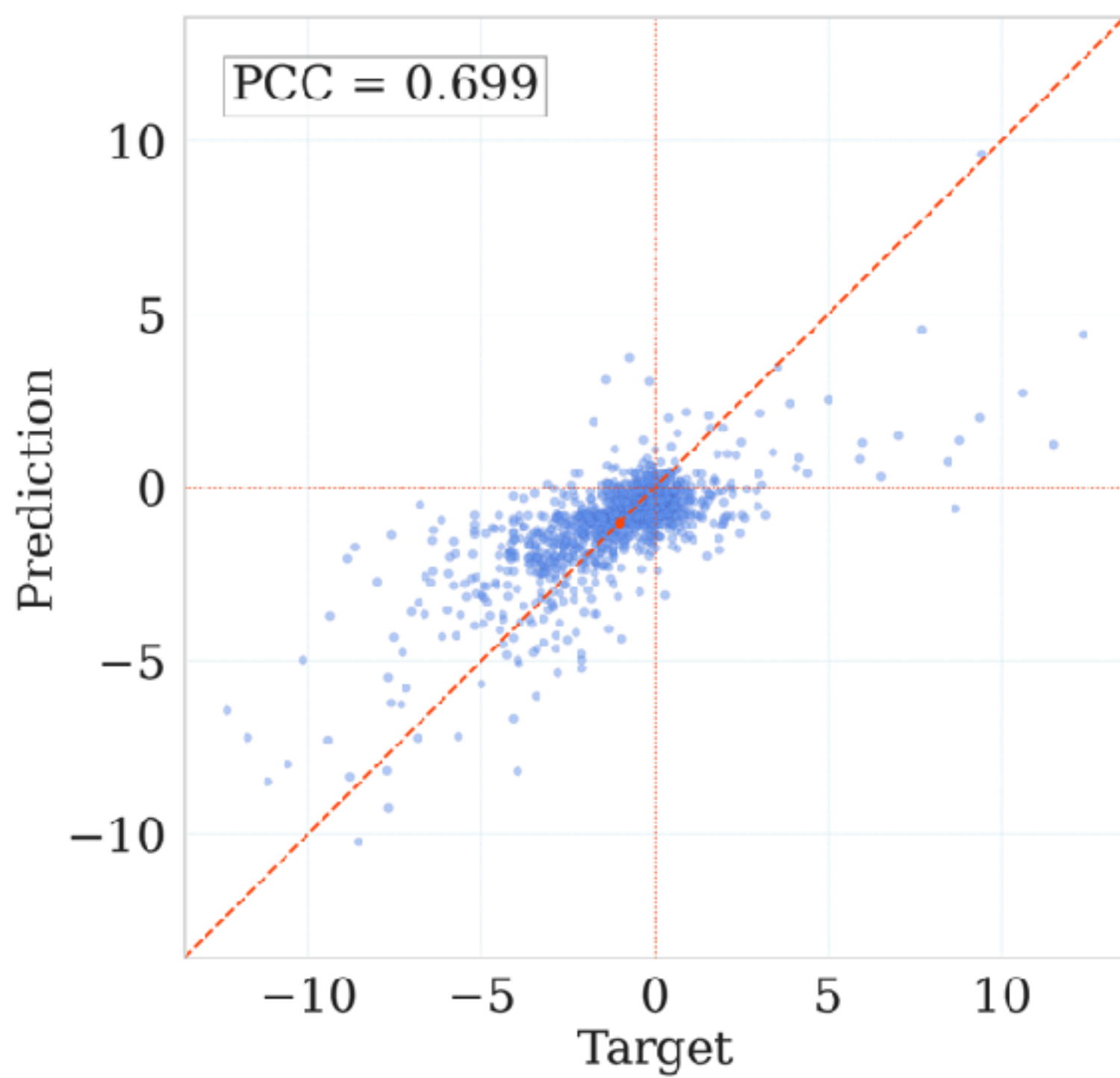
Test: 2



BEST

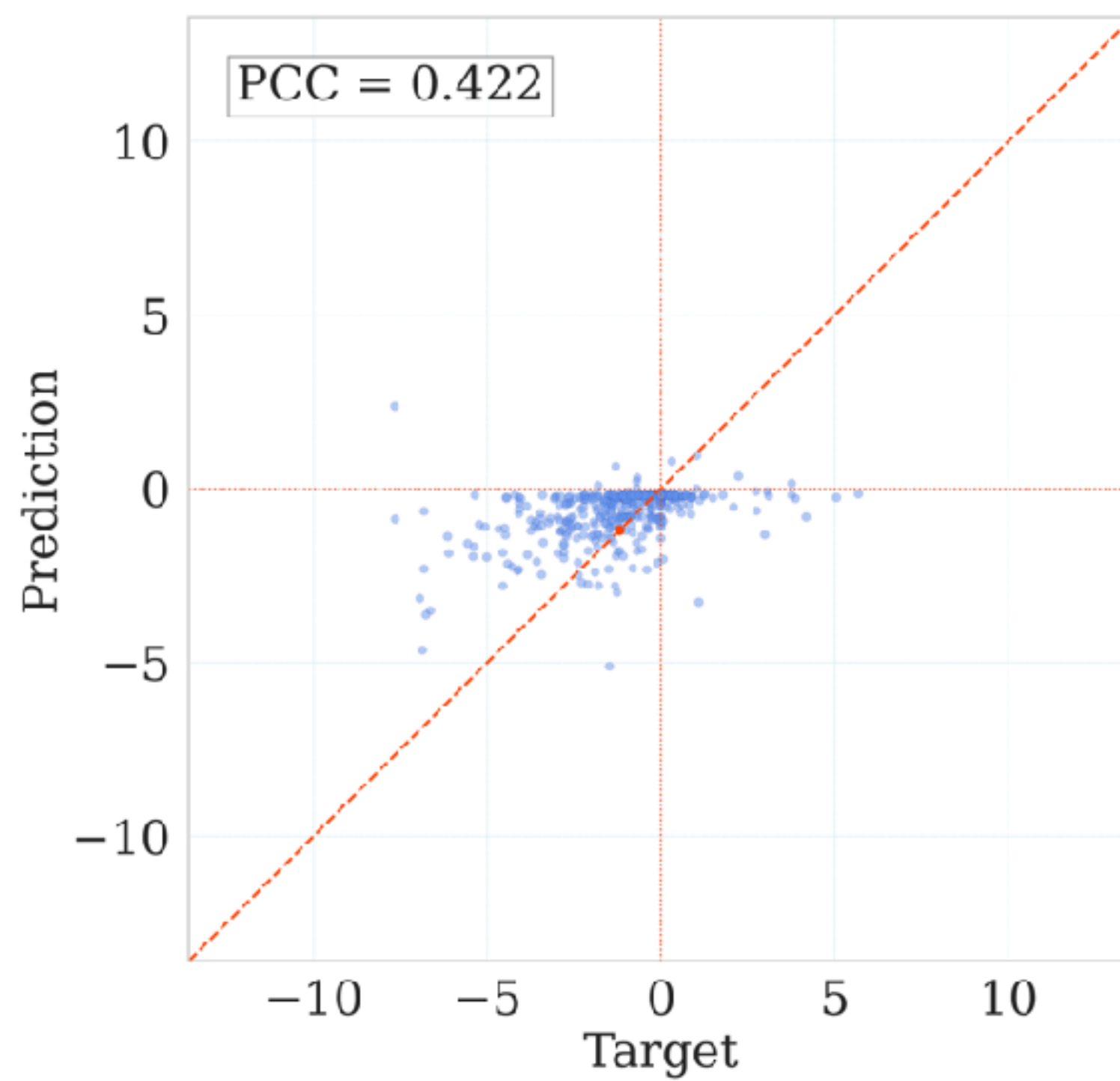
Run 6
Last Epoch: 766

Training: 0 1 4

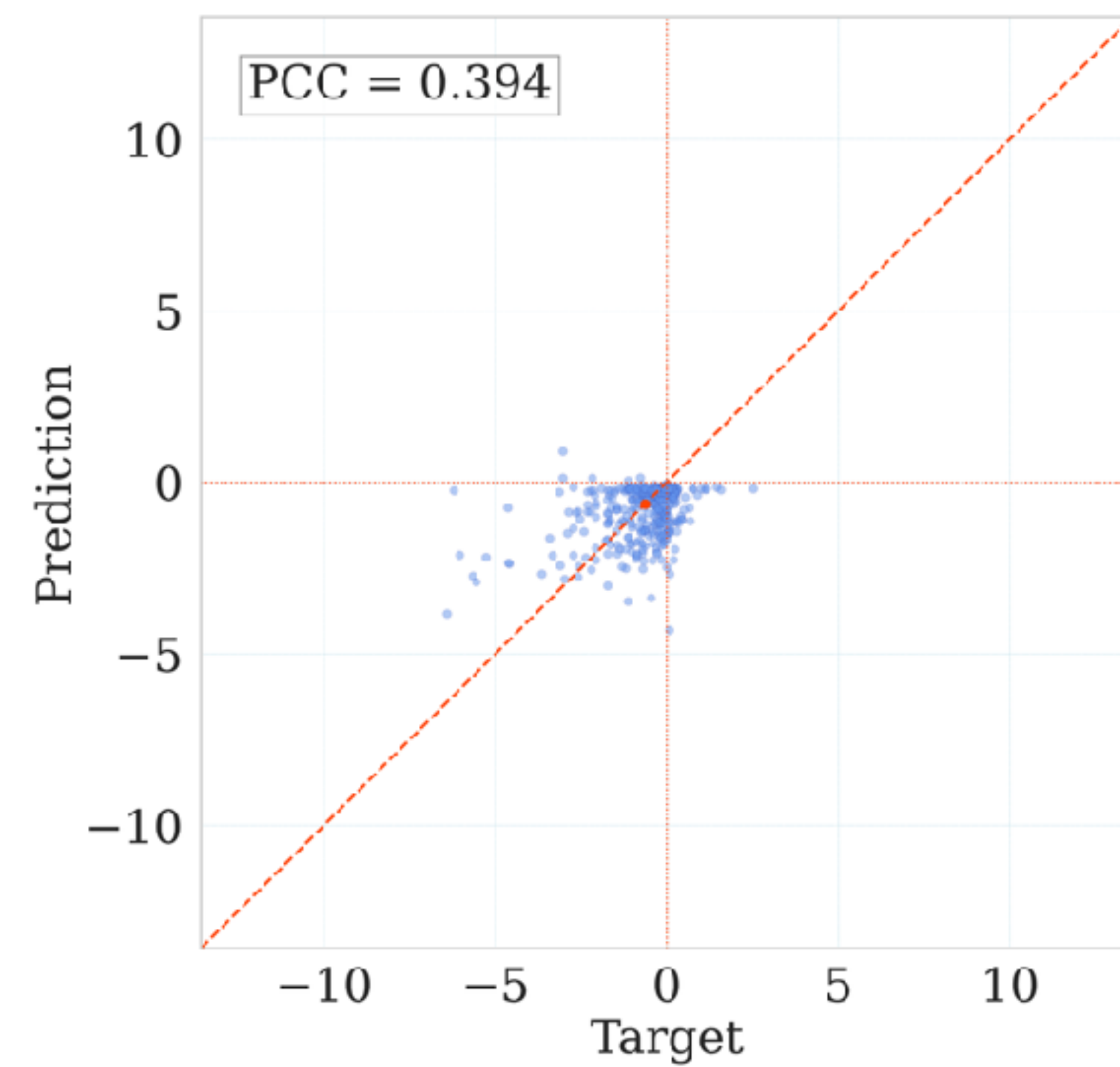


WORST

Validation: 3

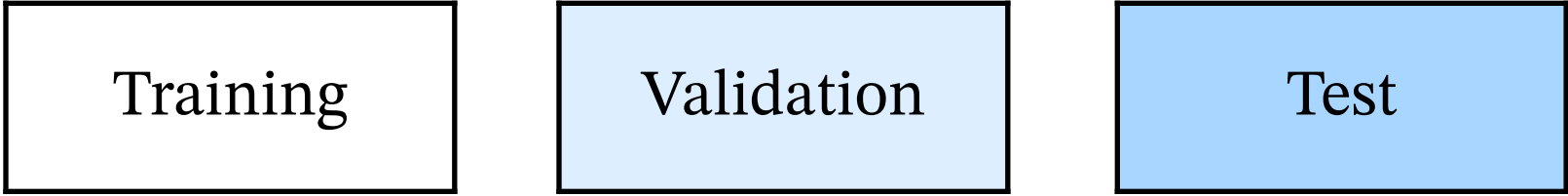


Test: 2



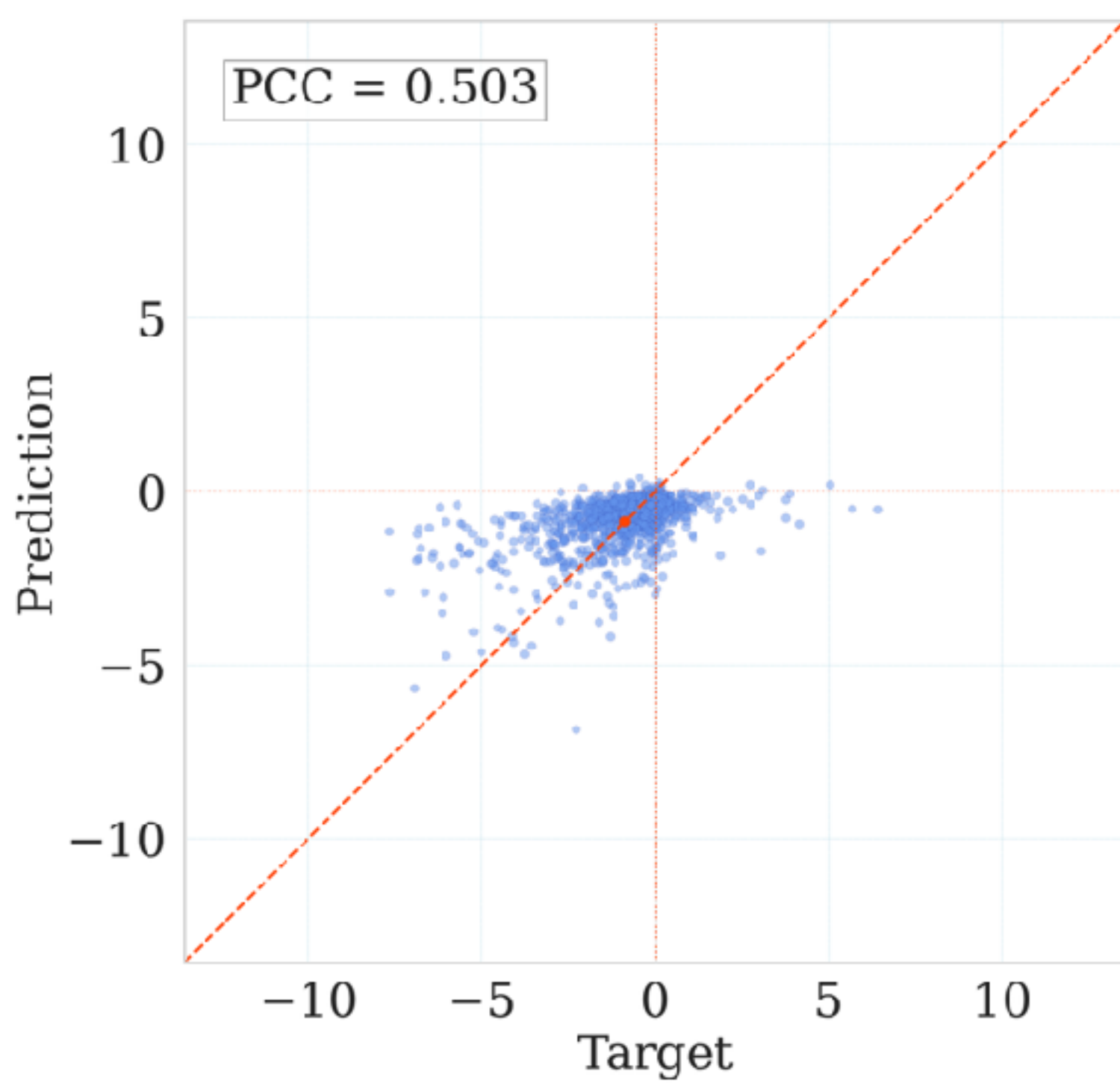
Run 7

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

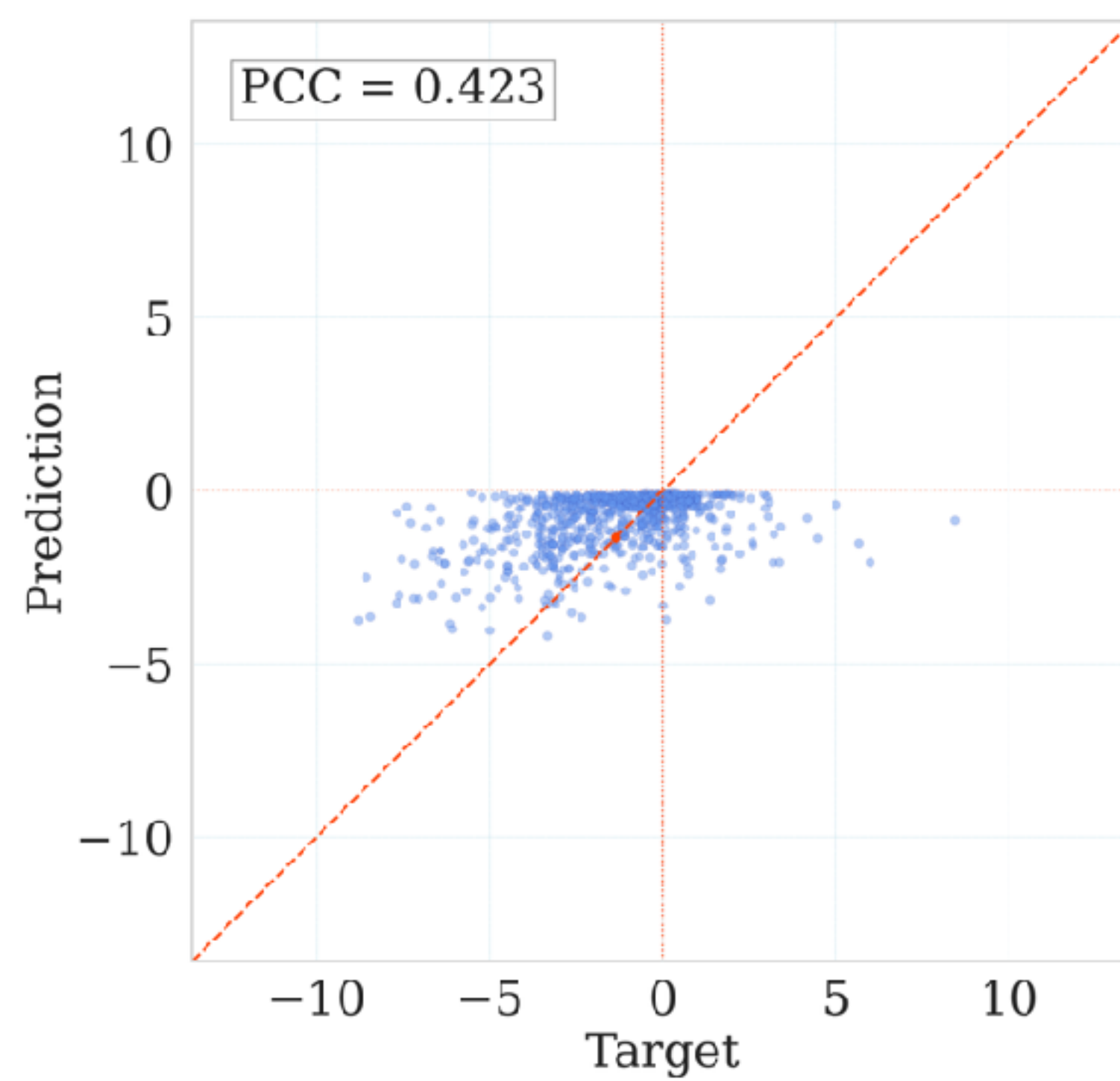


Run 7
Best Epoch: 28

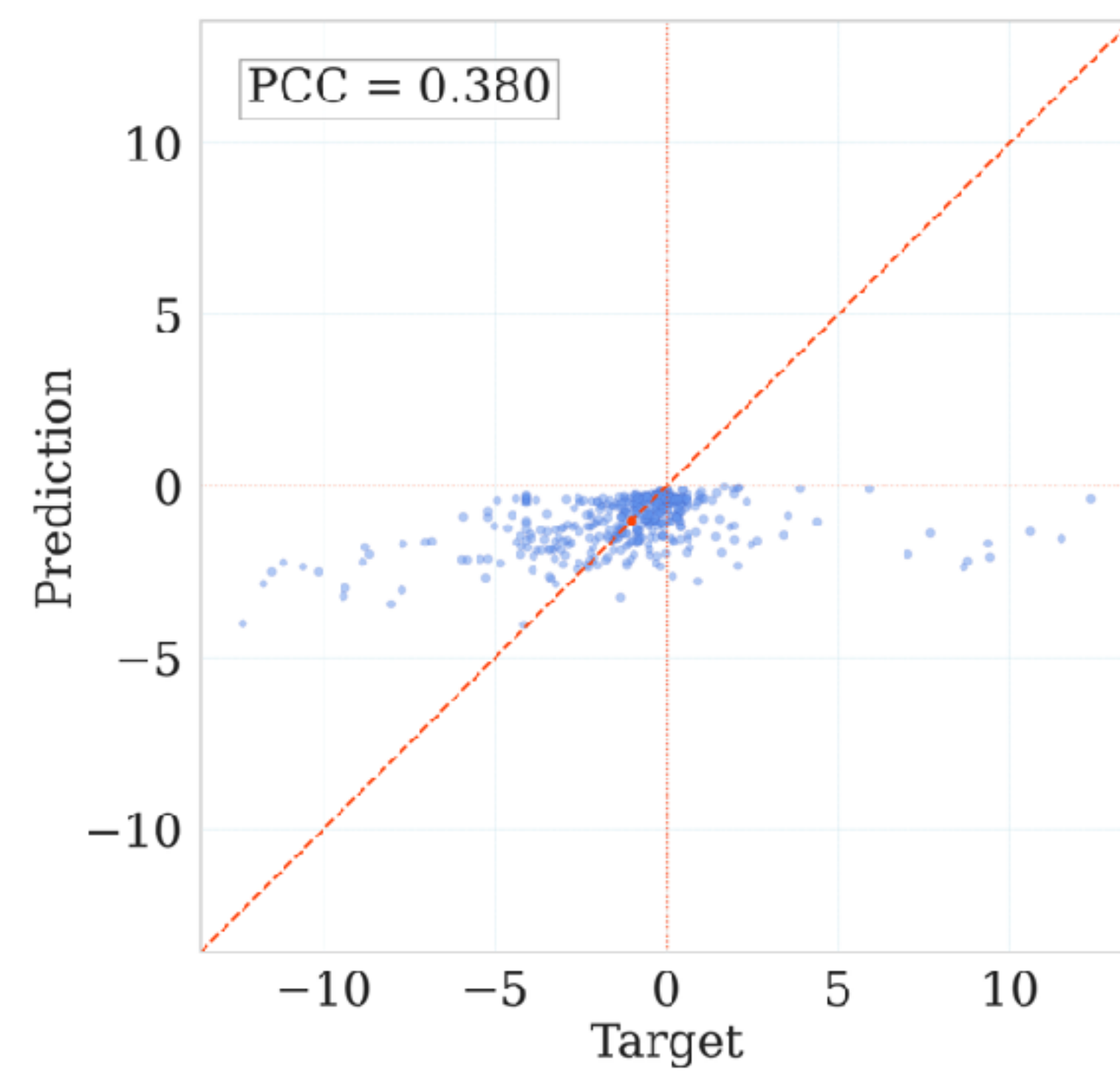
Training: 0 2 3



Validation: 1

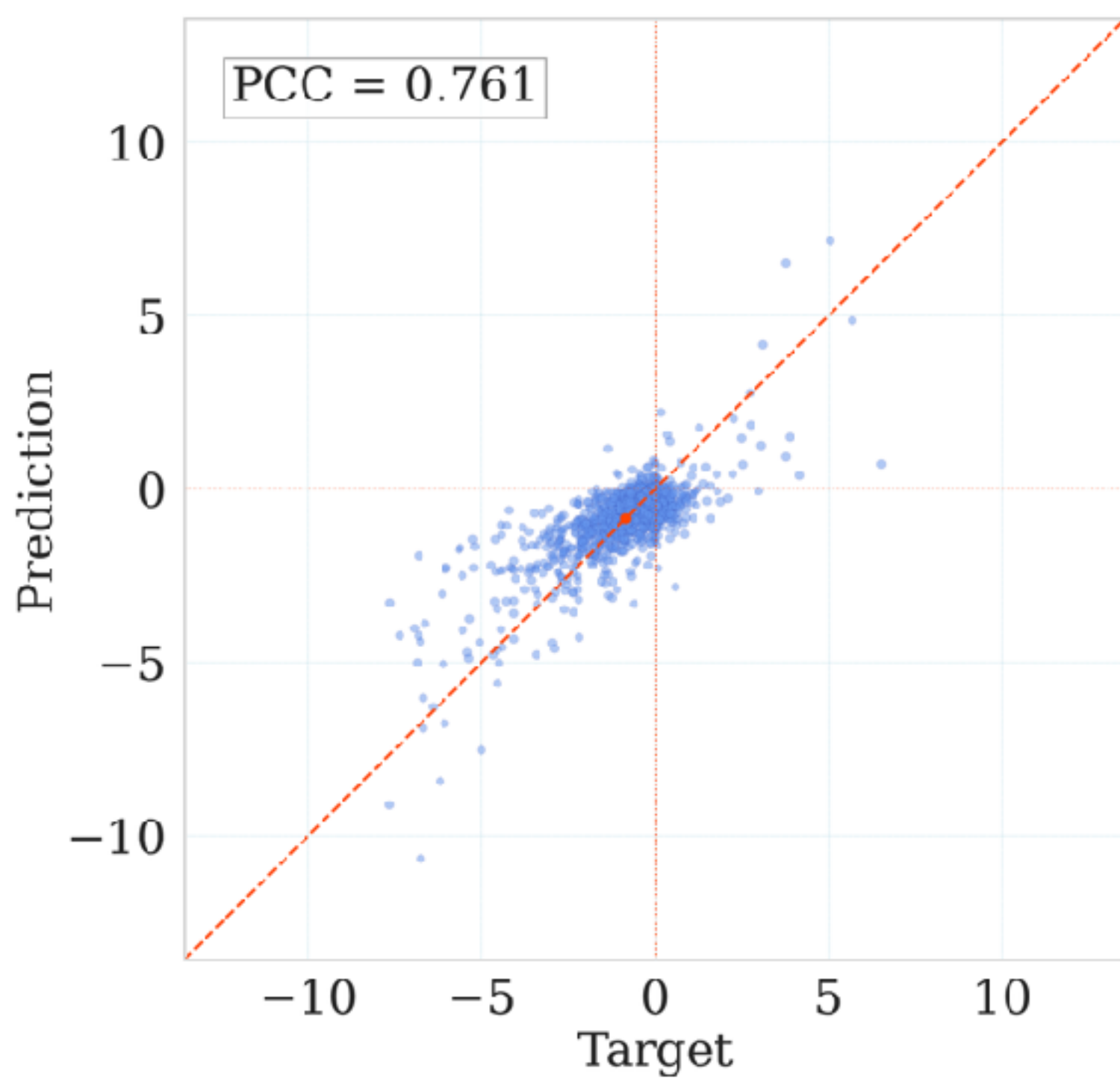


Test: 4

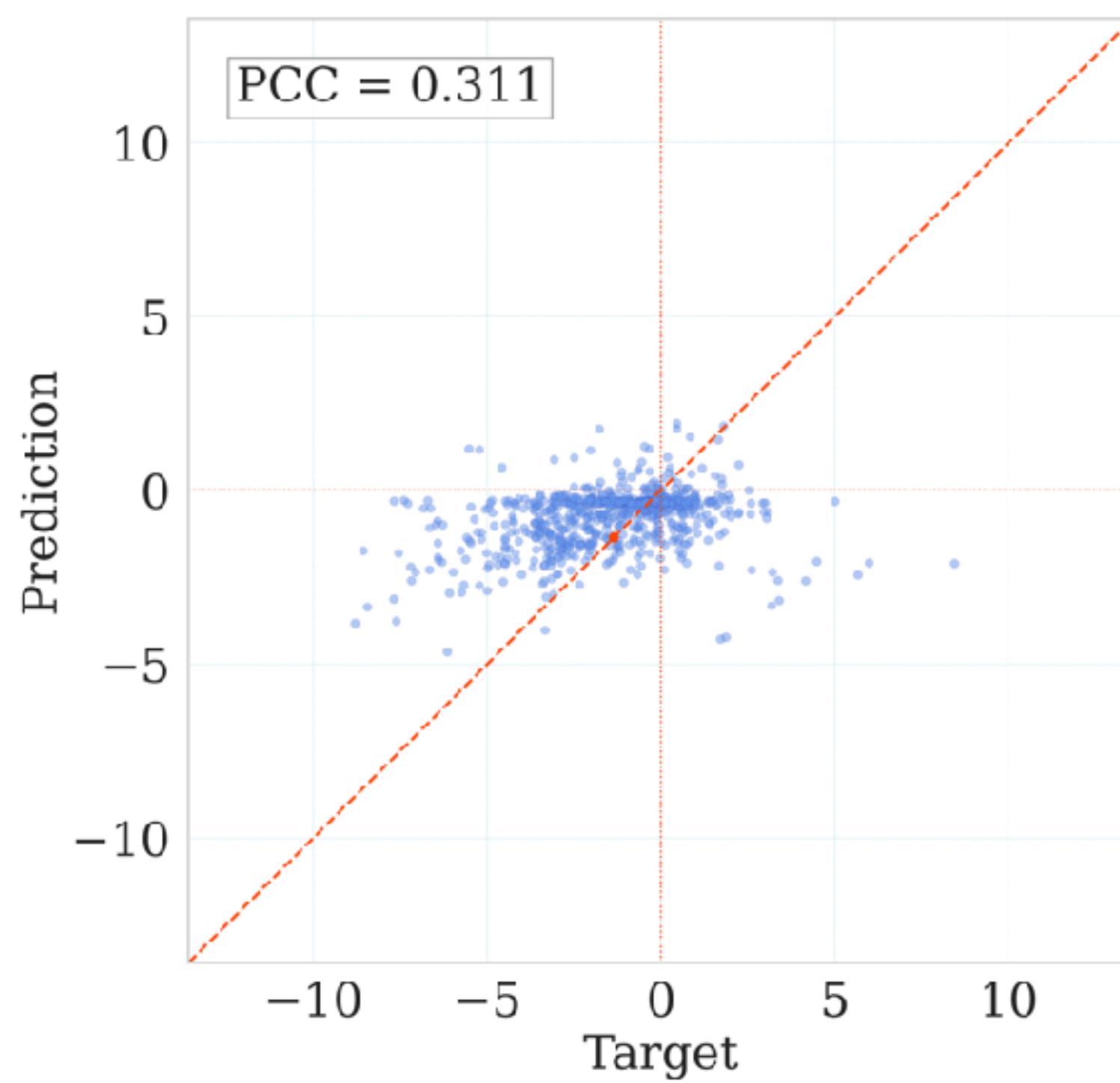


Run 7
Last Epoch: 601

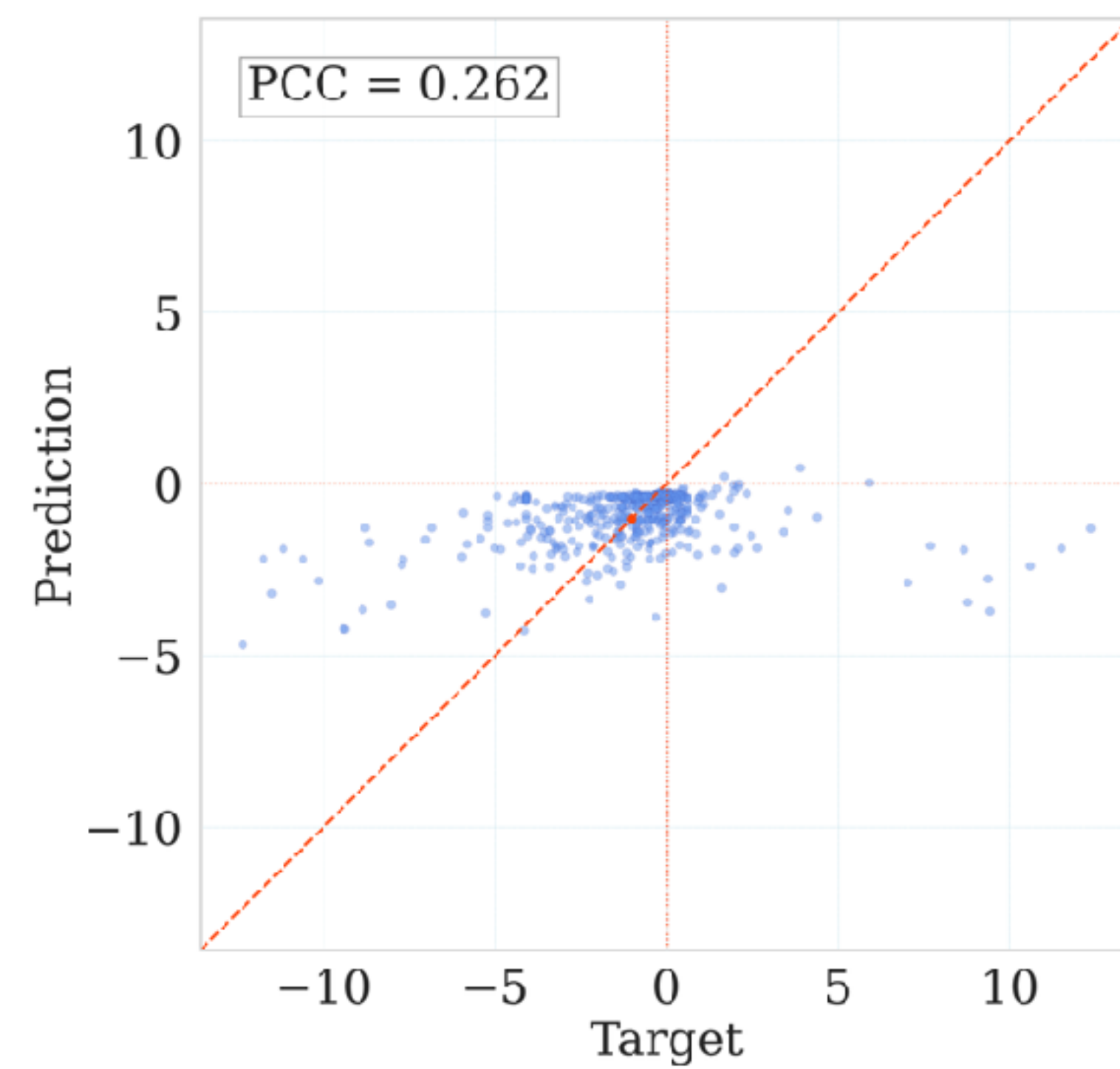
Training: 0 2 3



Validation: 1

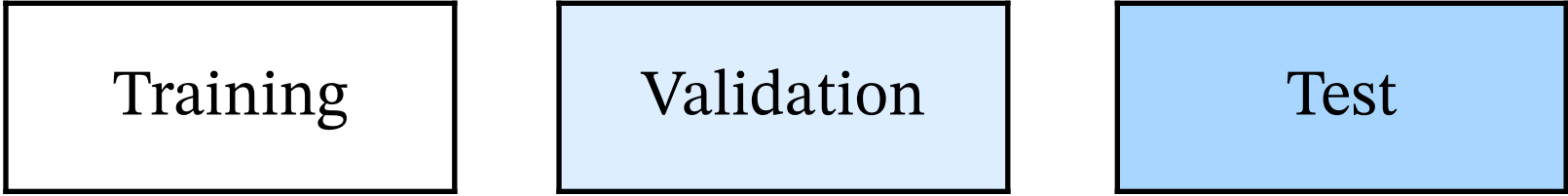


Test: 4



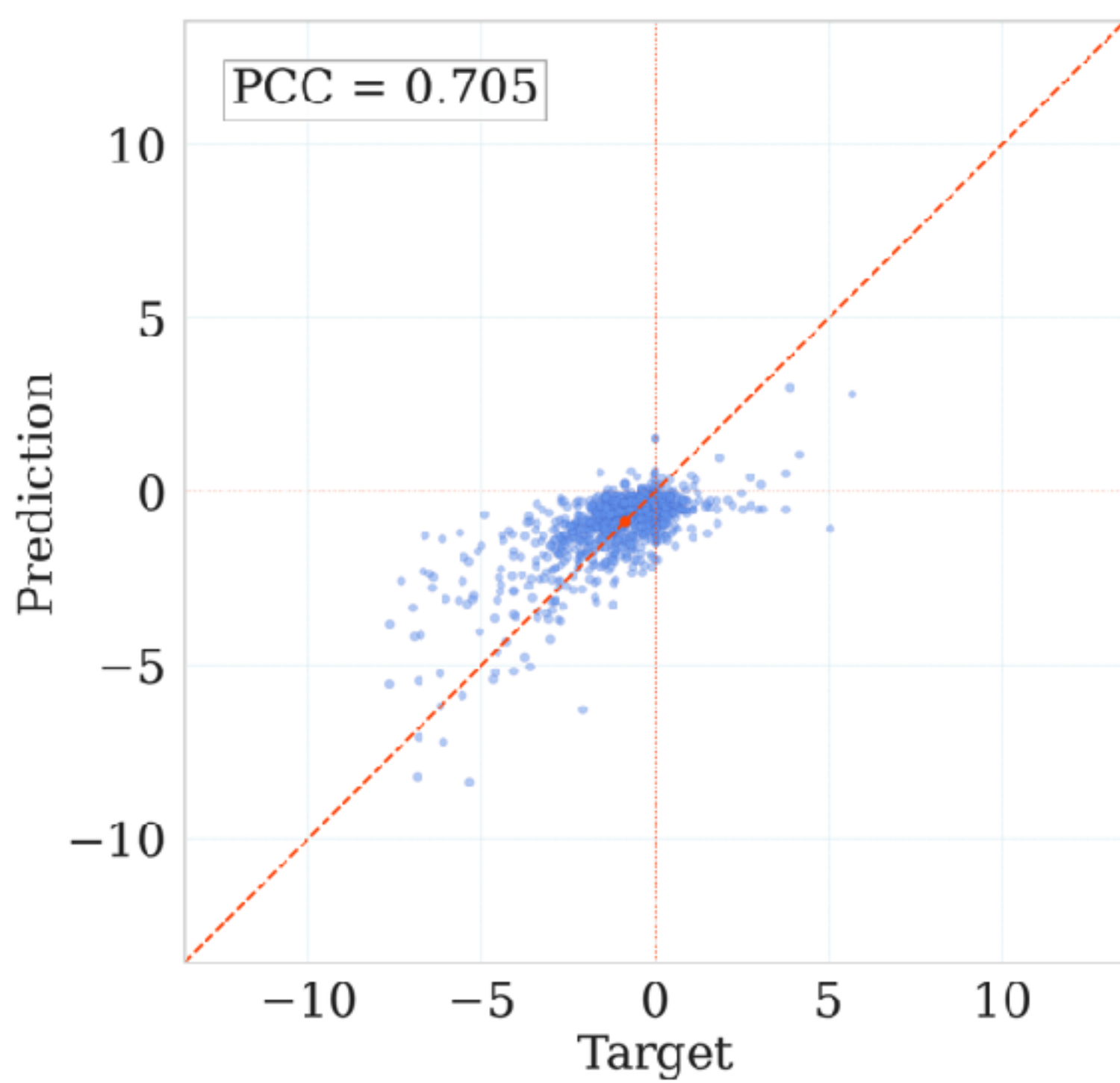
Run 8

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

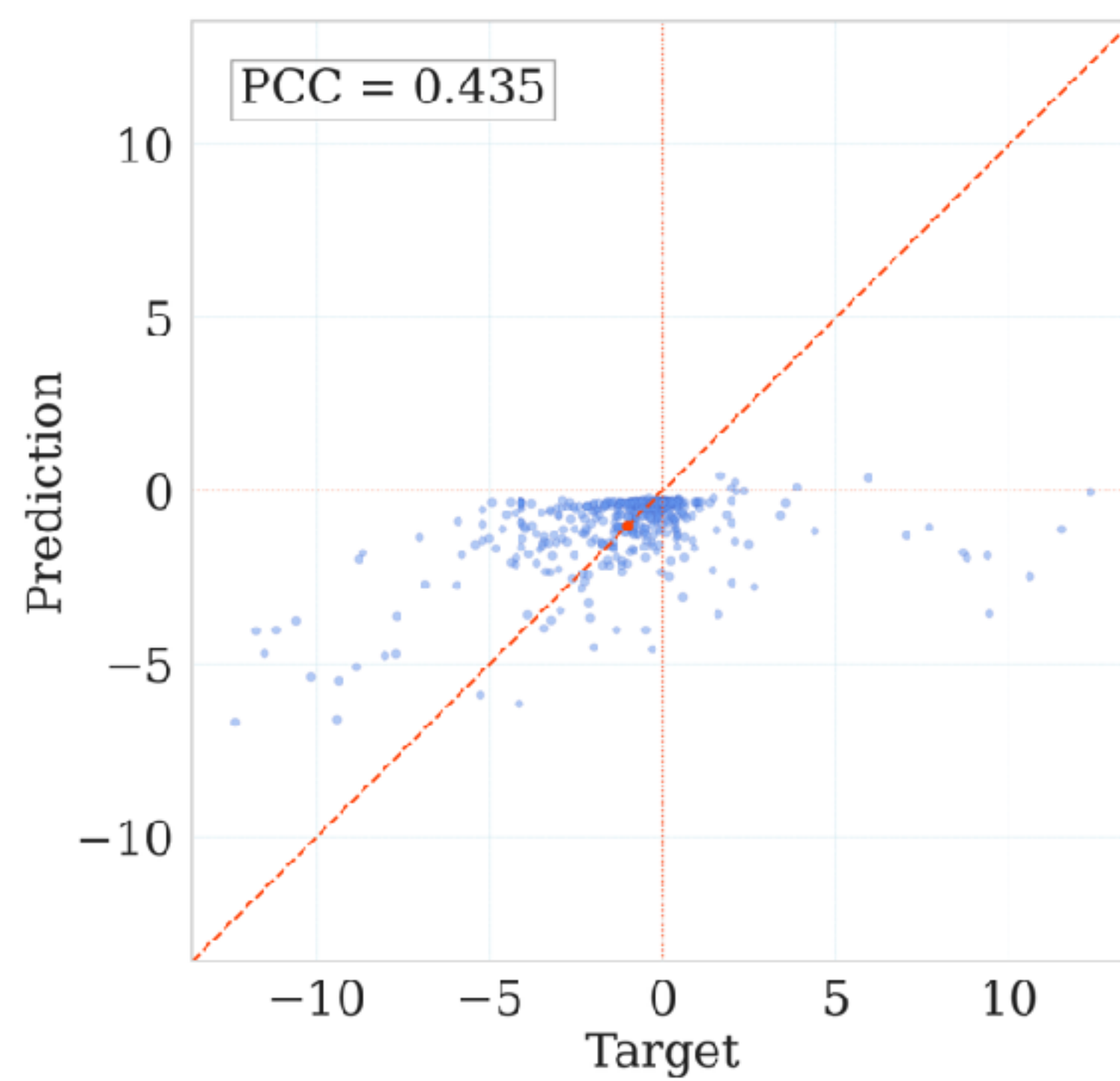


Run 8
Best Epoch: 248

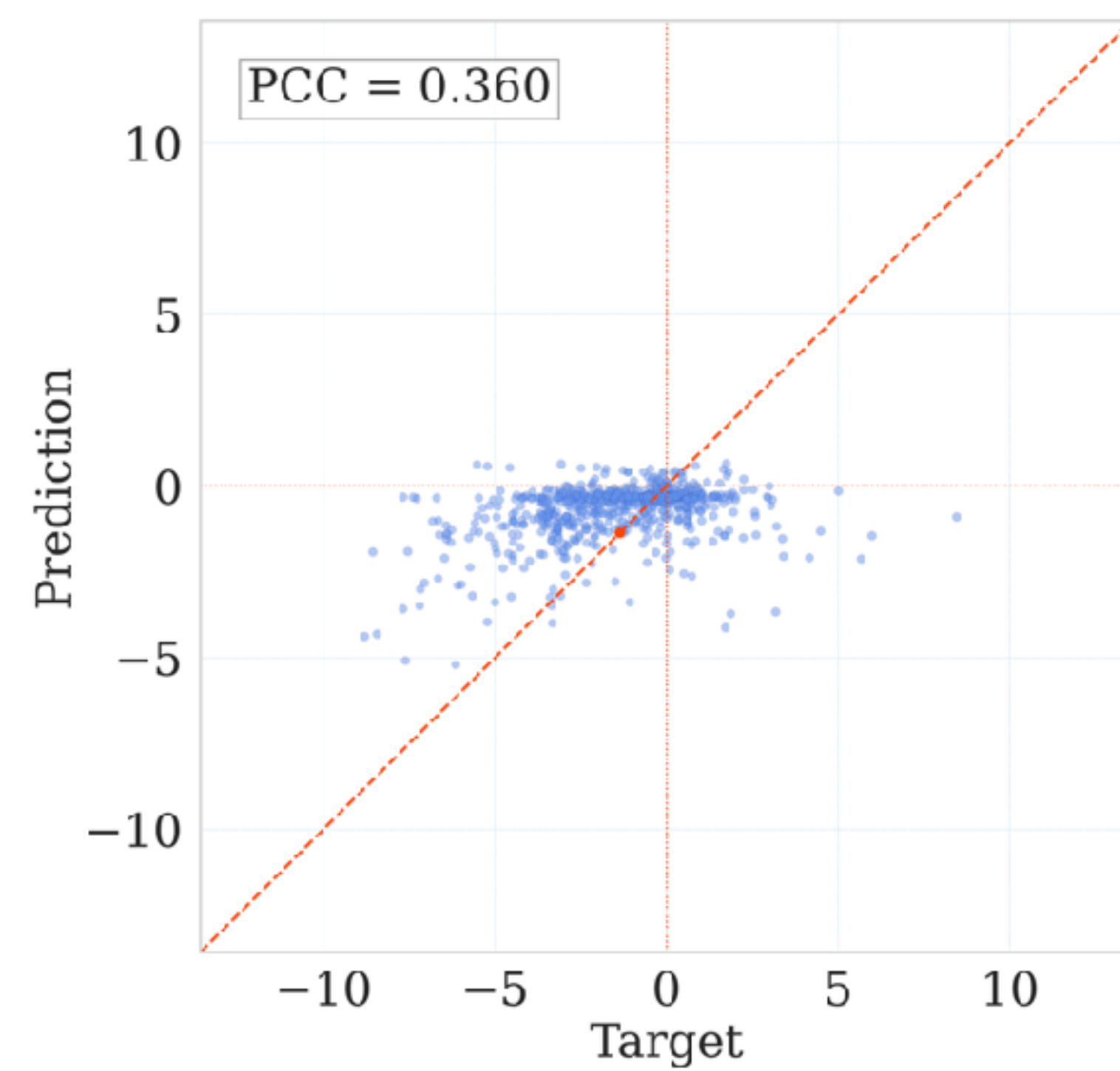
Training: 0 2 3



Validation: 4

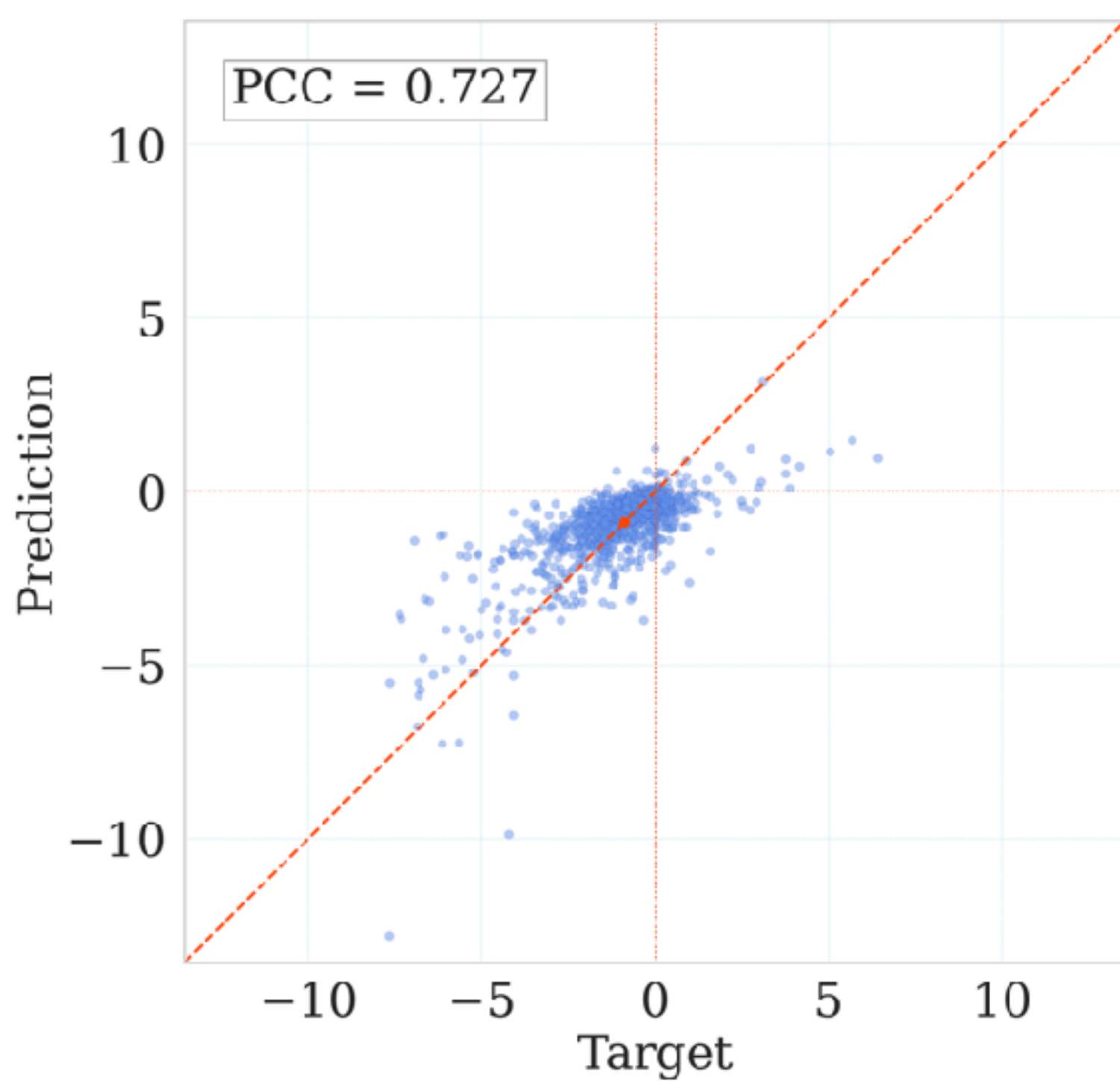


Test: 1

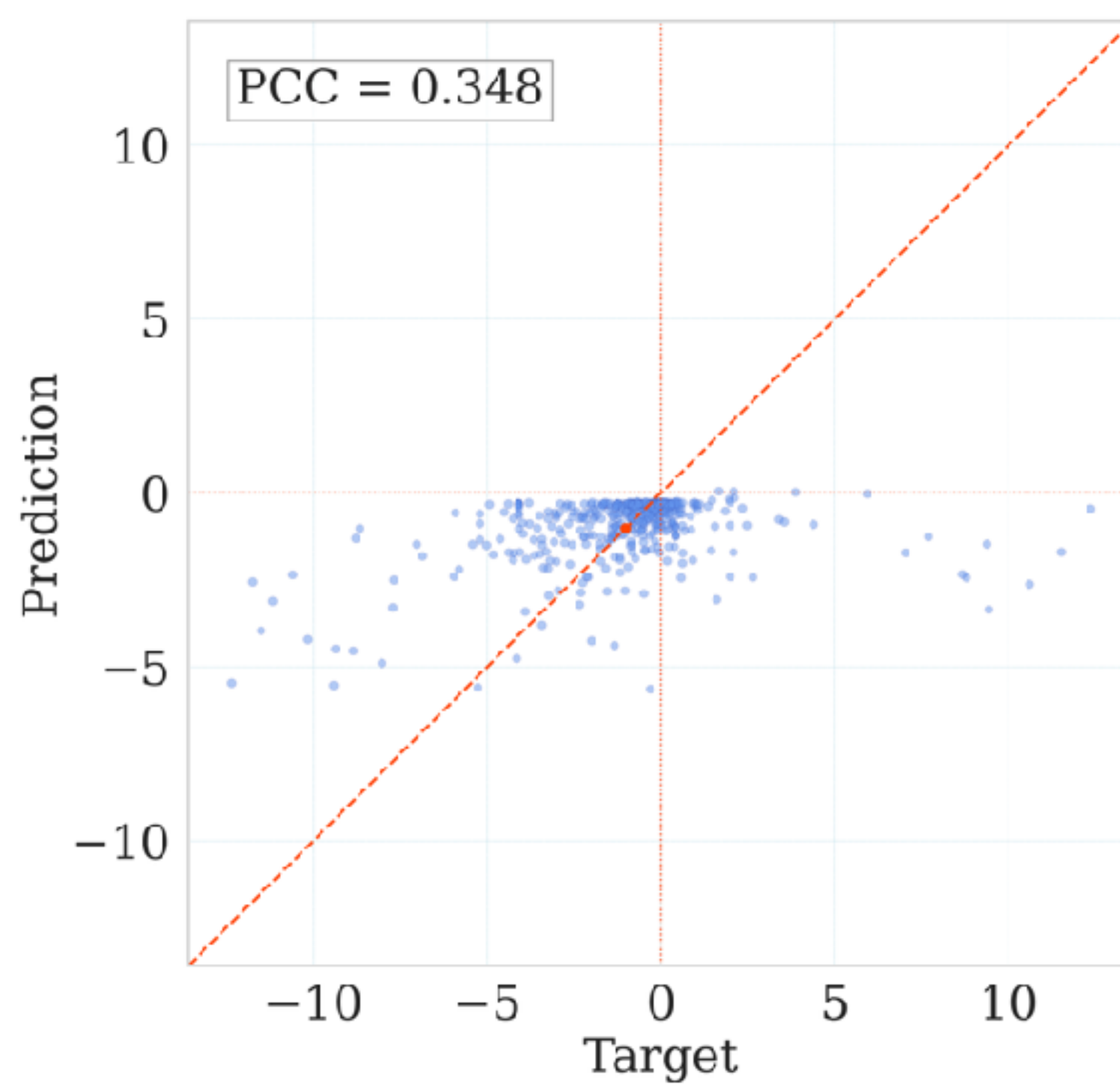


Run 8
Last Epoch: 638

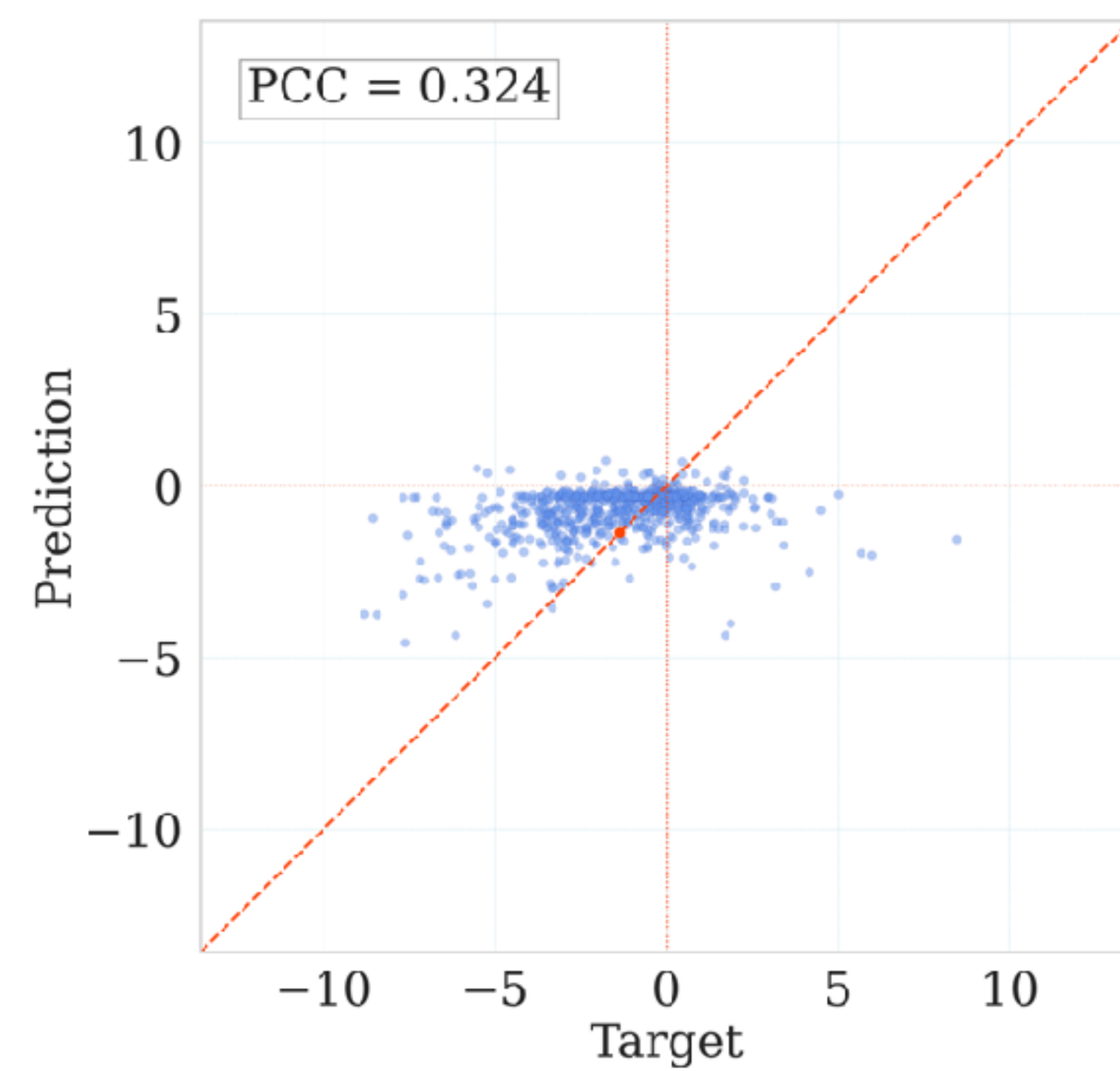
Training: 0 2 3



Validation: 4

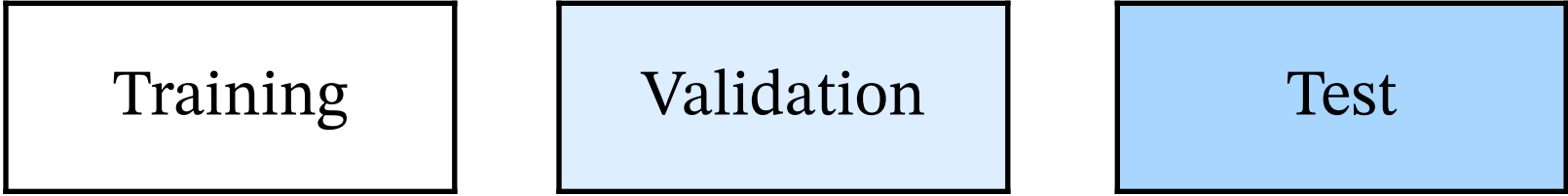


Test: 1



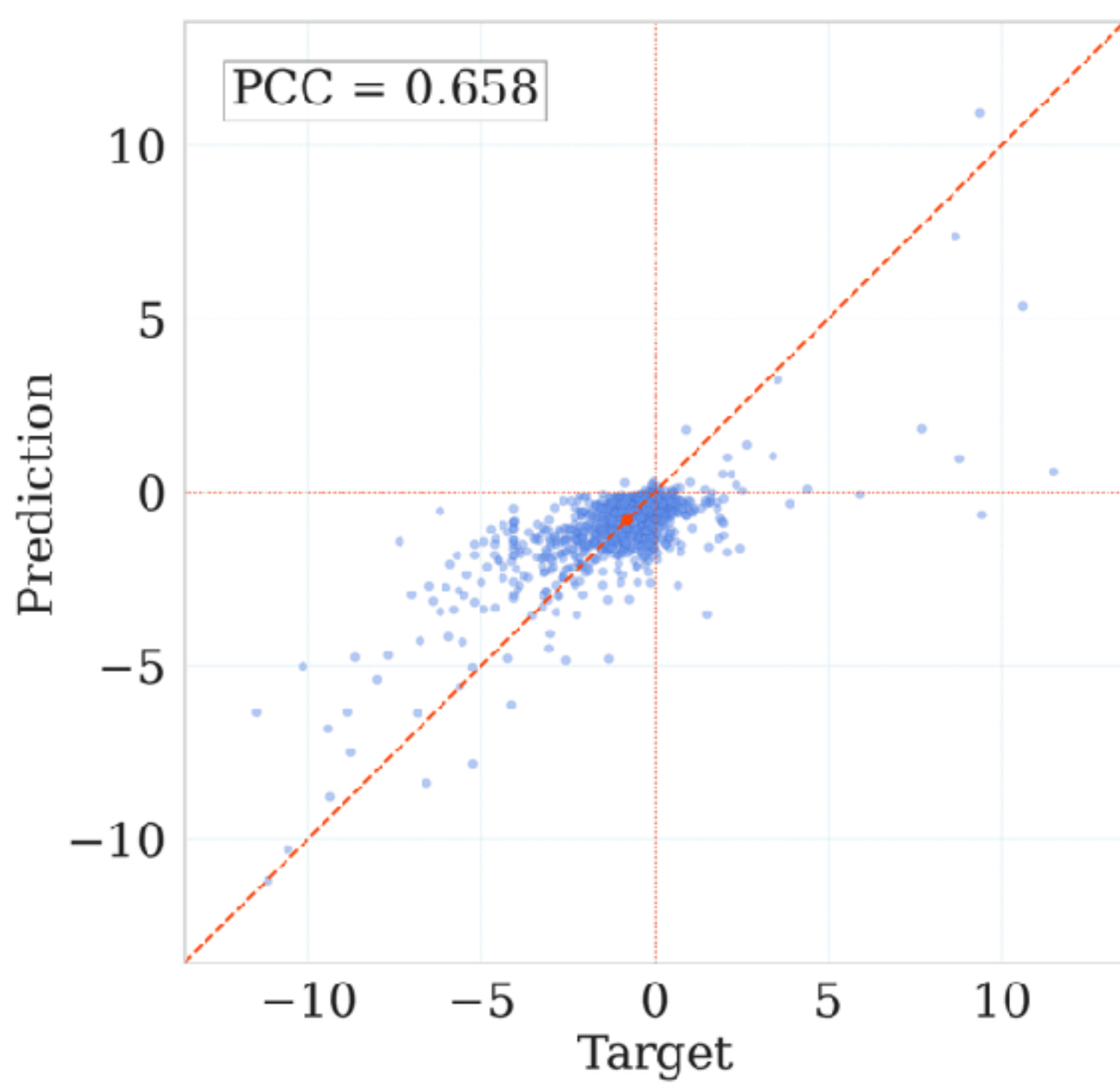
Run 9

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

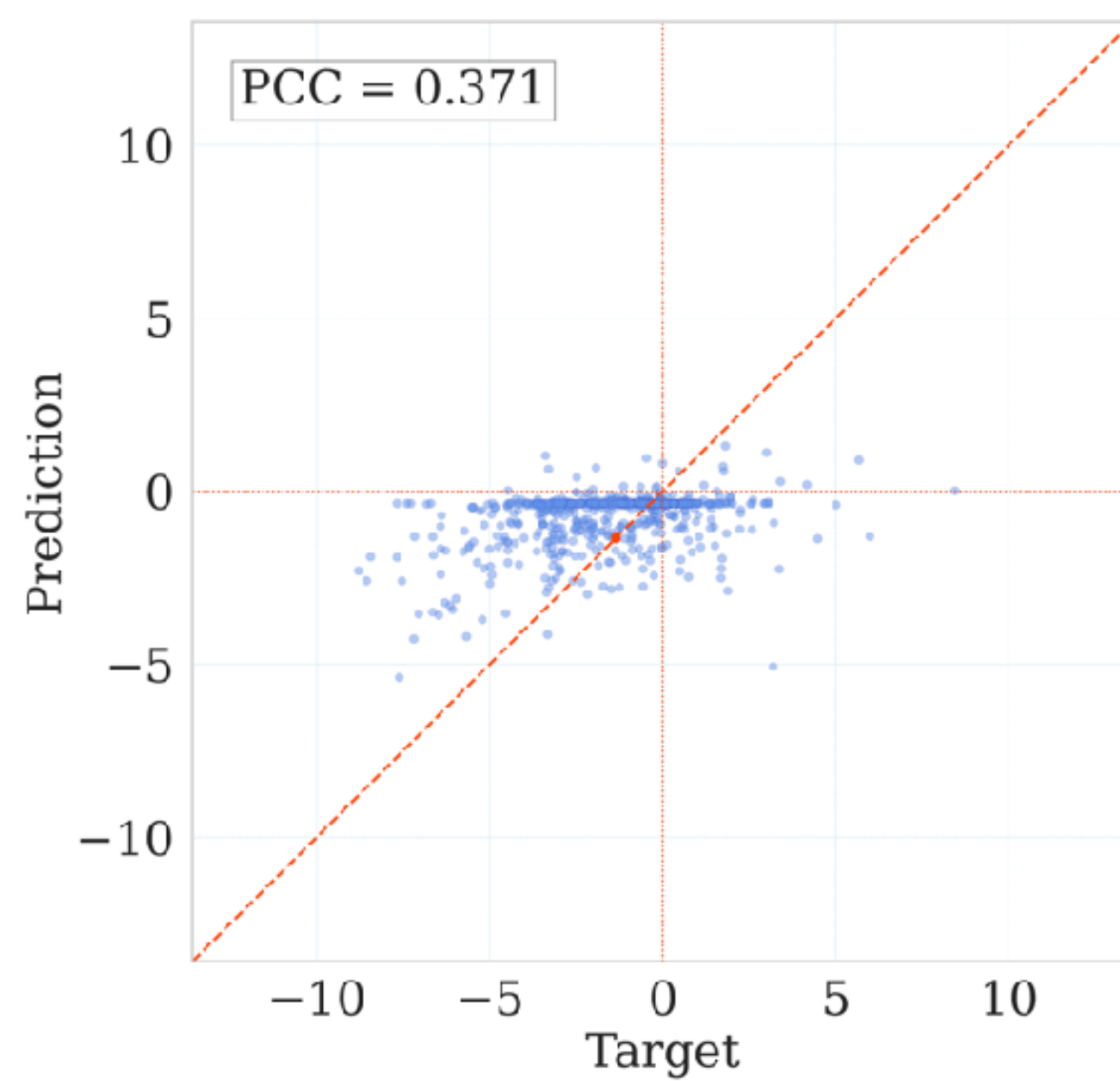


Run 9
Best Epoch: 467

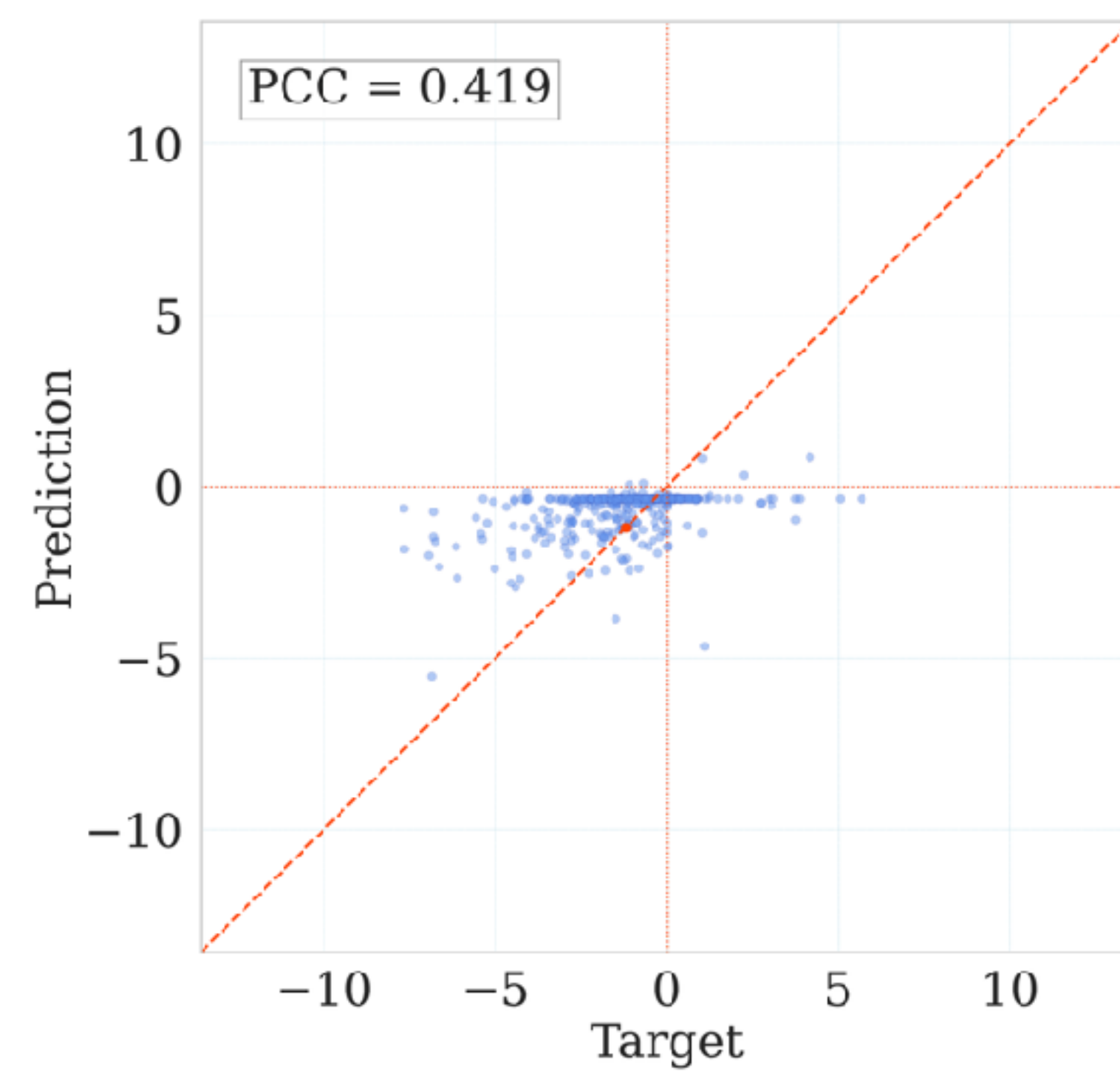
Training: 0 2 4



Validation: 1

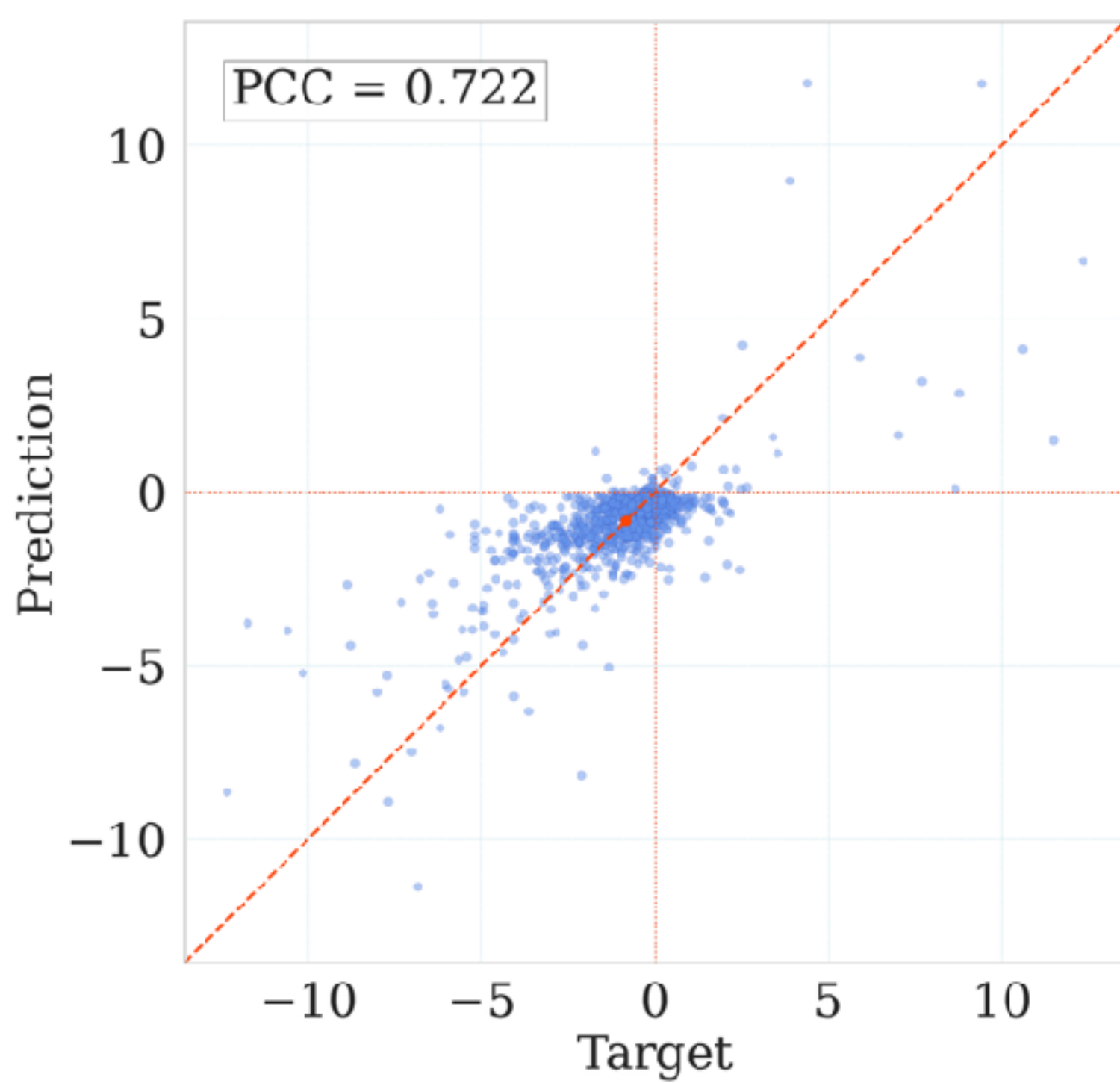


Test: 3

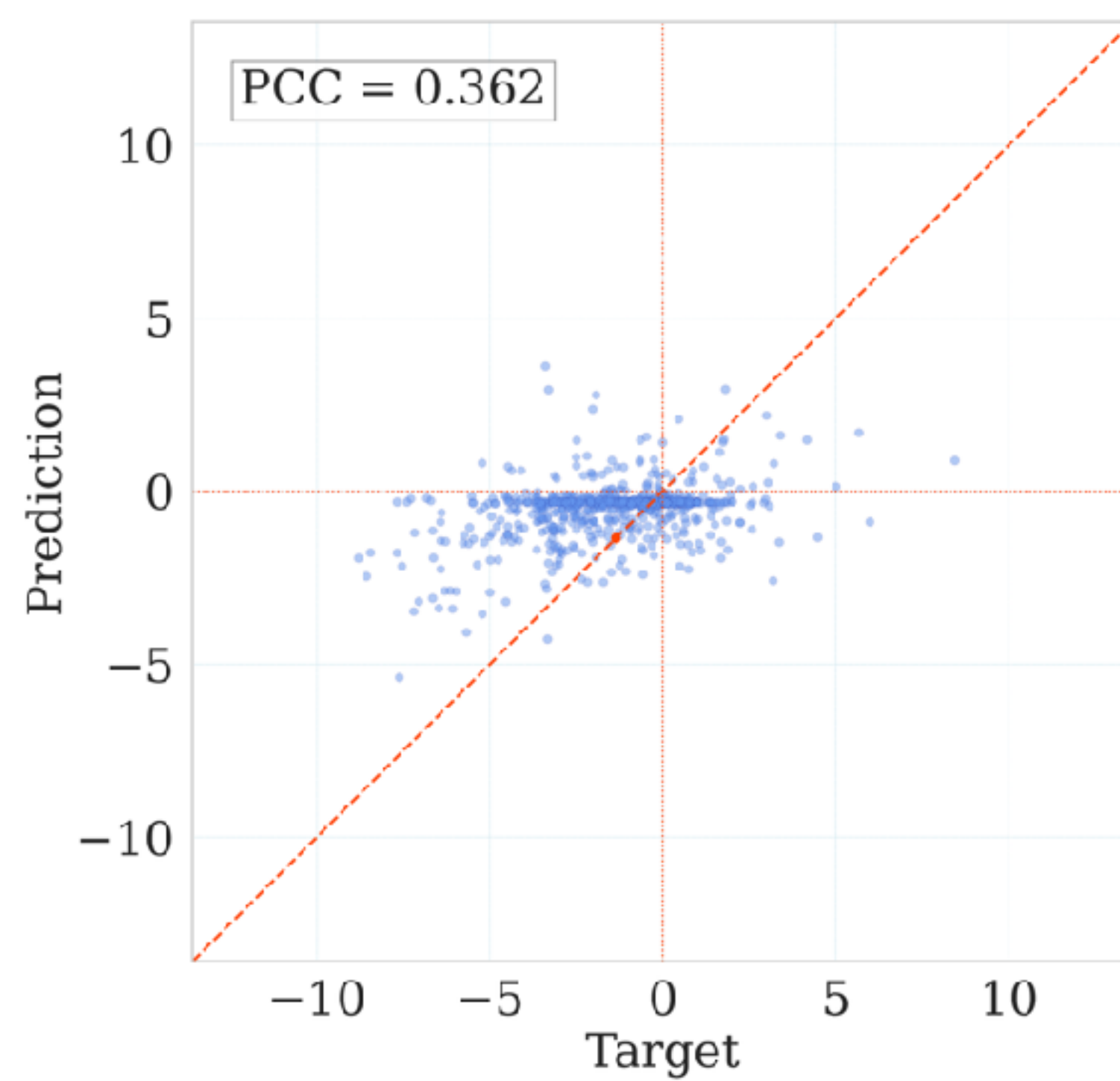


Run 9
Last Epoch: 661

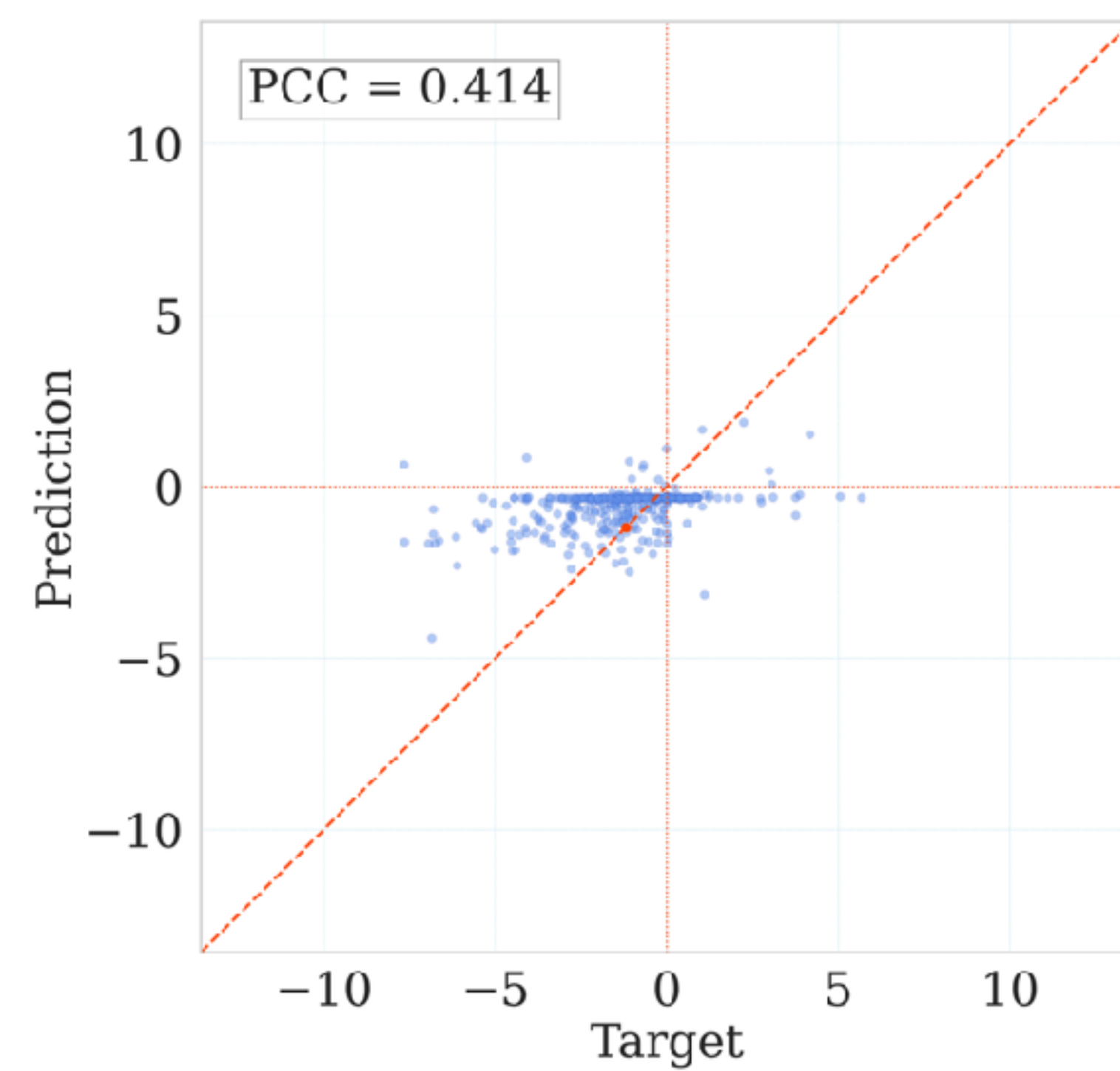
Training: 0 2 4



Validation: 1

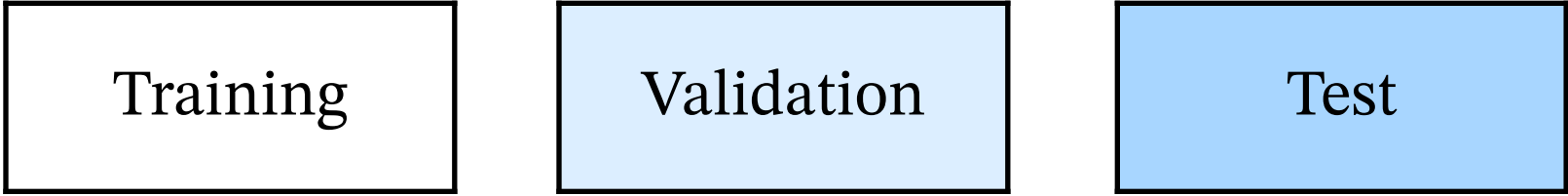


Test: 3



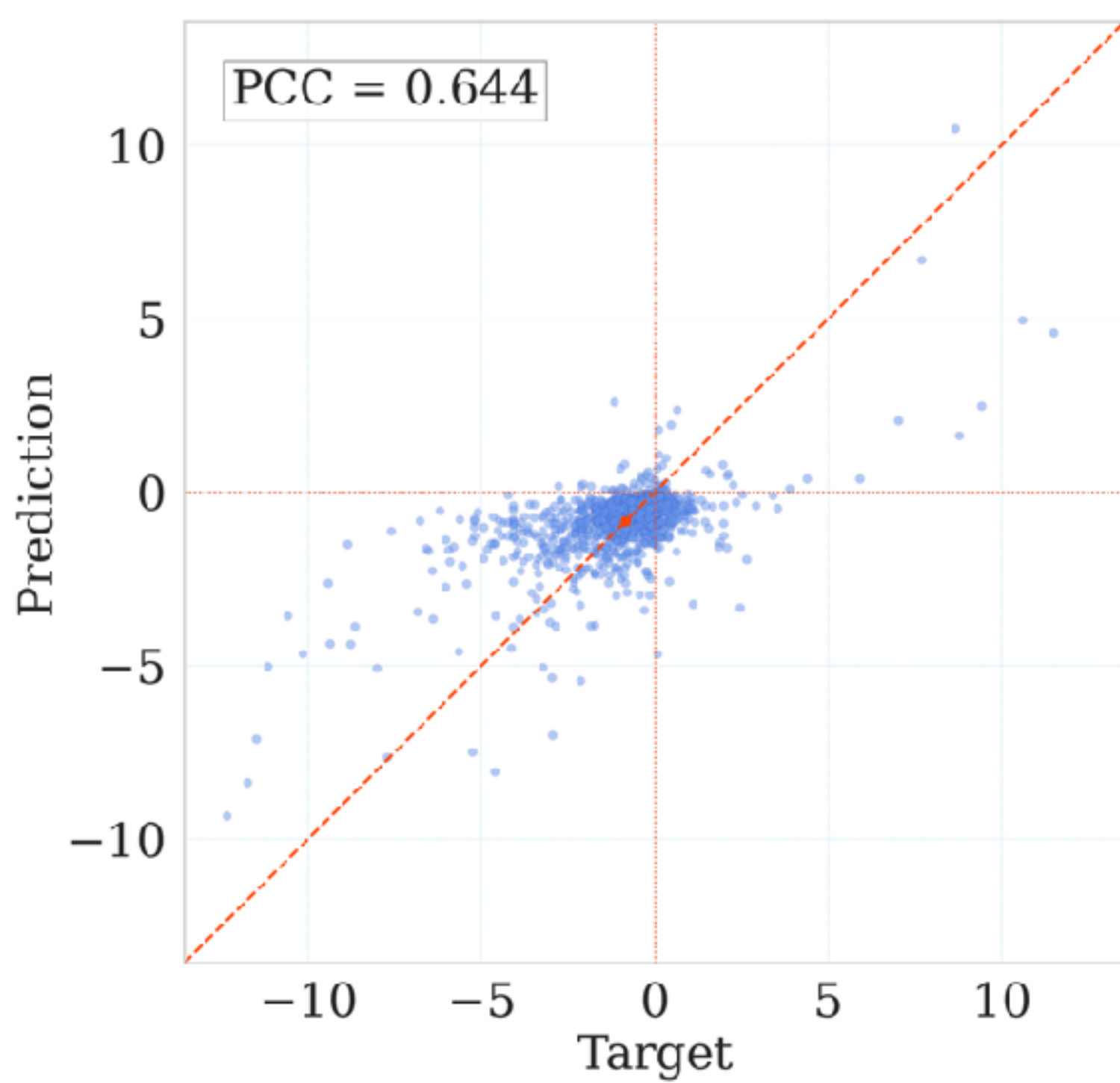
Run 10

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

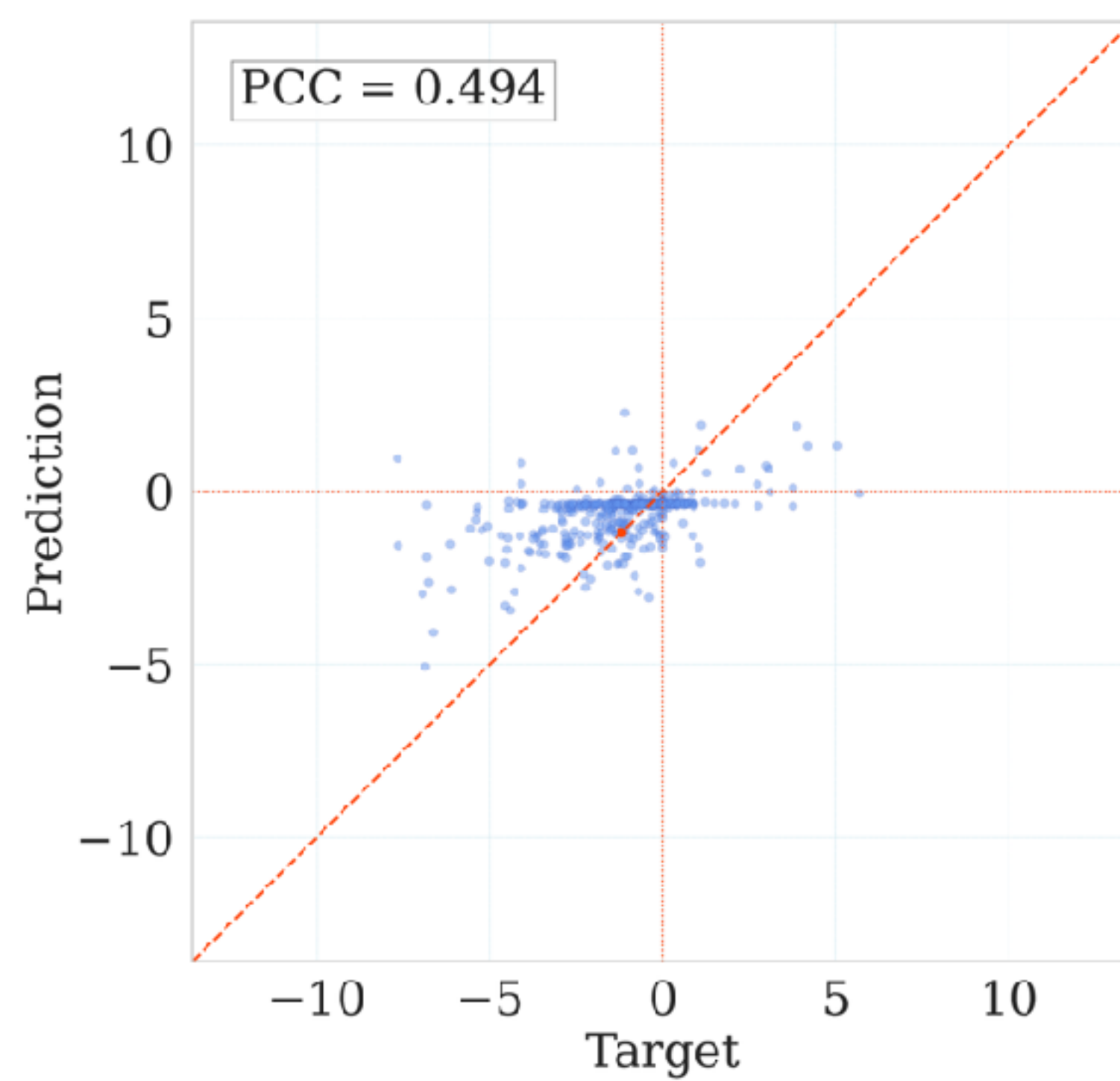


Run 10
Best Epoch: 81

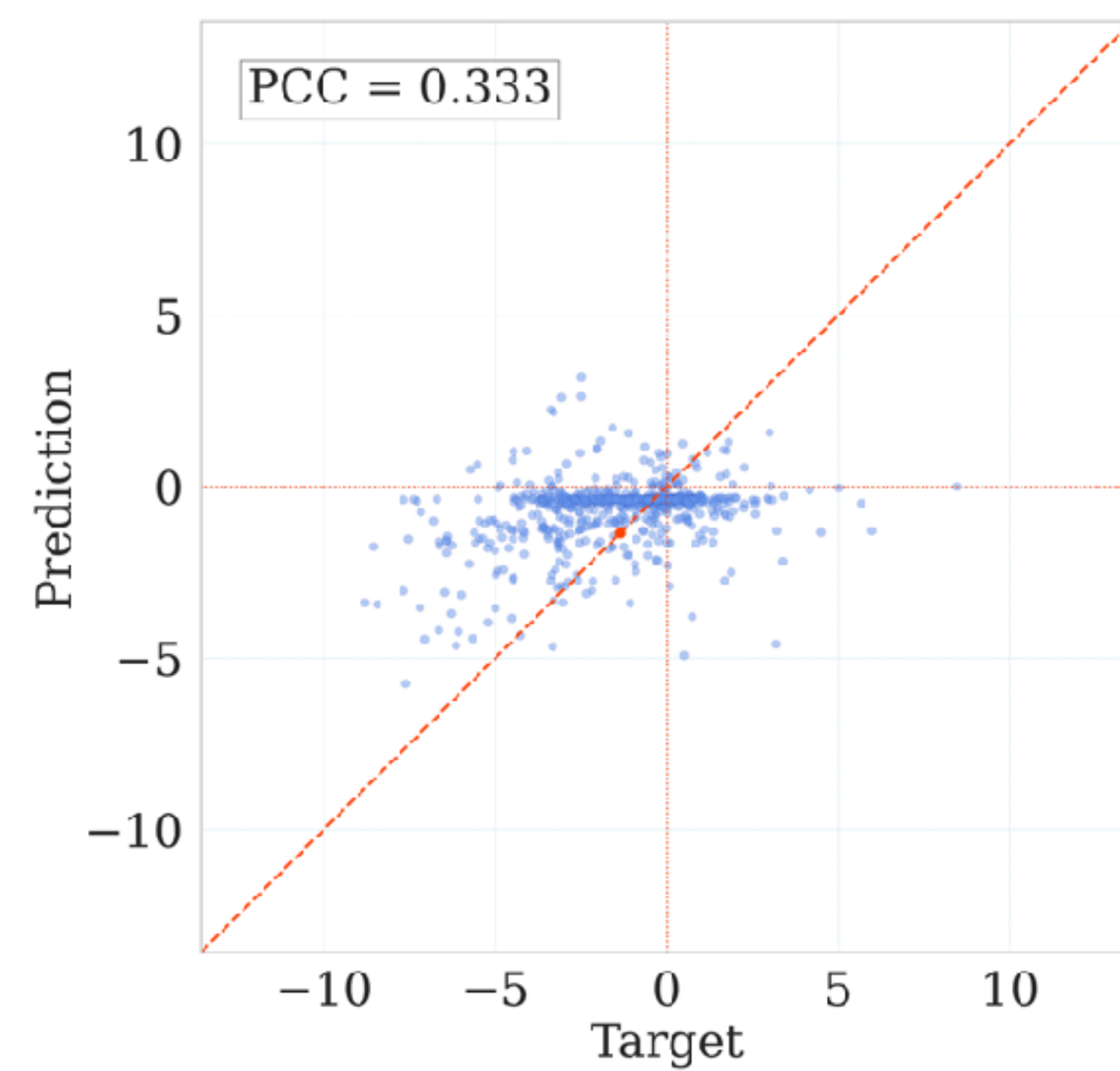
Training: 0 2 4



Validation: 3

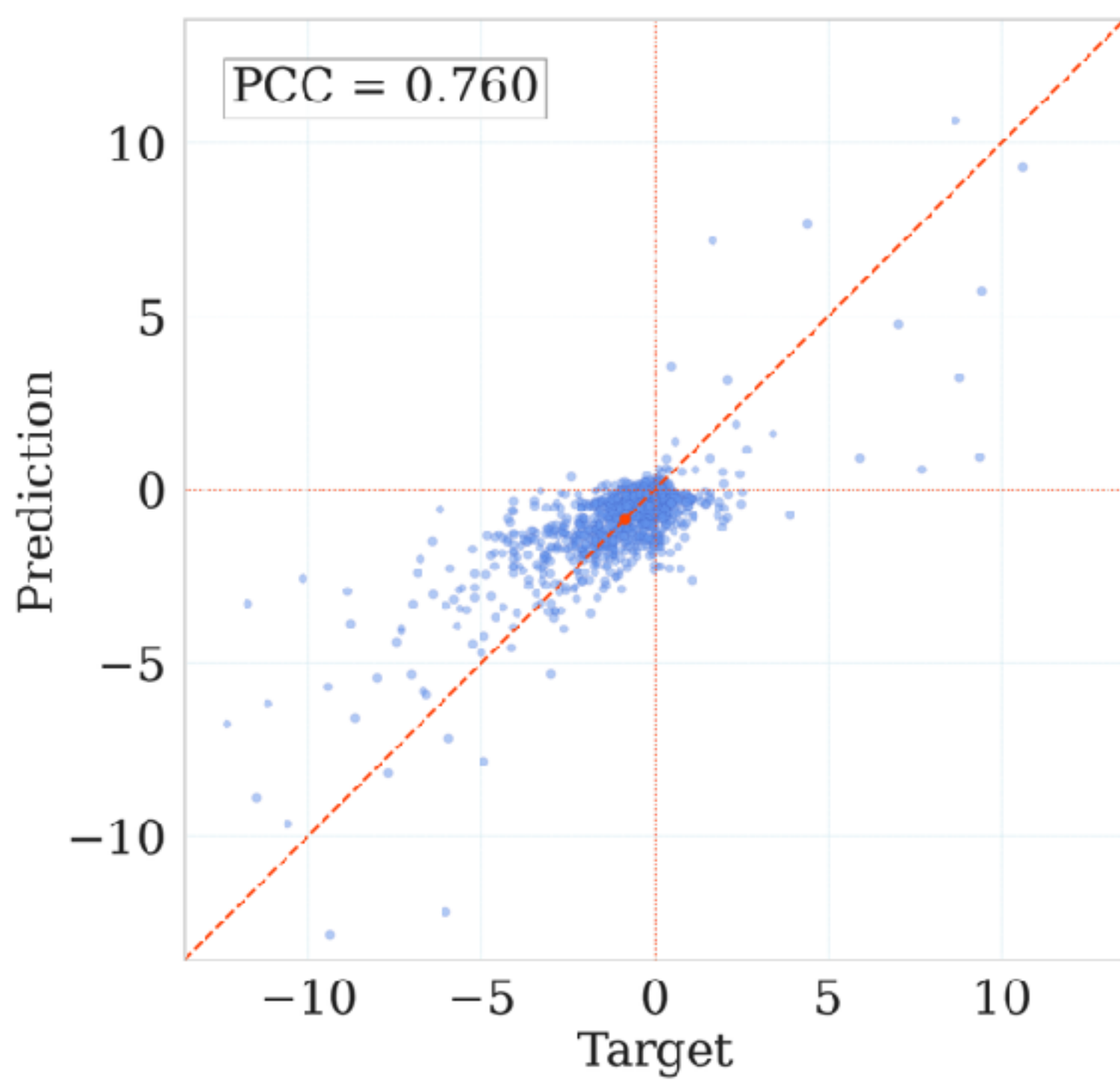


Test: 1

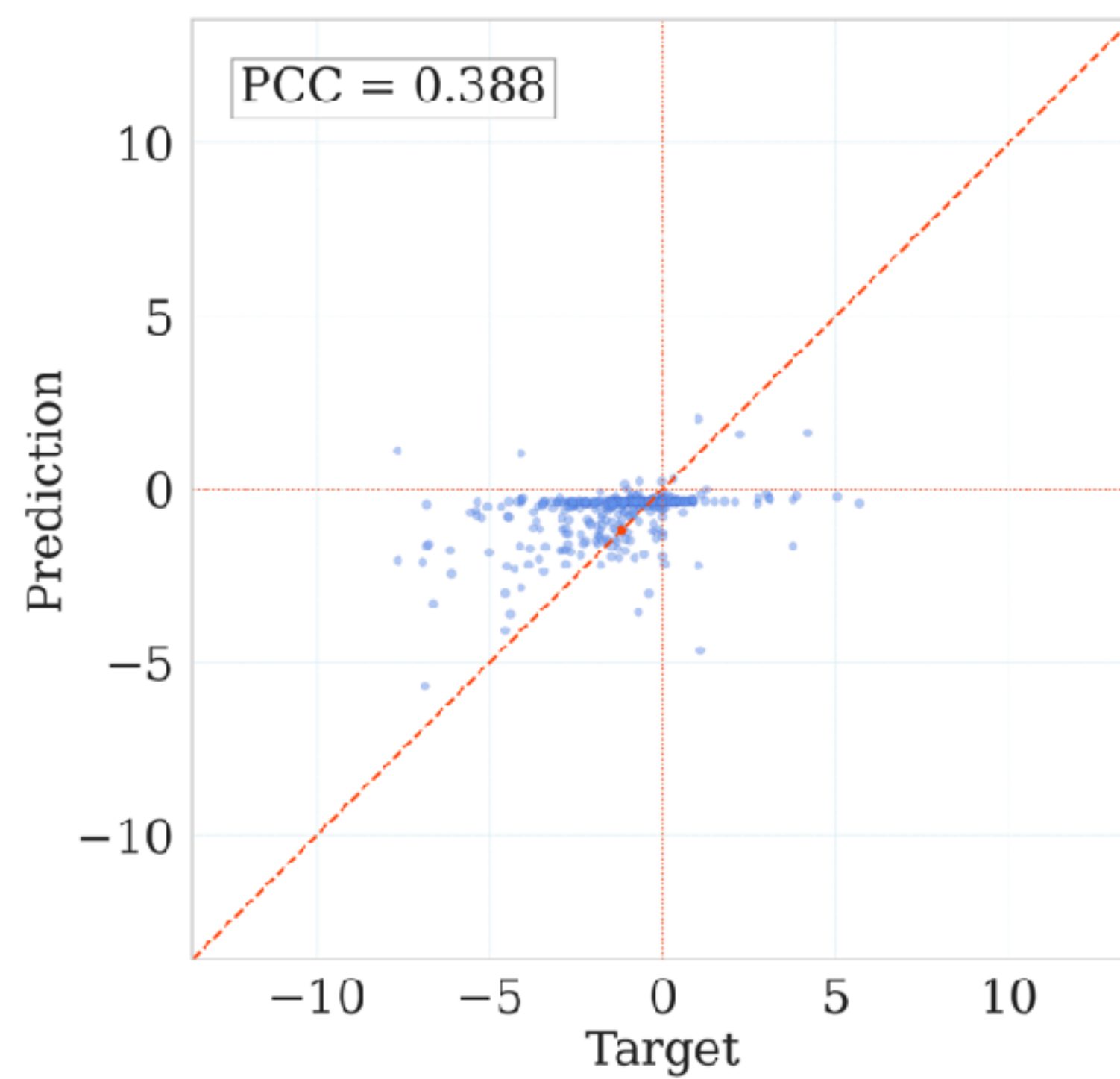


Run 10
Last Epoch: 685

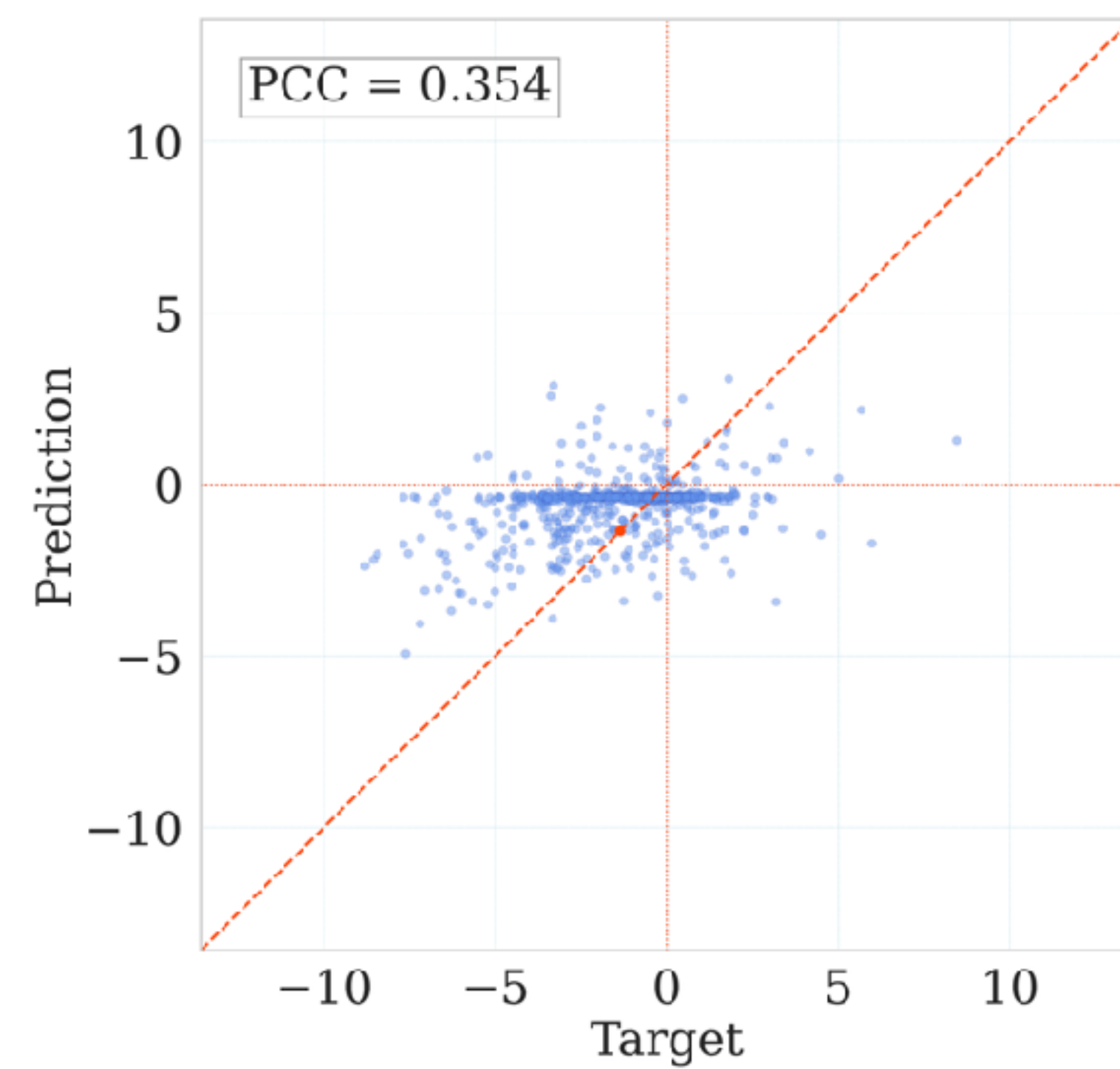
Training: 0 2 4



Validation: 3

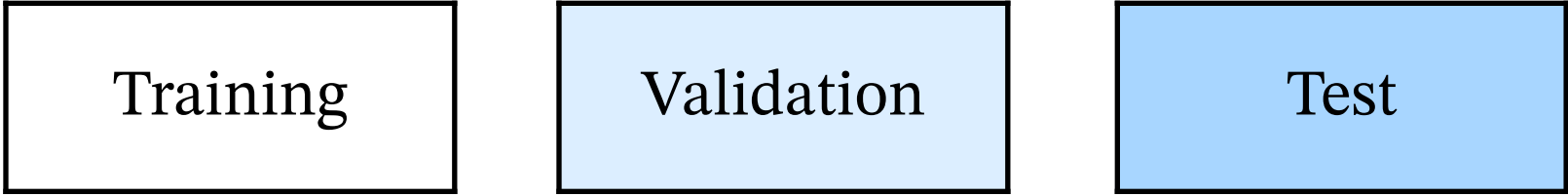


Test: 1



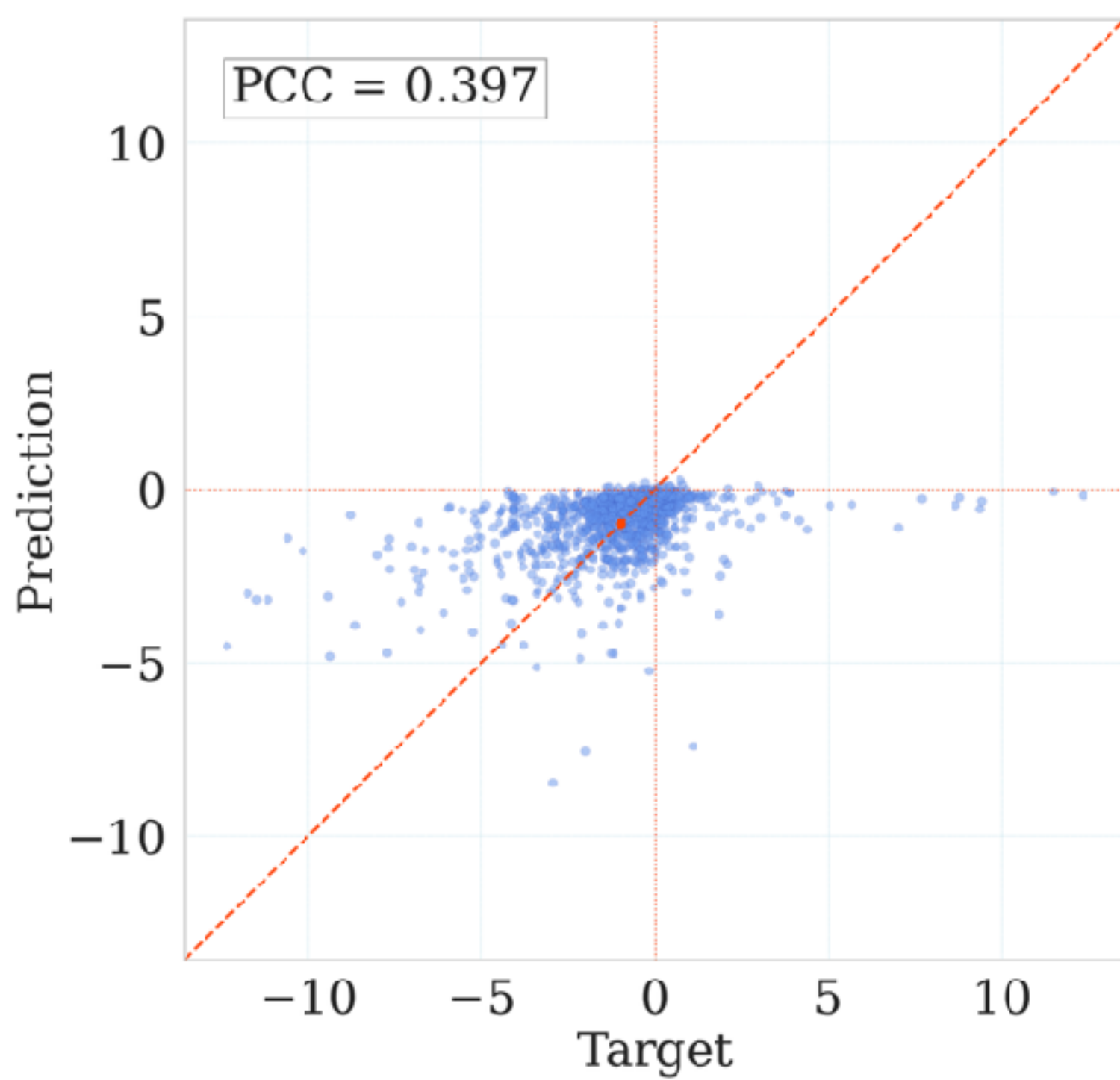
Run 11

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

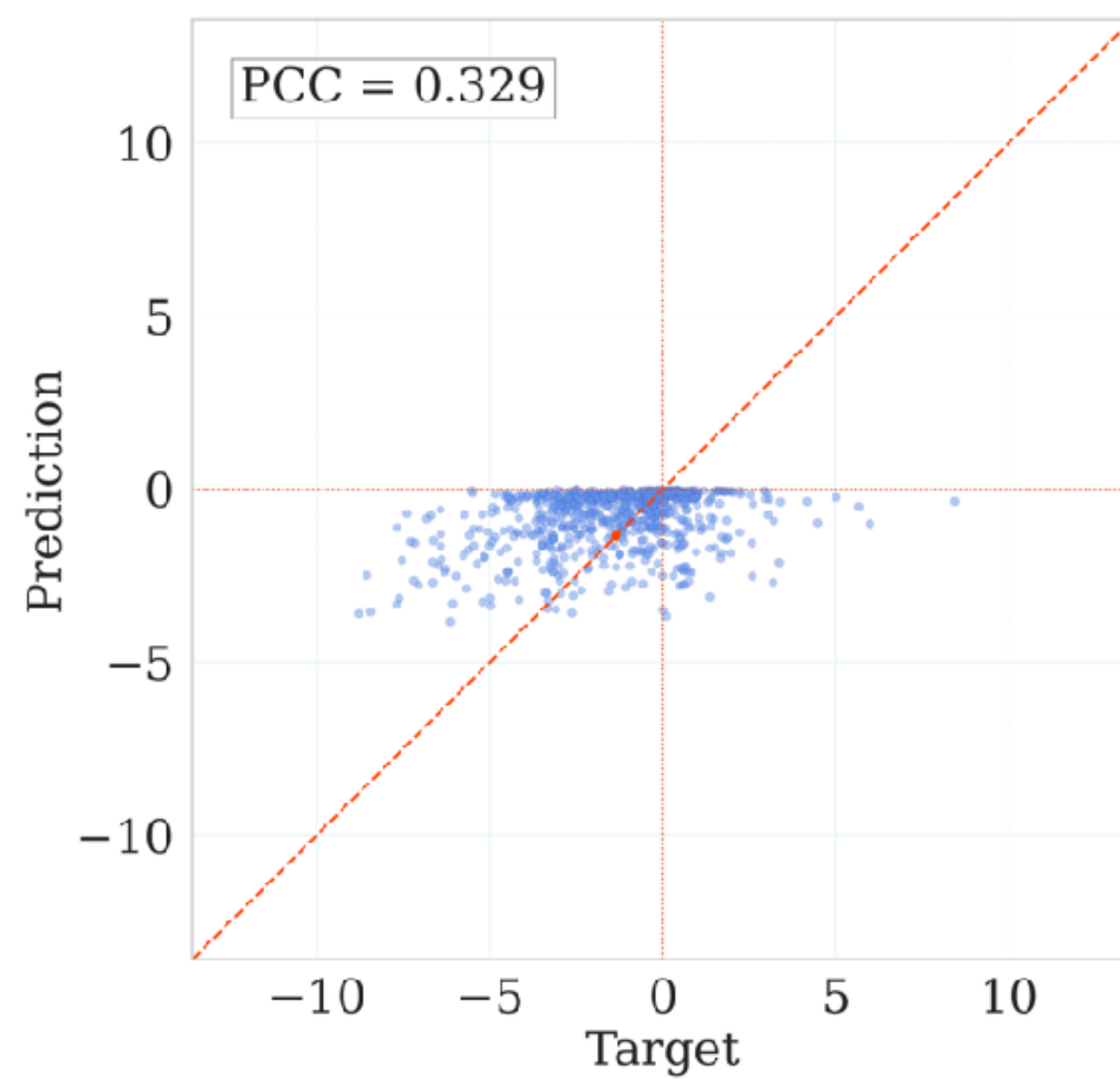


Run 11
Best Epoch: 17

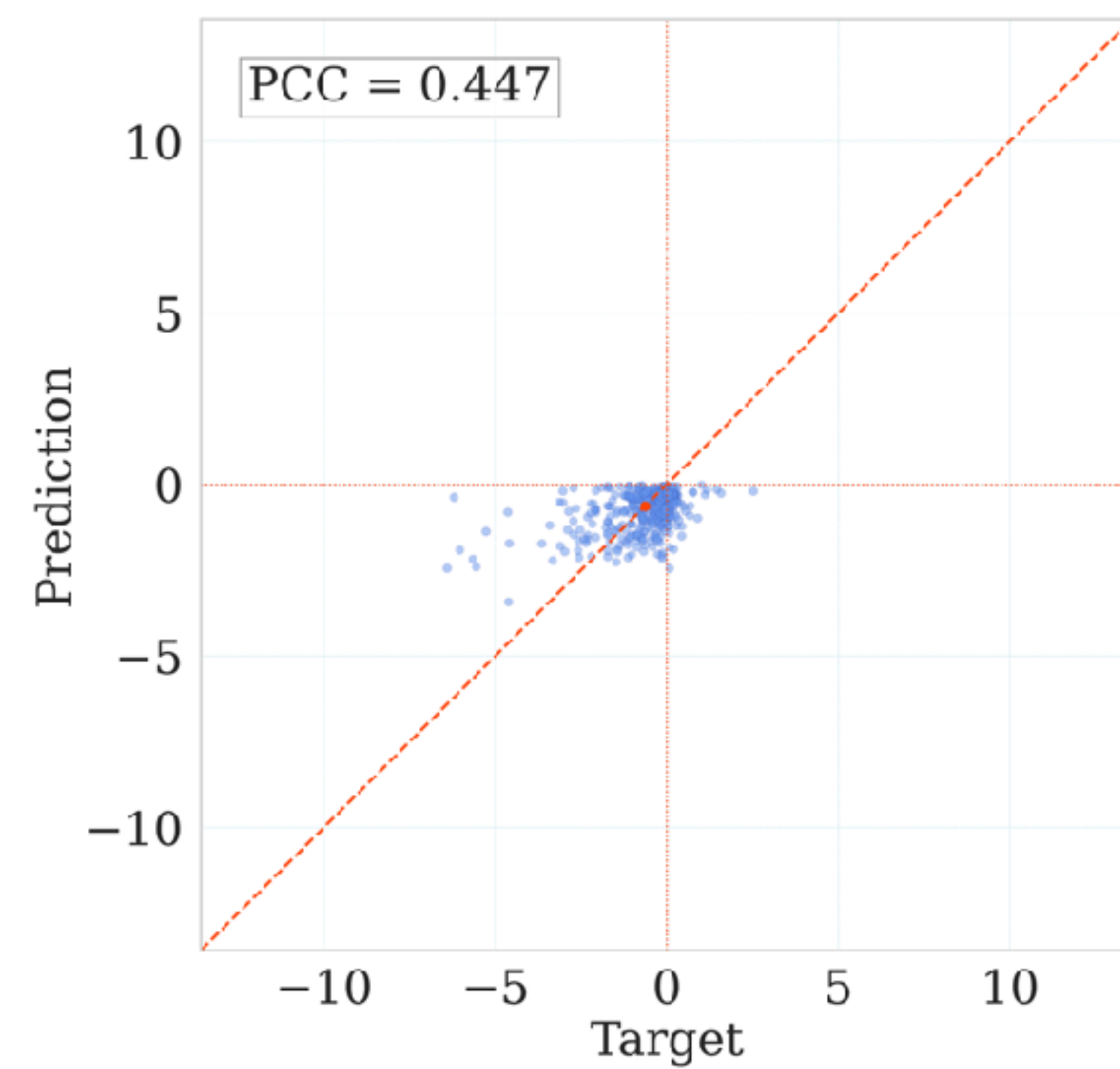
Training: 0 3 4



Validation: 1

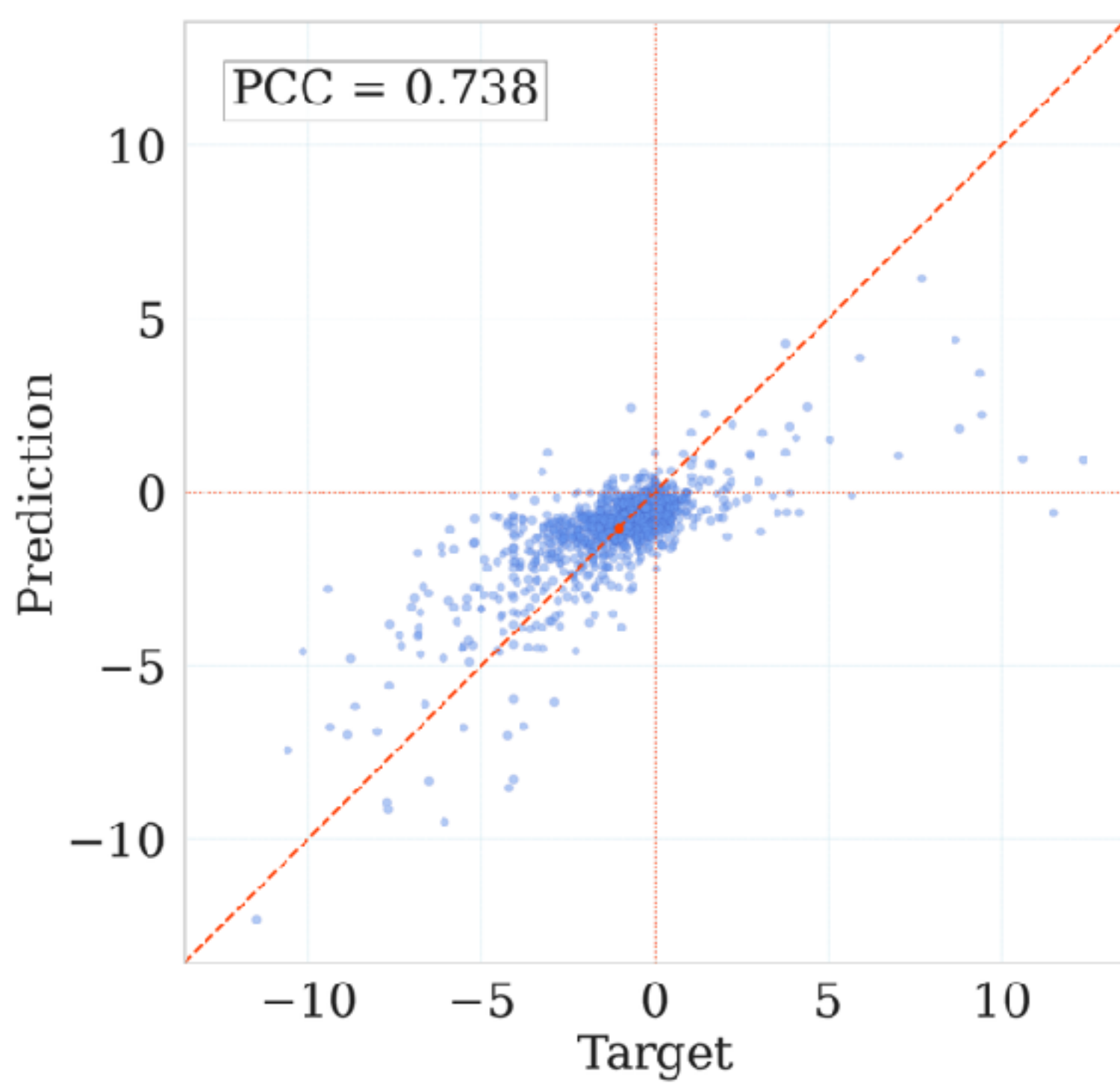


Test: 2

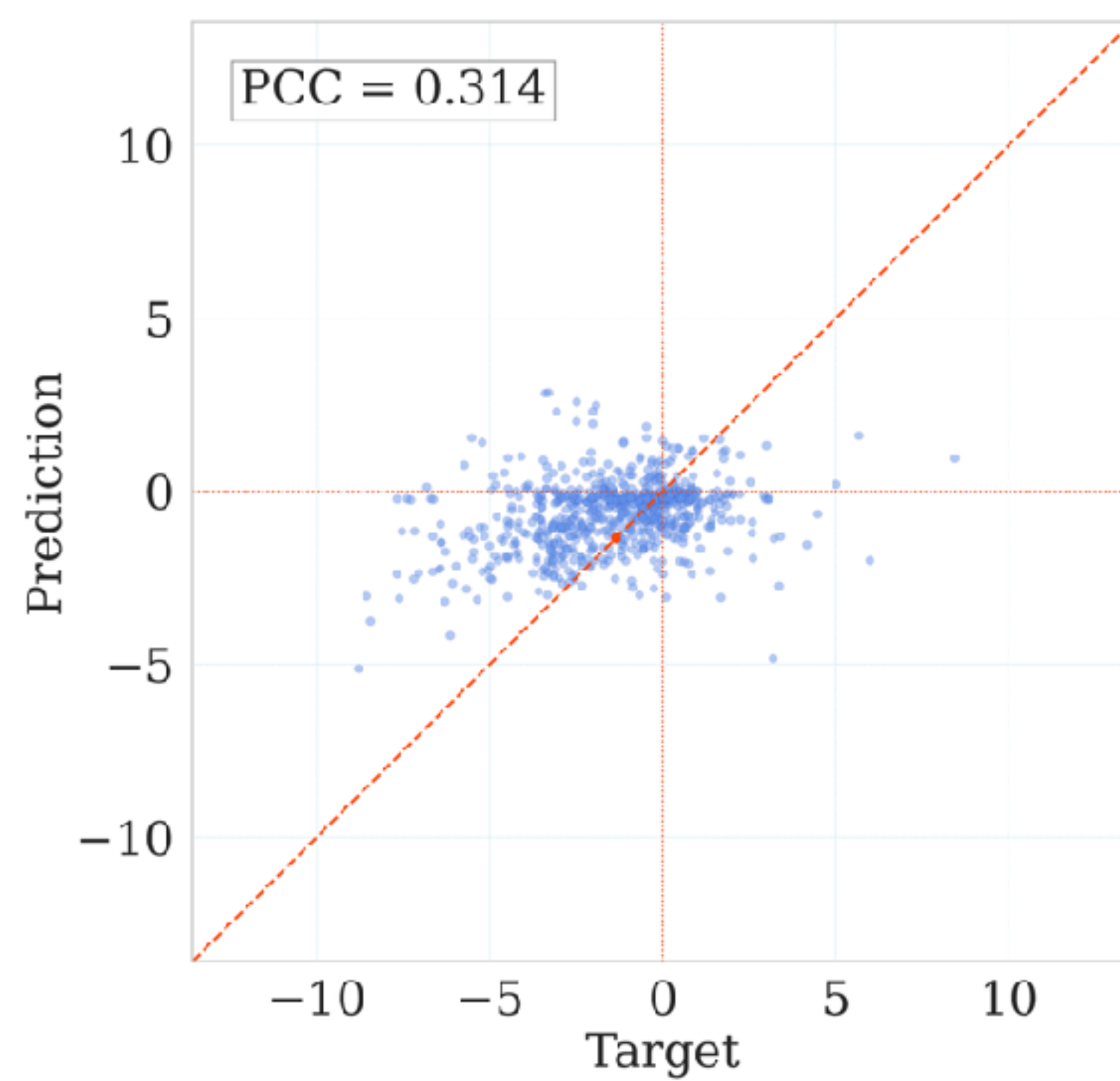


Run 11
Last Epoch: 622

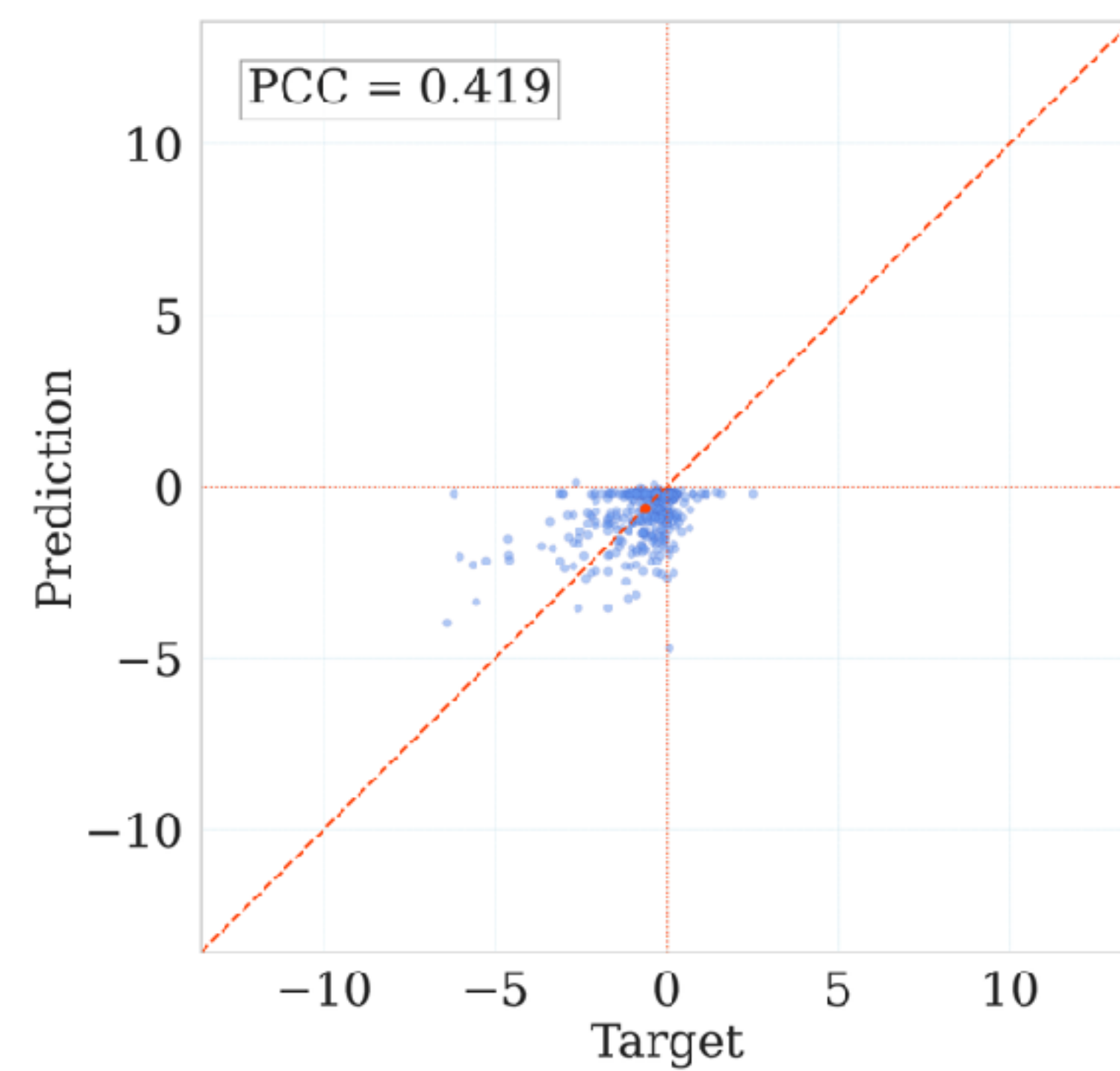
Training: 0 3 4



Validation: 1

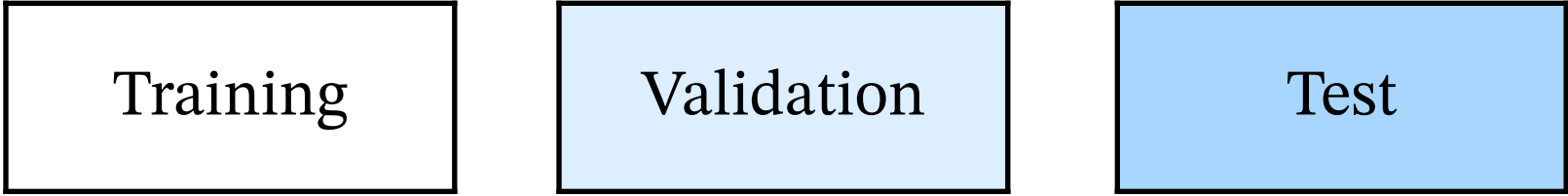


Test: 2



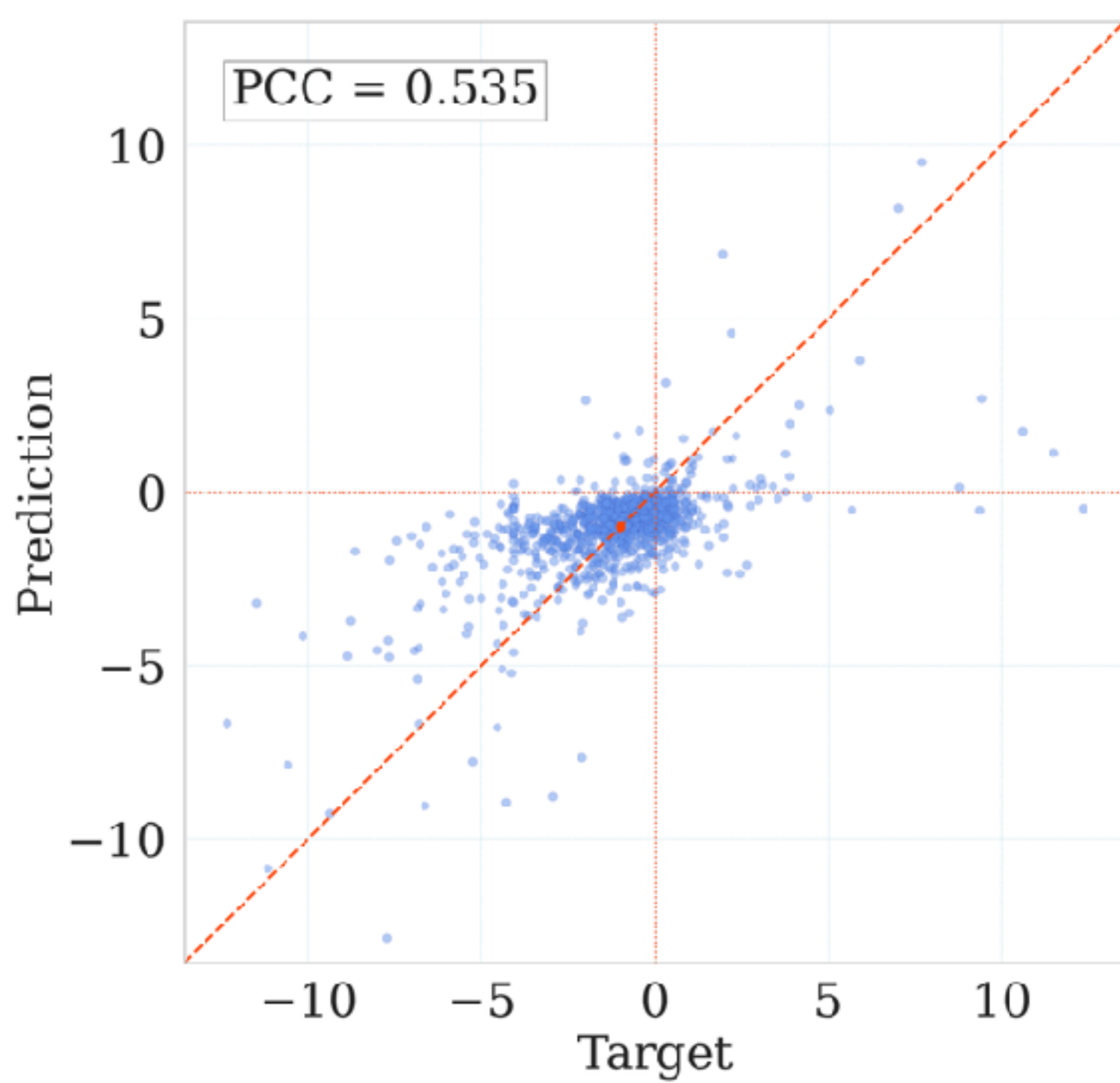
Run 12

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

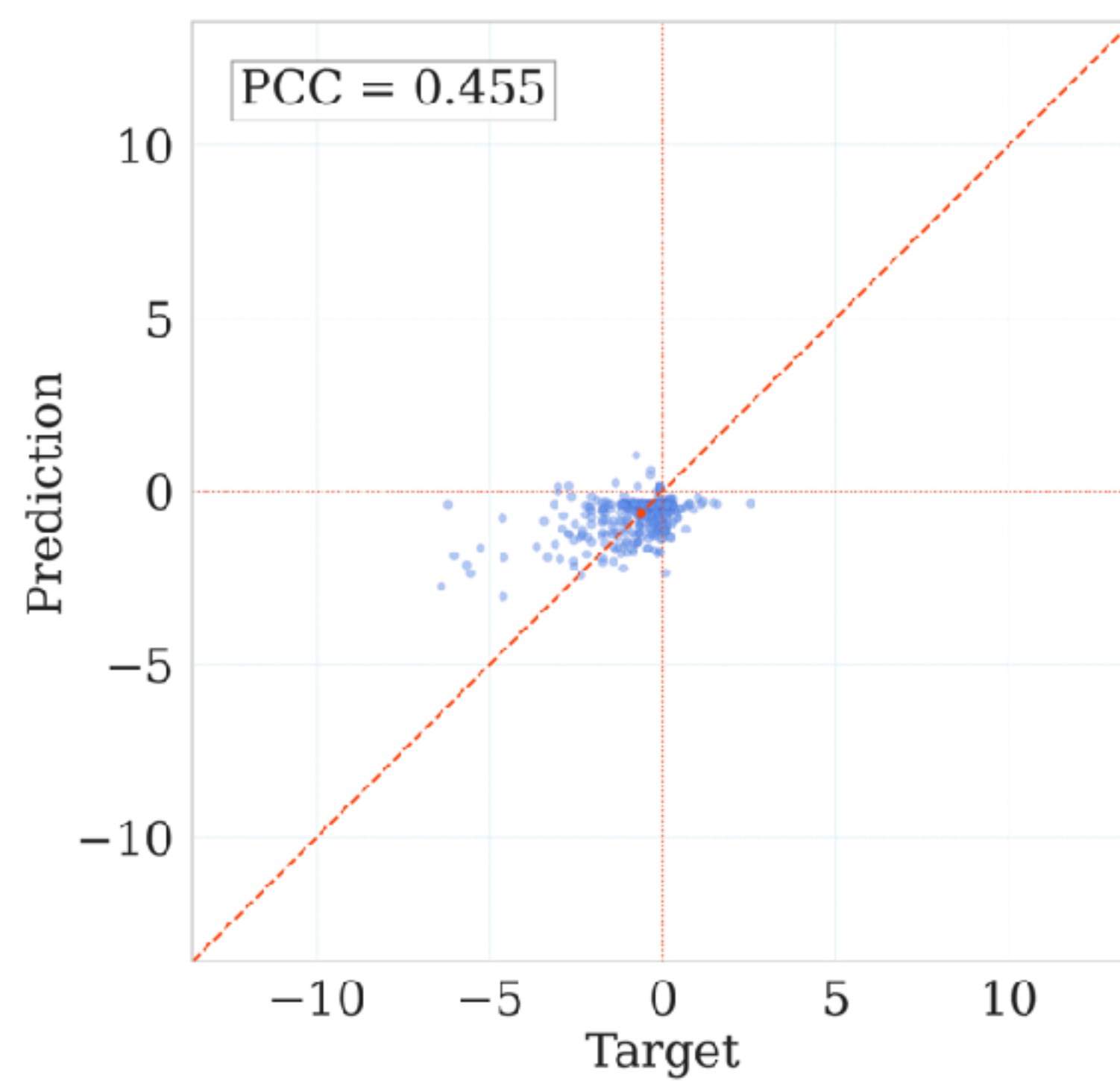


Run 12
Best Epoch: 72

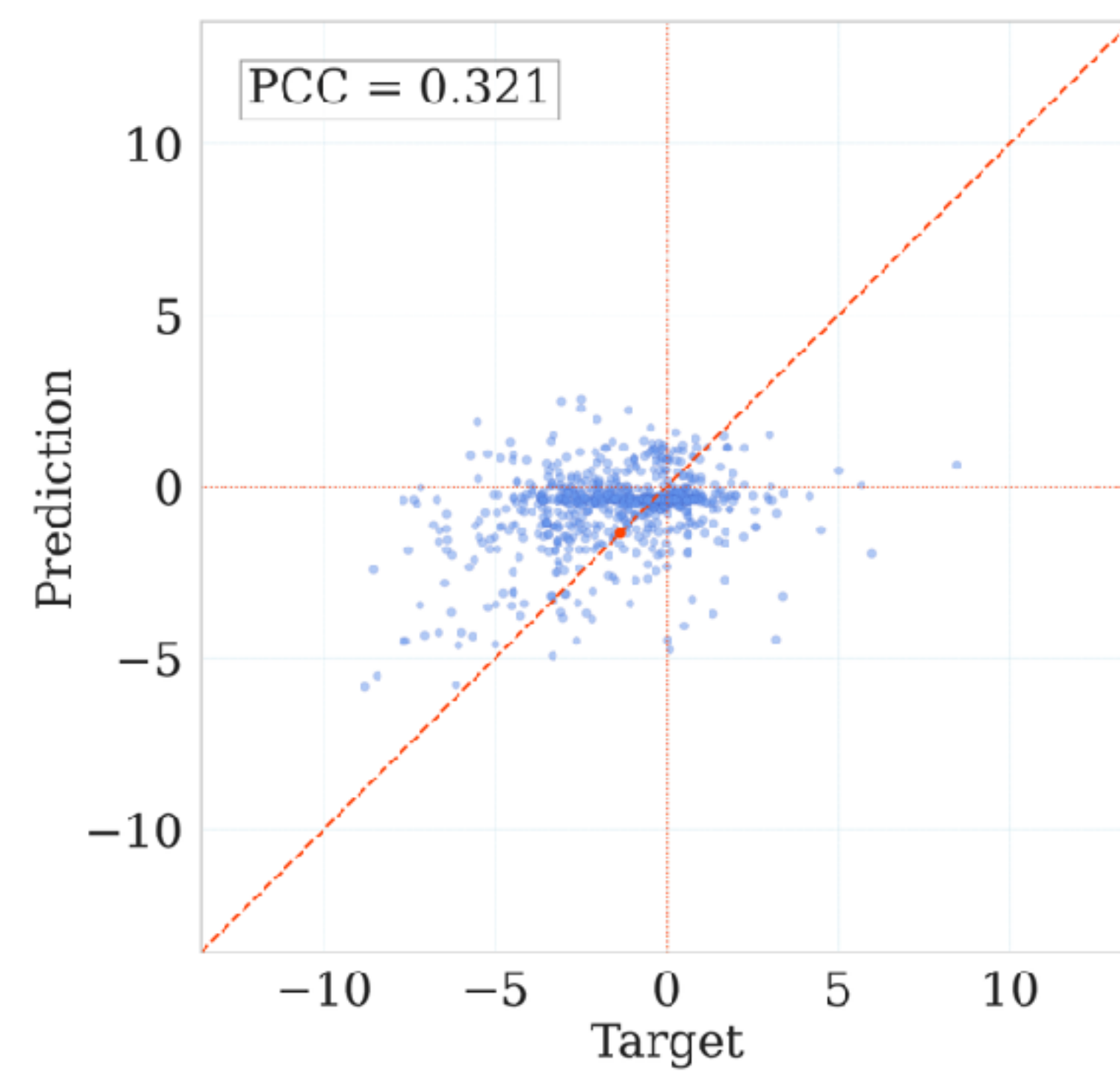
Training: 0 3 4



Validation: 2

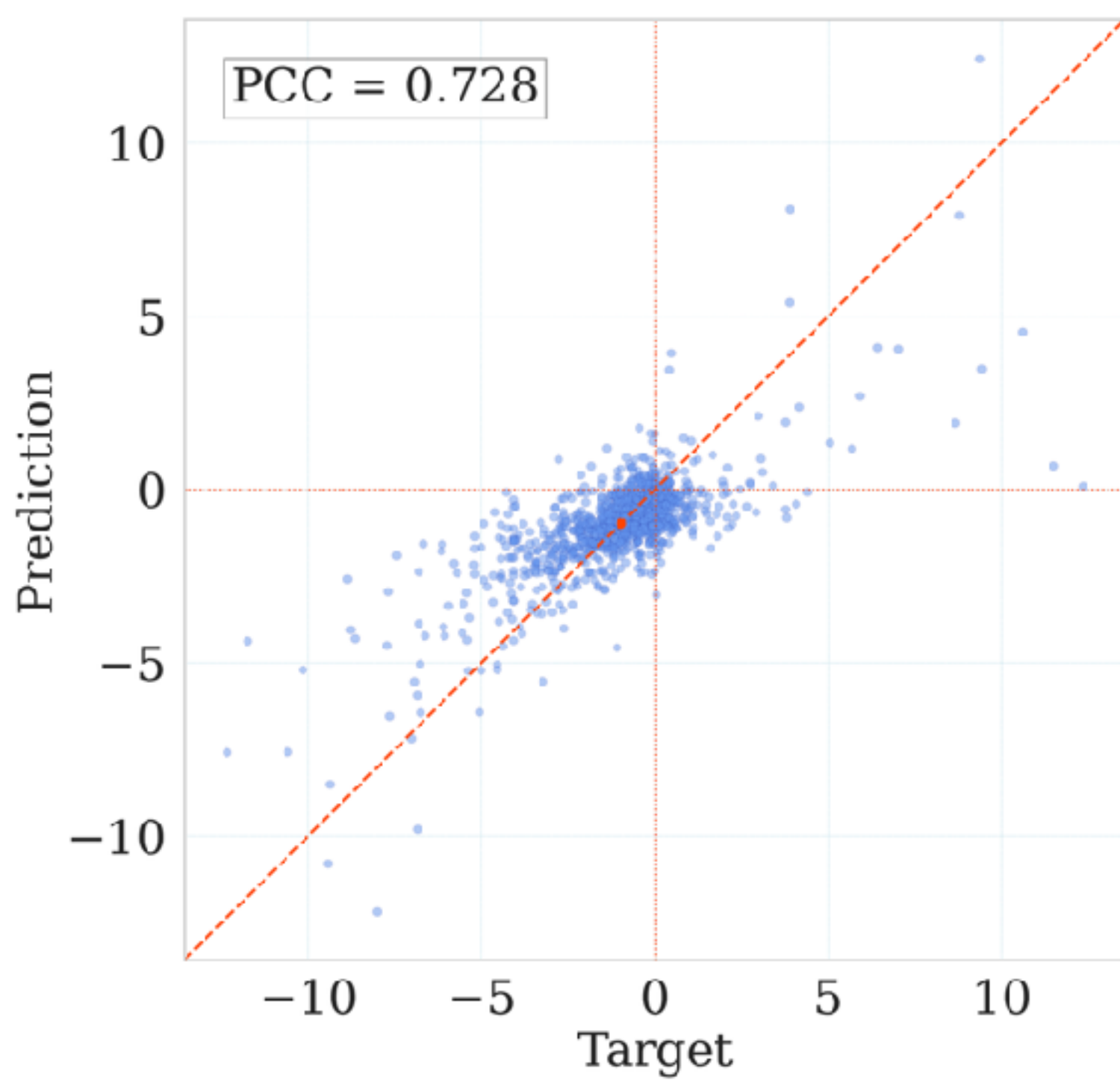


Test: 1

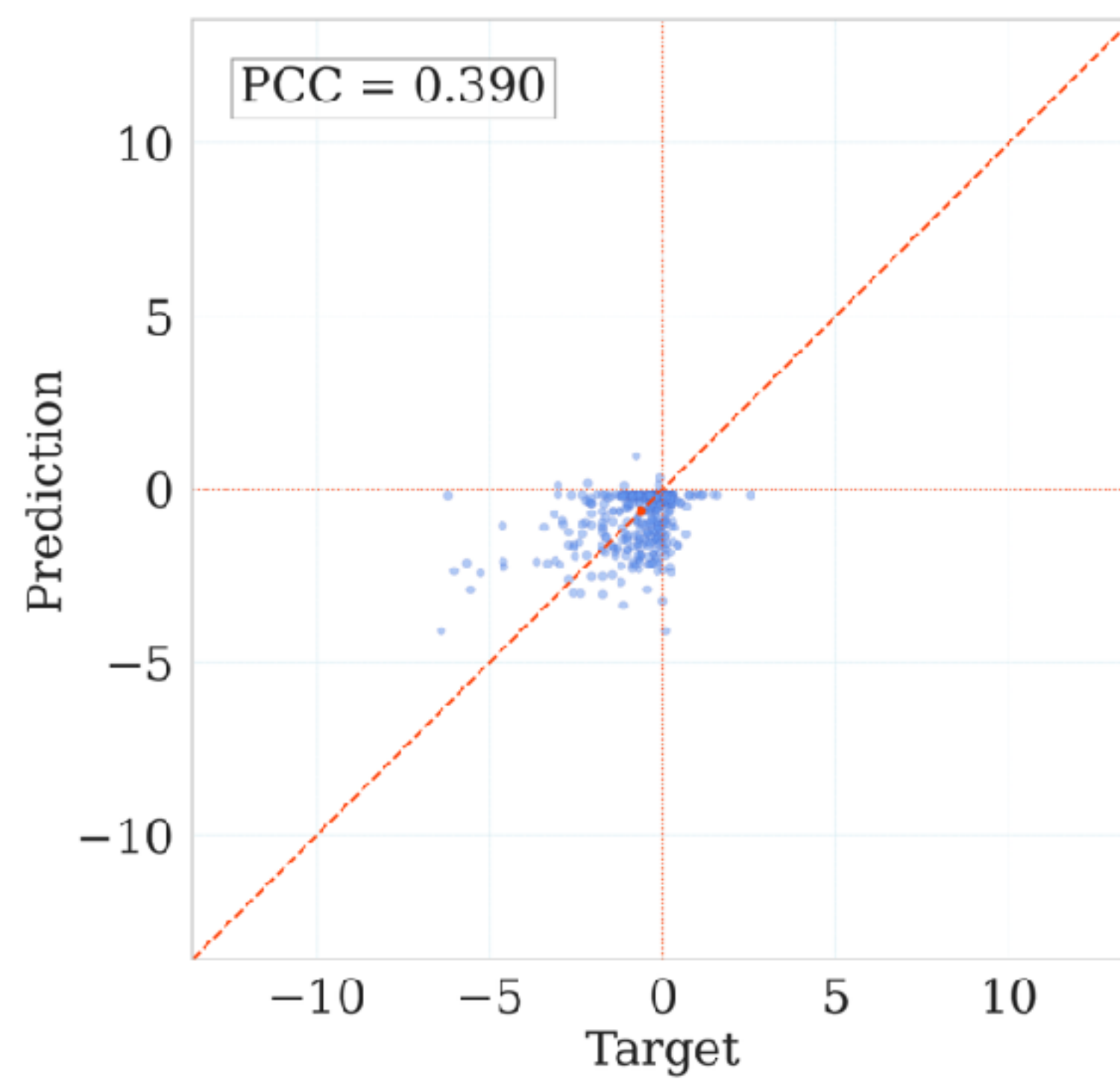


Run 12
Last Epoch: 639

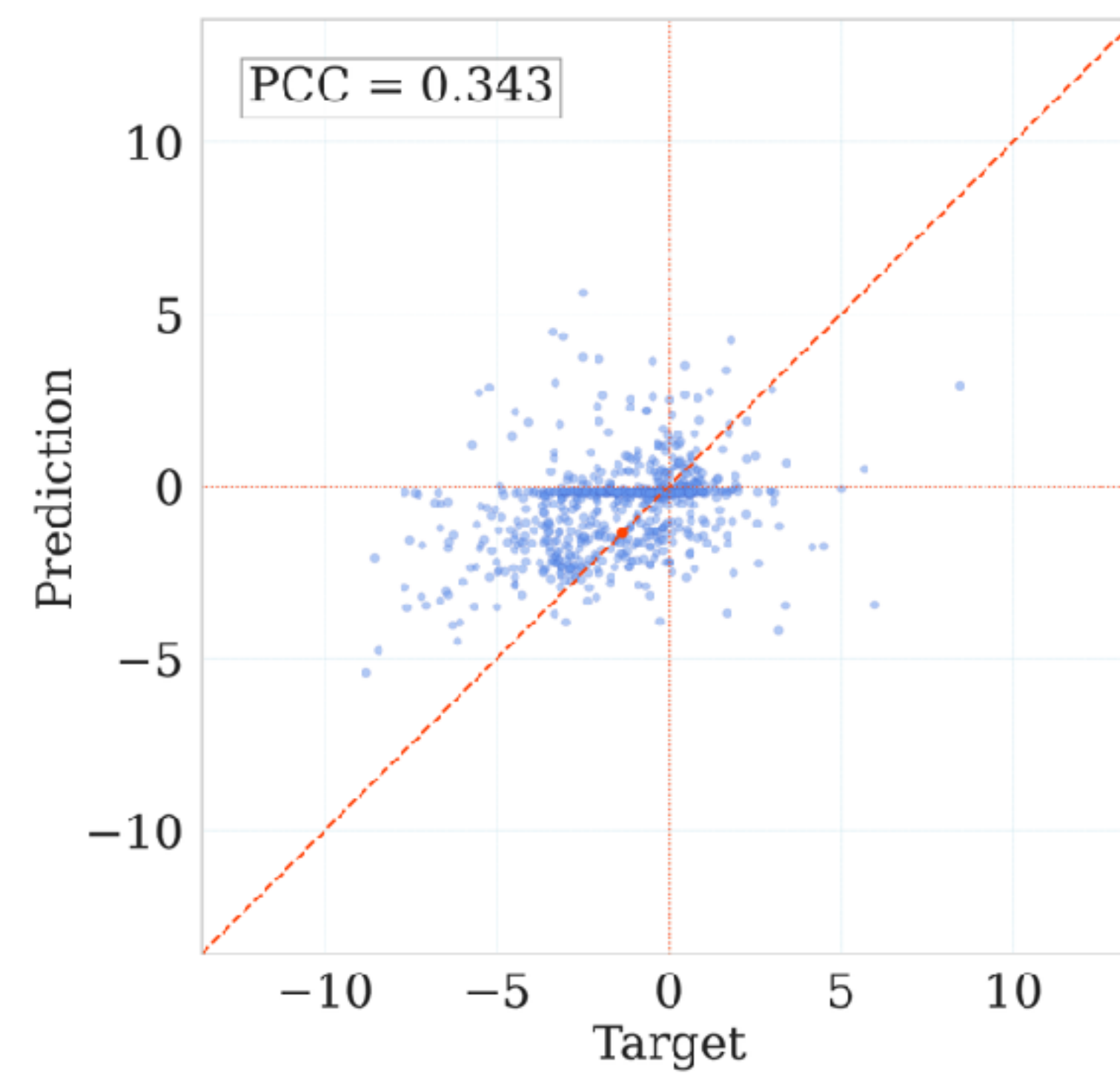
Training: 0 3 4



Validation: 2

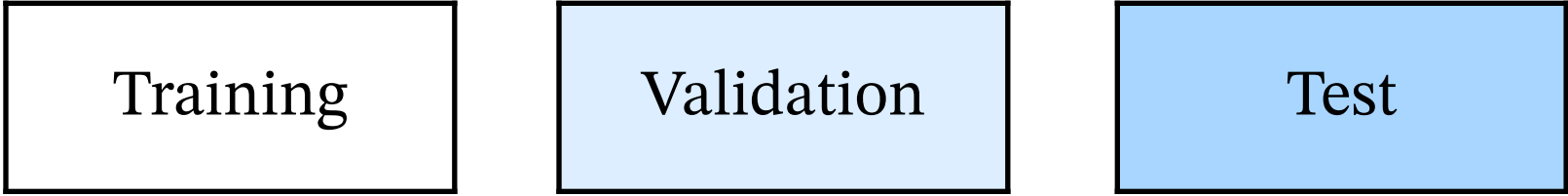


Test: 1



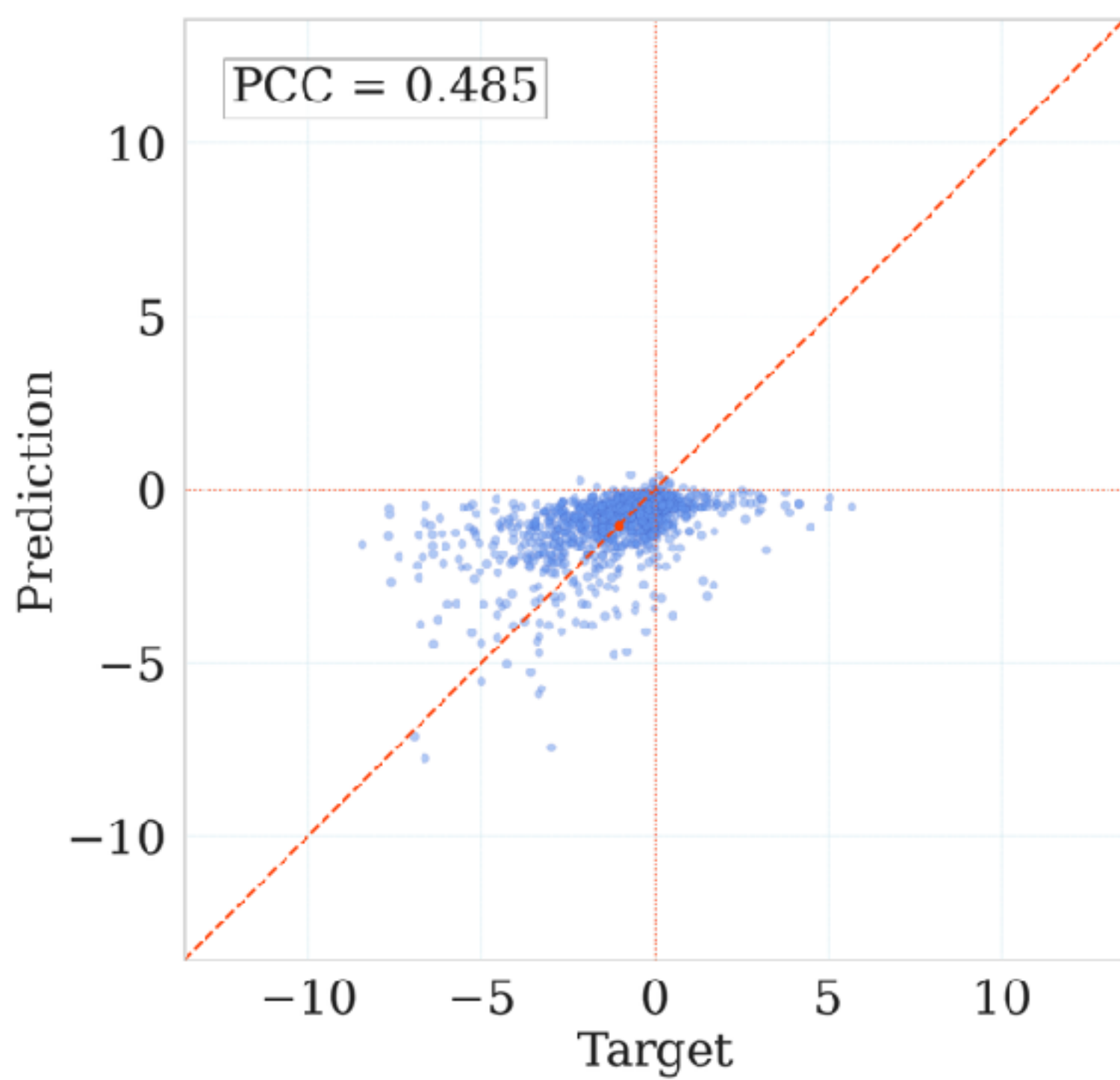
Run 13

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

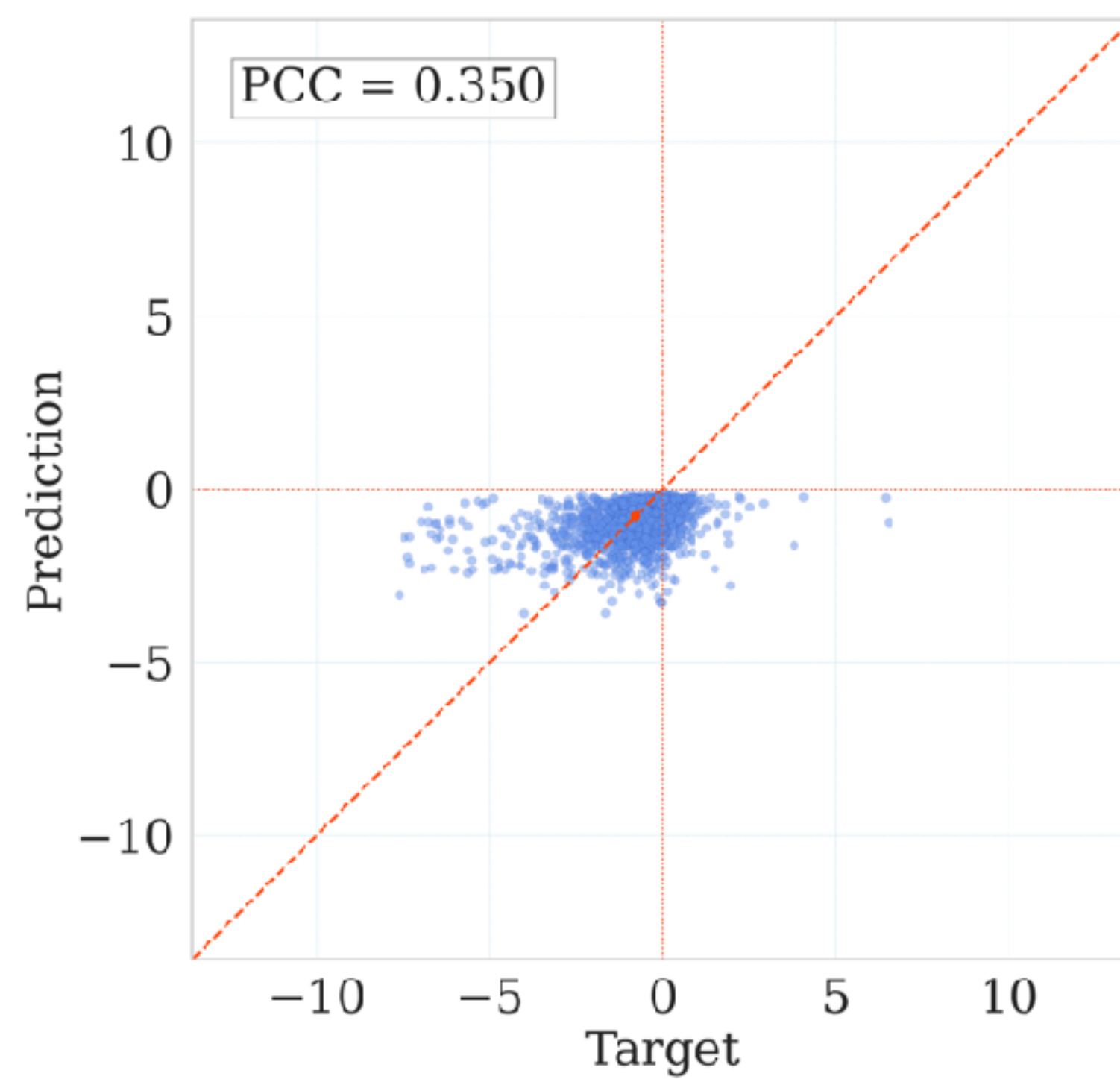


Run 13
Best Epoch: 21

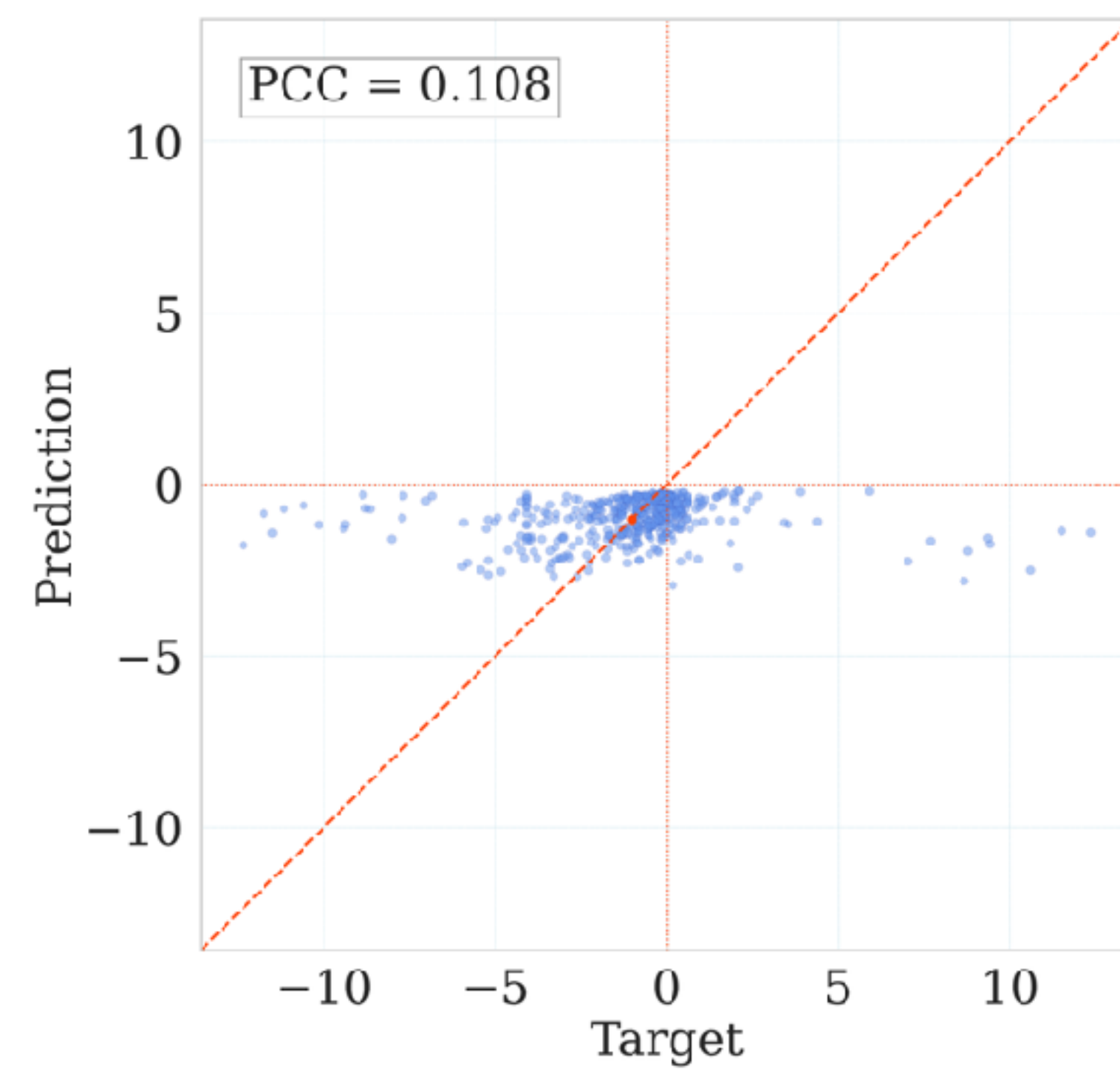
Training: 1 2 3



Validation: 0

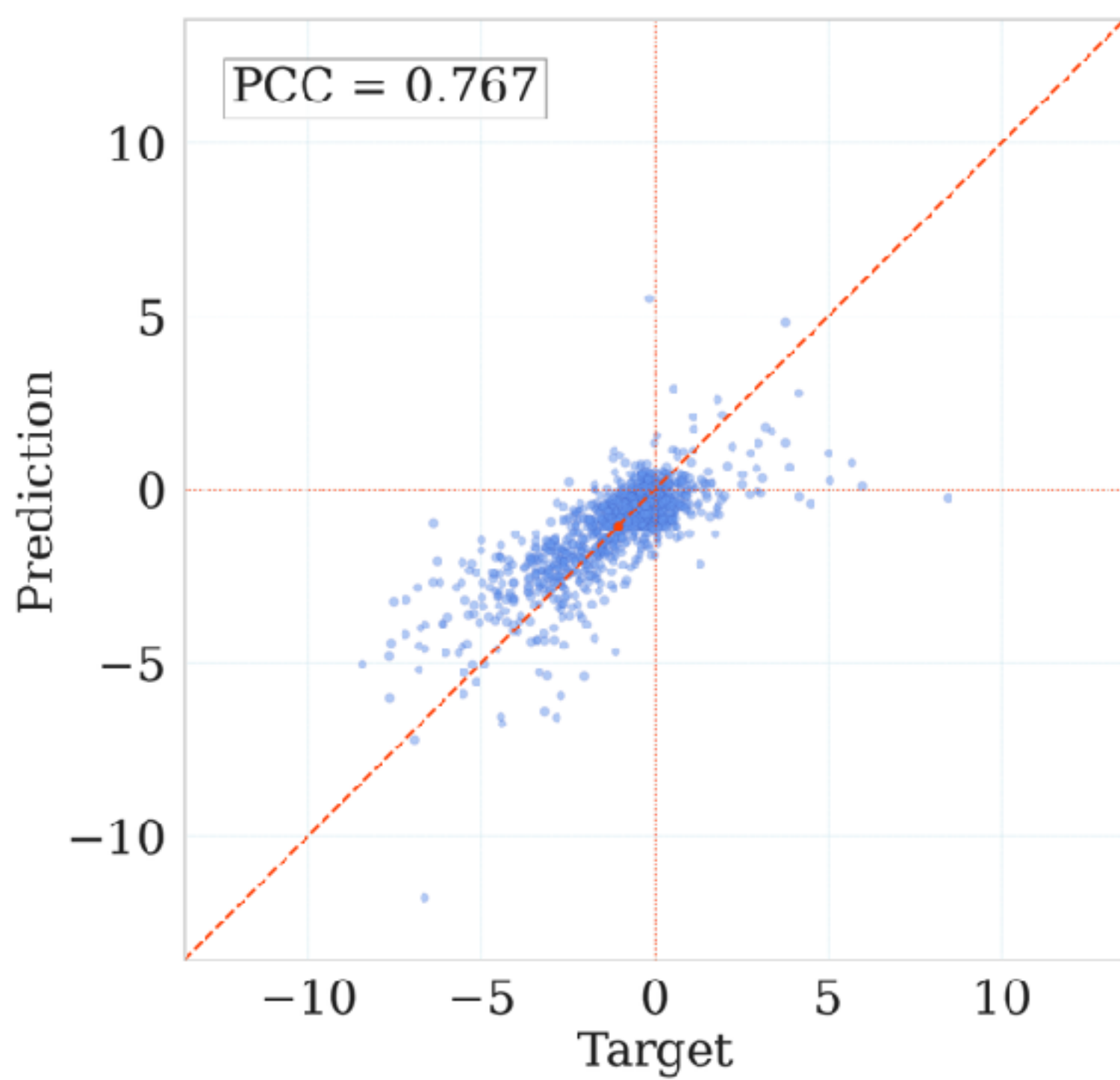


Test: 4

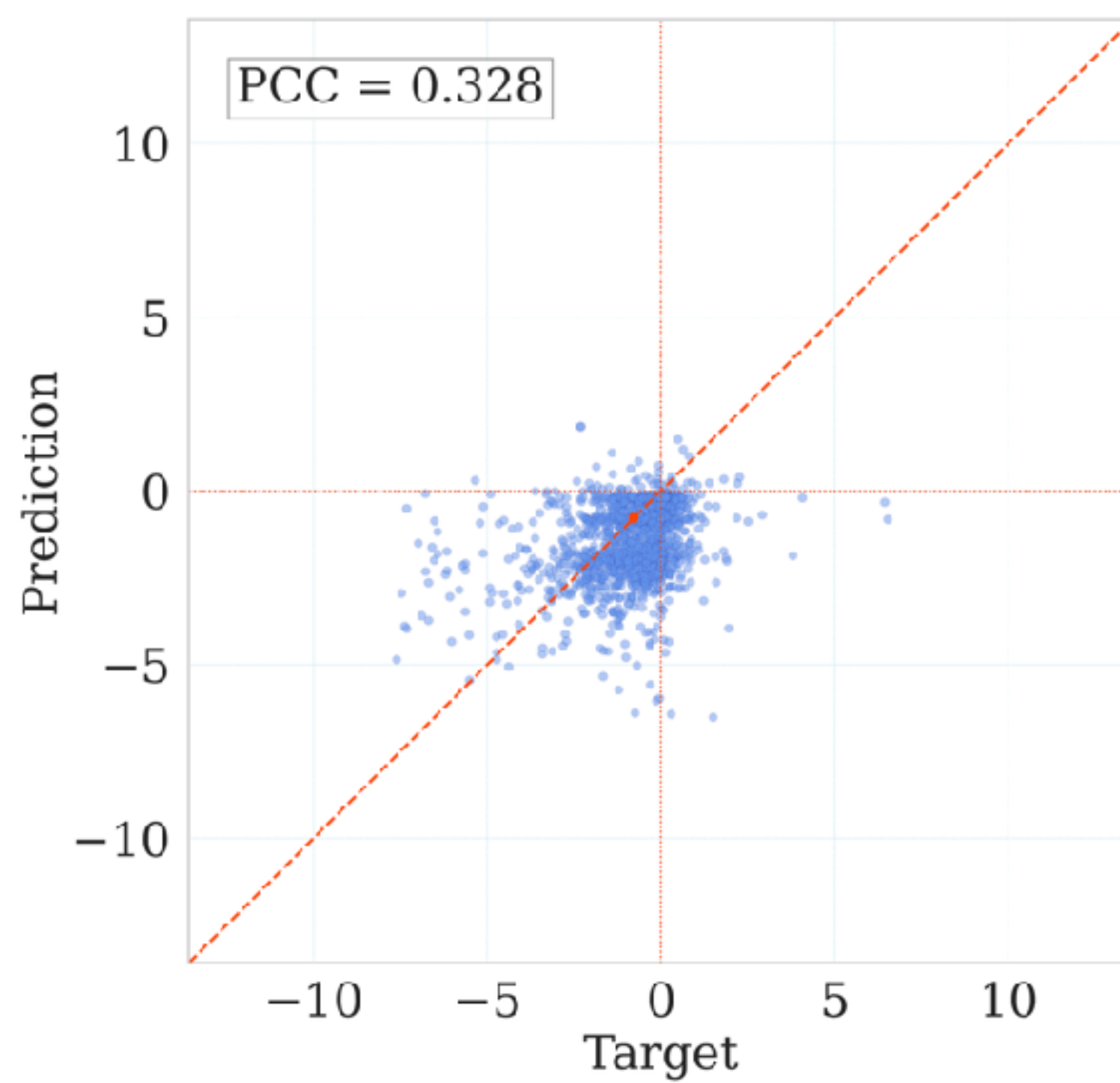


Run 13
Last Epoch: 698

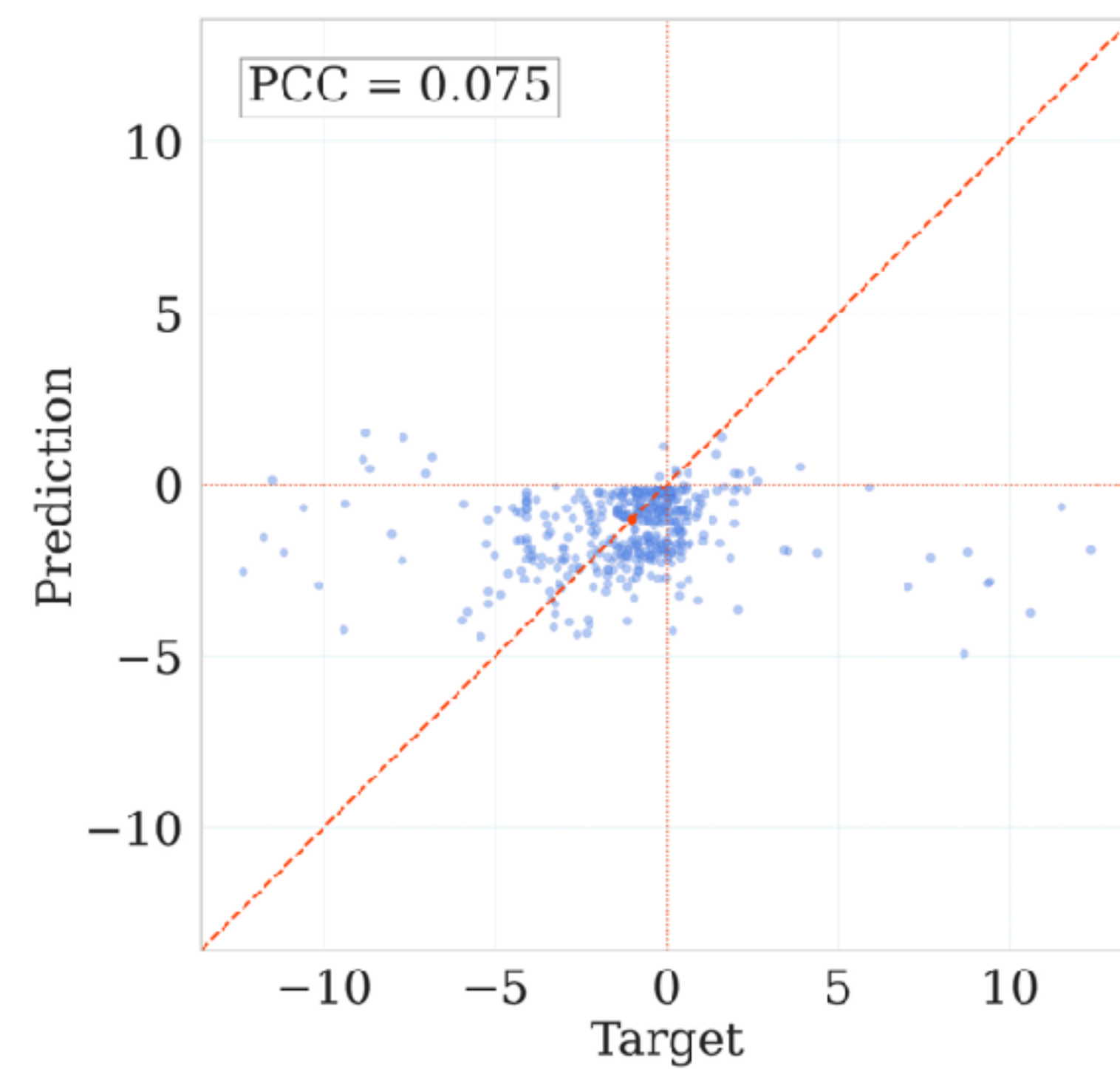
Training: 1 2 3



Validation: 0



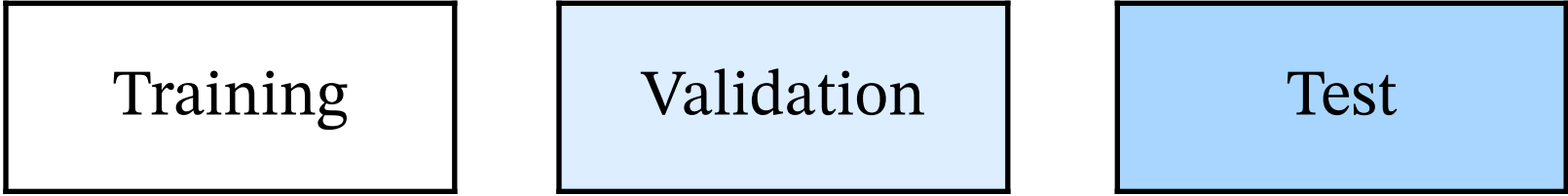
Test: 4



WORST

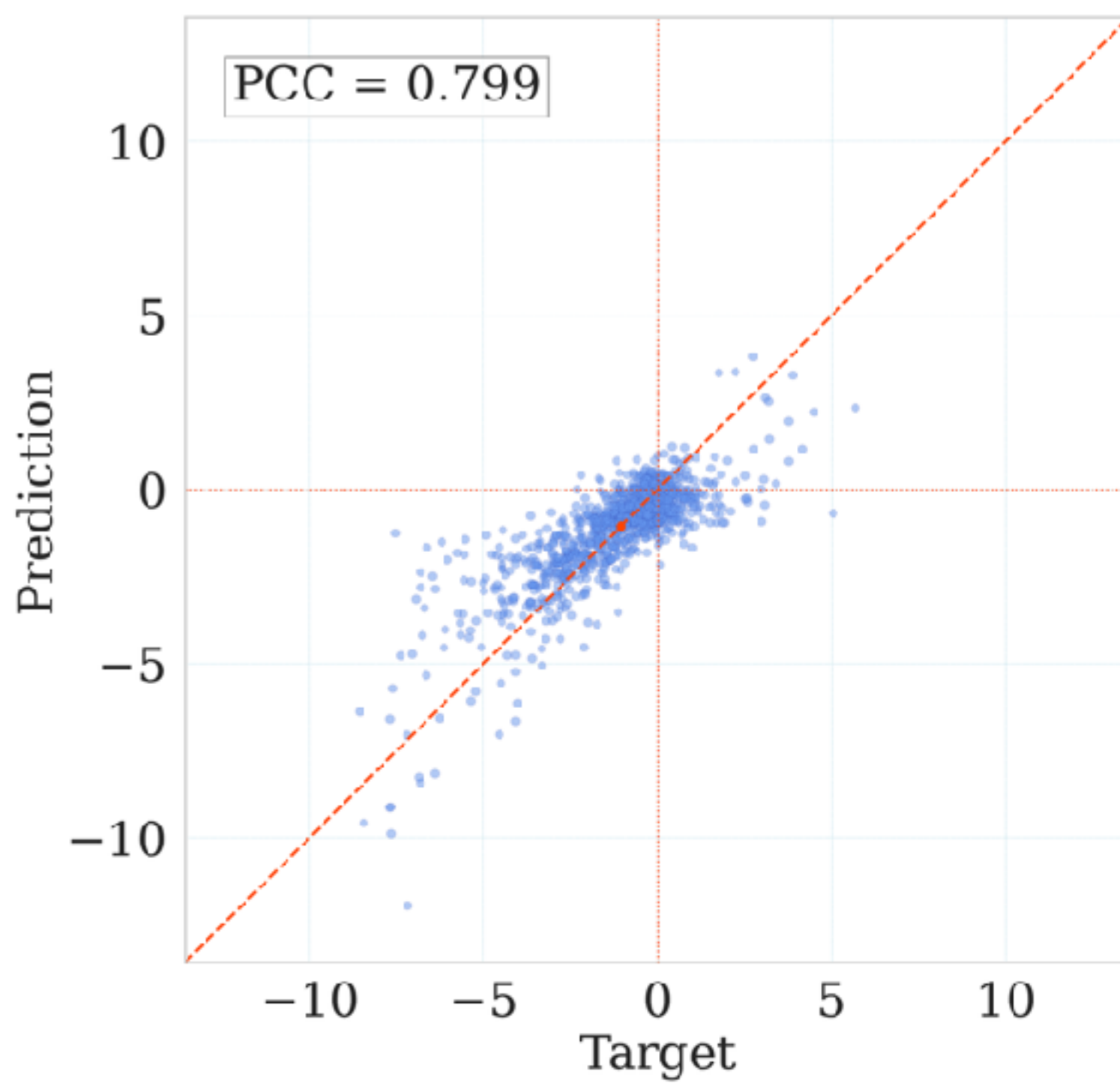
Run 14

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6



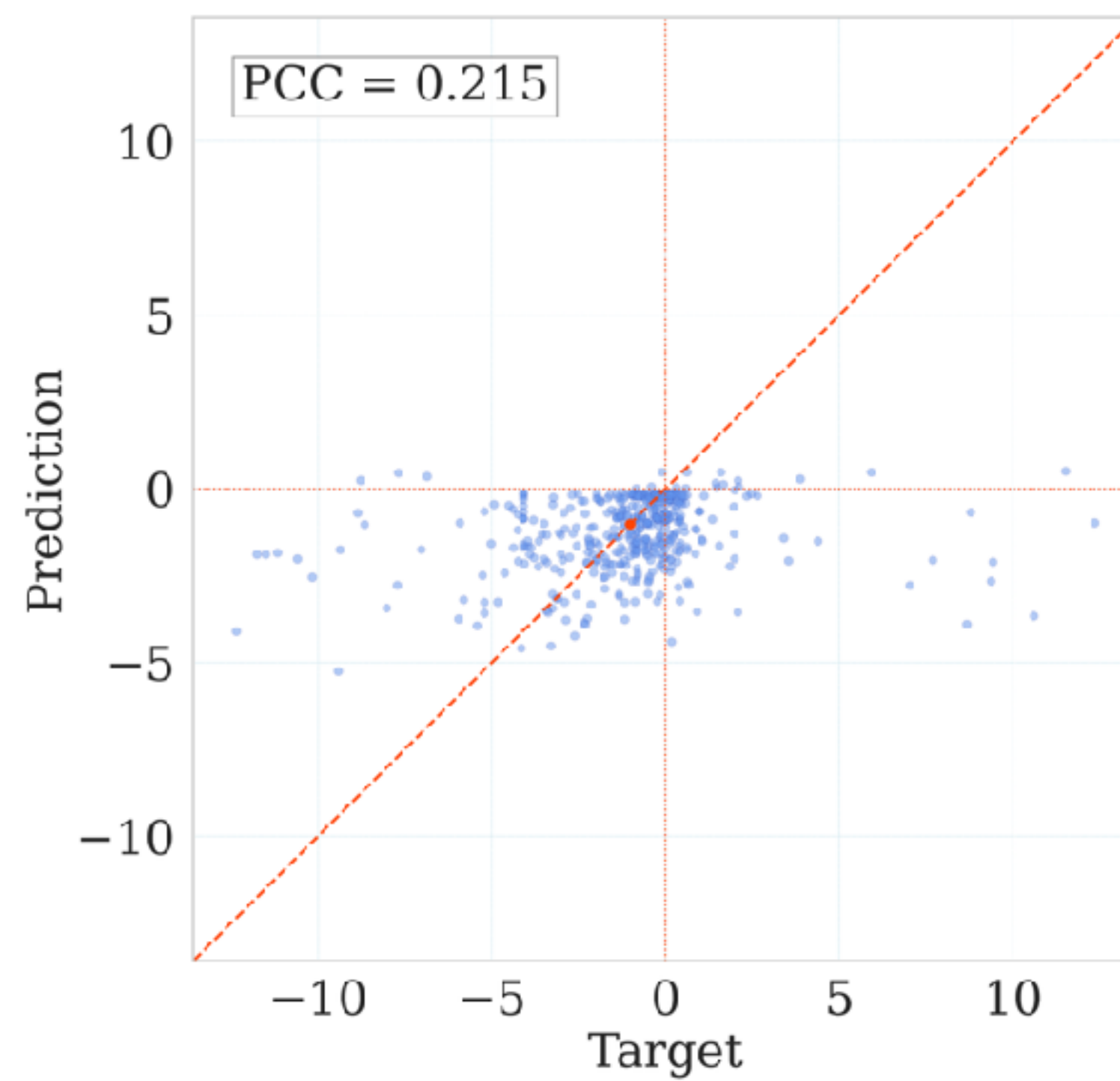
Run 14
Best Epoch: 638

Training: 1 2 3



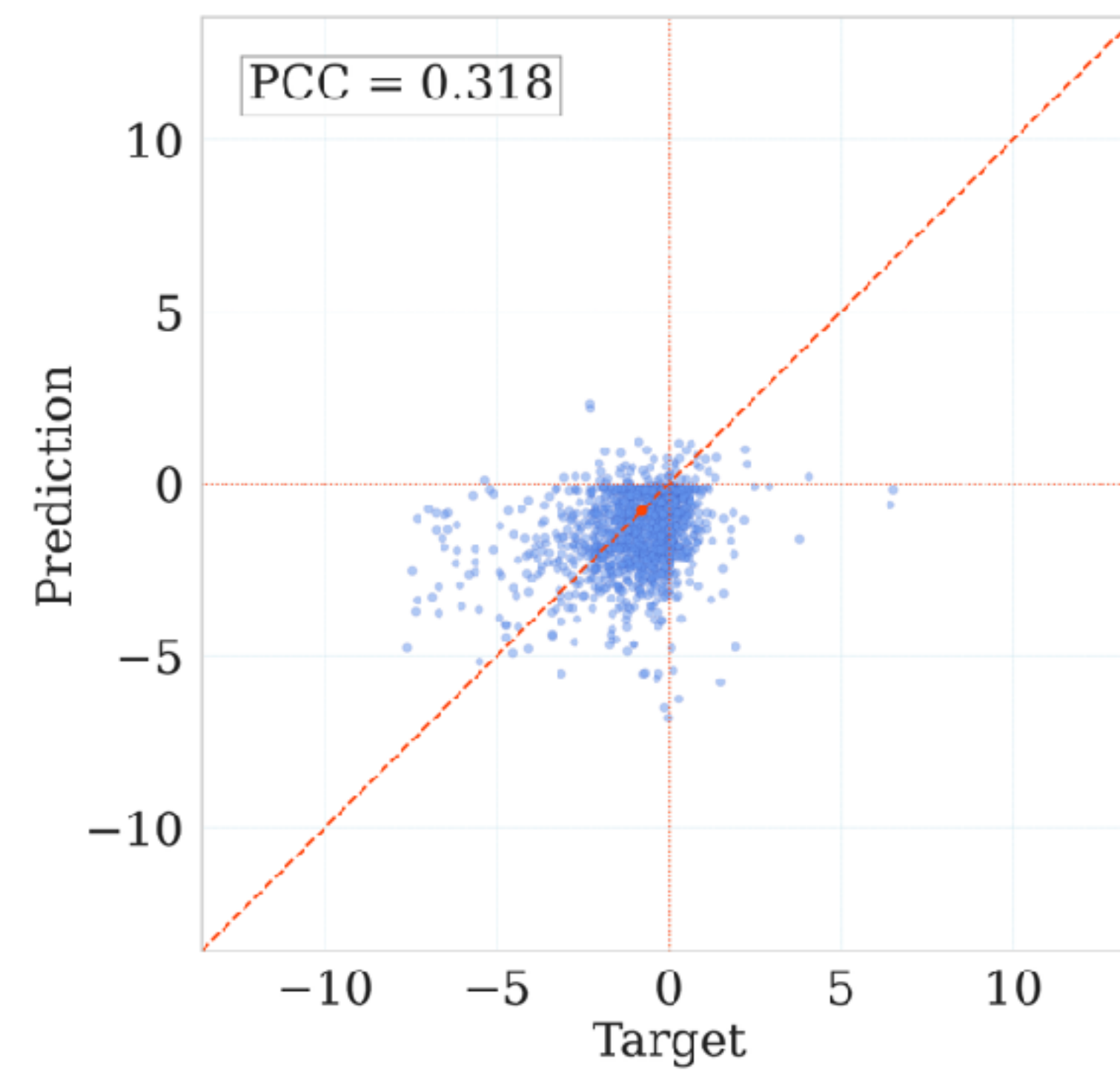
BEST

Validation: 4



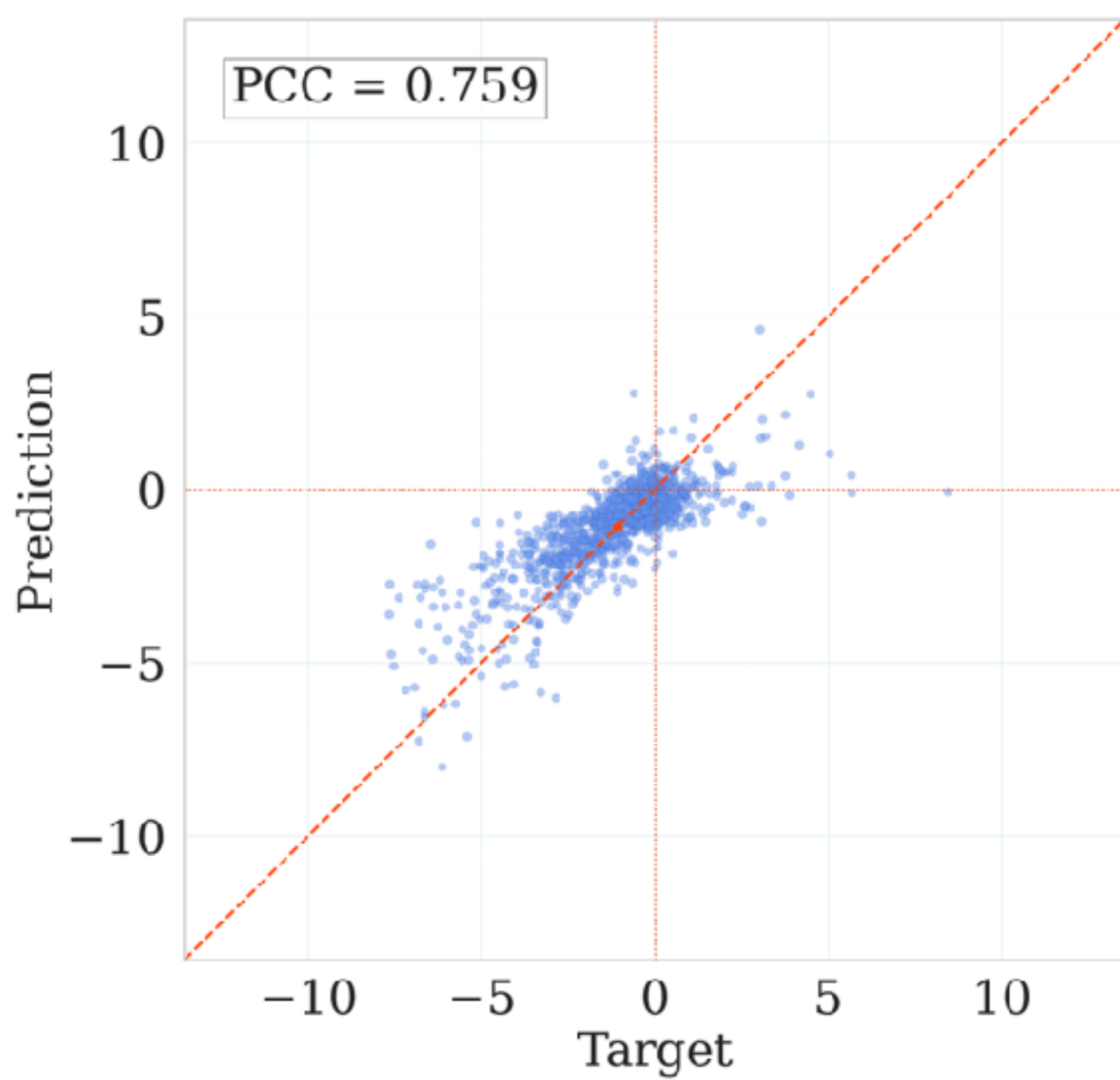
WORST

Test: 0

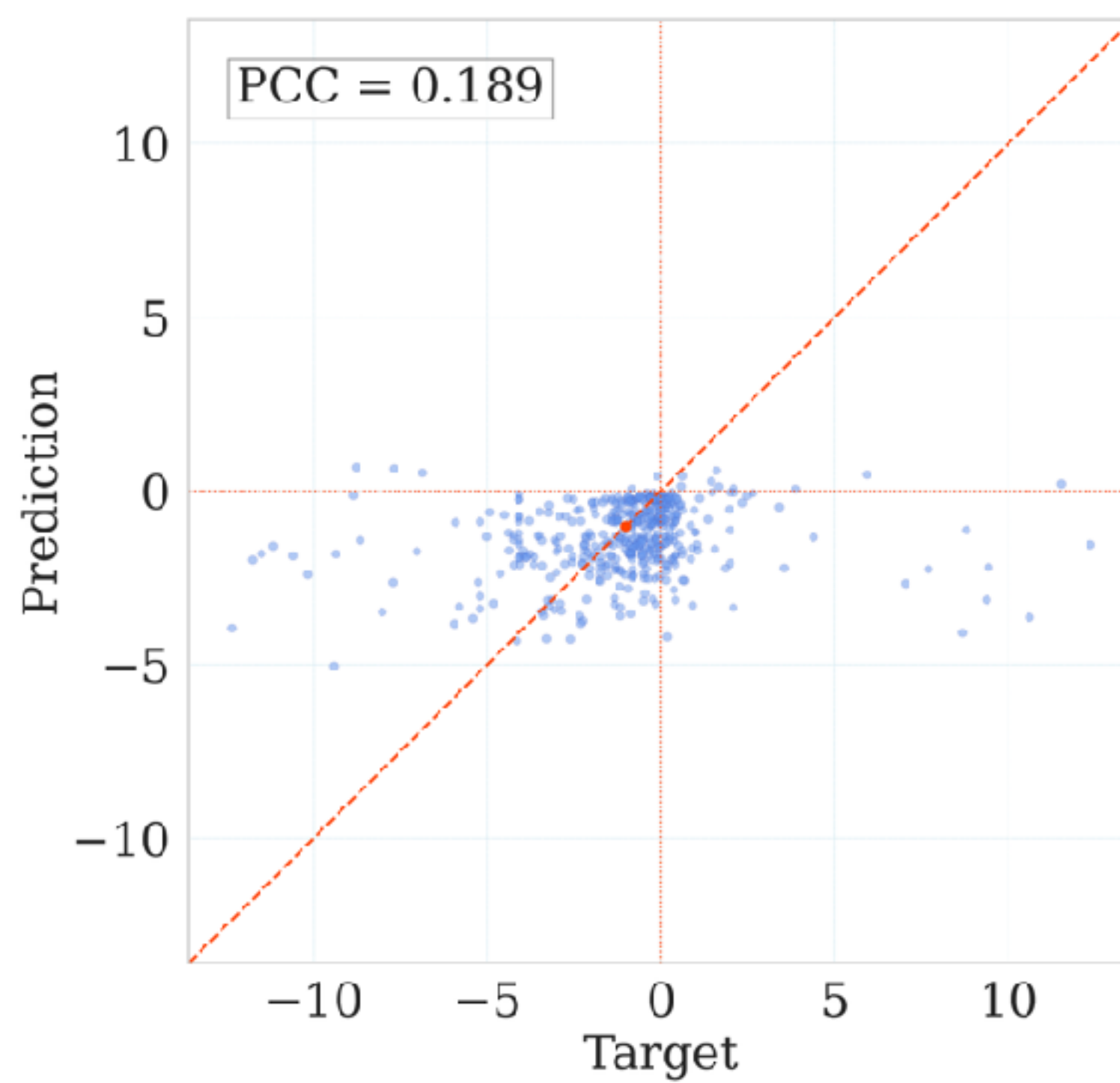


Run 14
Last Epoch: 738

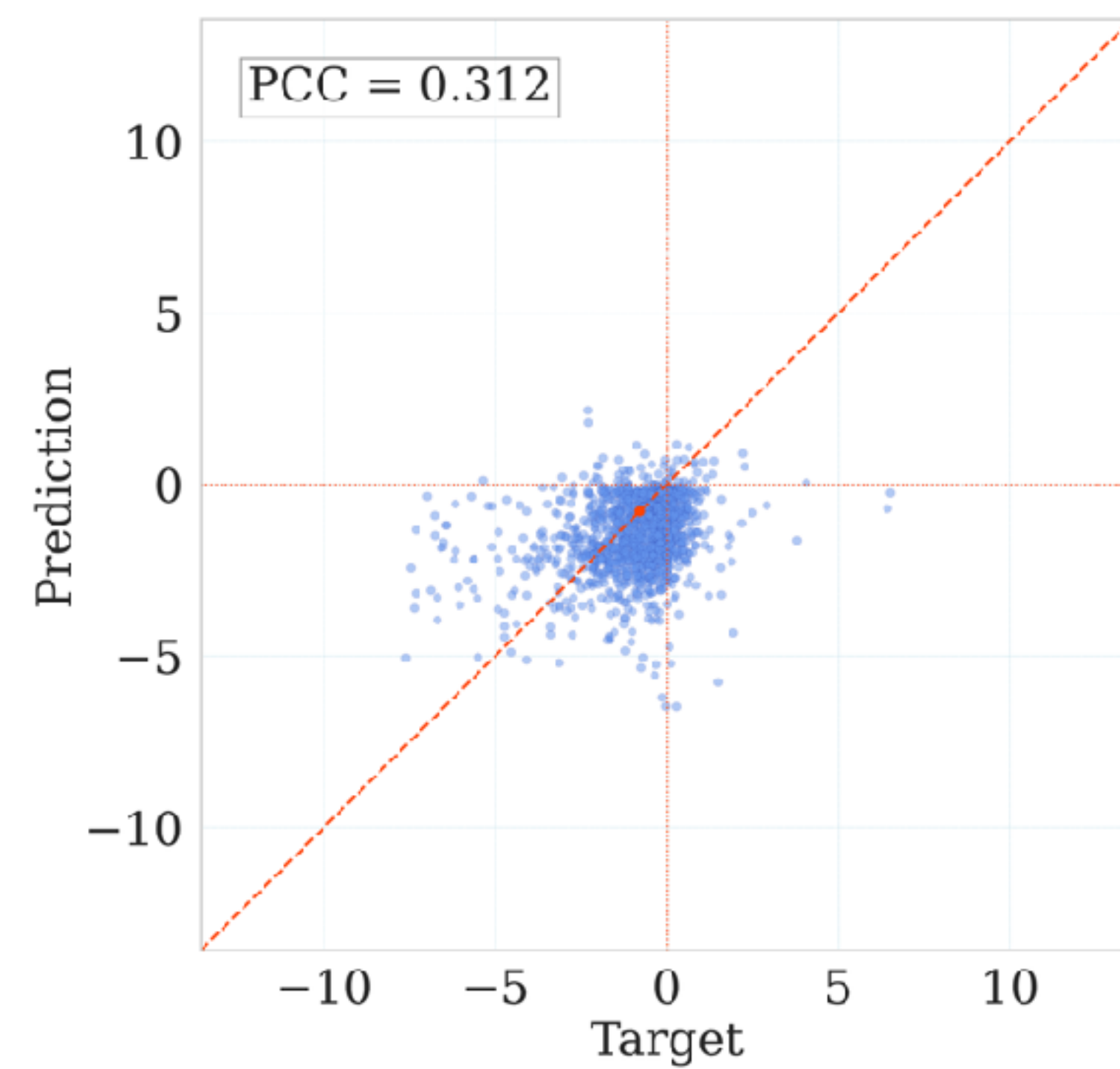
Training: 1 2 3



Validation: 4



Test: 0



Run 15

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

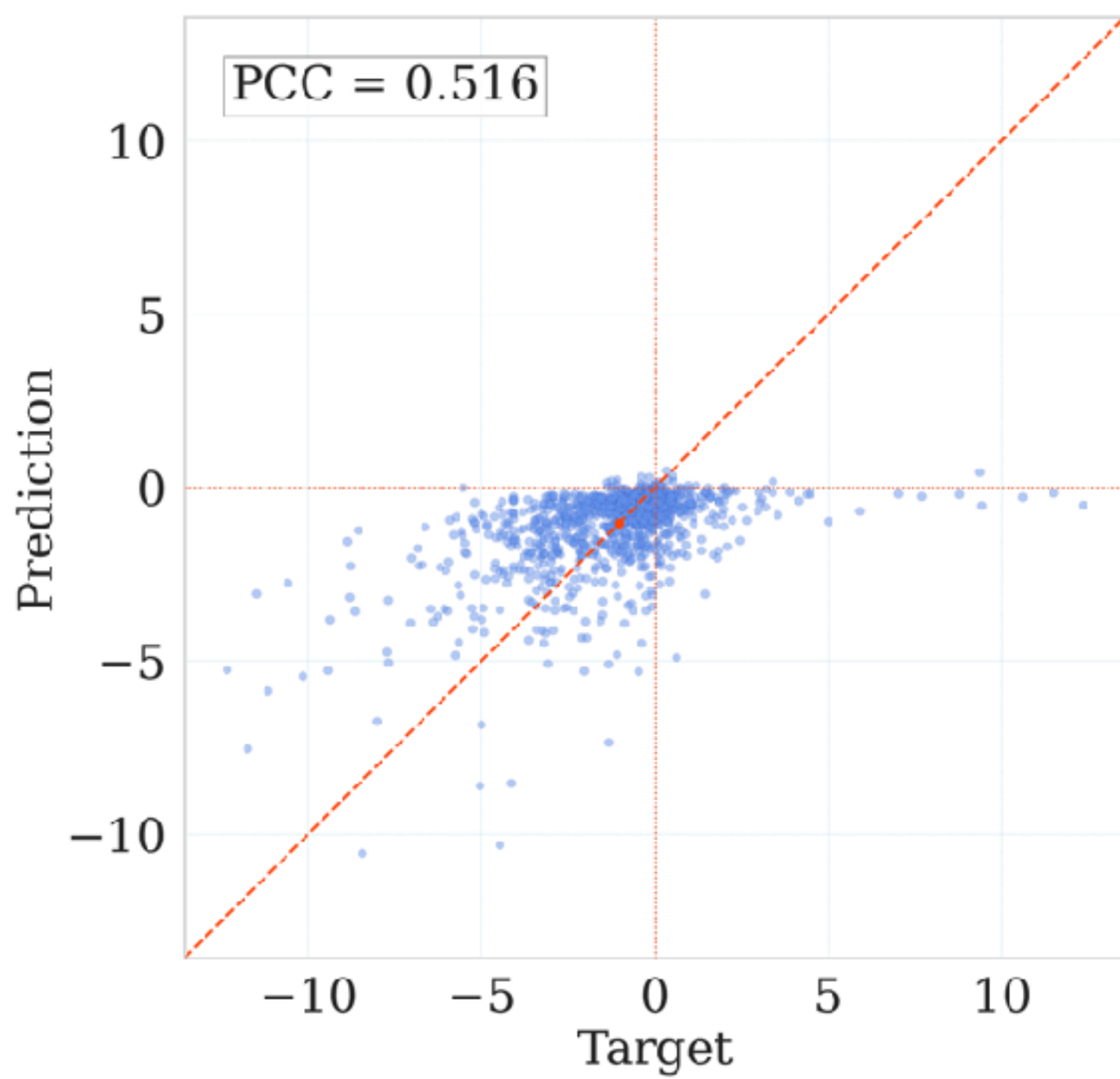
Training

Validation

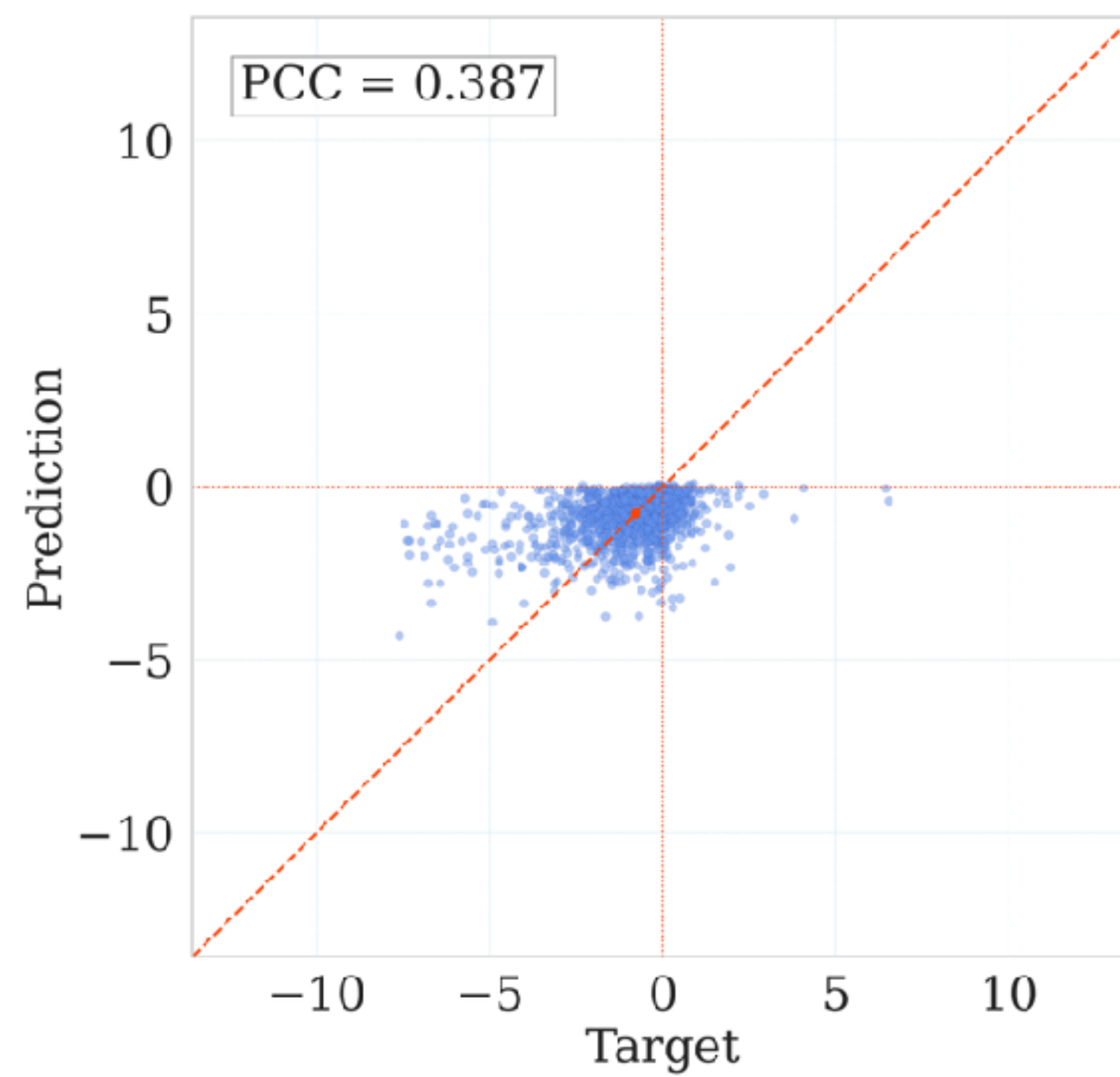
Test

Run 15
Best Epoch: 25

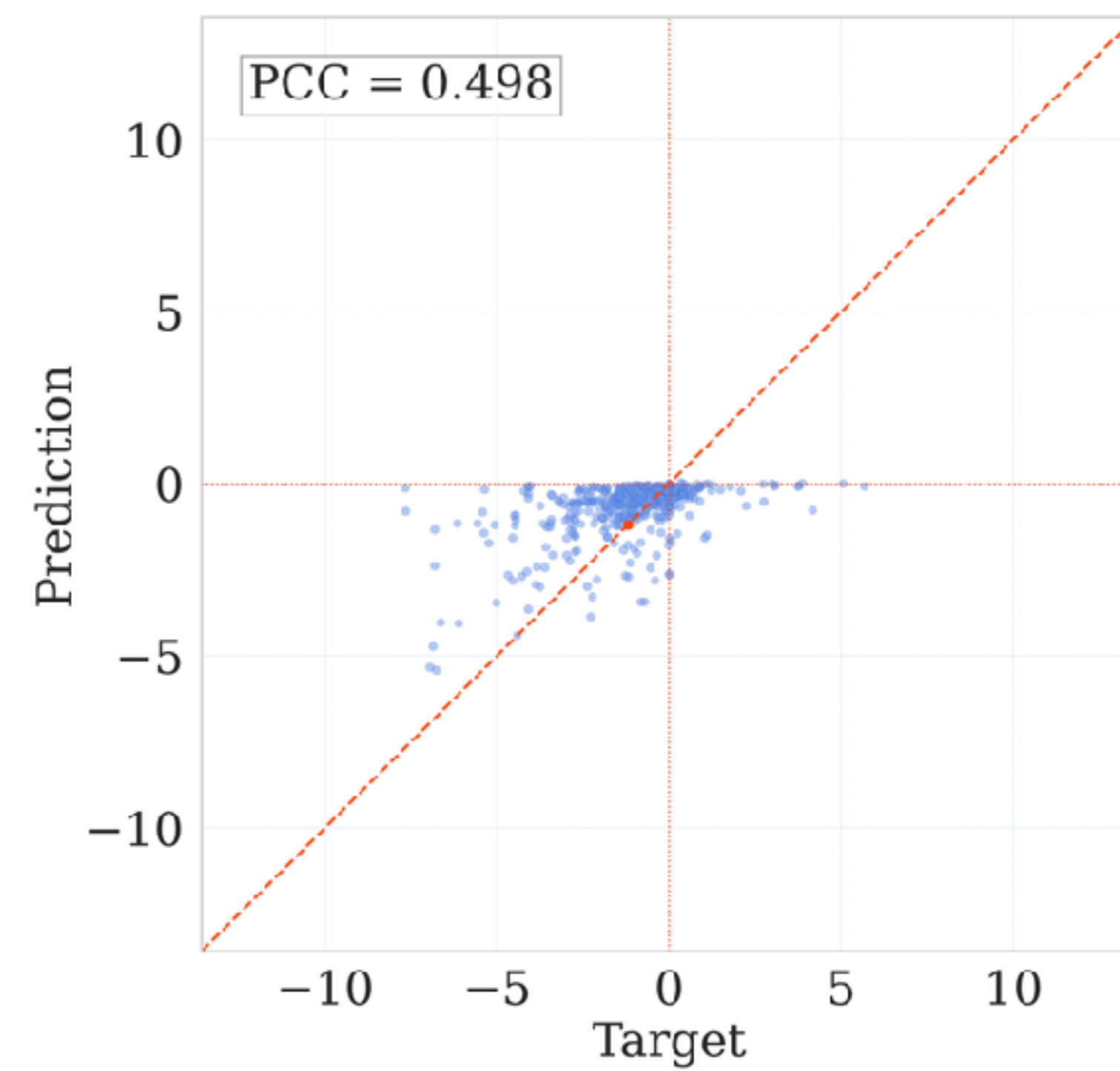
Training: 1 2 4



Validation: 0



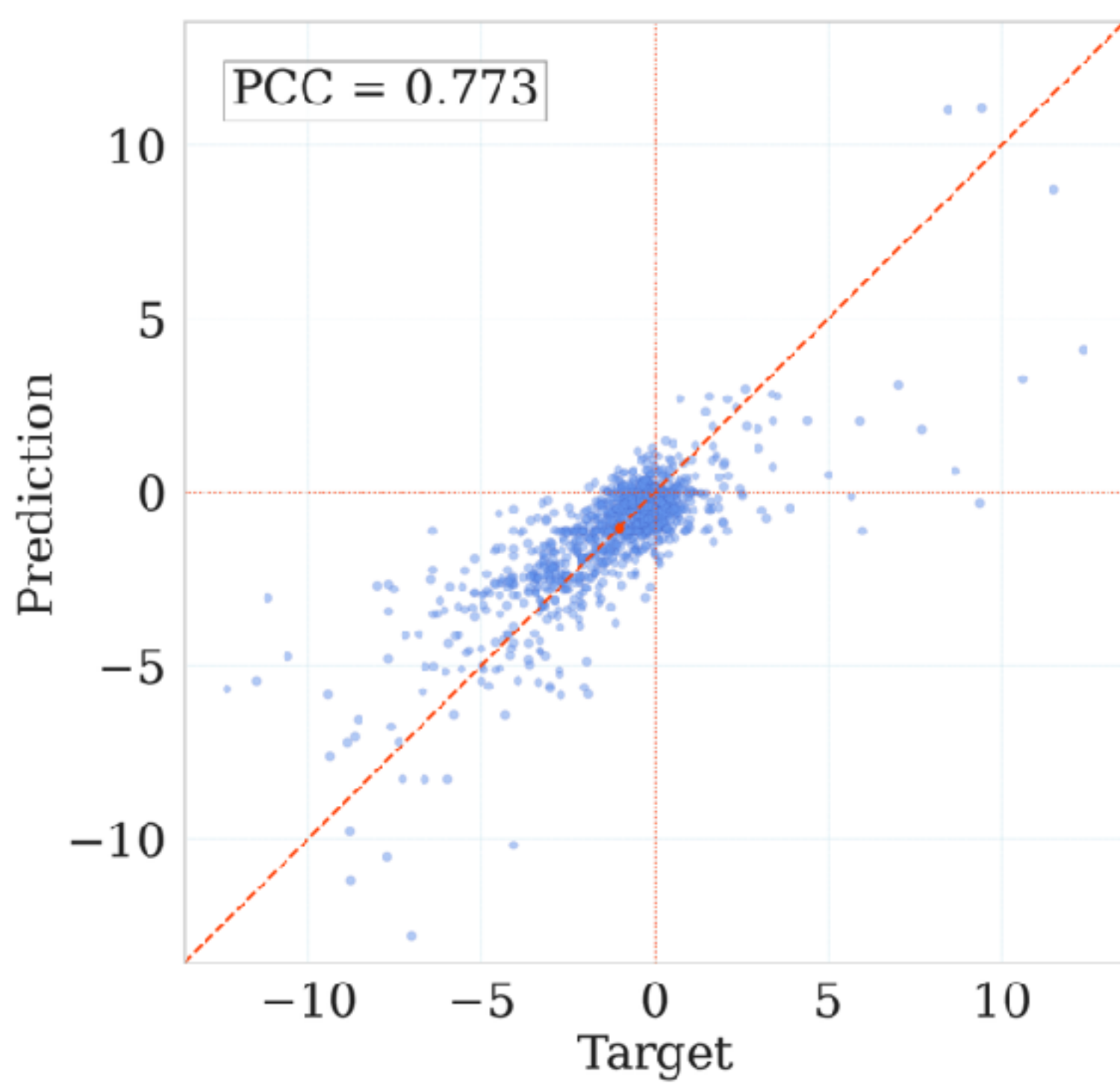
Test: 3



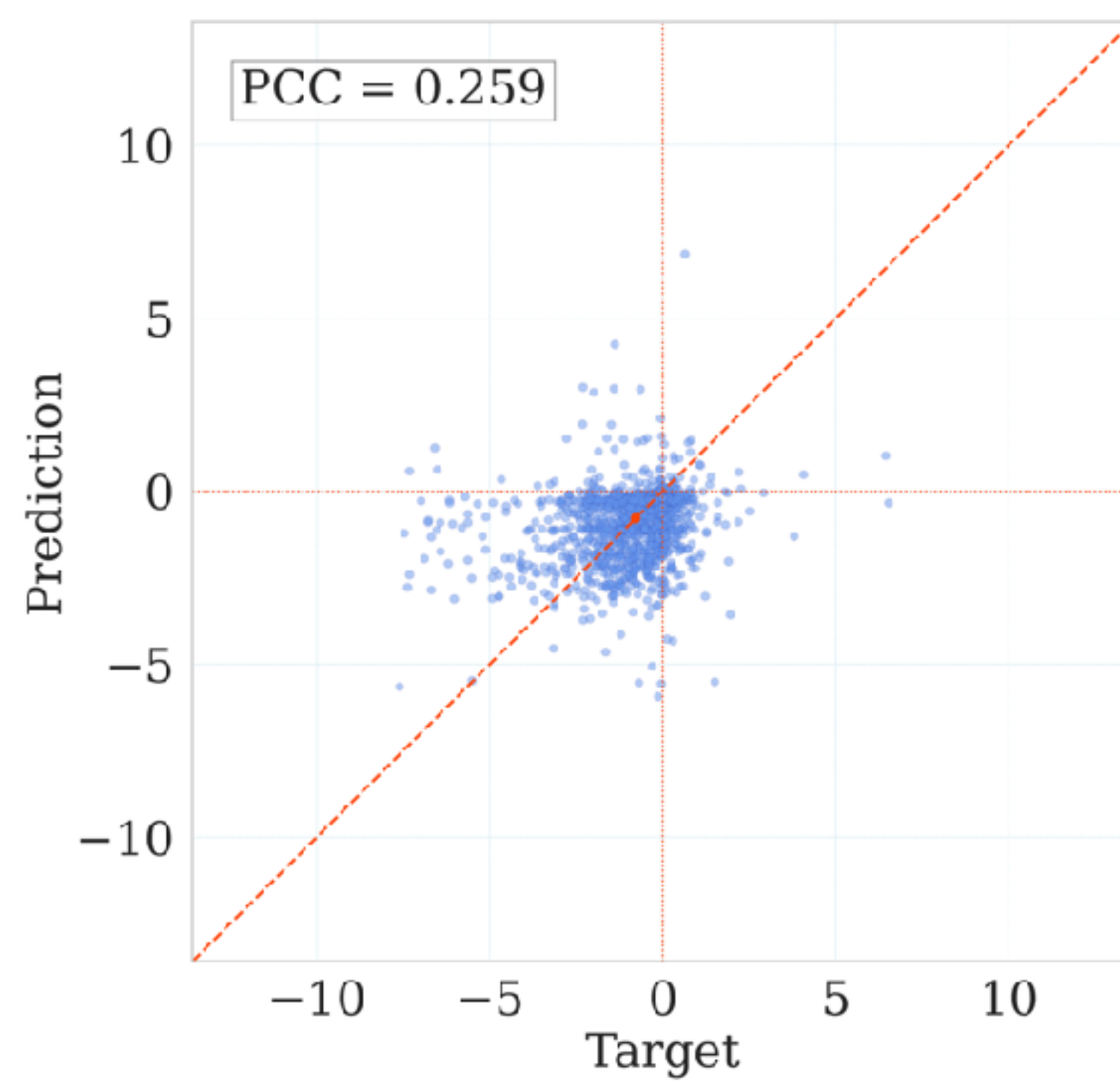
BEST

Run 15
Last Epoch: 662

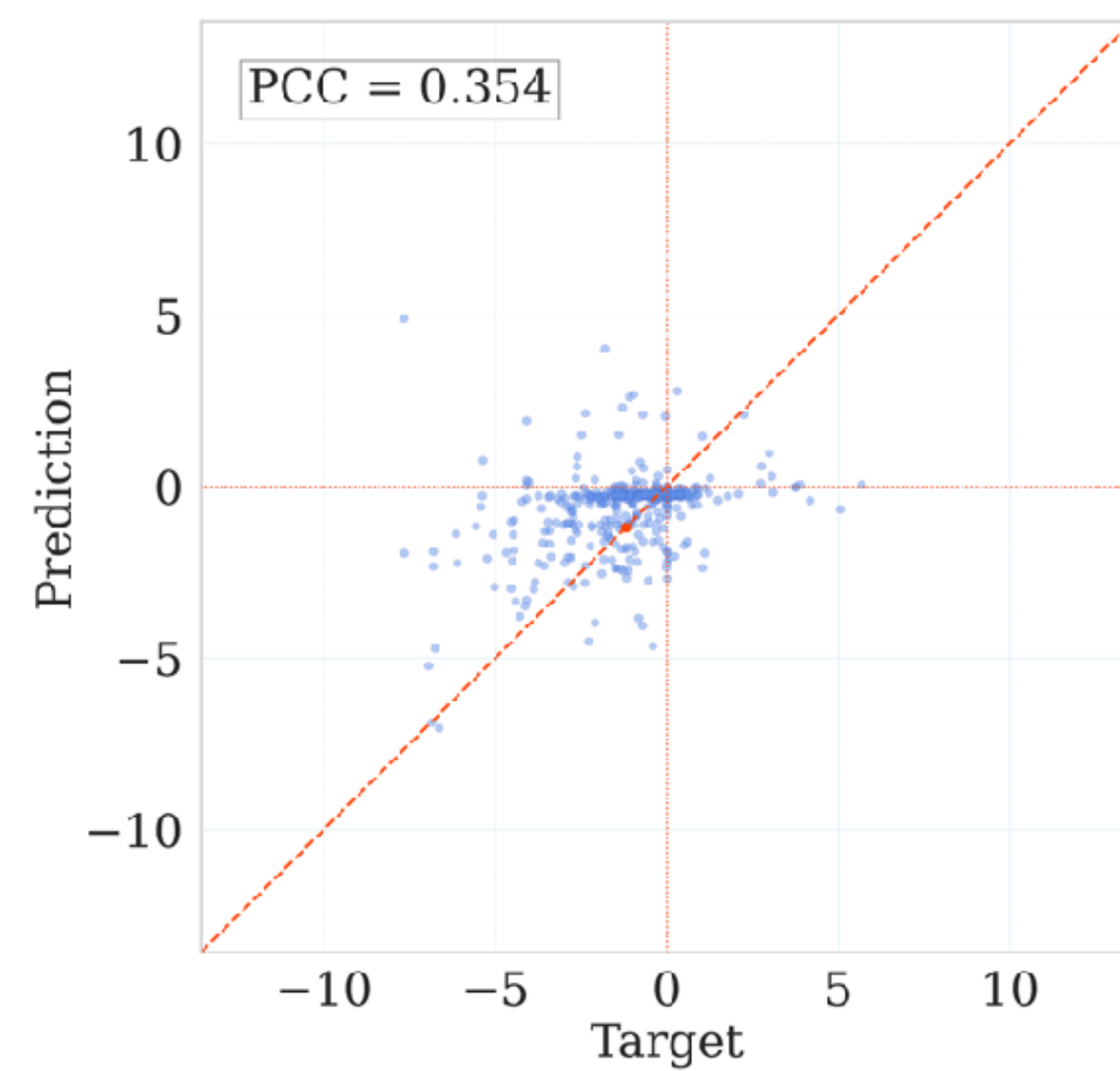
Training: 1 2 4



Validation: 0

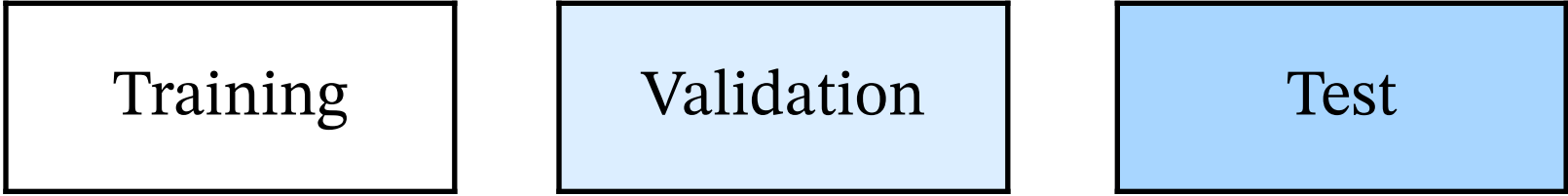


Test: 3



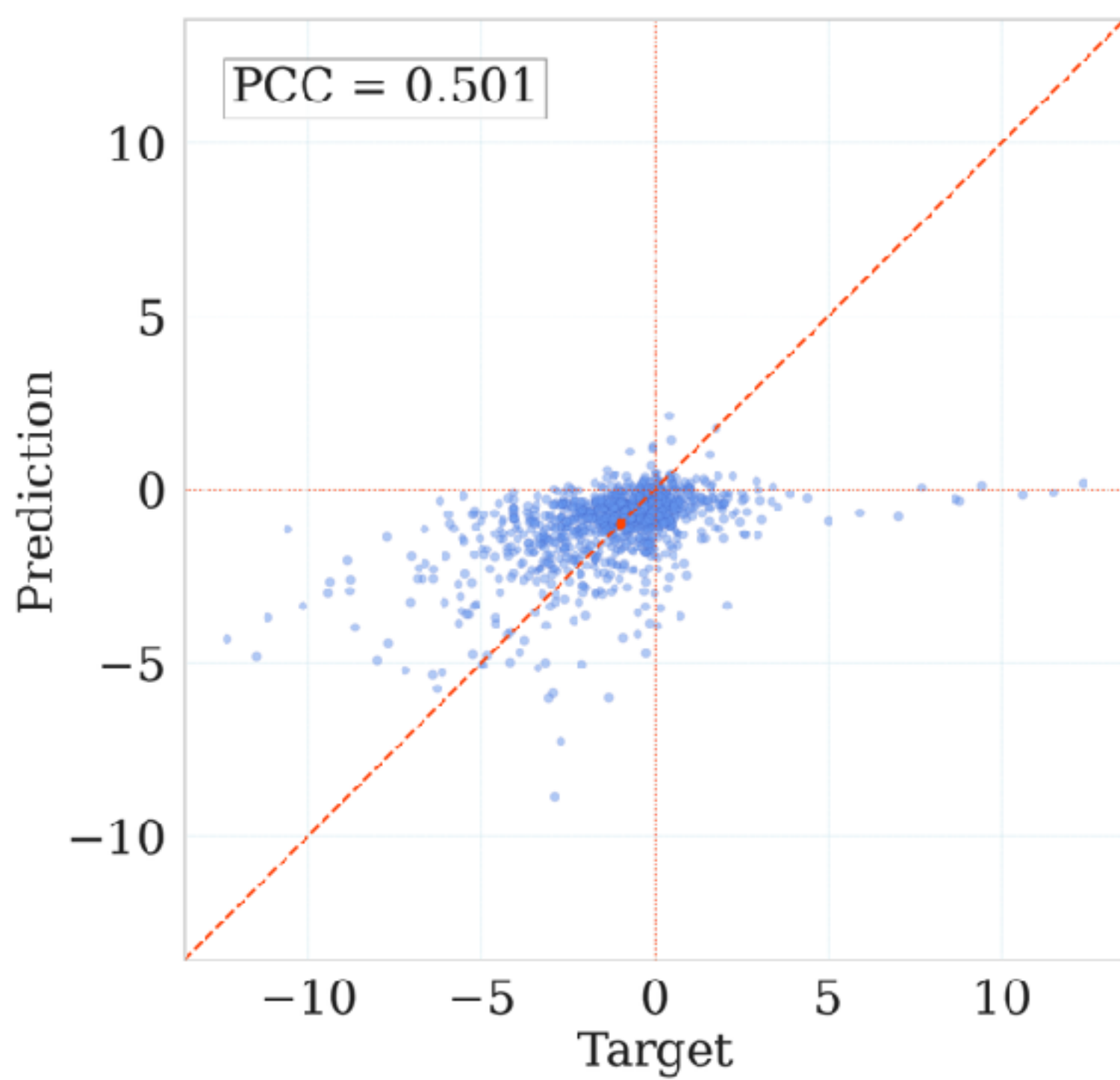
Run 16

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

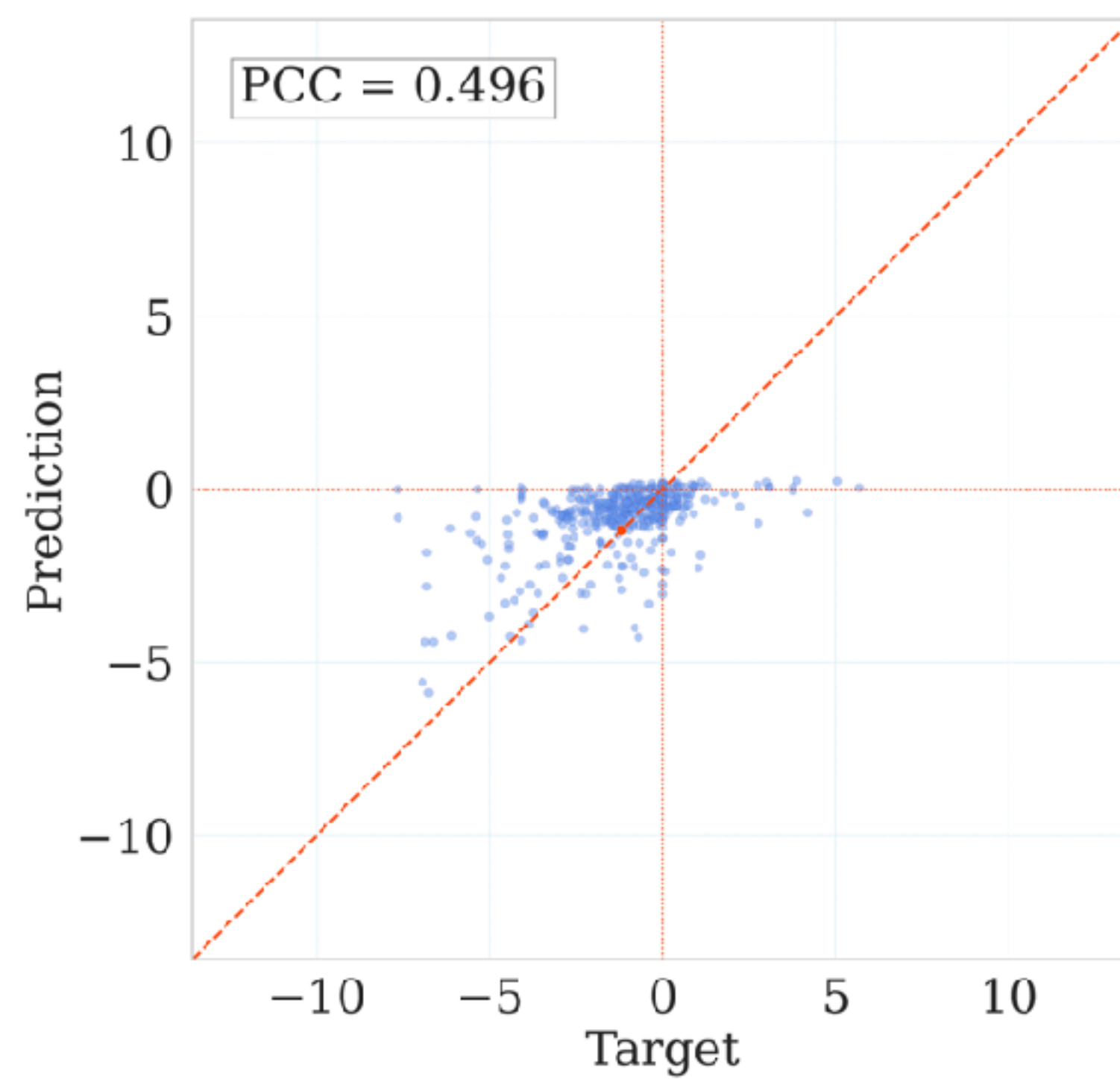


Run 16
Best Epoch: 24

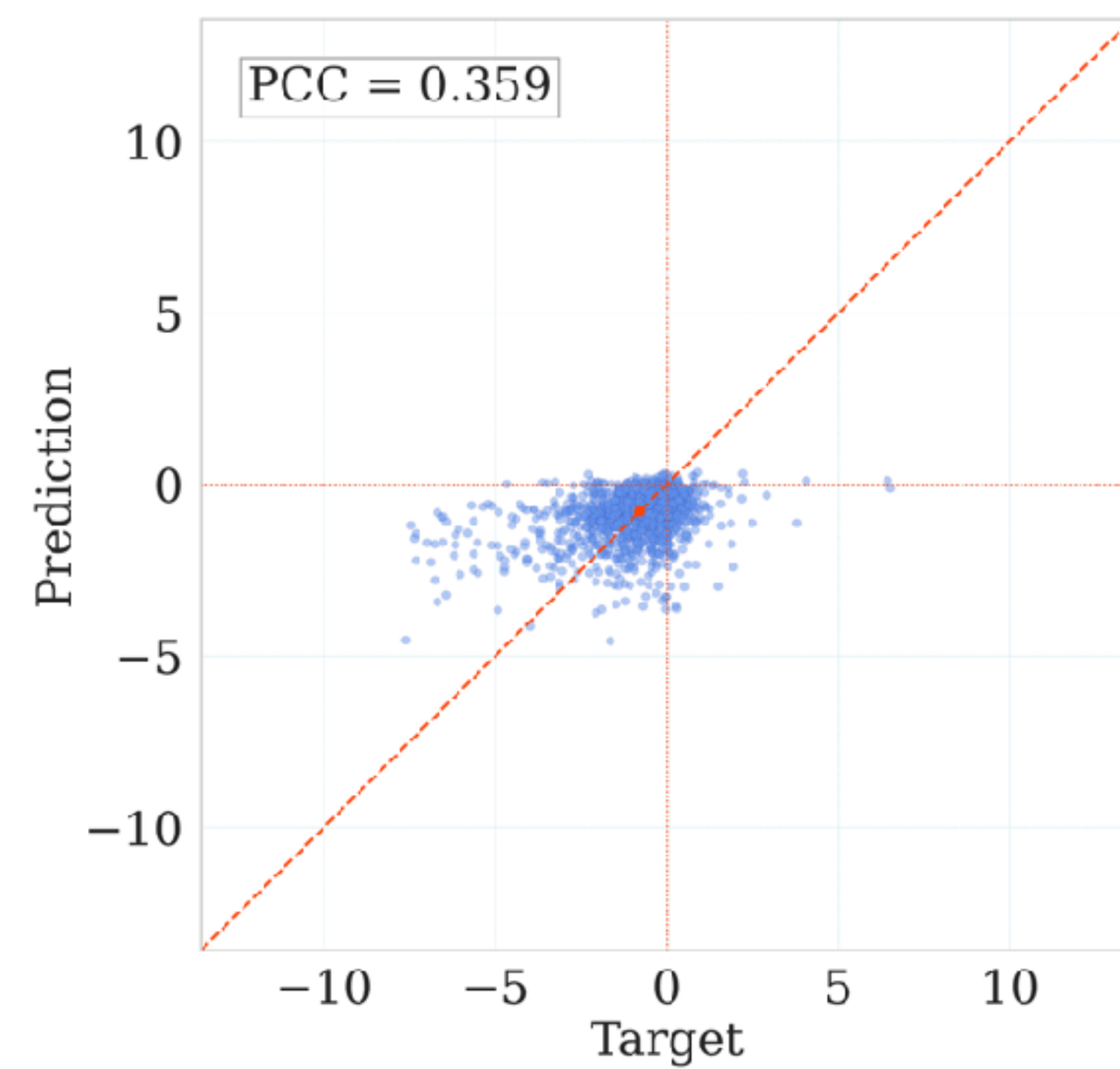
Training: 1 2 4



Validation: 3

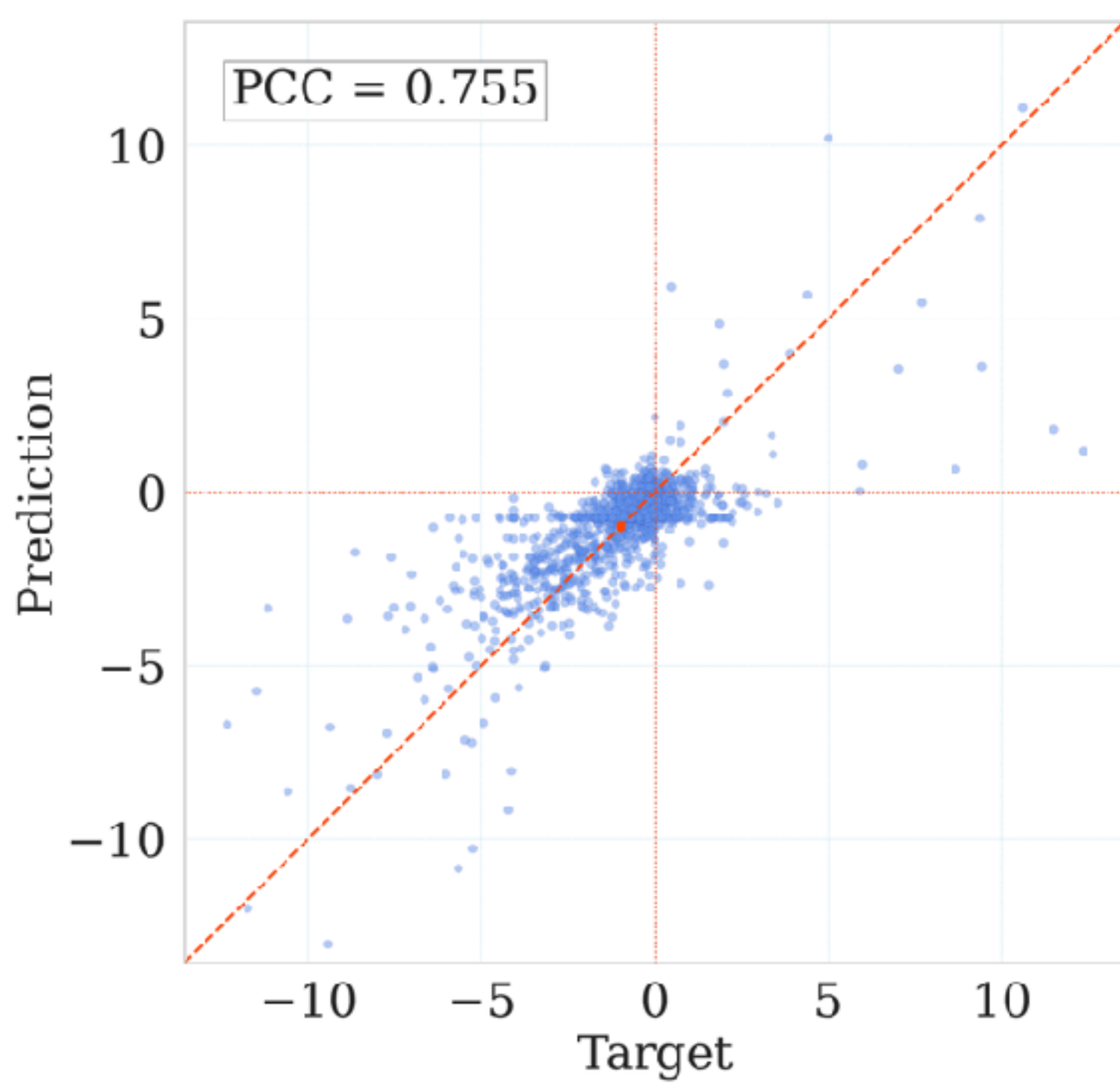


Test: 0



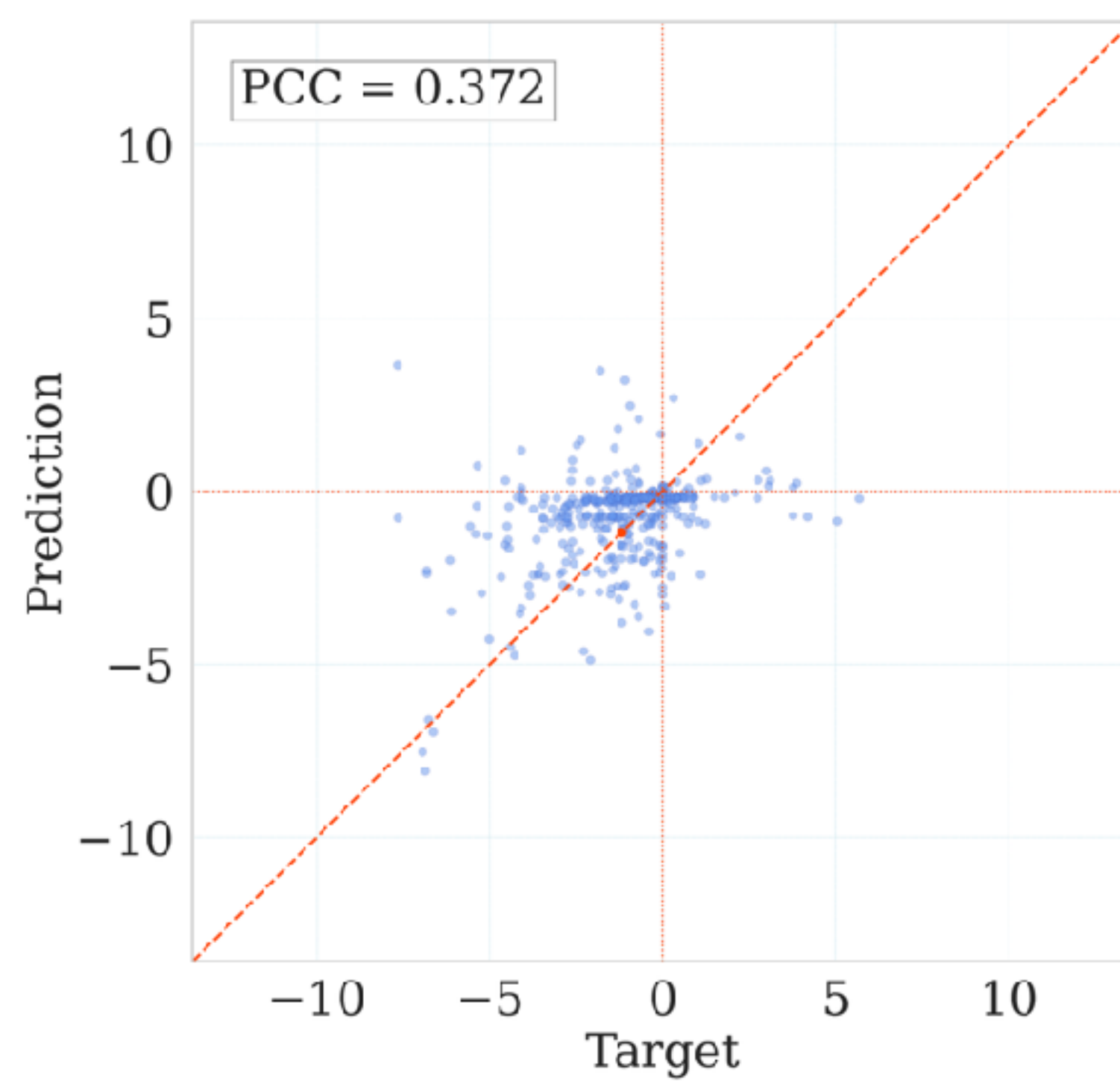
Run 16
Last Epoch: 641

Training: 1 2 4

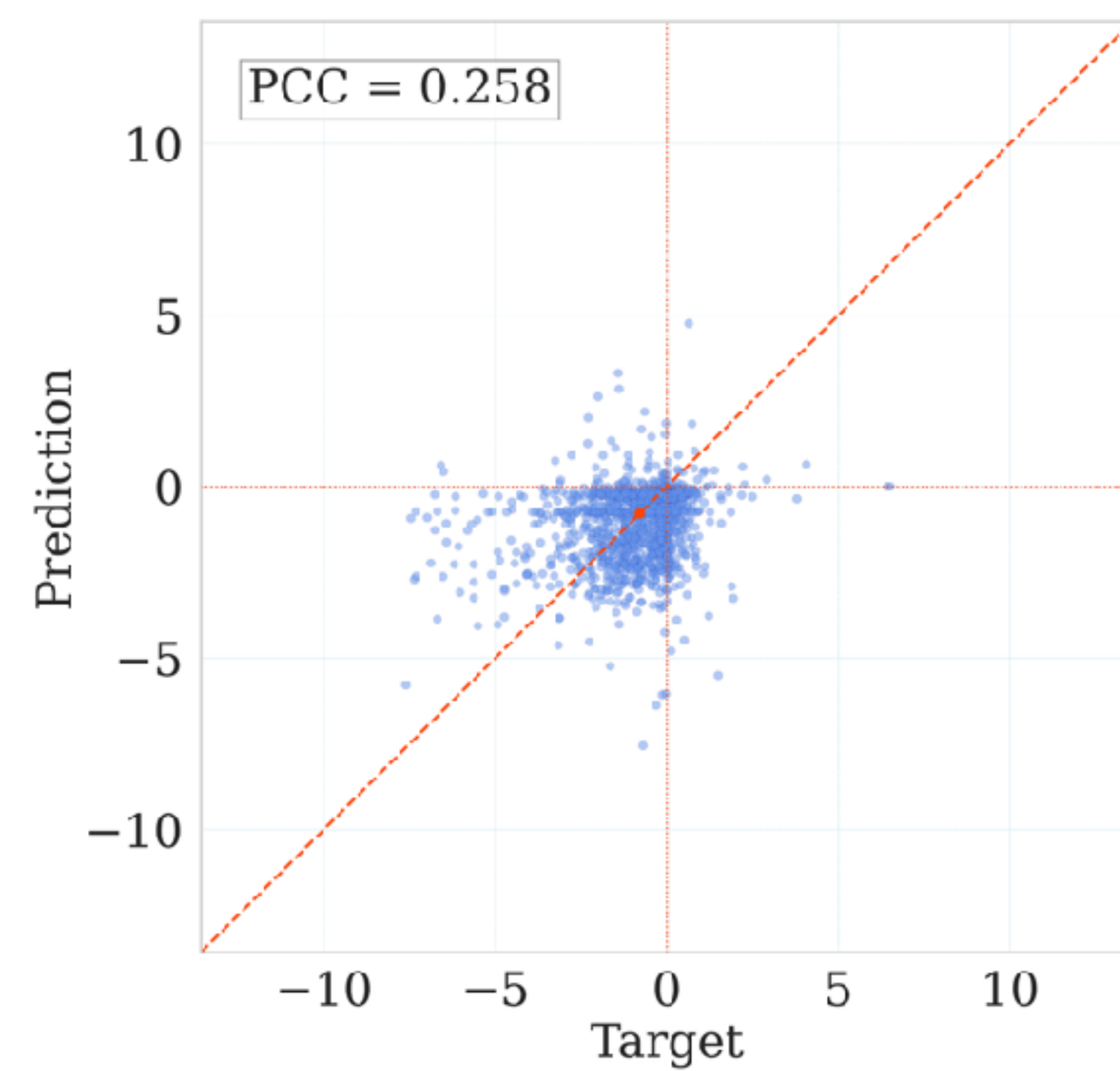


BEST

Validation: 3

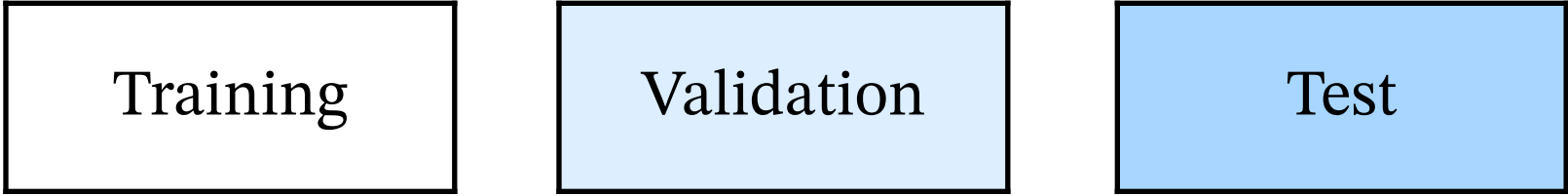


Test: 0



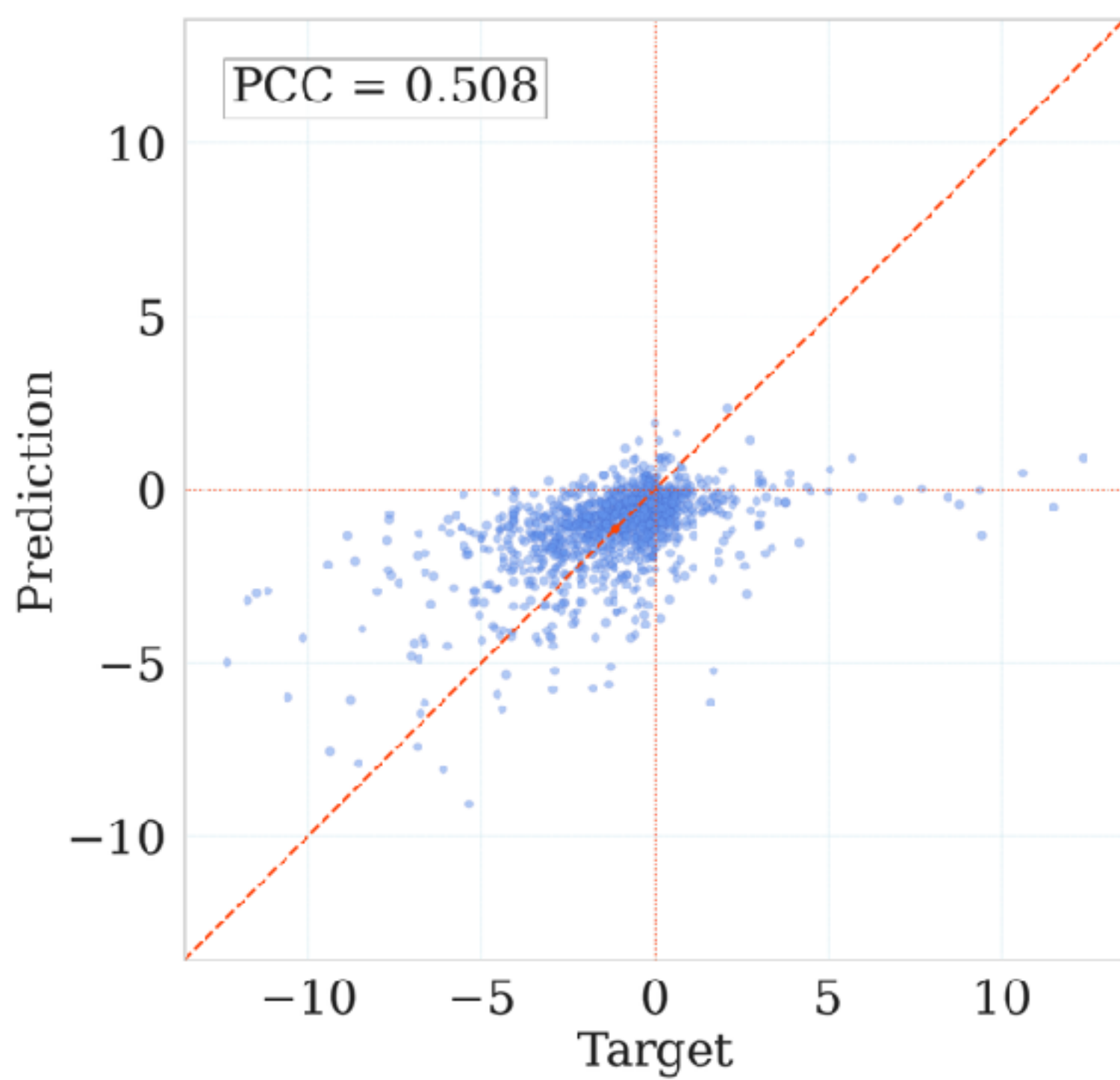
Run 17

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

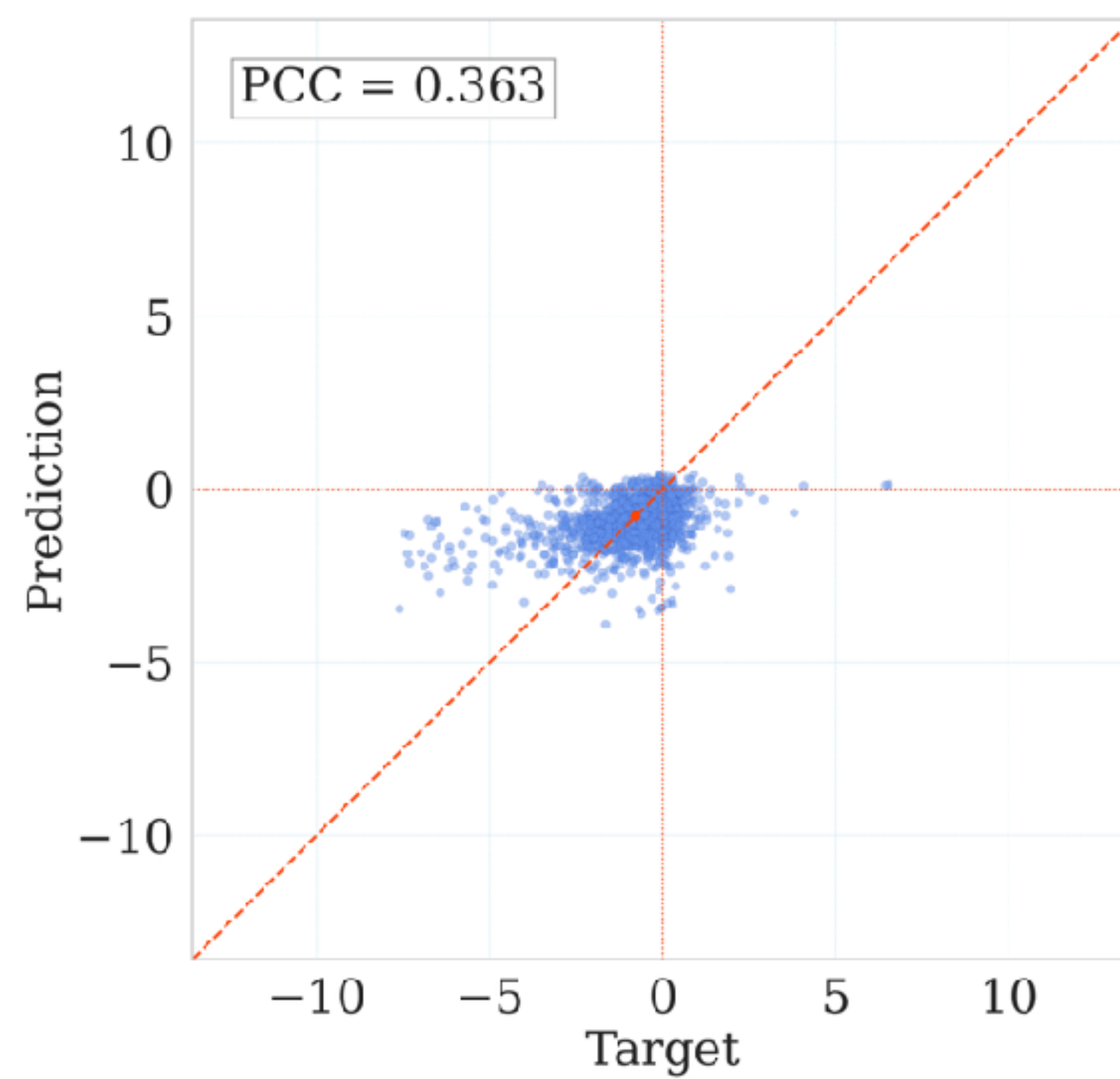


Run 17
Best Epoch: 27

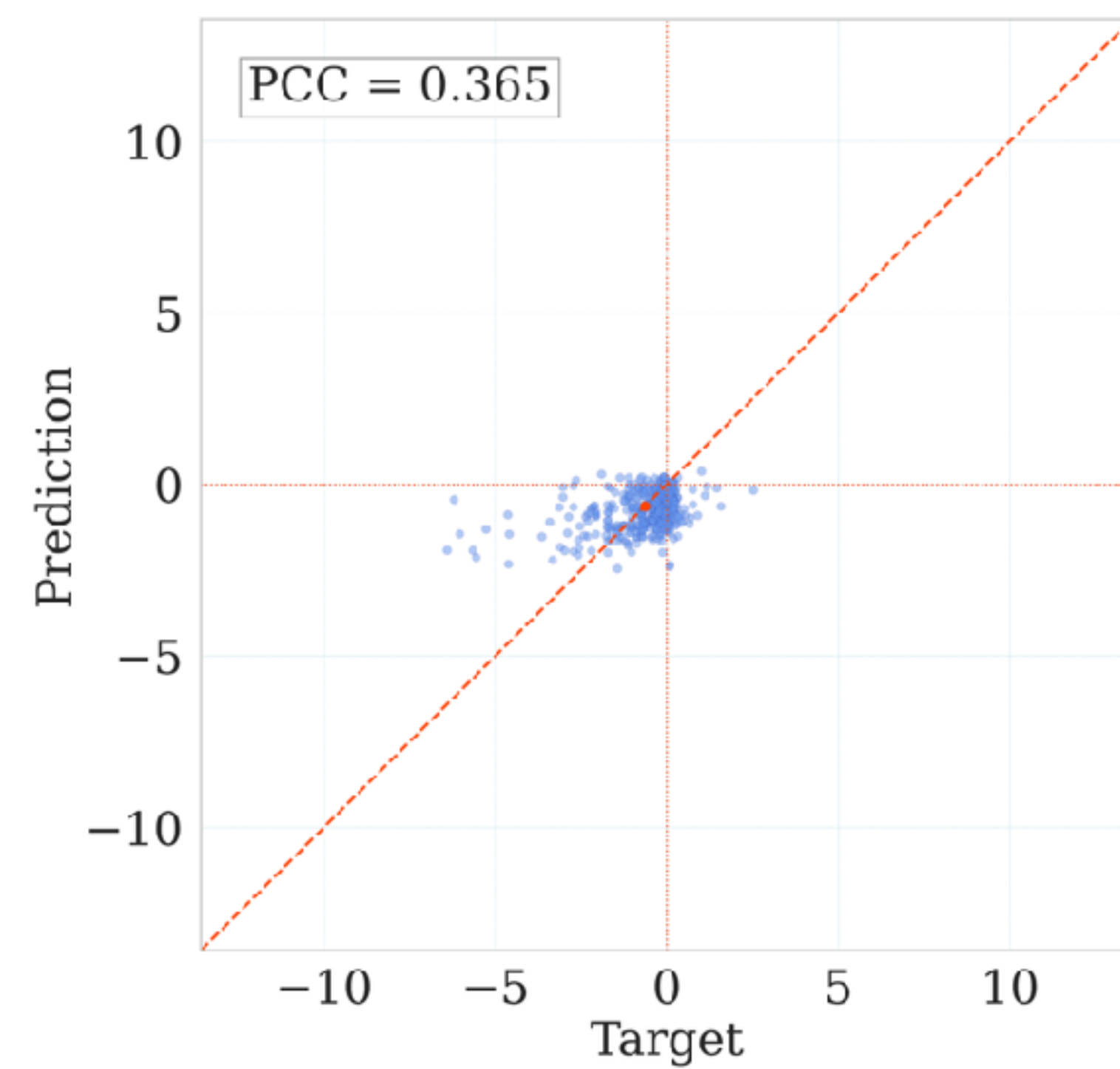
Training: 1 3 4



Validation: 0

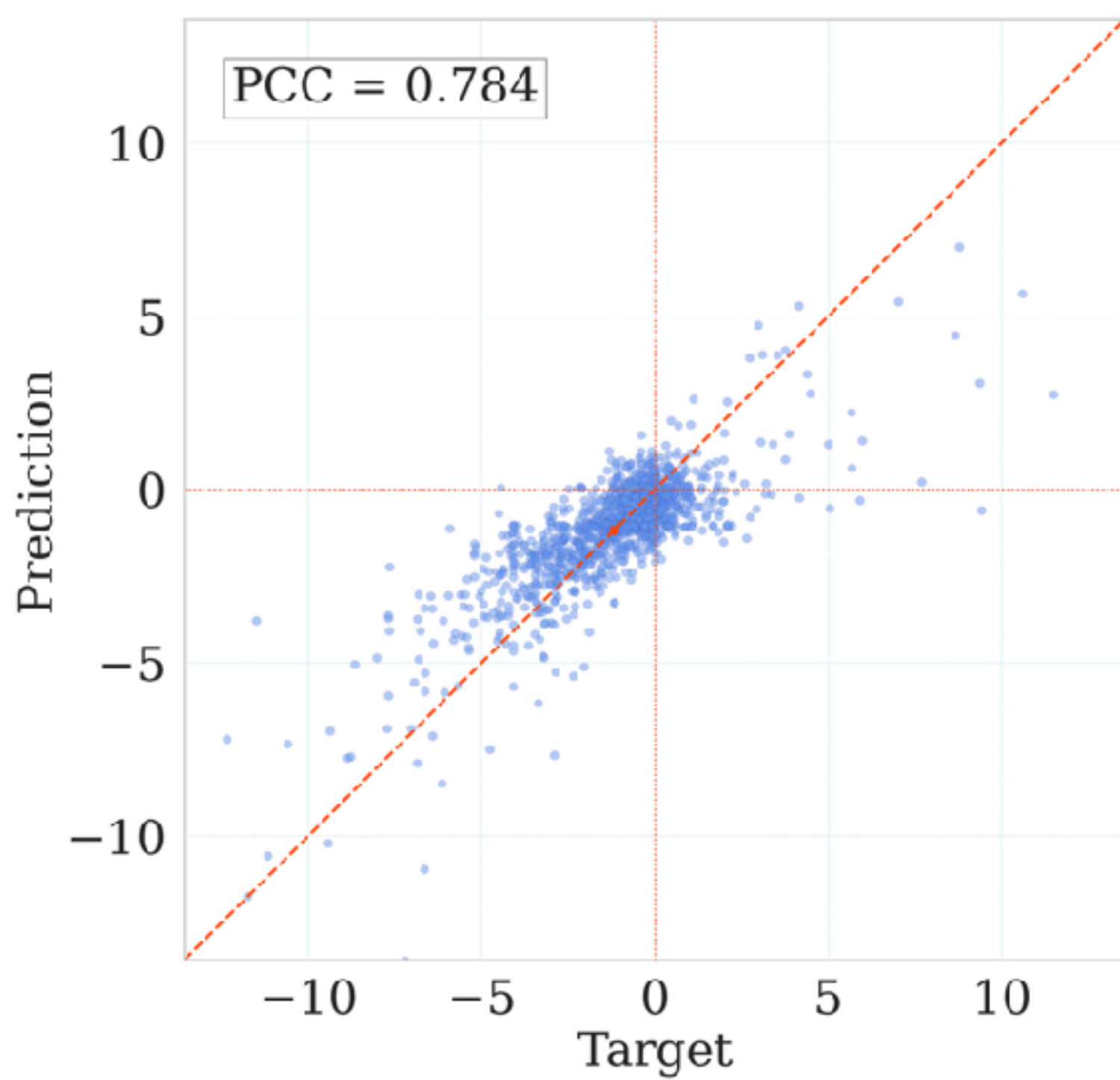


Test: 2

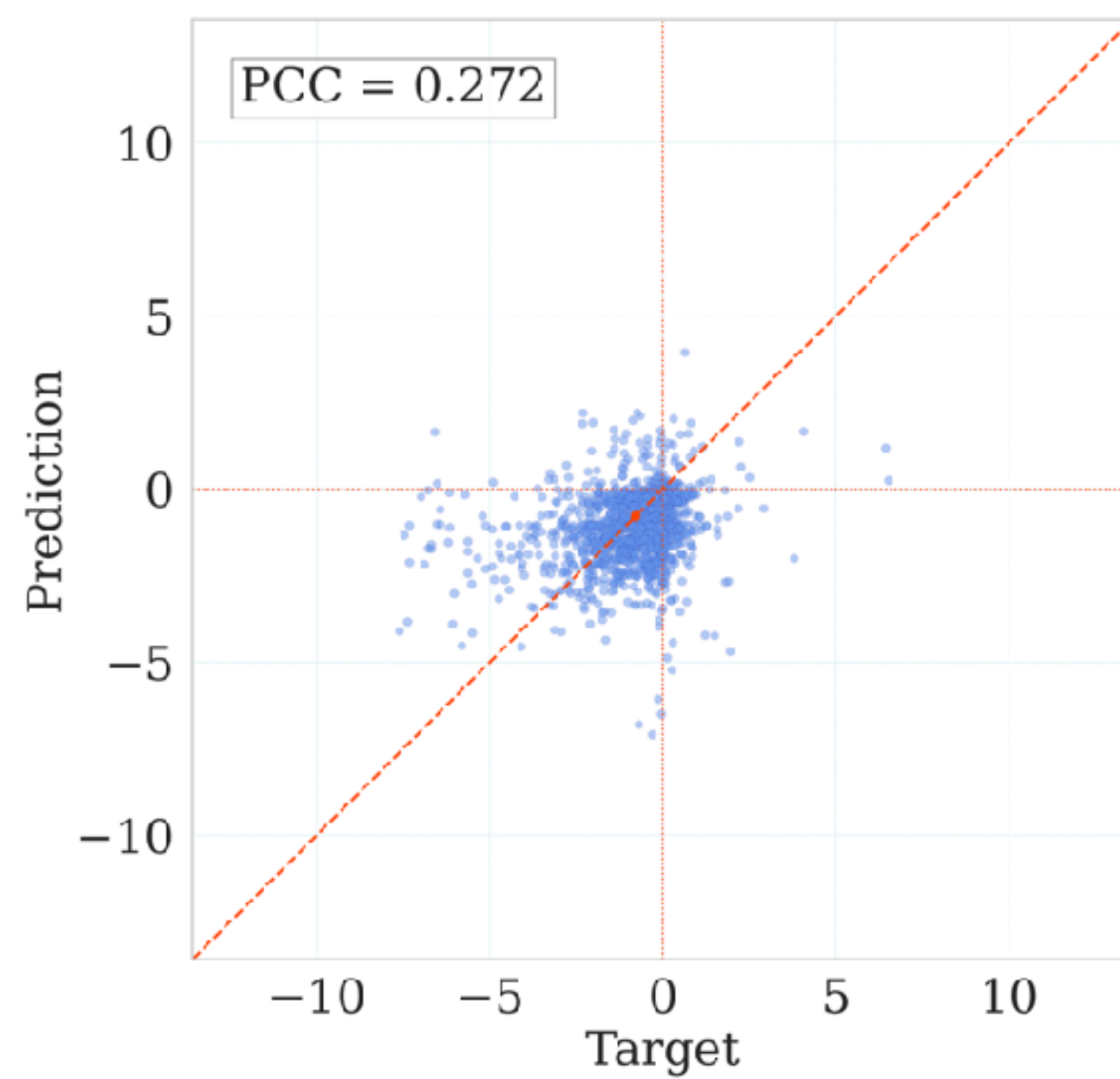


Run 17
Last Epoch: 621

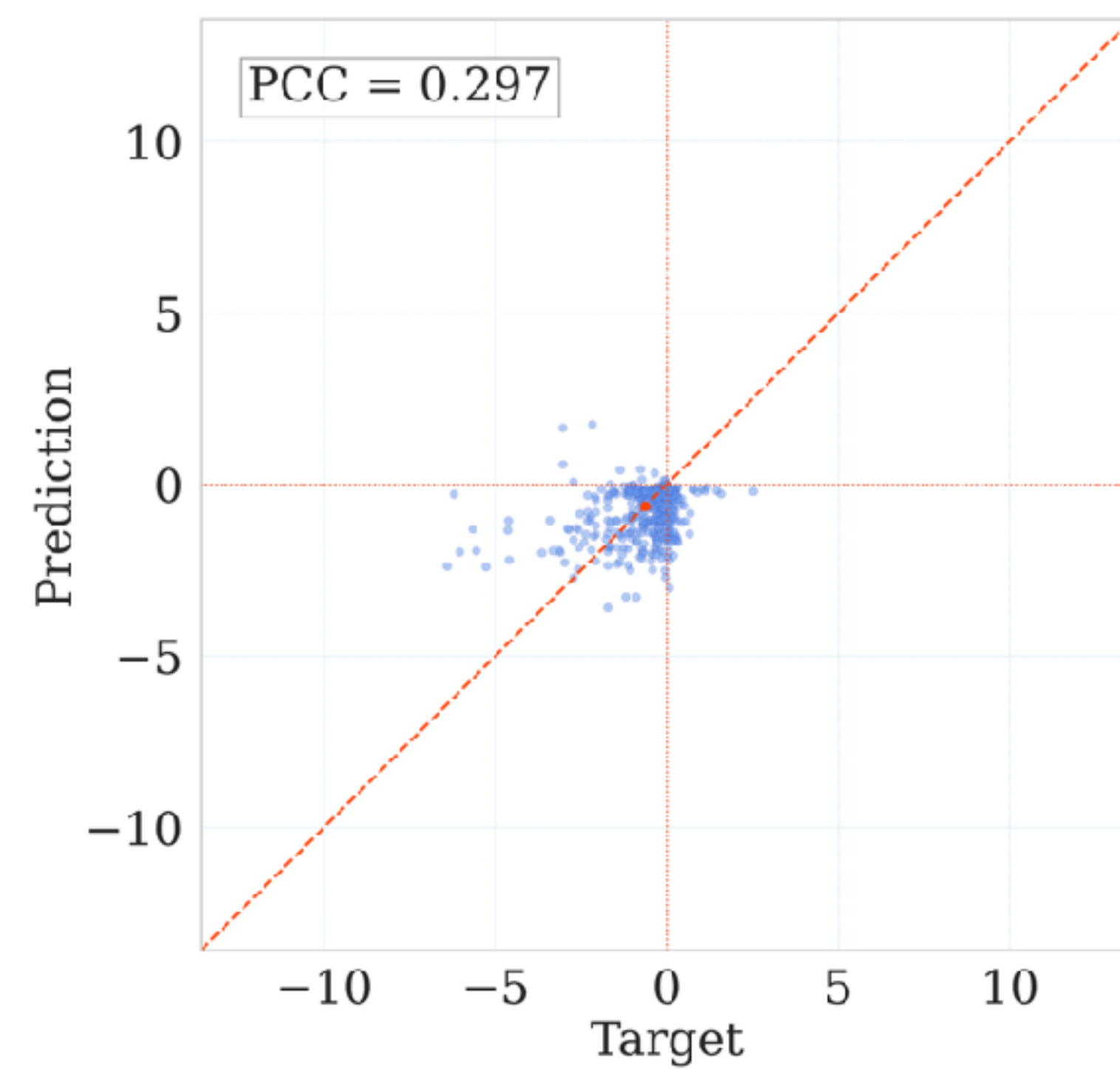
Training: 1 3 4



Validation: 0

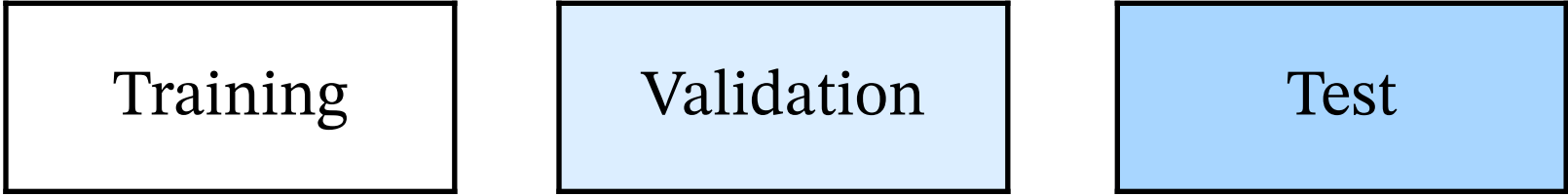


Test: 2



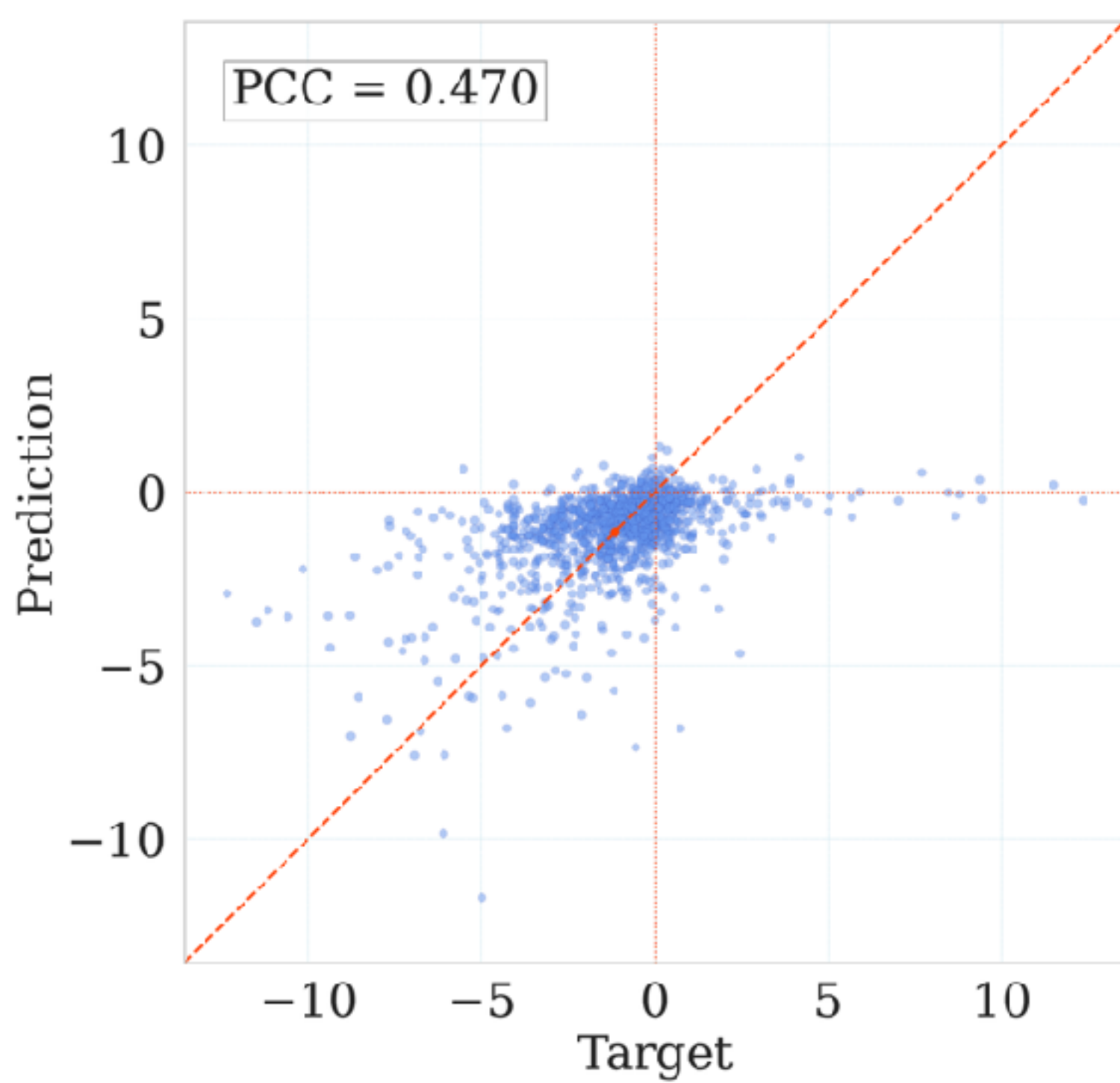
Run 18

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

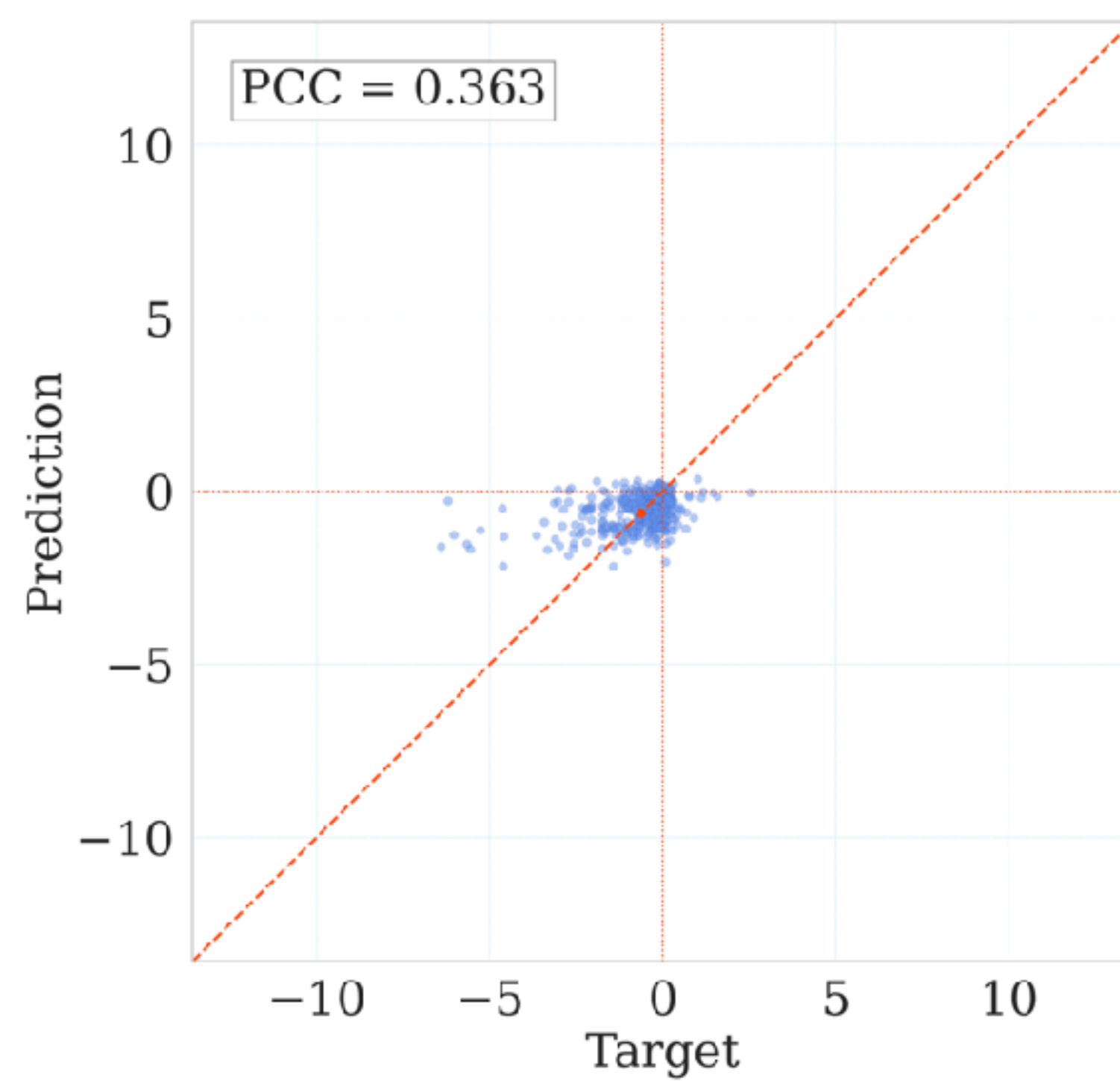


Run 18
Best Epoch: 28

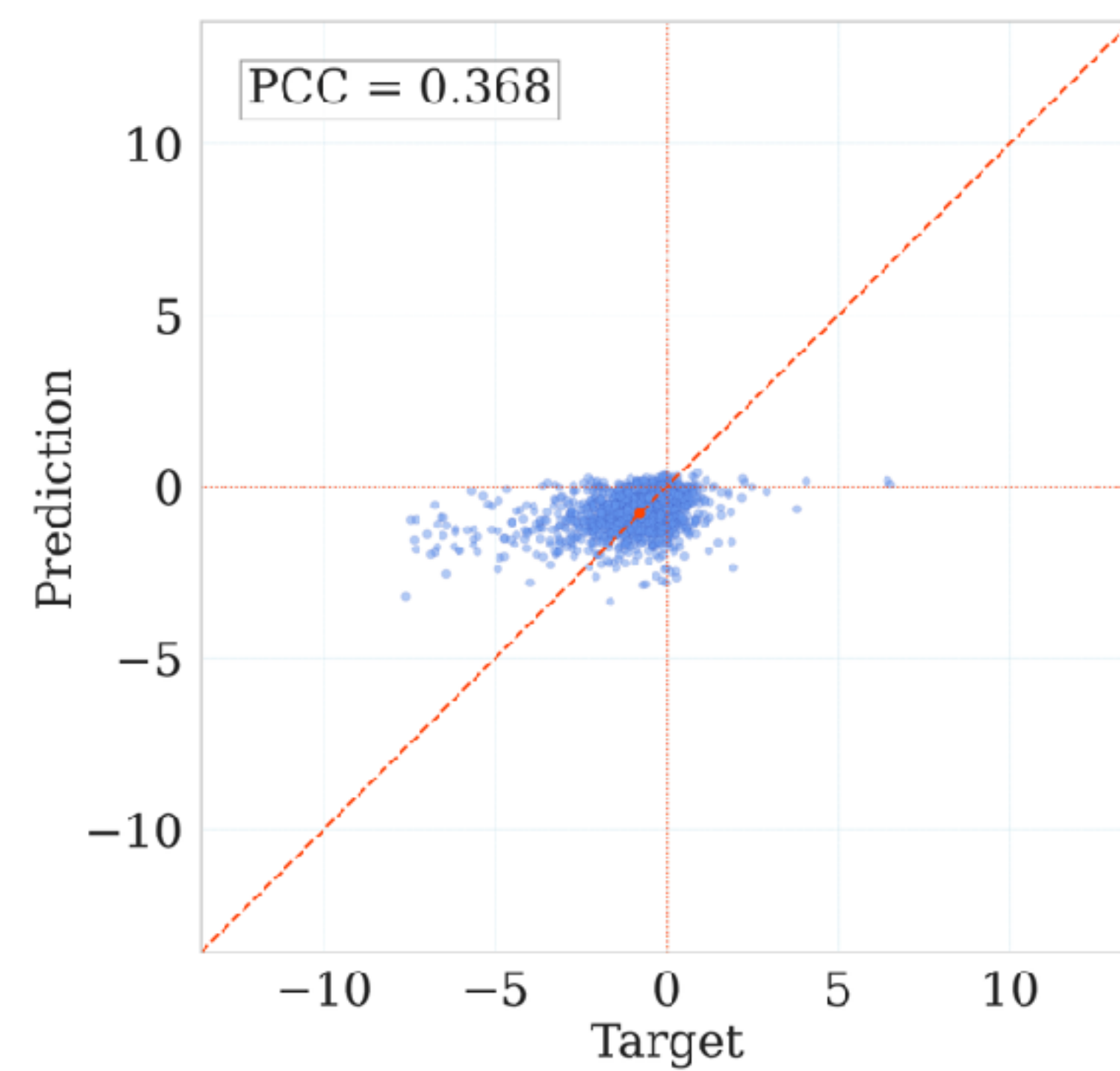
Training: 1 3 4



Validation: 2

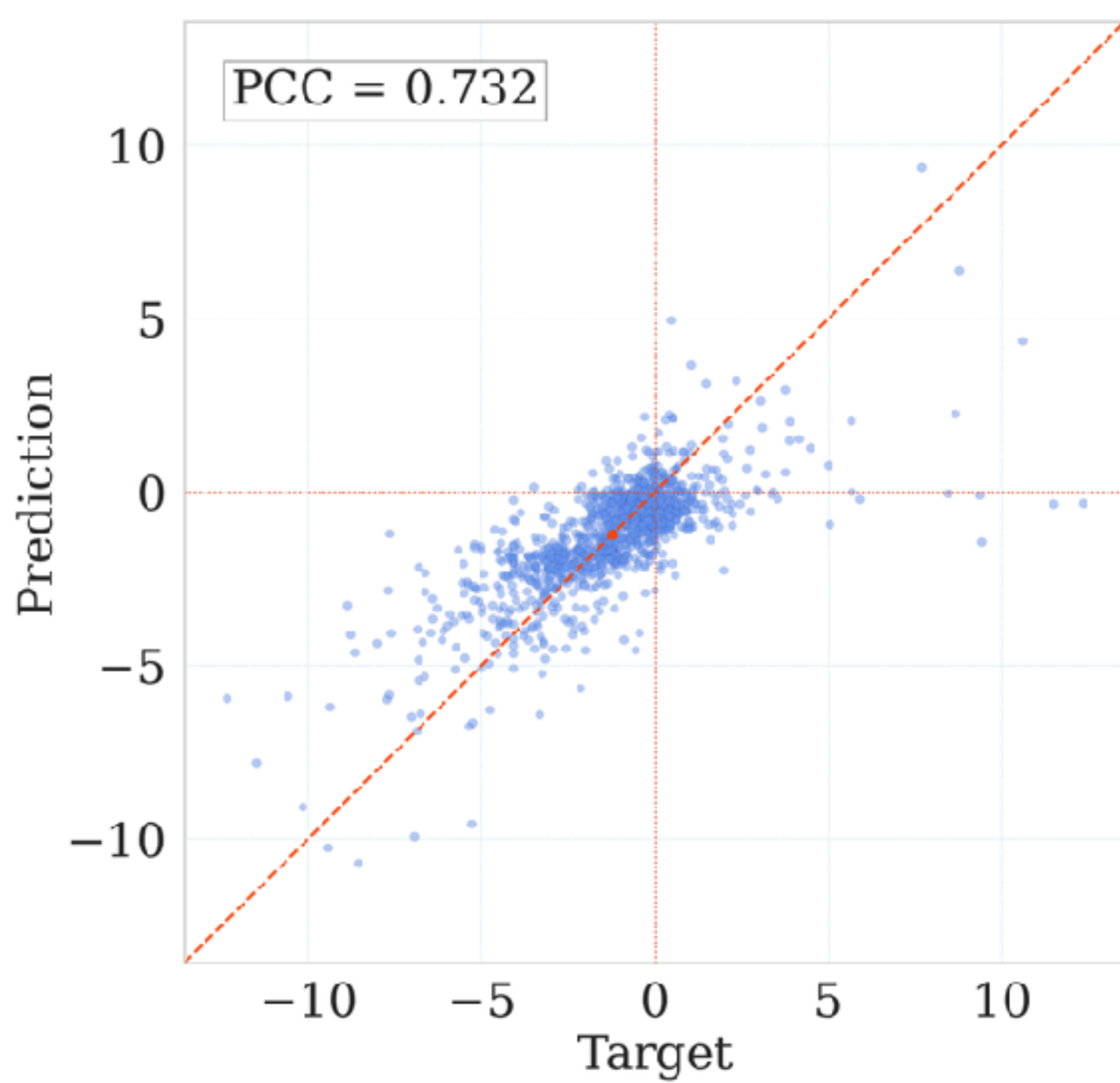


Test: 0

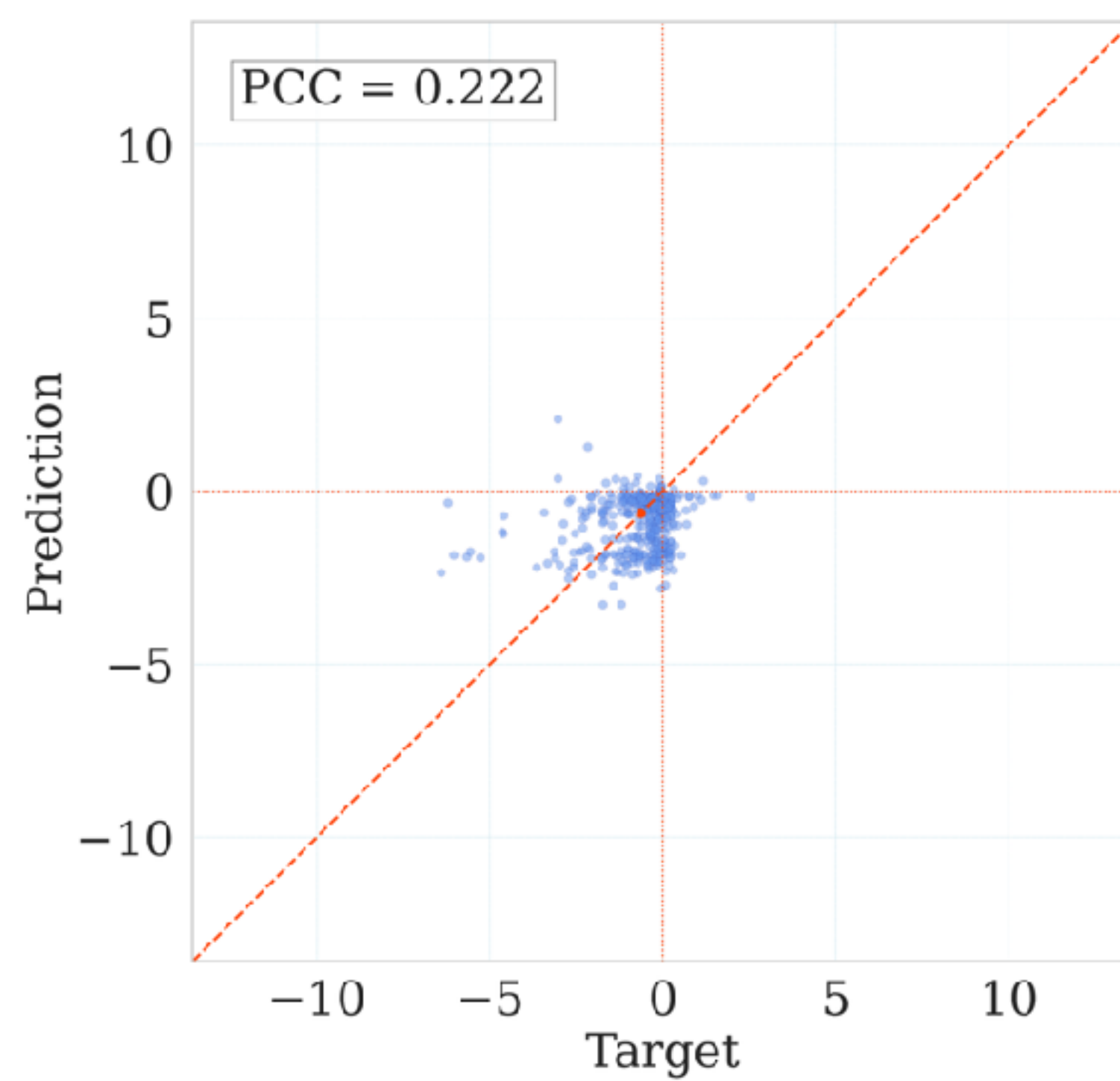


Run 18
Last Epoch: 648

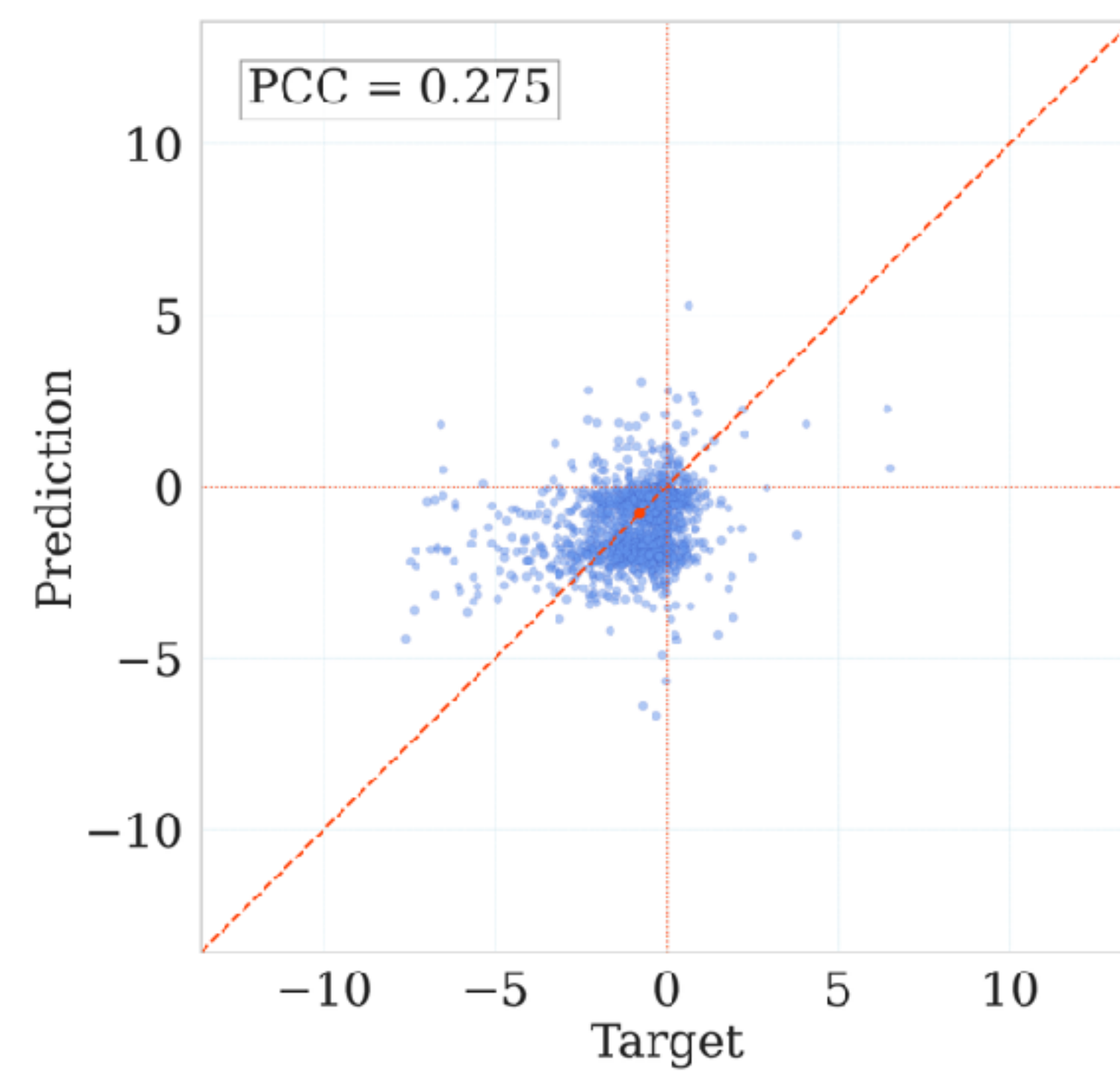
Training: 1 3 4



Validation: 2

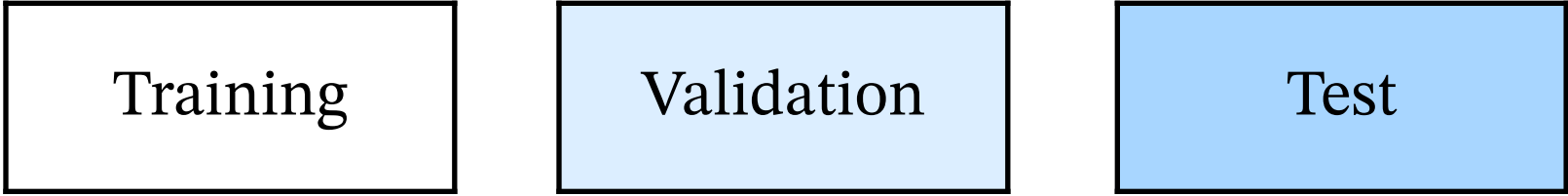


Test: 0



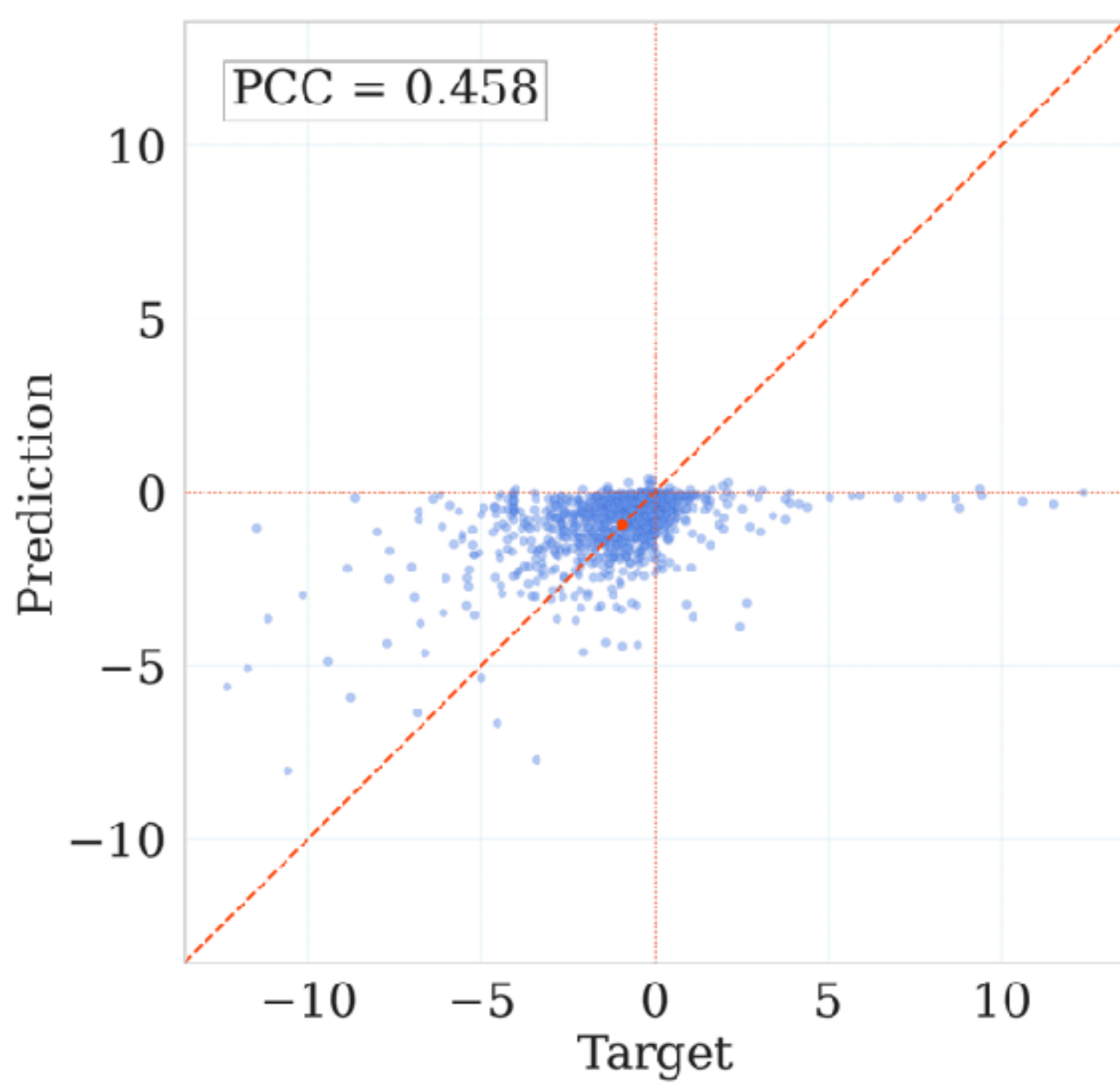
Run 19

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

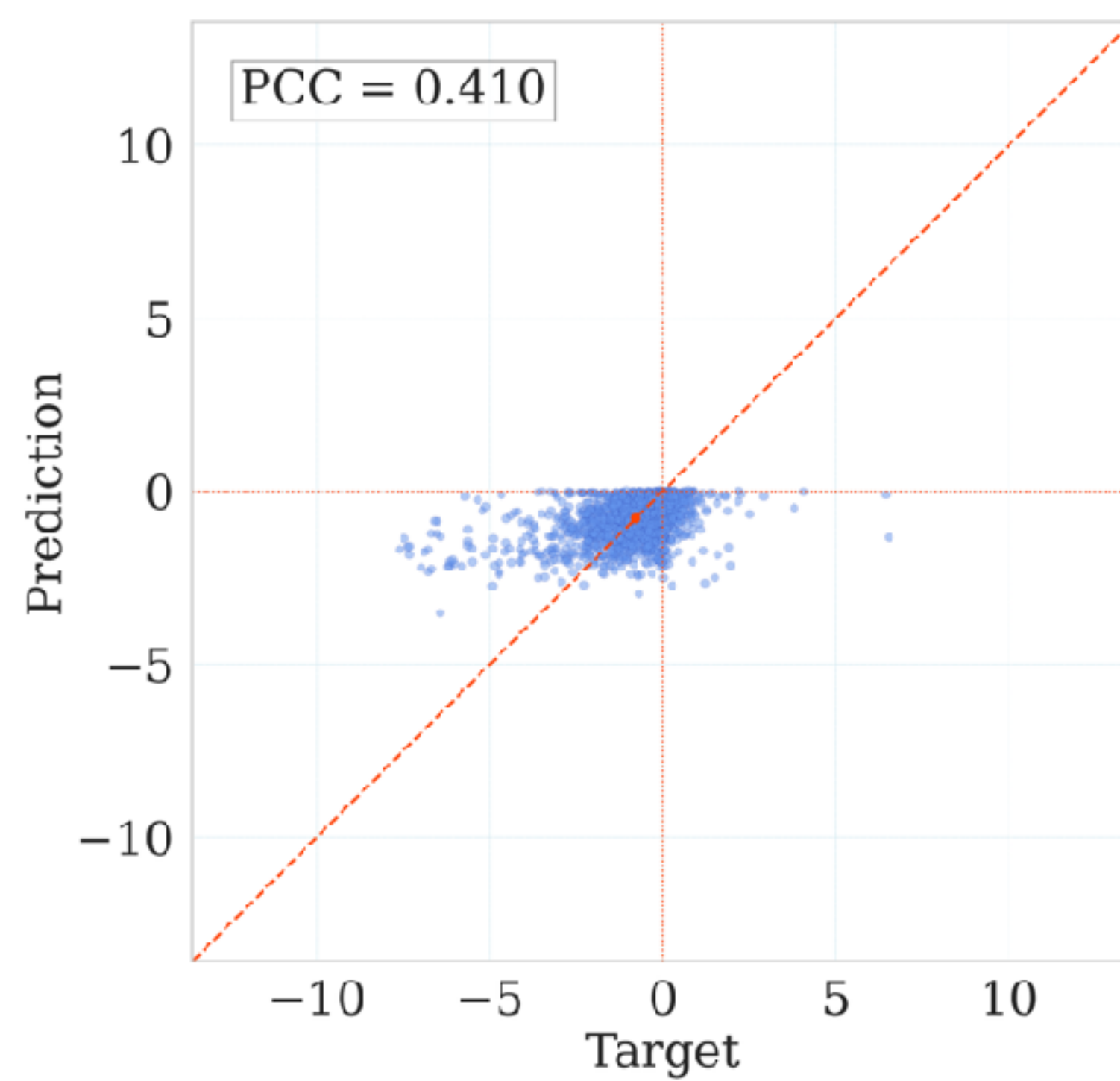


Run 19
Best Epoch: 15

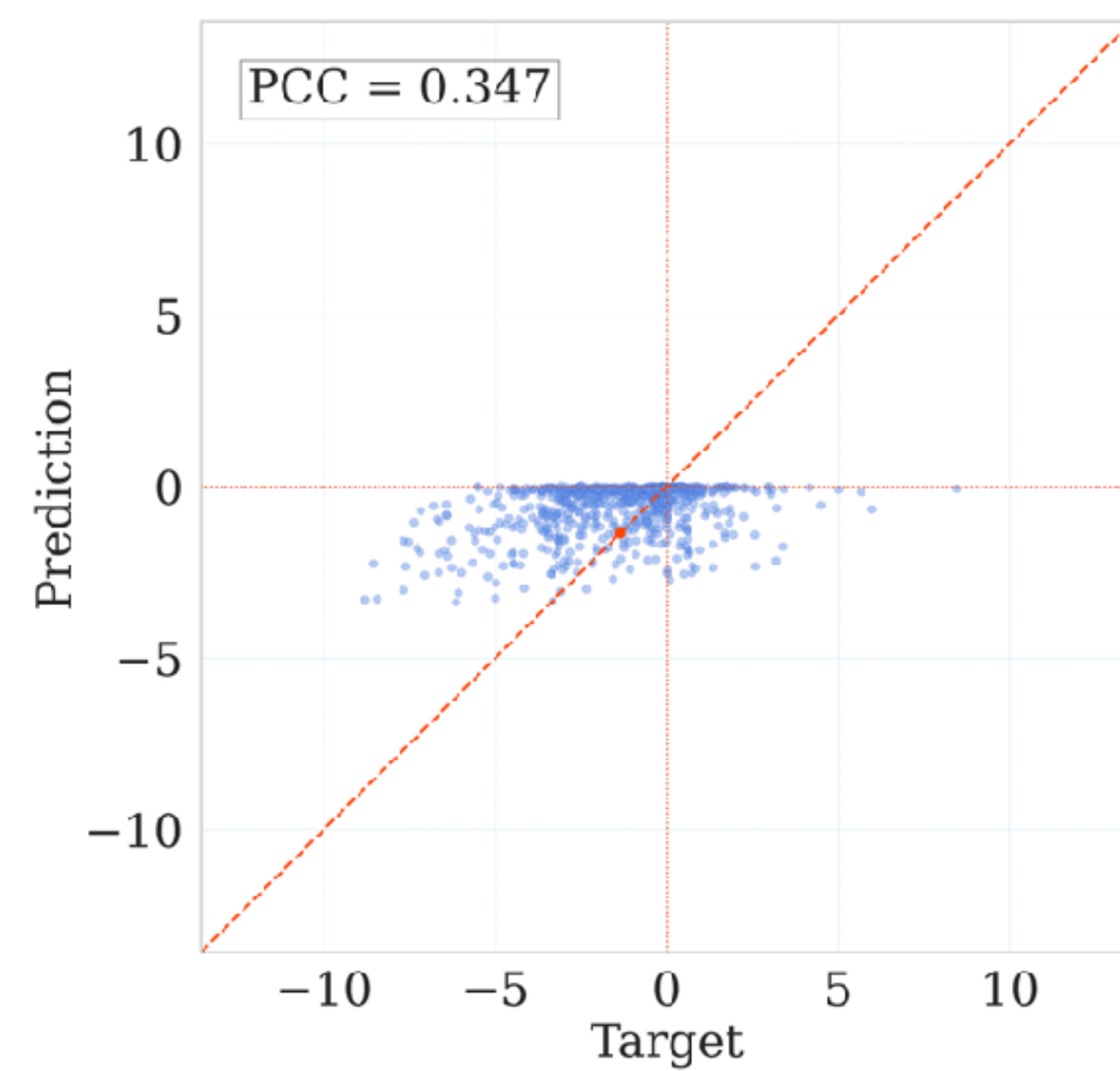
Training: 2 3 4



Validation: 0

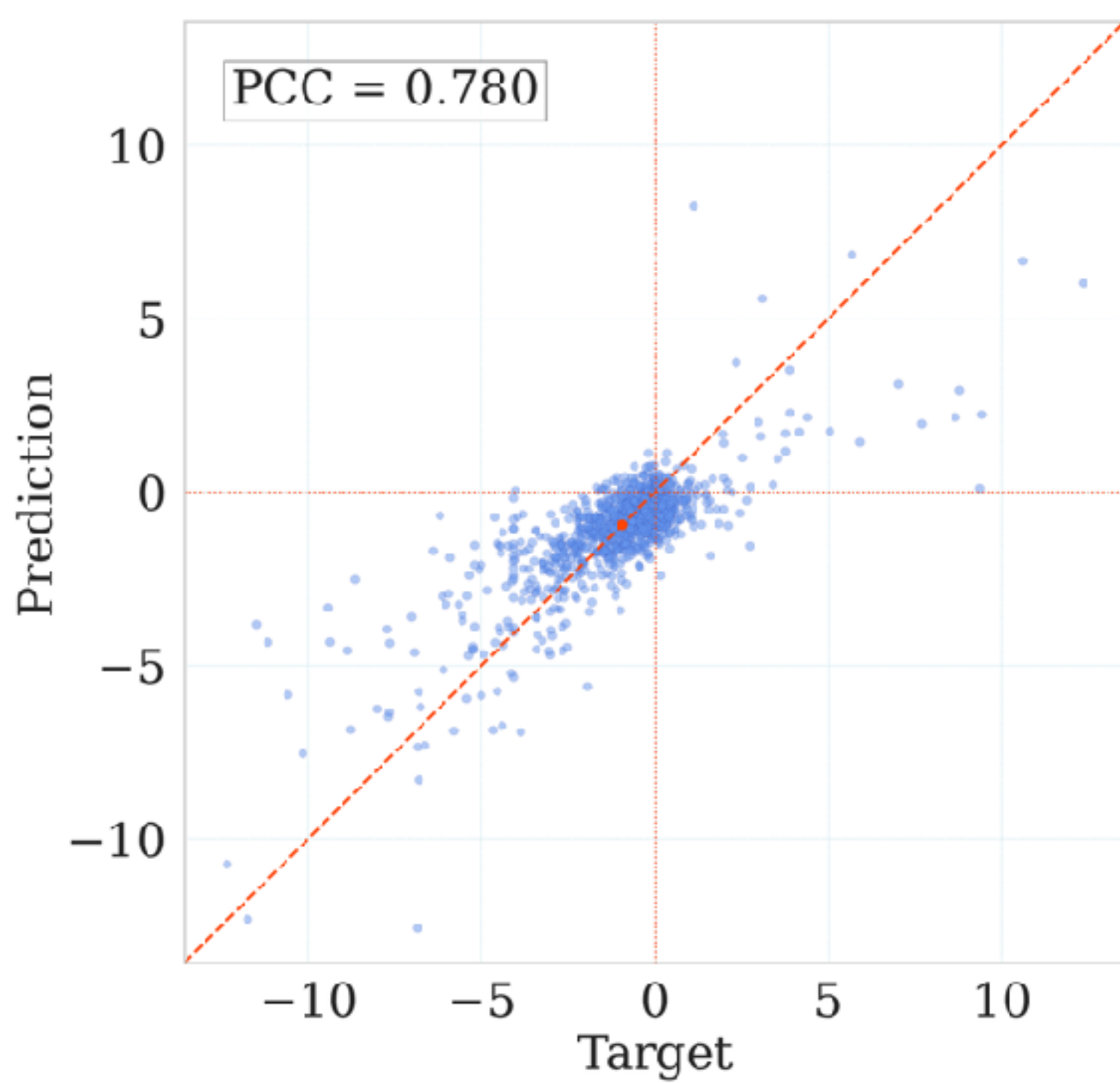


Test: 1

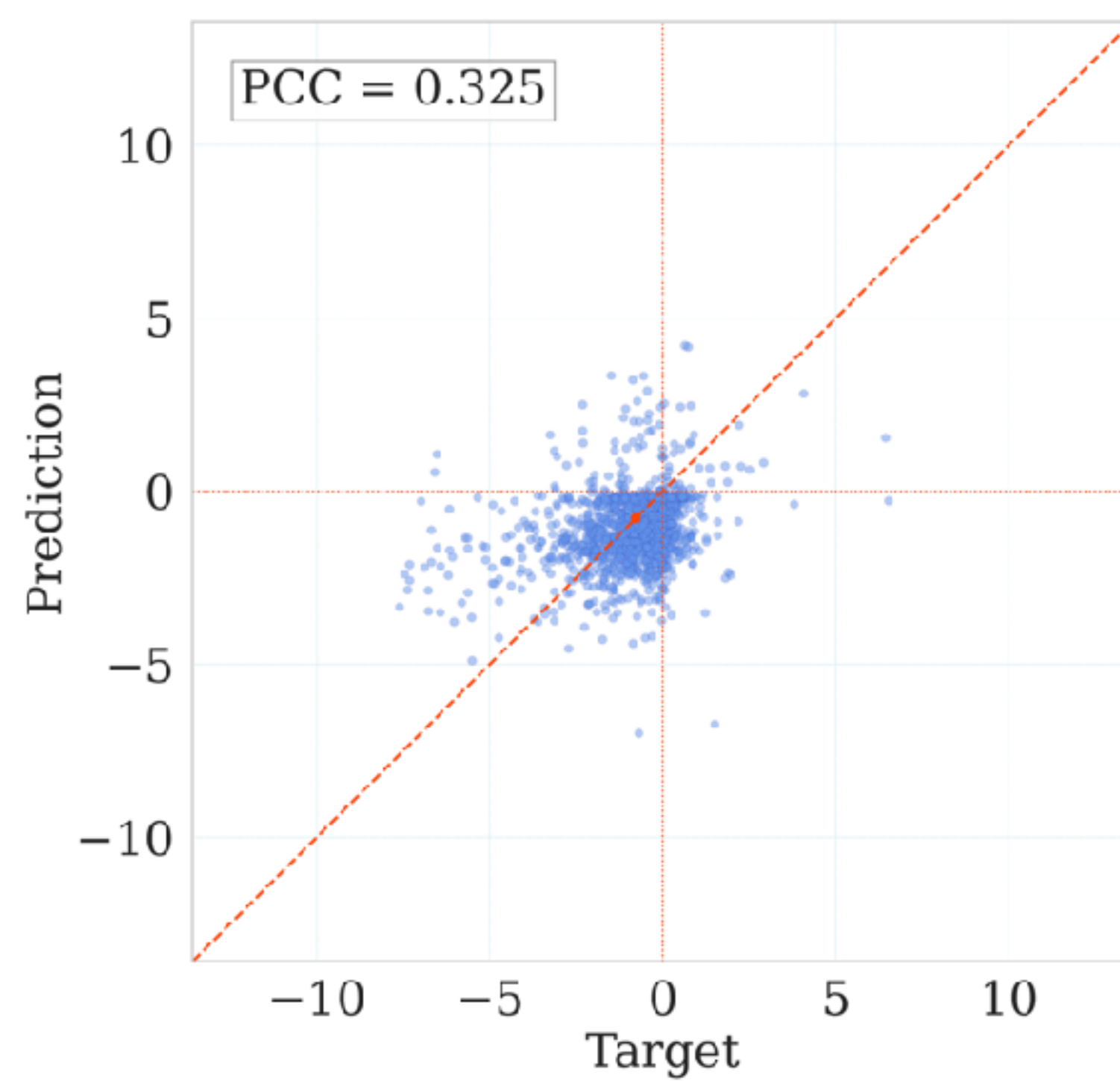


Run 19
Last Epoch: 608

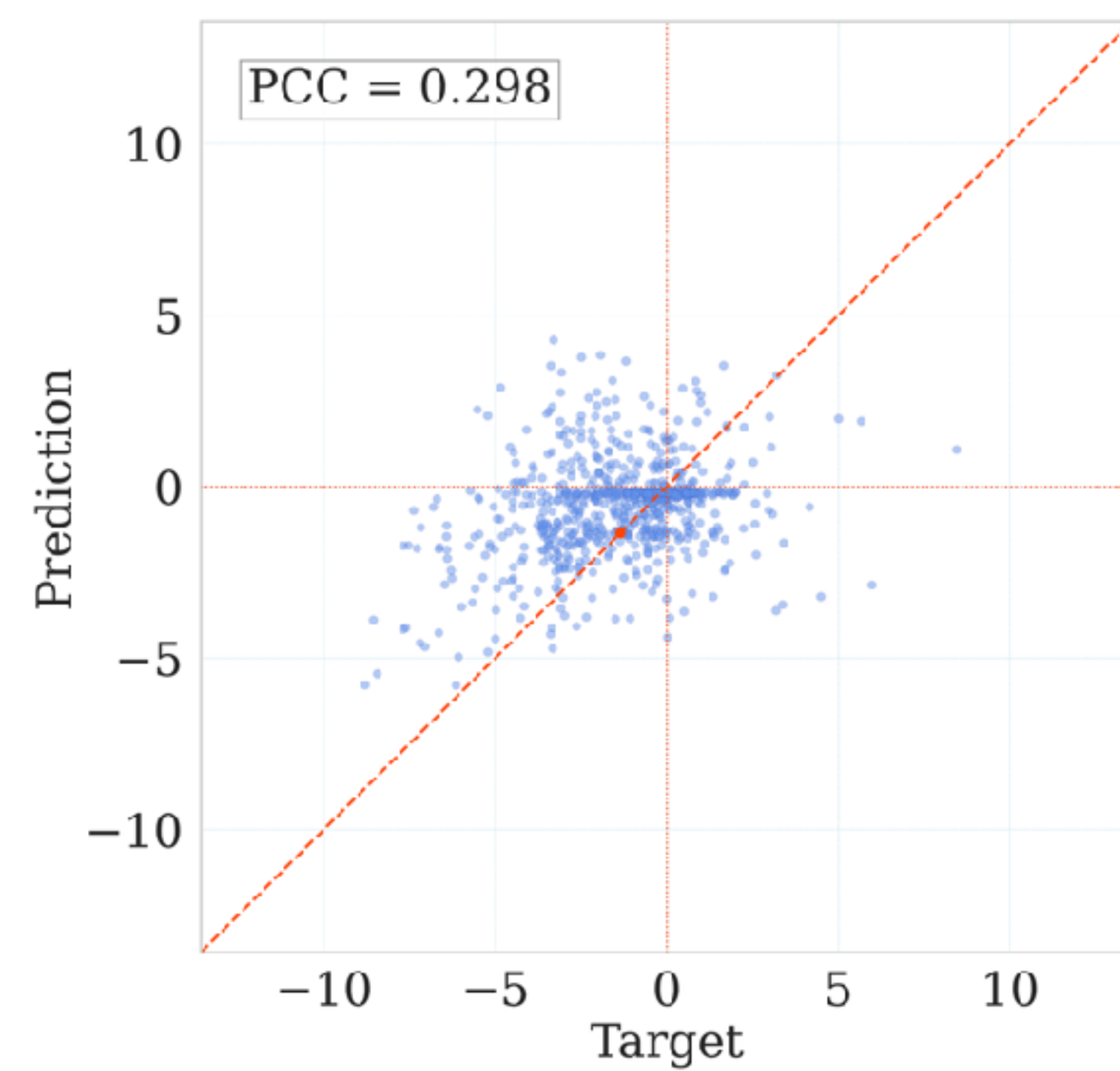
Training: 2 3 4



Validation: 0

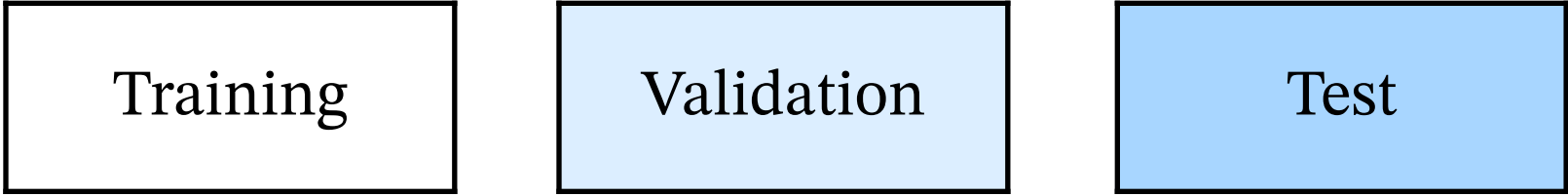


Test: 1



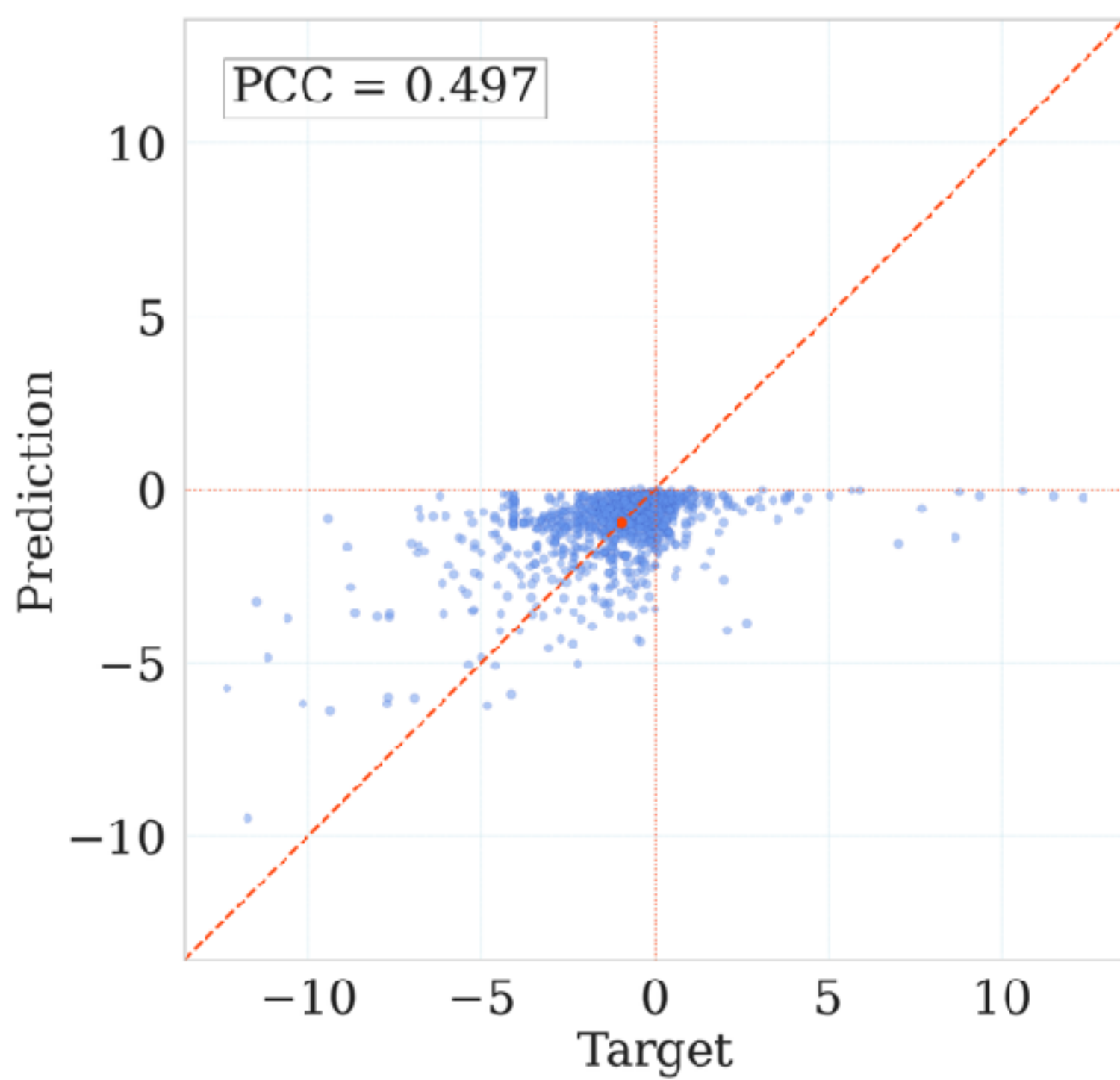
Run 20

Fold	SRVs	SRVs (Not Mapped on Uniprot)	Chains	Chains (Not Mapped on Uniprot)	Uniprot IDs	CCs
Training_1	1659	319	171	57	58	1
Training_2	779	0	20	0	2	1
Training_3	437	22	35	3	22	11
Training_4	414	10	64	2	24	7
Training_5	458	0	38	0	14	6

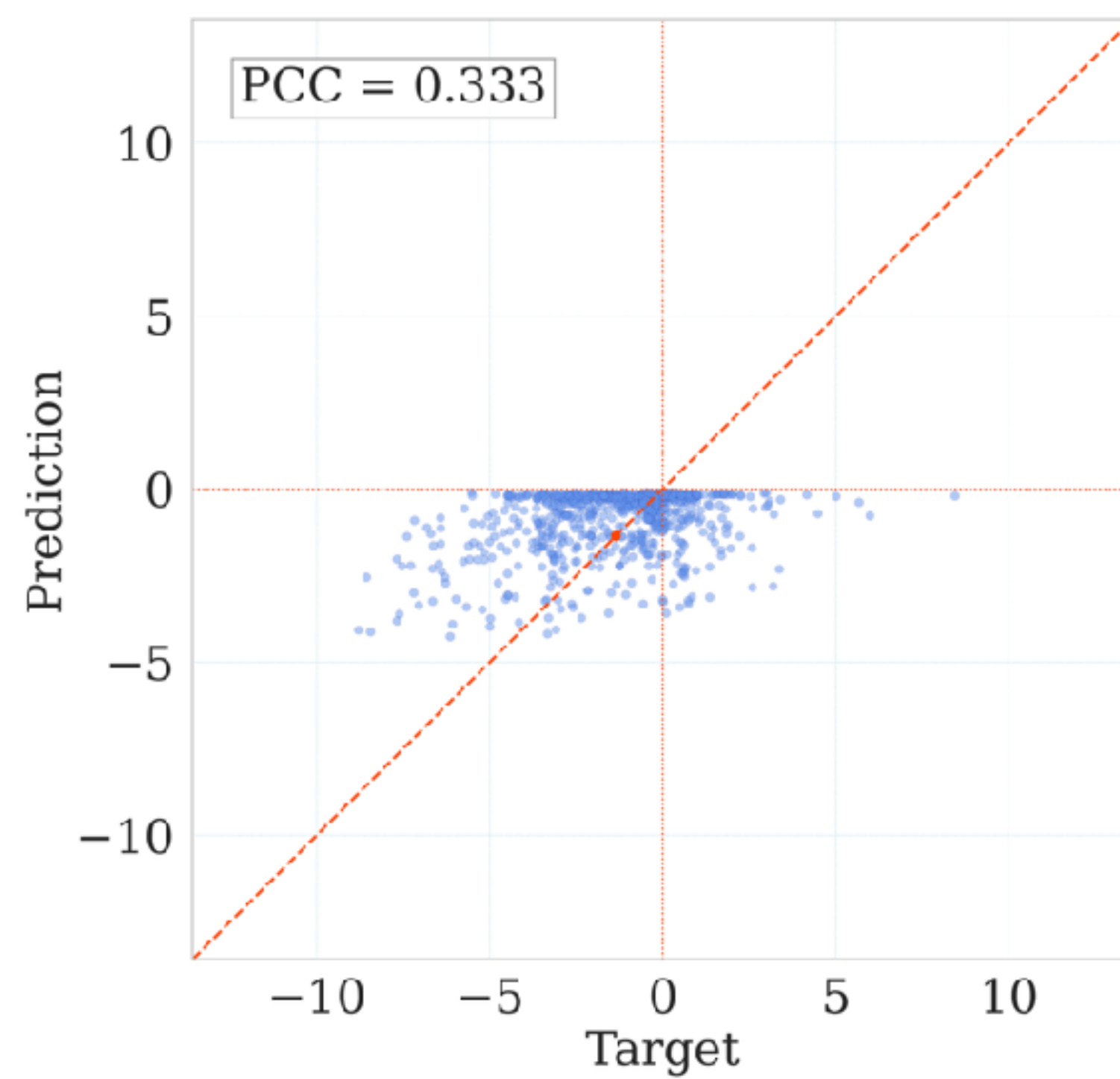


Run 20
Best Epoch: 16

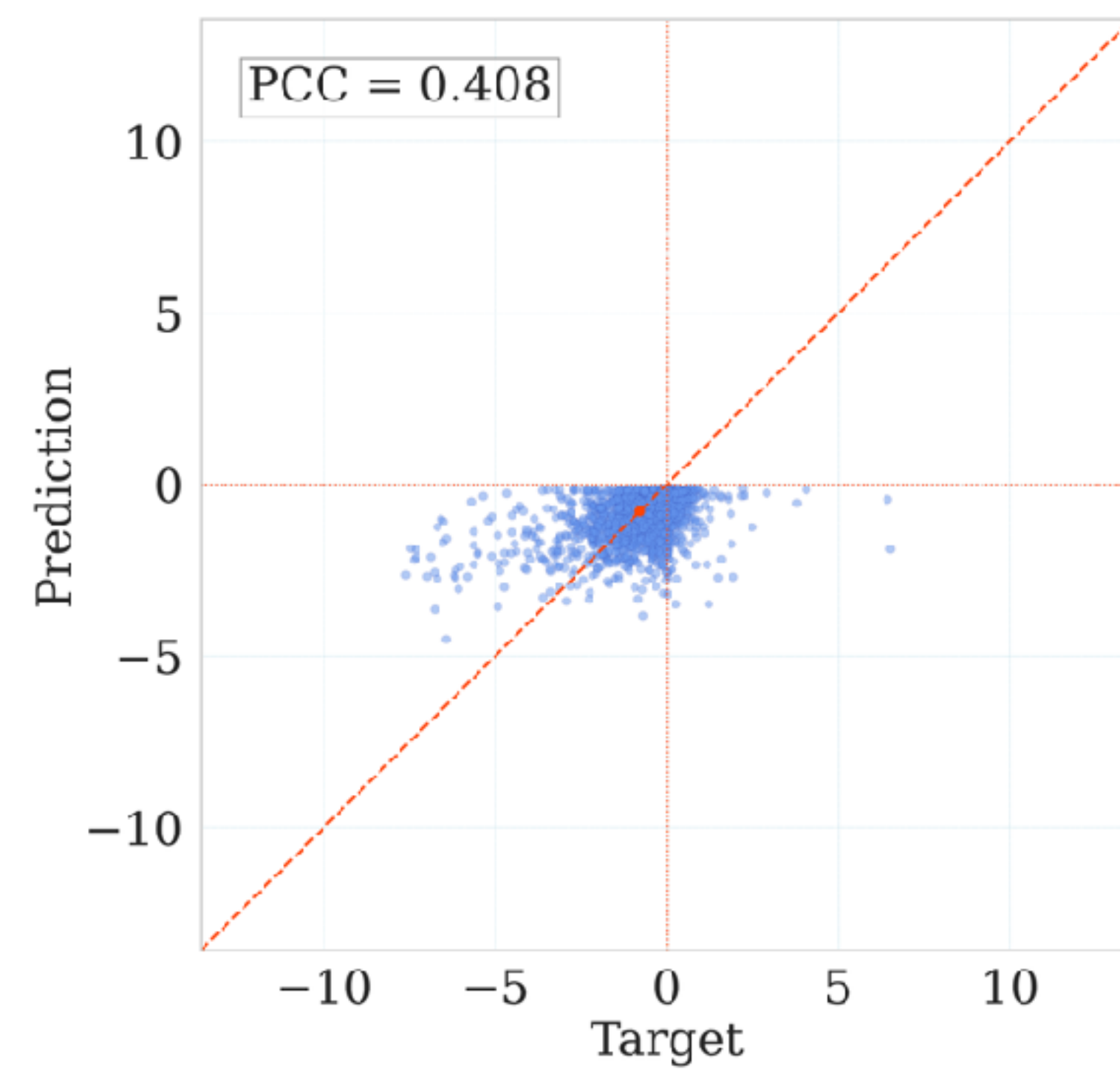
Training: 2 3 4



Validation: 1

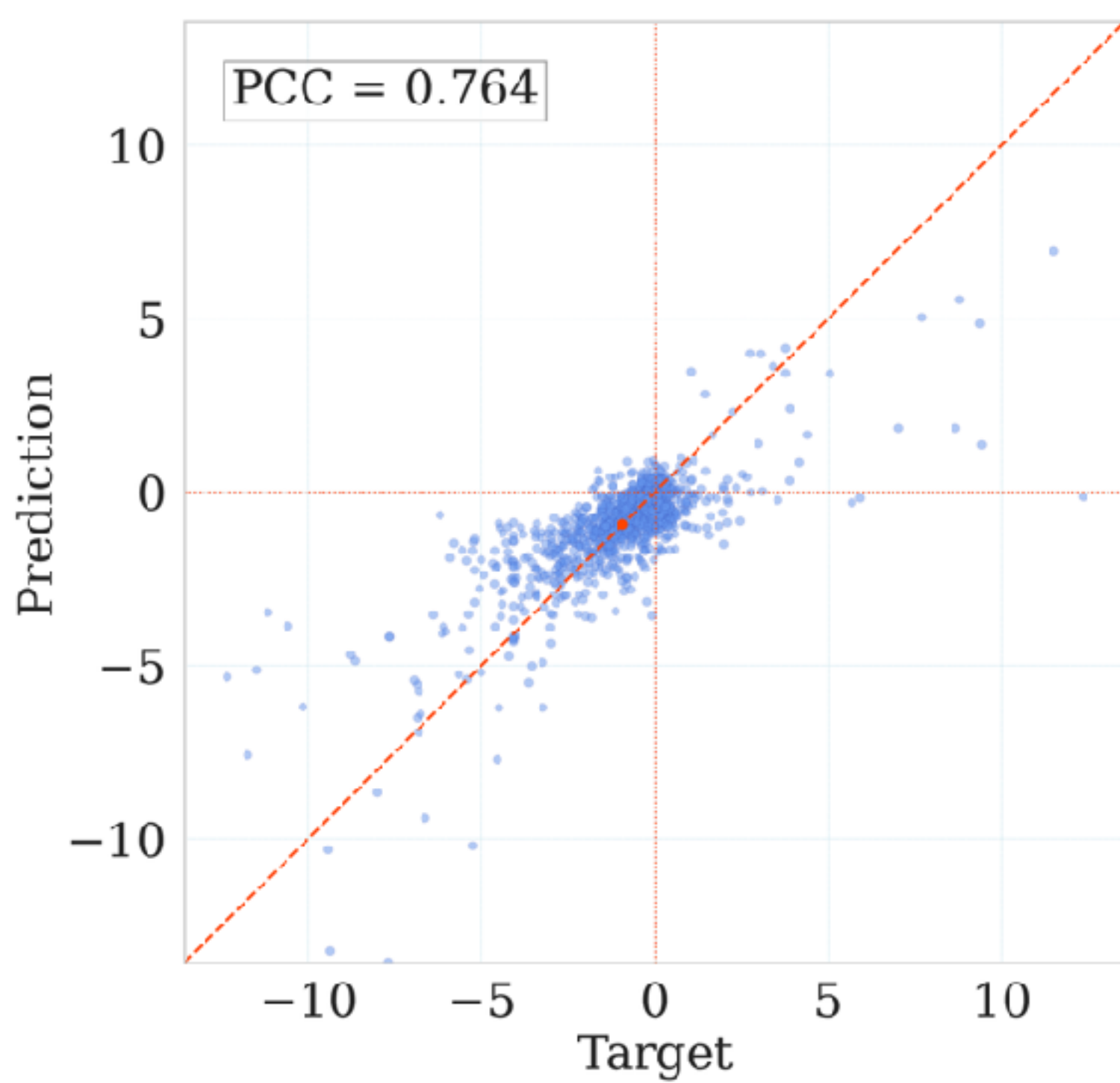


Test: 0

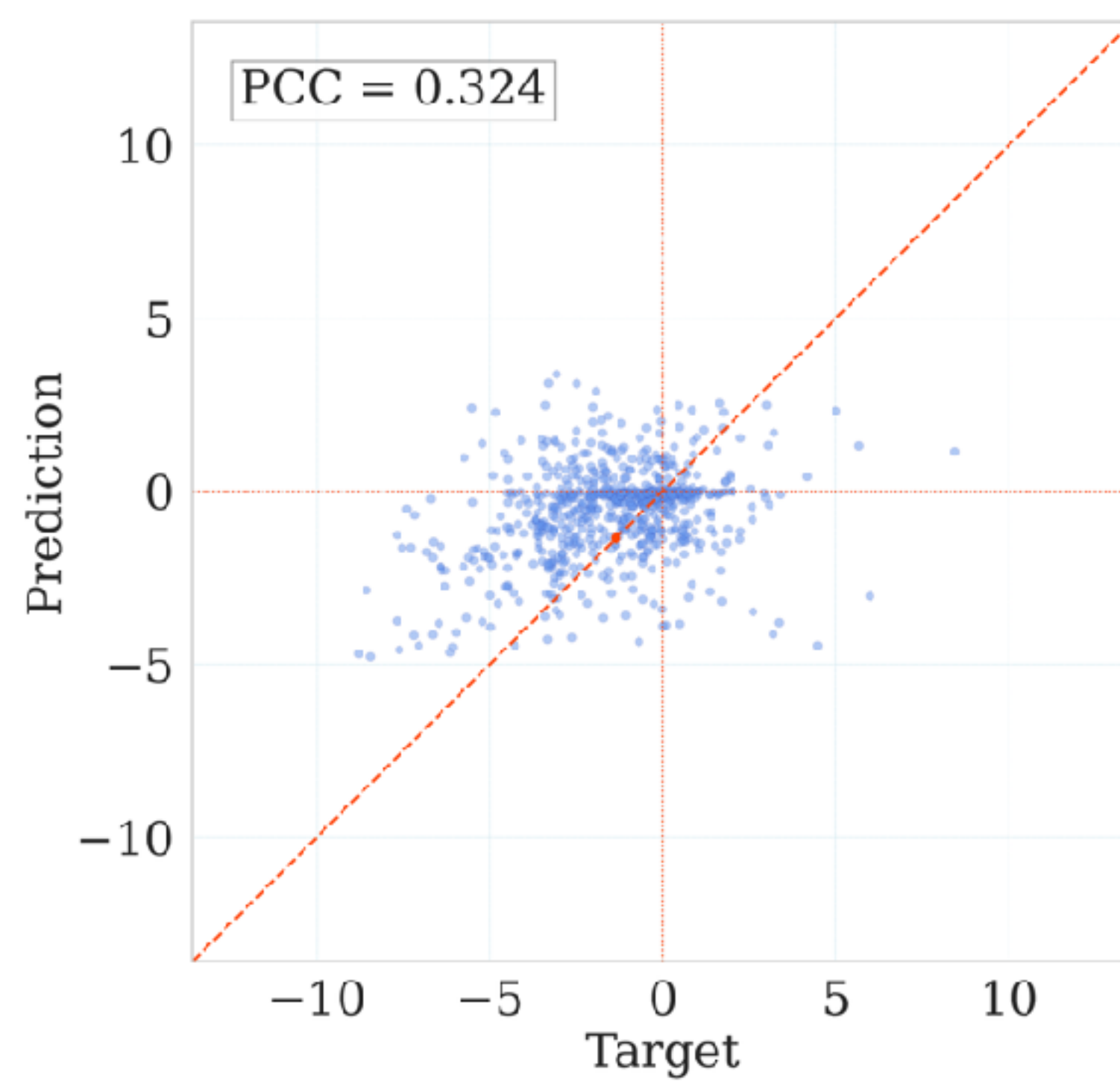


Run 20
Last Epoch: 622

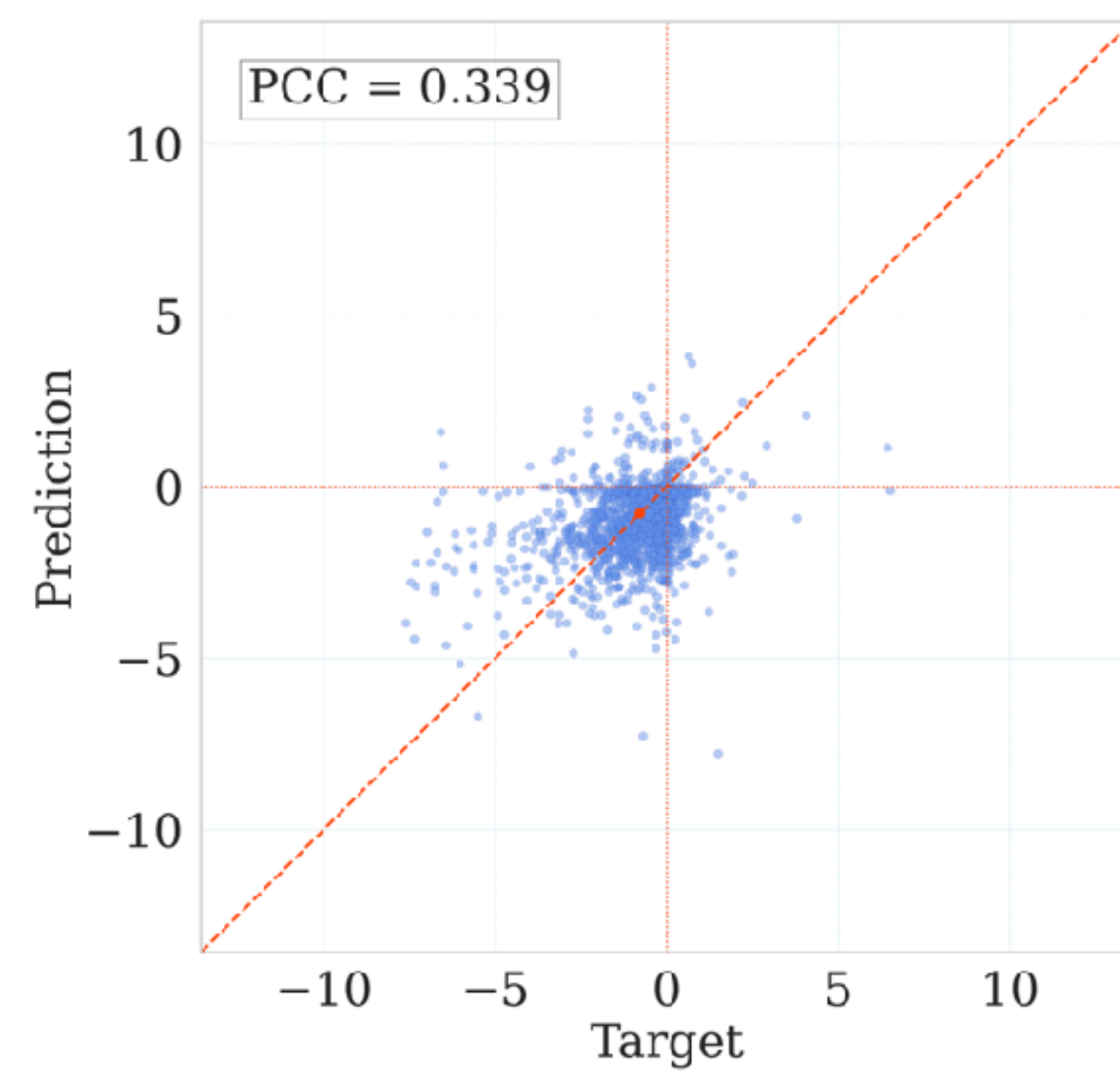
Training: 2 3 4



Validation: 1



Test: 0



Substitution patterns across folds

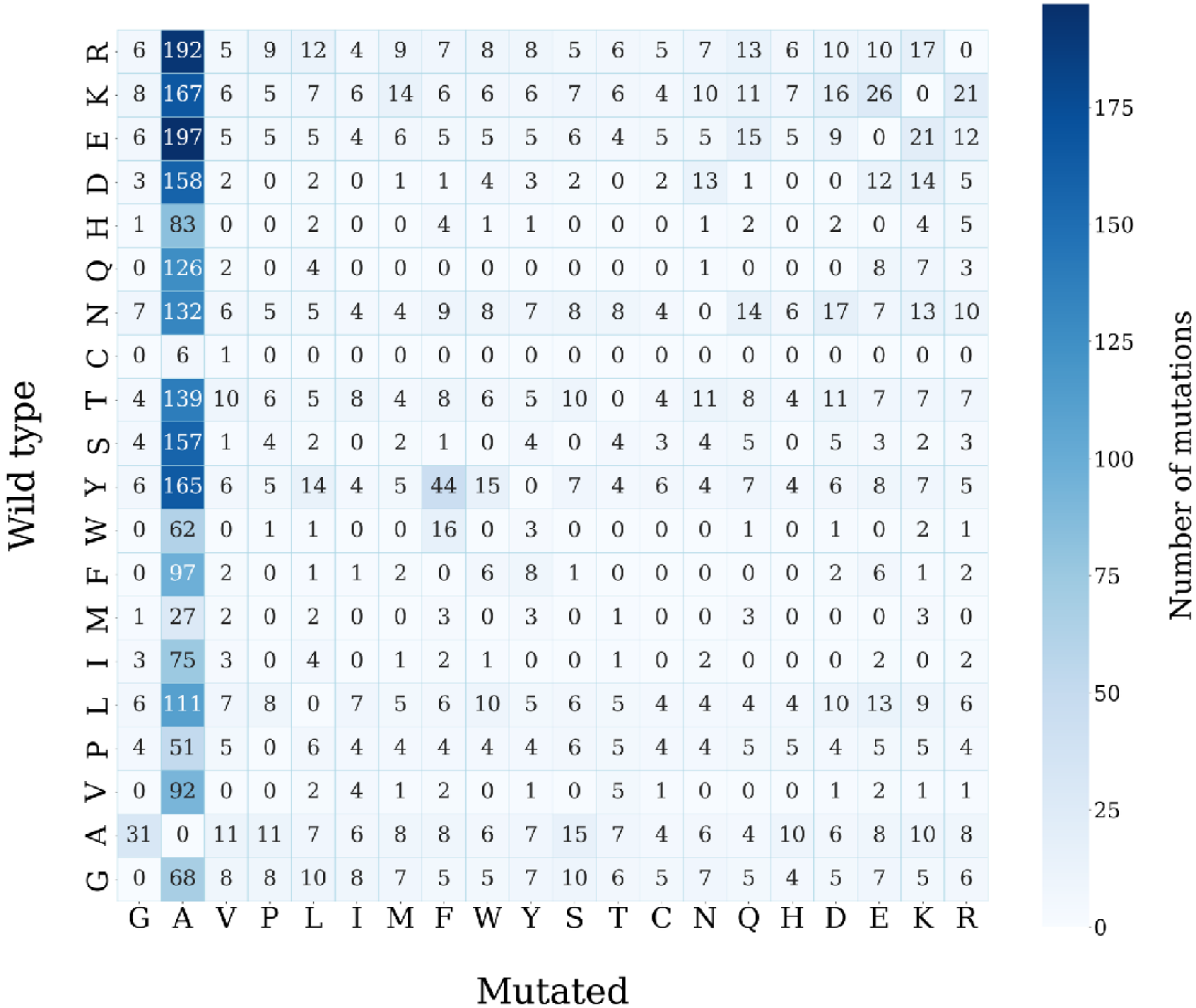
[mutation_heatmap](#)

	Train PCC (Best epoch)	Train PCC (Last epoch)	Validation PCC (Best epoch)	Validation PCC (Last epoch)	Test PCC (Best epoch)	Test PCC (Last epoch)
Max	0.799 (Run 14)	0.784 (Run 17)	0.521 (Run 6)	0.440 (Run 1)	0.498 (Run 15)	0.425 (Run 5)
Split	Tr: 1 2 3 Val: 4 Test: 0	Tr: 1 3 4 Val: 0 Test: 2	Tr: 0 1 4 Val: 3 Test: 2	Tr: 0 1 2 Val: 3 Test: 4	Tr: 1 2 4 Val: 0 Test: 3	Tr: 0 1 4 Val: 2 Test: 3
Min	0.371 (Run 5)	0.700 (Run 6)	0.215 (Run 14)	0.072 (Run 4)	0.068 (Run 1)	0.075 (Run 13)
Split	Tr: 0 1 4 Val: 2 Test: 3	Tr: 0 1 4 Val: 3 Test: 2	Tr: 1 2 3 Val: 4 Test: 0	Tr: 0 1 3 Val: 4 Test: 2	Tr: 0 1 2 Val: 3 Test: 4	Tr: 1 2 3 Val: 3 Test: 4

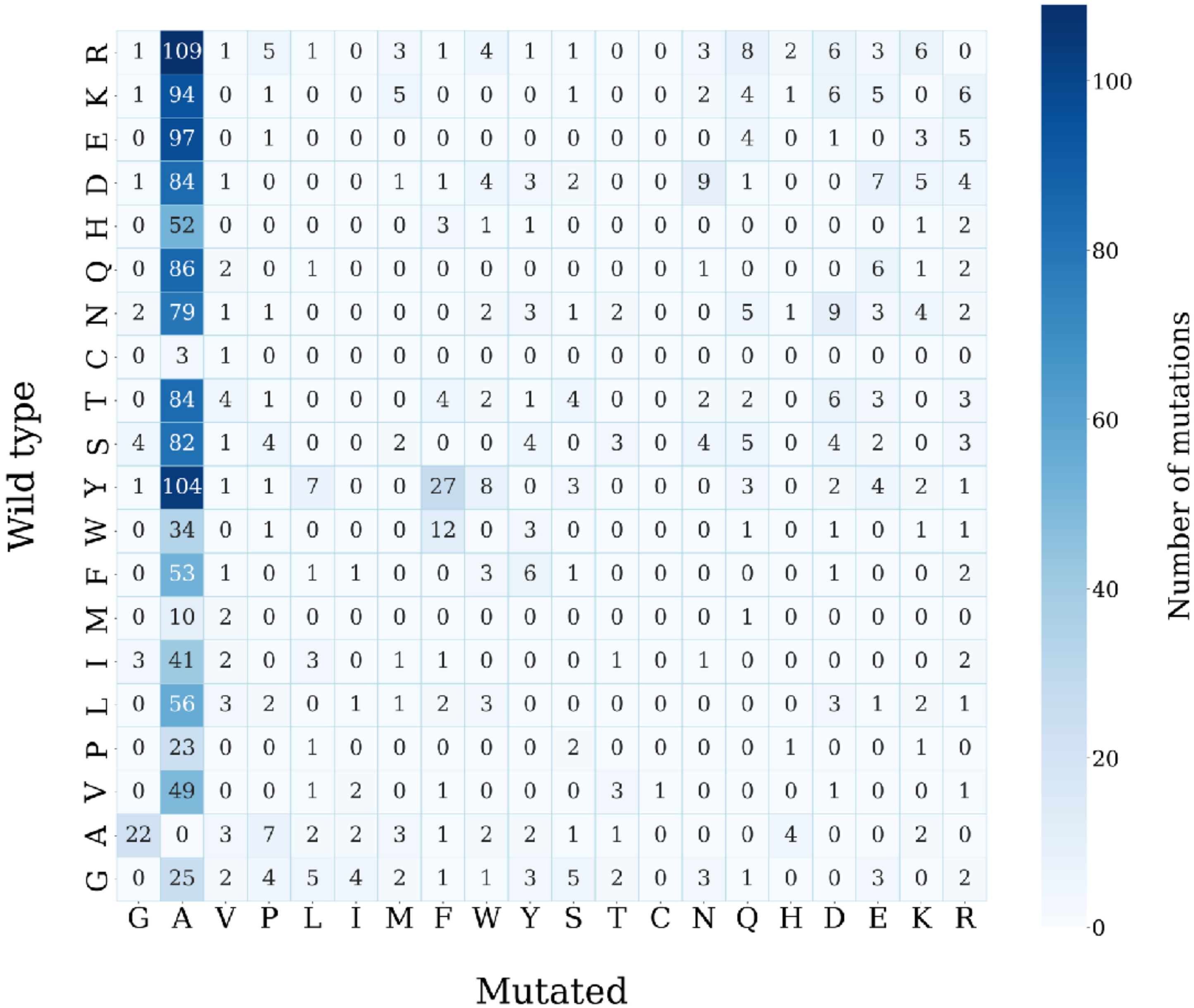
Fold	SRVs	DDG_mean	std	max	min	SRVs_to_alanine	DDG_mean_alanine	std_alanine	max_alanine	min_alanine	Ratio_SRVs_to_alanine
0	1659	-0.788	1.338	6.518	-7.608	1165	-0.830	1.284	1.89	-7.473	0.702
1	779	-1.358	2.143	8.446	-8.802	36	-1.135	1.856	2.572	-5.475	0.046
2	437	-0.636	1.081	2.529	-6.41	355	-0.632	1.064	1.112	-6.41	0.812
3	414	-1.194	1.839	5.682	-7.666	187	-1.373	1.358	0.889	-7.651	0.452
4	458	-1.019	2.715	12.352	-12.352	362	-1.119	1.553	2.076	-10.159	0.790

Fold	Destabilizing	Stabilizing	Destabilizing_A	Stabilizing_A
0	1205	368	858	232
1	579	197	26	10
2	342	89	281	72
3	312	71	163	18
4	334	117	282	73

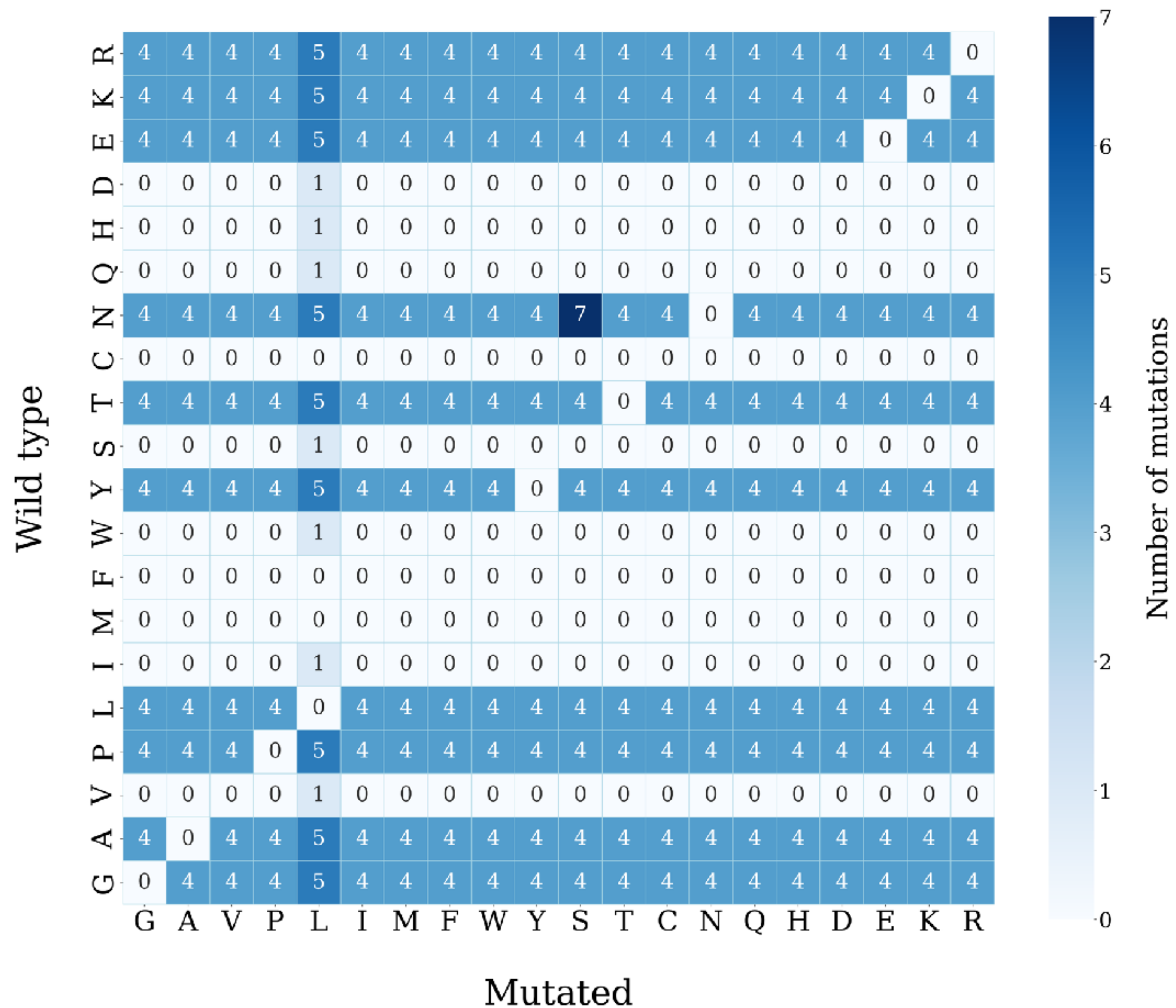
Mutation matrix (Training set)



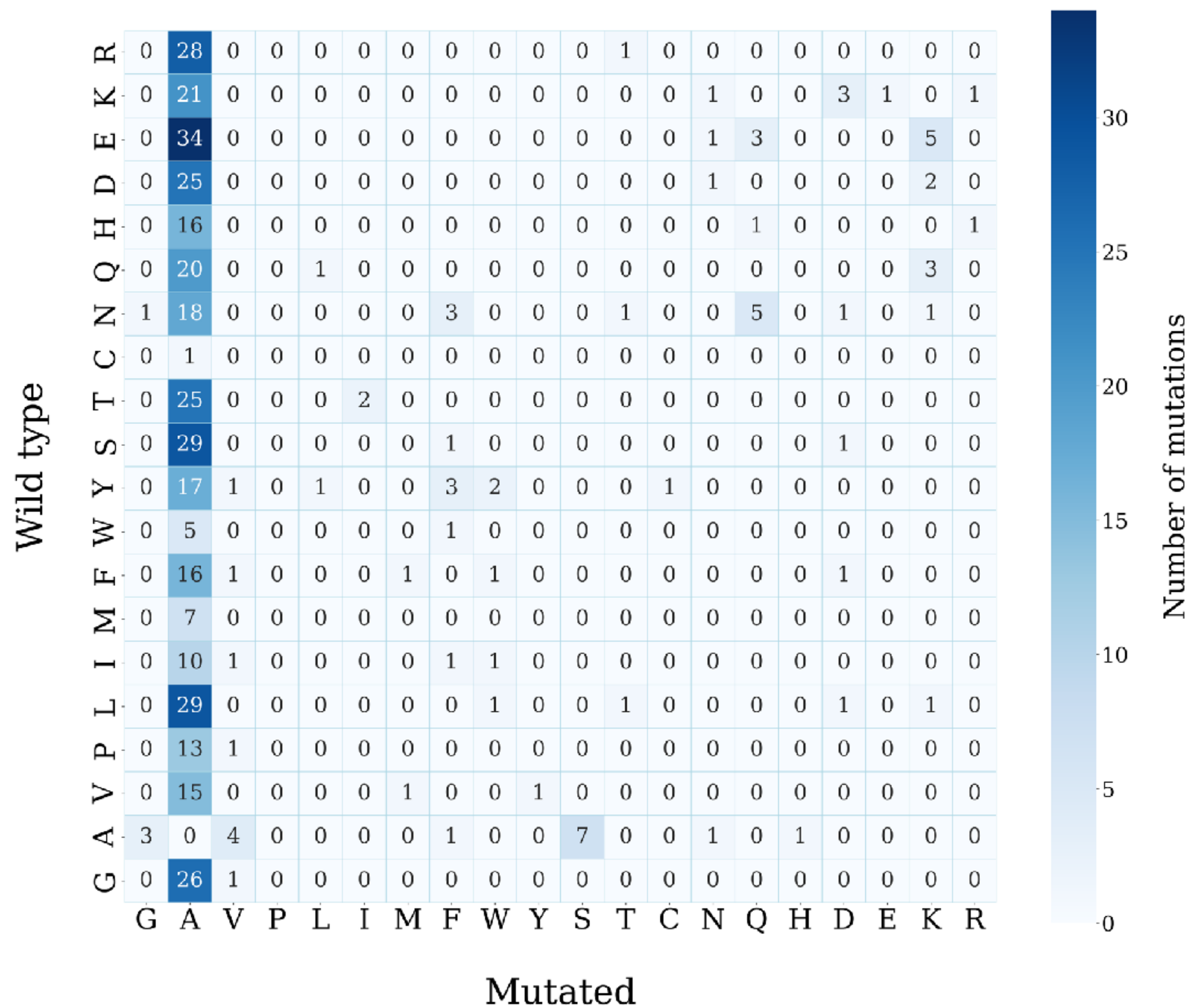
Mutation matrix (fold 1)



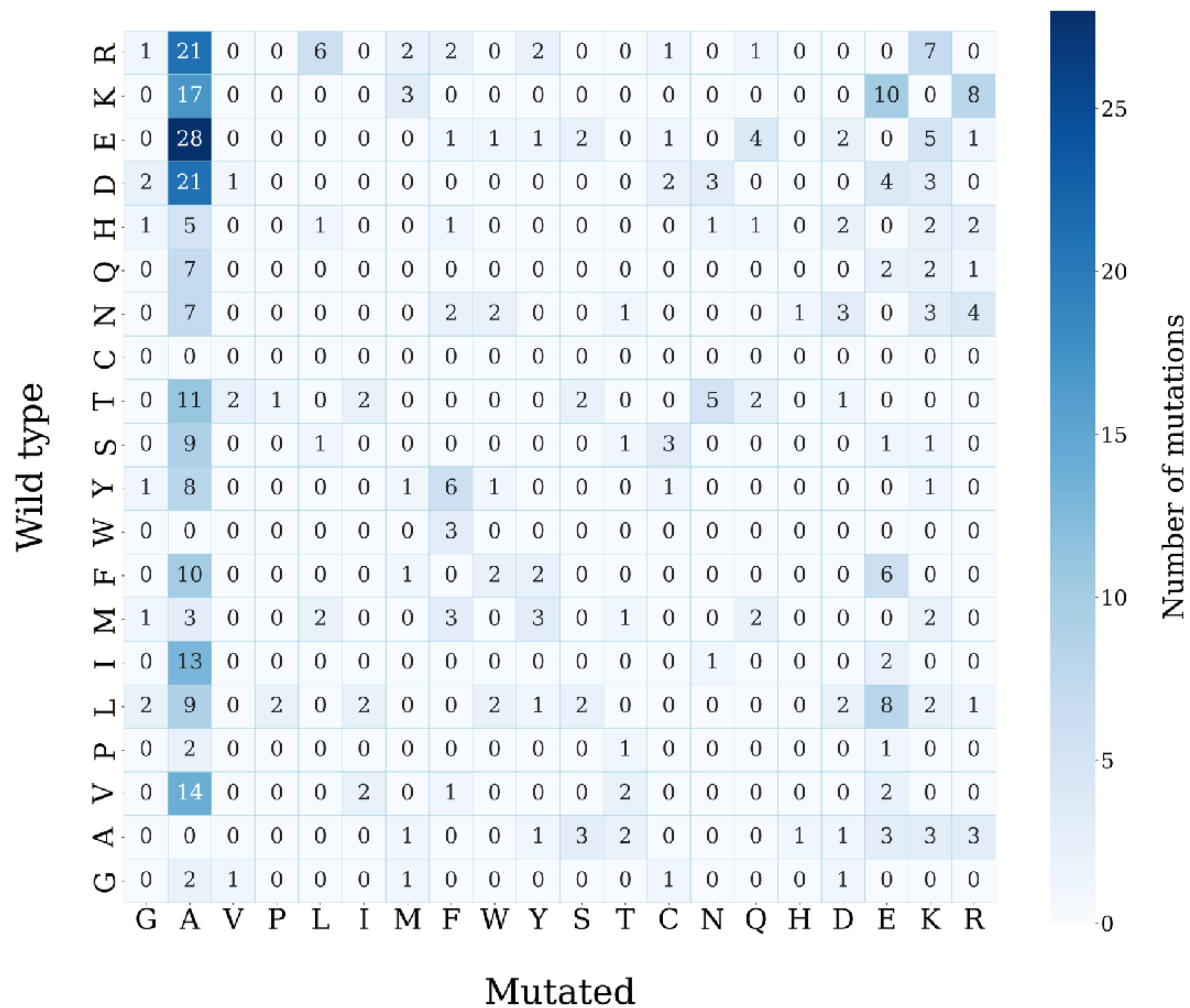
Mutation matrix (fold 2)



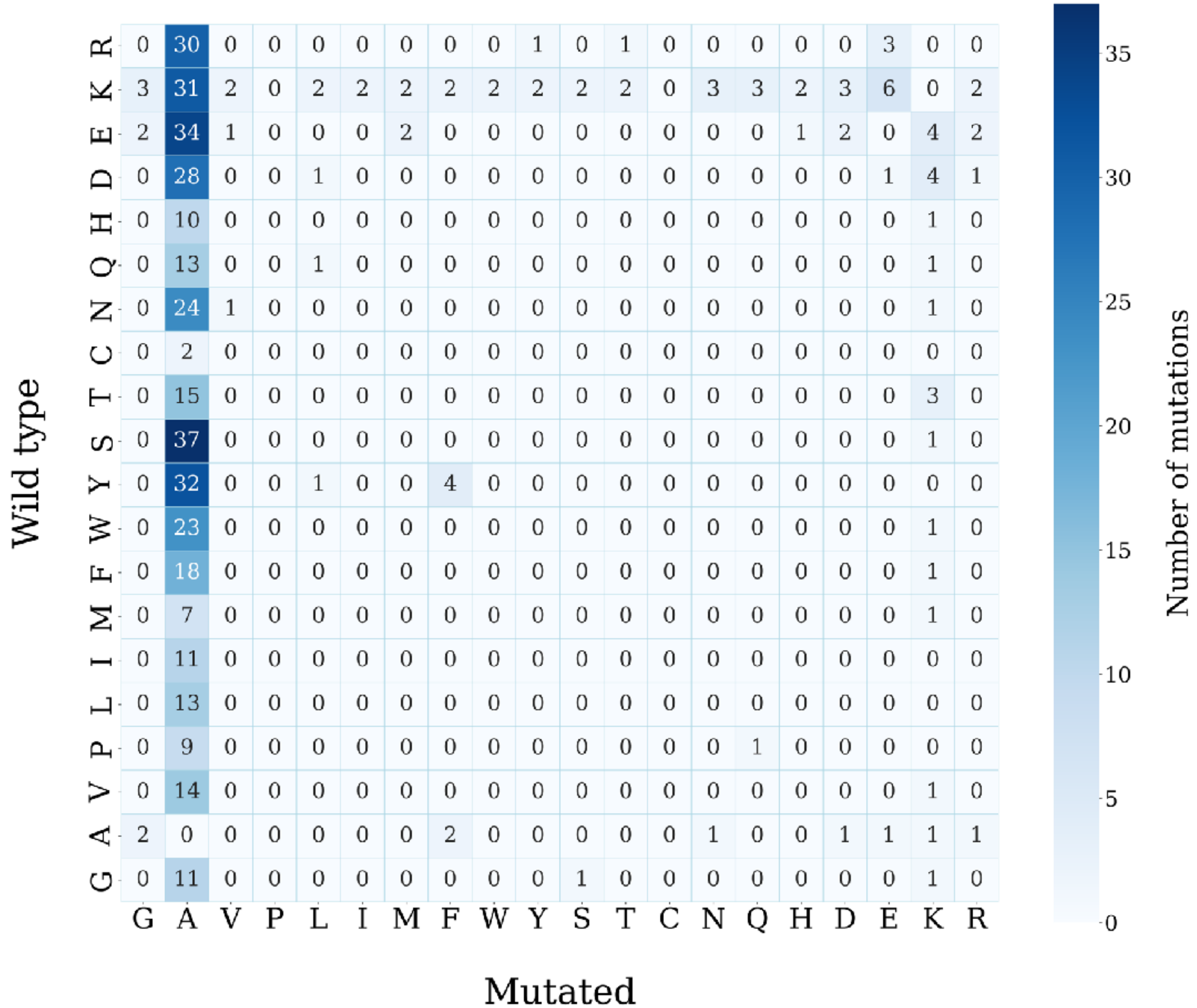
Mutation matrix (fold 3)



Mutation matrix (fold 4)

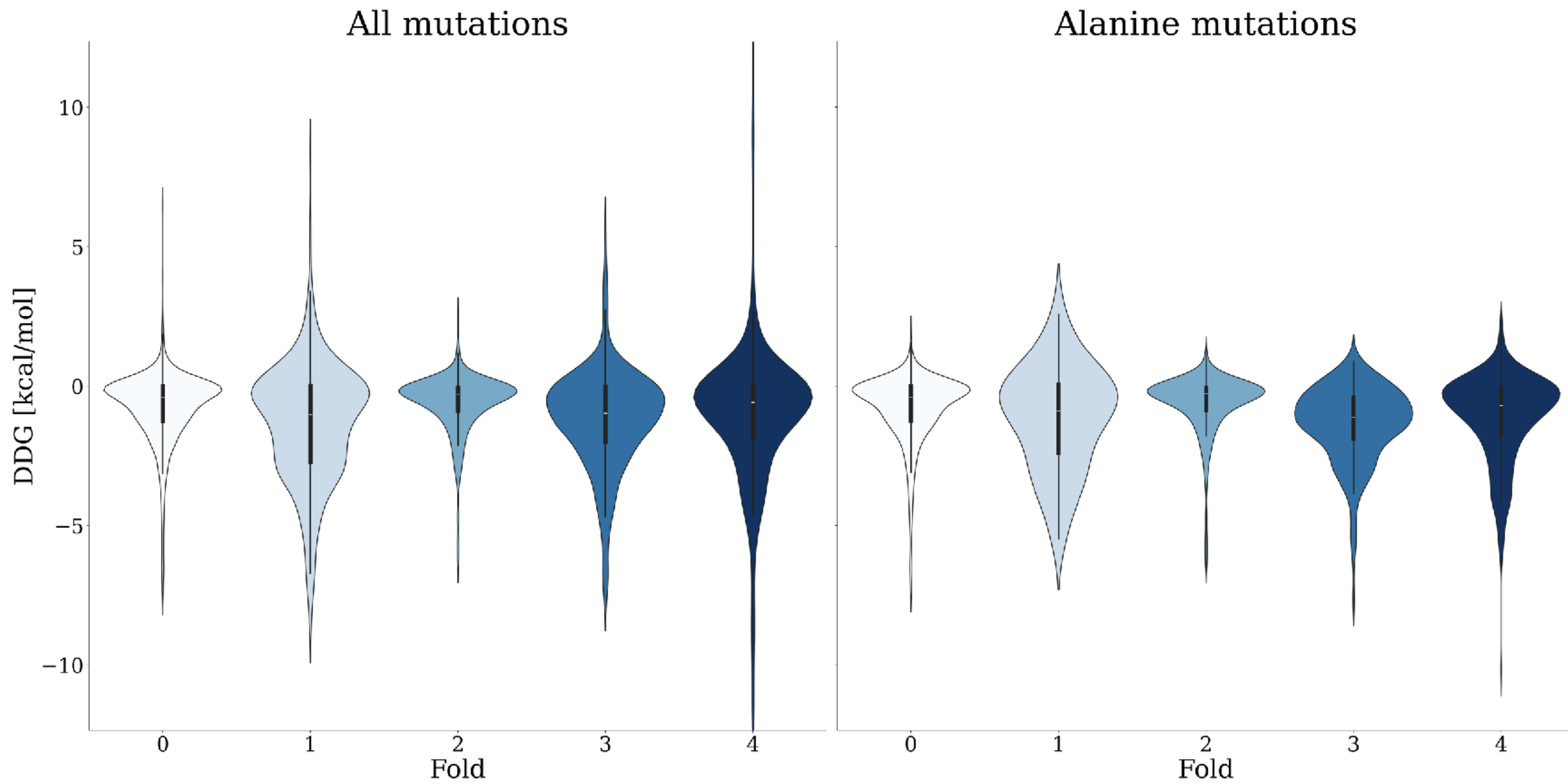


Mutation matrix (fold 5)



Alanine mutant distribution by DDG range

Fold	SRVs	DDG_mean	std	max	min	SRVs_to_alanine	DDG_mean_alanine	std_alanine	max_alanine	min_alanine	Ratio_SRVs_to_alanine
0	1659	-0.788	1.338	6.518	-7.608	1165	-0.830	1.284	1.89	-7.473	0.702
1	779	-1.358	2.143	8.446	-8.802	36	-1.135	1.856	2.572	-5.475	0.046
2	437	-0.636	1.081	2.529	-6.41	355	-0.632	1.064	1.112	-6.41	0.812
3	414	-1.194	1.839	5.682	-7.666	187	-1.373	1.358	0.889	-7.651	0.452
4	458	-1.019	2.715	12.352	-12.352	362	-1.119	1.553	2.076	-10.159	0.790



DDG mutation distribution (Training set)

Wild type

R	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	6	2	3	5	3	4	8	17	10	12	13	34	41	28	2	0	2	0	0	0	0
K	1	0	0	0	0	0	0	0	1	0	0	2	1	0	1	1	0	0	1	3	1	3	2	1	9	6	11	10	18	26	45	17	4	1	0	1	0	0	1
E	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	2	1	0	3	3	3	4	6	15	17	21	23	60	24	6	4	0	0	0	1	0
D	0	0	0	0	0	0	1	0	0	1	0	0	1	0	1	0	1	1	0	4	1	1	7	4	8	10	12	18	13	17	24	19	8	3	1	2	0	0	0
H	0	0	0	0	0	0	0	2	0	0	0	1	0	0	0	0	0	0	0	1	2	3	0	0	7	0	10	10	9	21	6	8	3	0	0	0	0	0	
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	2	0	1	3	6	5	12	9	17	39	23	8	0	0	0	0	0	
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	2	0	0	1	2	5	3	5	10	10	10	18	31	19	8	2	2	1	1	0	0
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	4	0	0	0	0	0	0	0	
T	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	1	2	0	1	2	2	4	4	9	9	16	25	37	20	4	1	0	0	0	0	
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	8	9	17	19	48	38	11	4	0	0	0	0	0	
Y	0	0	0	0	0	0	0	1	0	1	0	1	1	2	2	2	5	0	2	2	14	6	8	4	8	12	13	14	10	21	19	9	4	1	0	0	2	1	0
W	0	0	0	0	0	0	0	0	0	0	1	1	1	2	0	1	3	3	0	1	1	2	3	4	1	3	9	8	5	6	5	0	1	0	0	1	0	0	
F	0	0	0	0	0	0	0	0	0	1	0	0	0	2	1	0	1	0	7	2	3	3	3	5	7	4	6	8	10	8	19	4	2	1	0	0	0	0	
M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	1	1	1	2	6	4	7	1	1	0	0	0	0	0	
I	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	0	1	1	0	3	2	1	4	6	3	5	9	8	20	8	0	1	0	0	0	0	
L	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0	2	4	1	9	3	3	3	7	17	13	17	19	9	1	0	0	0	0	0	
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	2	1	2	7	3	10	9	11	3	0	0	0	0	0	
V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	2	2	3	4	4	13	11	7	27	10	5	2	0	0	0	0	
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
G	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	1	1	1	1	0	0	0	0	0	2	0	5	5	5	13	15	9	4	1	3	0	0	0	0

Number of mutations

DDG bins

(-10.16, -9.841]
 (-9.841, -9.522]
 (-9.522, -9.204]
 (-9.204, -8.886]
 (-8.886, -8.568]
 (-8.568, -8.249]
 (-8.249, -7.931]
 (-7.931, -7.613]
 (-7.613, -7.295]
 (-7.295, -6.976]
 (-6.976, -6.658]
 (-6.658, -6.34]
 (-6.34, -6.021]
 (-6.021, -5.703]
 (-5.703, -5.385]
 (-5.385, -5.067]
 (-5.067, -4.748]
 (-4.748, -4.43]
 (-4.43, -4.112]
 (-4.112, -3.794]
 (-3.794, -3.475]
 (-3.475, -3.157]
 (-3.157, -2.839]
 (-2.839, -2.52]
 (-2.52, -2.202]
 (-2.202, -1.884]
 (-1.884, -1.566]
 (-1.566, -1.247]
 (-1.247, -0.929]
 (-0.929, -0.611]
 (-0.611, -0.292]
 (-0.292, 0.0258]
 (0.0258, 0.344]
 (0.344, 0.662]
 (0.662, 0.981]
 (0.981, 1.299]
 (1.299, 1.617]
 (1.617, 1.935]
 (1.935, 2.254]
 (2.254, 2.572]

DDG mutation distribution (fold 1)

Wild type

R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	1	2	2	2	6	10	4	7	5	21	24	18	1	0	1	0	0	0	0
K	0	0	0	0	0	0	0	0	1	0	0	2	1	0	0	1	0	0	0	2	1	3	1	1	3	4	2	7	11	14	26	11	2	0	0	1	0	0	0
E	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	1	1	0	1	2	1	2	3	5	6	9	10	36	11	4	3	0	0	0	0	0
D	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	3	2	6	6	7	10	9	10	16	8	3	1	0	0	0	0	0
H	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0	6	0	6	6	5	11	4	6	3	0	0	0	0	0
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	2	2	5	9	6	11	29	14	6	0	0	0	0	0	
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	2	0	0	0	0	2	2	3	9	5	7	8	20	12	5	1	1	0	0	0	0
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	2	0	0	0	0	0	0	0	
T	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	2	3	2	5	6	10	12	25	12	4	1	0	0	0	0	
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	6	8	11	27	20	5	1	0	0	0	0	0	
Y	0	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	3	0	1	0	8	1	4	2	3	9	10	11	5	19	14	5	2	1	0	0	1	0	0
W	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	3	2	0	0	0	0	3	2	1	1	6	3	4	5	1	0	0	0	0	1	0	0	
F	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	2	1	1	0	1	2	2	1	4	4	7	7	12	4	2	1	0	0	0	0	
M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	1	0	1	3	2	0	0	0	0	0	0	0	
I	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	1	1	3	4	1	3	5	3	11	6	0	1	0	0	0	0	
L	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1	0	0	4	2	2	0	4	11	4	7	14	5	0	0	0	0	0	0	
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	1	4	0	4	6	5	1	0	0	0	0	0	
V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	4	7	6	3	12	8	3	2	0	0	0	0	
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
G	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	4	1	7	4	2	1	0	2	0	0	0	

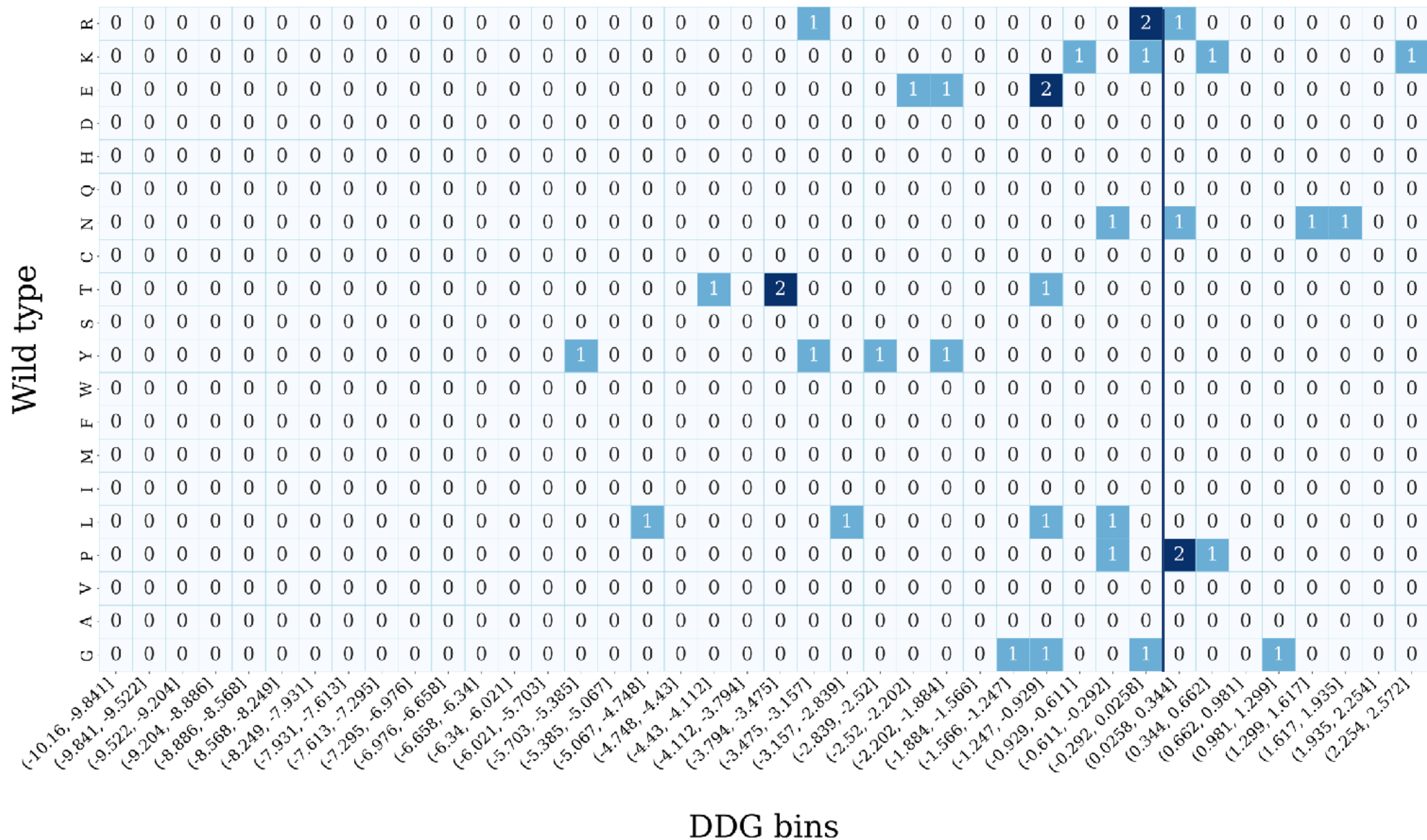
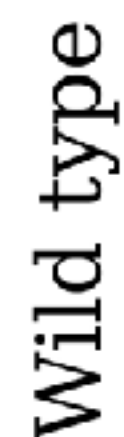
Number of mutations

DDG bins

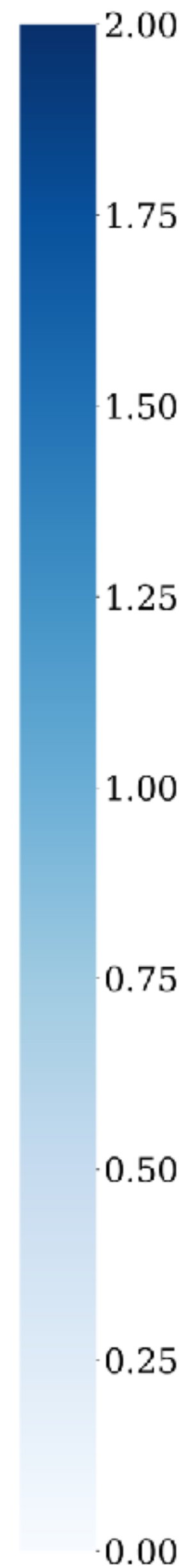
(-10.16, -9.841]
 (-9.841, -9.522]
 (-9.522, -9.204]
 (-9.204, -8.886]
 (-8.886, -8.568]
 (-8.568, -8.249]
 (-8.249, -7.931]
 (-7.931, -7.613]
 (-7.613, -7.295]
 (-7.295, -6.976]
 (-6.976, -6.658]
 (-6.658, -6.34]
 (-6.34, -6.021]
 (-6.021, -5.703]
 (-5.703, -5.385]
 (-5.385, -5.067]
 (-5.067, -4.748]
 (-4.748, -4.43]
 (-4.43, -4.112]
 (-4.112, -3.794]
 (-3.794, -3.475]
 (-3.475, -3.157]
 (-3.157, -2.839]
 (-2.839, -2.52]
 (-2.52, -2.202]
 (-2.202, -1.884]
 (-1.884, -1.566]
 (-1.566, -1.247]
 (-1.247, -0.929]
 (-0.929, -0.611]
 (-0.611, -0.292]
 (-0.292, 0.0258]
 (0.0258, 0.344]
 (0.344, 0.662]
 (0.662, 0.981]
 (0.981, 1.299]
 (1.299, 1.617]
 (1.617, 1.935]
 (1.935, 2.254]
 (2.254, 2.572]



DDG mutation distribution (fold 2)



Number of mutations

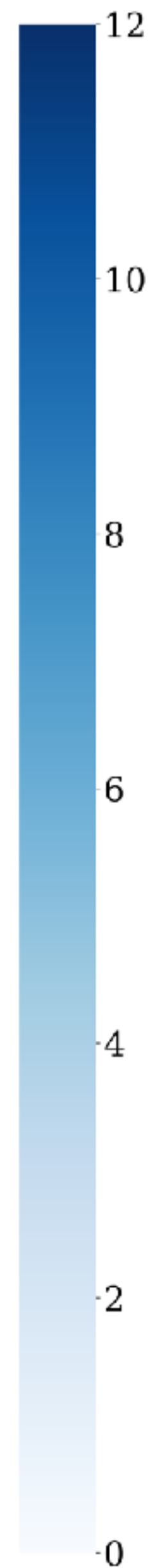


DDG mutation distribution (fold 3)

Wild type

R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	4	1	1	1	7	9	4	0	0	0	0	0	0
K	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	5	4	8	2	0	0	0	0	0	
E	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	5	4	12	7	0	1	0	0	0	0	
D	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	2	3	2	7	5	0	1	0	0	0	
H	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	2	1	3	8	0	1	0	0	0	0		
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	1	1	0	2	1	2	4	7	0	0	0	0	0		
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	0	6	4	3	0	1	1	0	0	
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0		
T	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	2	3	7	6	5	0	0	0	0		
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	3	3	11	9	0	0	0	0	0		
Y	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	0	0	2	0	3	1	0	3	3	0	0	0	0	0		
W	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0	0			
F	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	1	0	0	1	0	0	0	2	3	1	0	1	2	0	3	0	0	0	0	0		
M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	4	0	0	0	0	0		
I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	6	1	0	0	0	0		
L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	1	3	1	5	7	5	3	1	0	0	0	0		
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	4	3	3	0	0	0	0	0		
V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	3	1	7	0	1	0	0	0		
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	0	1	5	8	7	1	1	0	0	0		

Number of mutations



DDG bins

(-10.16, -9.841]
 (-9.841, -9.522]
 (-9.522, -9.204]
 (-9.204, -8.886]
 (-8.886, -8.568]
 (-8.568, -8.249]
 (-8.249, -7.931]
 (-7.931, -7.613]
 (-7.613, -7.295]
 (-7.295, -6.976]
 (-6.976, -6.658]
 (-6.658, -6.34]
 (-6.34, -6.021]
 (-6.021, -5.703]
 (-5.703, -5.385]
 (-5.385, -5.067]
 (-5.067, -4.748]
 (-4.748, -4.43]
 (-4.43, -4.112]
 (-4.112, -3.794]
 (-3.794, -3.475]
 (-3.475, -3.157]
 (-3.157, -2.839]
 (-2.839, -2.52]
 (-2.52, -2.202]
 (-2.202, -1.884]
 (-1.884, -1.566]
 (-1.566, -1.247]
 (-1.247, -0.929]
 (-0.929, -0.611]
 (-0.611, -0.292]
 (-0.292, 0.0258]
 (0.0258, 0.344]
 (0.344, 0.662]
 (0.662, 0.981]
 (0.981, 1.299]
 (1.299, 1.617]
 (1.617, 1.935]
 (1.935, 2.254]
 (2.254, 2.572]

DDG mutation distribution (fold 4)

Wild type

Wild type	R	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	3	1	0	2	2	4	2	3	0	0	1	1	0	0	0	0	0	0		
	K	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	3	2	3	1	1	3	0	1	1	0	0	0	0	0	0		
	E	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	1	4	4	2	7	4	2	1	0	0	0	0	0	0		
	D	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	3	1	0	1	2	3	1	0	0	2	3	1	0	0	0	0		
	H	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	1	1	0	0	0	0	0			
	Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	1	1	0	2	0	0	0	0	0	0	0			
	N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	2	2	0	1	0	0	0	0	0			
	C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
	T	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	1	1	3	0	1	1	2	0	0	0	0	0	0	0			
	S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	2	1	2	0	1	0	0	0	0			
	Y	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	2	2	0	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0		
	W	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	F	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	1	2	2	0	1	1	0	0	0	0	0	0		
	M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	0	0	0	0	0	0	0	0			
	I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	2	1	2	2	1	2	0	0	0	0	0	0	0		
	L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2	0	4	0	0	1	0	0	0	1	0	0	0	0	0	0	0		
	P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0		
	V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	4	1	2	5	1	0	0	0	0	0	0		
	A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0		
		(-10.16, -9.8411)	(-9.841, -9.5221)	(-9.522, -9.2041)	(-9.204, -8.8861)	(-8.886, -8.5681)	(-8.568, -8.2491)	(-8.249, -7.9311)	(-7.931, -7.6131)	(-7.613, -7.2951)	(-7.295, -6.9761)	(-6.976, -6.6581)	(-6.658, -6.341)	(-6.34, -6.0211)	(-6.021, -5.7031)	(-5.703, -5.3851)	(-5.385, -5.0671)	(-5.067, -4.7481)	(-4.748, -4.431)	(-4.43, -4.1121)	(-4.112, -3.7941)	(-3.794, -3.4751)	(-3.475, -3.1571)	(-3.157, -2.8391)	(-2.839, -2.521)	(-2.52, -2.2021)	(-2.202, -1.8841)	(-1.884, -1.5661)	(-1.566, -1.2471)	(-1.247, -0.9291)	(-0.929, -0.6111)	(-0.611, -0.2921)	(-0.292, 0.02581)	(0.0258, 0.3441)	(0.344, 0.6621)	(0.662, 0.9811)	(0.981, 1.2991)	(1.299, 1.6171)	(1.617, 1.9351)	(1.935, 2.2541)	(2.254, 2.5721)

Number of mutations

DDG bins

DDG mutation distribution (fold 5)

Wild type

R	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	2	0	1	1	2	4	6	6	4	0	0	1	0	0	0	0
K	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	3	0	4	2	0	5	10	3	0	1	0	0	0	0	0
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	0	1	2	6	1	5	2	8	4	1	0	0	0	0	1	0
D	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	3	0	0	1	1	2	0	1	3	0	5	1	4	2	0	1	2	0	0	0
H	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	1	3	0	2	1	0	0	0	0	0	0	
Q	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	4	4	2	2	0	0	0	0	0	
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	0	1	1	4	3	1	5	3	2	0	0	0	0	0	
C	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	
T	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	2	5	4	3	0	0	0	0	0	0	
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	2	4	4	8	9	5	3	0	0	0	0	0
Y	0	0	0	0	0	0	0	0	0	0	0	0	1	0	2	1	1	0	1	0	3	2	1	2	2	1	2	0	4	2	2	1	2	0	0	0	1	1	0
W	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0	1	1	2	0	2	0	2	3	4	1	1	3	0	0	0	0	0	0	0	0
F	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	2	2	2	0	2	1	0	1	1	0	3	0	0	0	0	0	0	0	0
M	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	2	0	1	1	1	0	0	0	0	0	0
I	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	2	0	0	0	0	1	0	2	2	1	1	0	0	0	0	0	0
L	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1	1	0	4	4	1	0	1	0	0	0	0	0	0	0
P	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	1	0	1	1	0	0	0	0	0
V	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	1	0	1	2	0	1	1	1	3	1	1	0	0	0	0	0	0
A	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0	0	0	0	0	0	0	0	1	0	3	1	1	0	2	0	0	0	0	0

Number of mutations

DDG bins

(-10.16, -9.841]
 (-9.841, -9.522]
 (-9.522, -9.204]
 (-9.204, -8.886]
 (-8.886, -8.568]
 (-8.568, -8.249]
 (-8.249, -7.931]
 (-7.931, -7.613]
 (-7.613, -7.295]
 (-7.295, -6.976]
 (-6.976, -6.658]
 (-6.658, -6.341]
 (-6.341, -6.021]
 (-6.021, -5.703]
 (-5.703, -5.385]
 (-5.385, -5.067]
 (-5.067, -4.748]
 (-4.748, -4.431]
 (-4.431, -4.112]
 (-4.112, -3.794]
 (-3.794, -3.475]
 (-3.475, -3.157]
 (-3.157, -2.839]
 (-2.839, -2.521]
 (-2.521, -2.202]
 (-2.202, -1.884]
 (-1.884, -1.566]
 (-1.566, -1.247]
 (-1.247, -0.929]
 (-0.929, -0.611]
 (-0.611, -0.292]
 (-0.292, 0.0258]
 (0.0258, 0.344]
 (0.344, 0.662]
 (0.662, 0.981]
 (0.981, 1.299]
 (1.299, 1.617]
 (1.617, 1.935]
 (1.935, 2.254]
 (2.254, 2.572]

